

Program overview

12-Sep-2023 11:08

Year 2023/2024
Organization Architecture and the Built Environment
Education Master Architecture, Urbanism and Building Sciences

Code	Omschrijving	ECTS	p1	p2	p3	p4	p5
Master AUBS							
variant Architecture							
AD	Architecture and Dwelling						
MSc 1 Architecture & Dwelling	MSc 1 Architecture and Dwelling						
AR1A061	Delft Lectures on Architectural Design and Research Methods	5					
AR1A066	Delft Lectures on Architectural History and Theory	5					
AR1A080	Building Engineering Studios	10					
AR1AD014	Fundamentals of Housing Design	10					
MSc 2 AD	MSc 2						
Compulsory Choice							
AR2A011	Architectural History Thesis	5					
AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5					
25 ECTS Electives							
25 ects of electives, at least 10 for an approved Design Project							
MSc 2 Elective Design Studios							
AR0139	MEGA	15					
AR0142	EXTREME technology	15					
AR0143	Sustainability project - design and elaboration	15					
AR0144	BK-Launch studio	15					
AR0149	ON SITE: Landscape architectonic explorations	15					
AR0167	Architecture and Urban Design	15					
AR0177	The Why Factory MSc2 Design Studio	15					
AR0194	Bucky Lab A	15					
AR0216	Towards an inclusive living environment	15					
AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15					
AR0227	Infrastructure and Environment Design	10					
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15					
AR0897	Van Gezel tot Meester	20					
AR2AA015	Architectural Design Studio	15					
AR2AA016	Architectural Design Studio	15					
AR2AA017	Architectural Design Studio	15					
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15					
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15					
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15					
AR2AT021	Architectural Technicities Design Studio	15					
AR2BO010	Borders and Territories International Design Studio	15					
AR2CP011	MSc2 Complex Projects Design and Research Studio	15					
AR2DC010	Architectural Design Crossovers Studio	15					
AR2FO010	The Delta Shelter	15					
AR2FST010	Studio 'High-Rise Culture'	15					
AR2HA020 (for A and U	Urban Archipelago	15					
AR keuzevakken 2023							
AR Electives 2023							
AR0049	Urban Geography	5					
AR0051	Online Digital Portfolio	6					
AR0089 (for U, LA, MBE	Design in Process, Process in Design	15					
AR0095	Social Inequality in the City, Diversity and Design	5					
AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5					
AR0106	Architectural Ethnography	5					
AR0107	Housing Studies: An open intersectional Archive	5					
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5					
AR0109	City of Innovations Project	5					
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5					
AR0113	Tools of the Architect	5					

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
AR0117	Didactic coaching skills for architecture and the built environment	5	
AR0118	Experiments in Drawing Theory	5	
AR0119	Figures	5	
AR0121	Analytical Models	5	
AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
AR0126	Bridge Design	5	
AR0131	Entrepreneurship in Architecture and the Built Environment	5	
AR0132	Zero-Energy Design	5	
AR0133	Technoledge Glass Structures	5	
AR0136	Making	5	
AR0139	MEGA	15	
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
AR0142	EXTREME technology	15	
AR0143	Sustainability project - design and elaboration	15	
AR0144	BK-Launch studio	15	
AR0145	Circular Product Design	5	
AR0149	ON SITE: Landscape architectonic explorations	15	
AR0167	Architecture and Urban Design	15	
AR0168	People, Movement and Public Space	5	
AR0169	Materialisation: The Future Envelope	5	
AR0171	Geo-design for a Circular Economy in Urban Region	5	
AR0172	Globalisation - Research on the Urban Impact	5	
AR0173	Globalisation Free Choice	10	
AR0175	Campus Utopias	5	
AR0177	The Why Factory MSc2 Design Studio	15	
AR0179	Value Capturing	5	
AR0185	Research Methods 3	5	
AR0187	Transition Landscapes	5	
AR0194	Bucky Lab A	15	
AR0195	Urbanism Elective Studio	10	
AR0196	Urbanism Elective Seminar	5	
AR0202	Computational Intelligence for Integrated Design	5	
AR0203	Eco-friendly Material Choices	5	
AR0215	Form & Inspiration	5	
AR0216	Towards an inclusive living environment	15	
AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
AR0227	Infrastructure and Environment Design	10	
AR0228	Infrastructure and Environment Method Module	5	
AR0570	Worlding Theories and Concepts	5	
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
AR0771	Beyond 3D Computer Visualisation	6	
AR0796	Ornamatics	5	
AR0805	An Archeology of Digital Design	5	
AR0815	Idiosyncratic Infrastructures II	5	
AR0825	Building Stories: The Heteronomy of Urban Design	5	
AR0835	Social Sustainability in Human Habitats	5	
AR0897	Van Gezel tot Meester	20	
AR2AA010	Architectural Research and Design Seminar	5	
AR2AA015	Architectural Design Studio	15	
AR2AA016	Architectural Design Studio	15	
AR2AA017	Architectural Design Studio	15	
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
AR2AP041	Public Building Seminar Composition and Perception	5	
AR2AT021	Architectural Technicities Design Studio	15	
AR2AT041	Architecture and Philosophy Lecture Seminar	5	
AR2BO010	Borders and Territories International Design Studio	15	
AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
AR2DC010	Architectural Design Crossovers Studio	15	
AR2FO010	The Delta Shelter	15	
AR2FST010	Studio 'High-Rise Culture'	15	
AR2HA011	Building Green: Past, Present, Future	5	
AR2HA020 (for A and U	Urban Archipelago	15	
AR2MET011	Designing with Others	15	
AR2MM001	Building Ideologies Freedom	15	
AR2UA010	The Living City	5	
AR2UA020	Urban Architecture MSc2 design studio	15	
AR8003TU	Legal and Governance	5	
ARB106	The Berlage Sessions	3	
ARB206	The Berlage Sessions	3	

students)

	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 Architecture and Dwelling				
Compulsory Choice MSc 3 and 4				
Advanced Housing Design				
	AR3A010	Research Plan	5	
	AR3AD100	Advanced Housing Design	55	
Global Housing				
	AR3A010	Research Plan	5	
	AR3AD105	Dwelling Graduation Studio: Global Housing	55	
Designing for Care				
Designing for Care in an Inclusive Environment				
	AR3A010	Research Plan	5	
	AR3AD110	Dwelling Graduation Studio: Designing for Care in an Inclusive Environment	55	
FSA				
MSc 1 FSA				
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	
	AR1A080	Building Engineering Studios	10	
	AR1FO010	Form, Structure and Aesthetics	10	
MSc 2 FSA				
MSc 2				
Compulsory Choice				
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
25 ECTS Electives				
25 ects of electives, at least 10 for an approved Design Project				
MSc 2 Elective Design Studios				
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	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0194	Bucky Lab A	15	
	AR0216	Towards an inclusive living environment	15	
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	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA020 (for A and U	Urban Archipelago	15	
students)	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA020	Urban Architecture MSc2 design studio	15	
AR keuzevakken 2023				
AR Electives 2023				
	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
students only)	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
			5	

AR0109	City of Innovations Project		
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
AR0113	Tools of the Architect	5	
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
AR0117	Didactic coaching skills for architecture and the built environment	5	
AR0118	Experiments in Drawing Theory	5	
AR0119	Figures	5	
AR0121	Analytical Models	5	
AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
AR0126	Bridge Design	5	
AR0131	Entrepreneurship in Architecture and the Built Environment	5	
AR0132	Zero-Energy Design	5	
AR0133	Technoledge Glass Structures	5	
AR0136	Making	5	
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AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
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AR0143	Sustainability project - design and elaboration	15	
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AR0149	ON SITE: Landscape architectonic explorations	15	
AR0167	Architecture and Urban Design	15	
AR0168	People, Movement and Public Space	5	
AR0169	Materialisation: The Future Envelope	5	
AR0171	Geo-design for a Circular Economy in Urban Region	5	
AR0172	Globalisation - Research on the Urban Impact	5	
AR0173	Globalisation Free Choice	10	
AR0175	Campus Utopias	5	
AR0177	The Why Factory MSc2 Design Studio	15	
AR0179	Value Capturing	5	
AR0185	Research Methods 3	5	
AR0187	Transition Landscapes	5	
AR0194	Bucky Lab A	15	
AR0195	Urbanism Elective Studio	10	
AR0196	Urbanism Elective Seminar	5	
AR0202	Computational Intelligence for Integrated Design	5	
AR0203	Eco-friendly Material Choices	5	
AR0215	Form & Inspiration	5	
AR0216	Towards an inclusive living environment	15	
AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
AR0227	Infrastructure and Environment Design	10	
AR0228	Infrastructure and Environment Method Module	5	
AR0570	Worlding Theories and Concepts	5	
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
AR0771	Beyond 3D Computer Visualisation	6	
AR0796	Ornamatics	5	
AR0805	An Archeology of Digital Design	5	
AR0815	Idiosyncratic Infrastructures II	5	
AR0825	Building Stories: The Heteronomy of Urban Design	5	
AR0835	Social Sustainability in Human Habitats	5	
AR0897	Van Gezel tot Meester	20	
AR2AA010	Architectural Research and Design Seminar	5	
AR2AA015	Architectural Design Studio	15	
AR2AA016	Architectural Design Studio	15	
AR2AA017	Architectural Design Studio	15	
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
AR2AP041	Public Building Seminar Composition and Perception	5	
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AR2HA011	Building Green: Past, Present, Future	5	
AR2HA020 (for A and U	Urban Archipelago	15	
AR2MET011	Designing with Others	15	
AR2MM001	Building Ideologies Freedom	15	
AR2UA010	The Living City	5	

students)

	AR2UA020	Urban Architecture MSc2 design studio			
	AR8003TU	Legal and Governance	5		
	ARB106	The Berlage Sessions	3		
	ARB206	The Berlage Sessions	3		
	GEO1000	Python Programming for Geomatics	5		
	GEO1002	Geographical Information Systems (GIS) and Cartography	5		
	GEO5012	Land Administration	5		
	GEO5014	Geomatics as support for energy applications	5		
	GEO5015	Modelling wind and dispersion in urban environments	5		
	GEO5017	Machine Learning for the Built Environment	5		
	GEO5018	Ethics for the Data-driven City	5		
	MSc 3 and 4 Form, Structure and Aesthetics				
	AR3A010	Research Plan	5		
	AR3TA001	Technologies and Aesthetics	55		
	ADC	Architectural Design Crossovers			
	MSc1 ADC	MSc 1 Architectural Design Crossovers			
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5		
	AR1A066	Delft Lectures on Architectural History and Theory	5		
	AR1A080	Building Engineering Studios	10		
	AR1DC010	Architectural Design Crossovers Studio	10		
	MSc 2 ADC	MSc 2			
	Compulsory Choice				
	AR2A011	Architectural History Thesis	5		
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5		
	25 ECTS Electives	25 ects of electives, at least 10 for an approved Design Project			
	MSc 2 Elective Design Studios				
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	AR0177	The Why Factory MSc2 Design Studio	15		
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	AR2FST010	Studio 'High-Rise Culture'	15		
	AR2HA020 (for A and U	Urban Archipelago	15		
students)	AR2MET011	Designing with Others	15		
	AR2MM001	Building Ideologies Freedom	15		
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	AR keuzevakken 2023				
	AR0049	Urban Geography	5		
	AR0051	Online Digital Portfolio	6		
	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15		
students only)	AR0095	Social Inequality in the City, Diversity and Design	5		
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5		
	AR0106	Architectural Ethnography	5		
	AR0107	Housing Studies: An open intersectional Archive	5		
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	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
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	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	5	
	AR0195	Urbanism Elective Studio	15	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
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	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
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	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
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	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
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	AR2BO010	Borders and Territories International Design Studio	15	
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students)	AR2MET011	Designing with Others	15	
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	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	

	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
	MSc 3 and 4 ADC	MSc 3 and 4 Architectural Design Crossovers		
	AR3A010	Research Plan	5	
	AR3DC100	Architectural Design Crossovers Graduation Studio	55	
	AE	Architectural Engineering		
	MSc 1 AE	MSc 1 Architectural Engineering		
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
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	AR1A080	Building Engineering Studios	10	
	AR1AE011	EXTREME architecture	10	
	MSc2 AE	MSc 2		
	Compulsory Choice			
	AR2A011	Architectural History Thesis	5	
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	25 ECTS Electives	25 ects of electives, at least 10 for an approved MSc 2 Architecture design project		
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	AR0139	MEGA	15	
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	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	

	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
students)	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 AE		MSc 3 and 4 Architectural Engineering		

	AR3A010	Research Plan	5	
	AR3AE100	Architectural Engineering Graduation Studio	55	
A&PB		Architecture and Public Building		
MSc 1 Architecture and Public Building				
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	
	AR1A080	Building Engineering Studios	10	
	AR1AP012	Public Building Design Studio	10	
MSc 2 AP		MSc 2		
Compulsory Choice				
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
25 ECTS Electives		25 ects of electives, at least 10 for an approved Design Project		
MSc 2 Elective Design Studios				
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0194	Bucky Lab A	15	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0897	Van Gezel tot Meester	20	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA020 (for A and U	Urban Archipelago	15	
students)	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA020	Urban Architecture MSc2 design studio	15	
AR keuzevakken 2023		AR Electives 2023		
	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
students only)	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
			15	

	AR0143	Sustainability project - design and elaboration		
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
students)	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 Architecture and Public Building				
	AR3A010	Research Plan	5	
	AR3AP100	Public Building Graduation Studio	55	
CP	Complex projects			
MSc 1 CP	MSc 1 Complex projects			
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	

	AR1A080	Building Engineering Studios	10	
	AR1CP011	Complex Projects Design Studio	10	
	MSc 2 CP	MSc 2		
	Compulsory Choice			
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
	25 ECTS Electives	25 ects of electives, at least 10 for an approved Design Project		
	MSc 2 Elective Design Studios			
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0194	Bucky Lab A	15	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0897	Van Gezel tot Meester	20	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
students)	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR keuzevakken 2023	AR Electives 2023		
	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
students only)	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	

students)	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 CP		MSc 3 and 4 Complex projects		
AR3A010		Research Plan	5	
AR3CP100		Complex Projects Graduation Studio	55	
MAI		Methods of Analysis and Imagination		
MSc 1 MAI		MSc 1 Methods of Analysis and Imagination		
AR1A061		Delft Lectures on Architectural Design and Research Methods	5	
AR1A066		Delft Lectures on Architectural History and Theory	5	
AR1A080		Building Engineering Studios	10	
AR1MET015		Situating Architecture	10	
MSc 2 MET		MSc 2		
Compulsory Choice				
AR2A011		Architectural History Thesis	5	
AR2AT031		Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	

25 ECTS Electives		25 ects of electives, at least 10 for an approved Design Project		
MSc 2 Elective Design Studios				
students)	AR0139	MEGA	15	</

students)	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 MA		MSc 3 and 4 Methods and Analysis		
	AR3A010	Research Plan	5	
	AR3MET105	A Matter of Scale	55	
HA		Heritage and Architecture		
MSc 1 HA		MSc 1 Heritage and Architecture		
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	
	AR1A080	Building Engineering Studios	10	
	AR1AH010	Heritage and Architecture Design Studio: Architectonic Design	10	
MSc 2 AH (nw)		MSc 2		
Compulsory Choice				
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
25 ECTS Electives		25 ects of electives, at least 10 for an approved Design Project		
MSc 2 Elective Design Studios				
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	

students)	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0194	Bucky Lab A	15	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0897	Van Gezel tot Meester	20	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AT021	Architectural Technicians Design Studio	15	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
students)	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
AR keuzevakken 2023 AR Electives 2023				
students only)	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	

students)	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 HA		MSc 3 and 4 Heritage and Architecture		
MSc3 Adapting 20th century Heritage				
	AR3A010	Research Plan	5	
	AR3AH105	Graduation Studio Adapting 20th century Heritage	55	
MSc3 Revitalising Heritage				
	AR3A010	Research Plan	5	
	AR3AH115	Graduation Studio Revitalising Heritage	55	
Interiors Buildings Cities				
MSc 1 AI		MSc 1 Interiors Buildings Cities		
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	
	AR1A080	Building Engineering Studios	10	
	AR1AI013	Interiors Buildings Cities MSc1 Design Project	10	
MSc 2 AI		MSc 2		
Compulsory Choice				
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
25 ECTS Electives		25 ects of electives, at least 10 for an approved Design Project		
MSc 2 Elective Design Studios				
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	

students)	AR0194	Bucky Lab A		
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0897	Van Gezel tot Meester	20	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA020 (for A and U)	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA020	Urban Architecture MSc2 design studio	15	
AR keuzevakken 2023 AR Electives 2023				
students only)	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE)	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	

students)	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 AI		MSc 3 and 4 Interiors Buildings Cities		
	AR3A010	Research Plan	5	
	AR3AI100	Interiors Buildings Cities Graduation Project	55	
The Why Factory				
MSc 1 TWF		MSc 1 The Why Factory		
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	
	AR1A081	Building Engineering Studios	10	
	AR1TWF011	The Why Factory MSc1 Design Studio	10	
MSc 2 TWF (nw)		MSc 2		
Compulsory Choice				
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
25 ECTS Electives		25 ects of electives, at least 10 for an approved Design Project		
MSc 2 Elective Design Studios				
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0194	Bucky Lab A	15	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0897	Van Gezel tot Meester	20	

students)	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
students only)	AR2MM001	Building Ideologies Freedom	15	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR keuzevakken 2023			
	AR Electives 2023			
	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	

students)	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
BO		Borders and Territories		
MSc 1 BO		MSc 1 Borders and Territories		
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	
	AR1A080	Building Engineering Studios	10	
	AR1BO010	Borders and Territories Design Studio	10	
MSc 2 BO		MSc 2		
Compulsory Choice				
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
25 ECTS Electives		25 ects of electives, at least 10 for an approved Design Project		
MSc 2 Elective Design Studios				
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0194	Bucky Lab A	15	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0897	Van Gezel tot Meester	20	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	

students)	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA020	Urban Architecture MSc2 design studio	15	
AR keuzevakken 2023 AR Electives 2023				
	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
students only)	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	

students)	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
MSc 3 and 4 BO				
MSc 3 and 4 Borders and Territories				
	AR3A010	Research Plan	5	
	AR3BO100	Borders and Territories Graduation Studio	55	
Urban Architecture				
MSc 1 UA		MSc 1 Urban Architecture		
	AR1A061	Delft Lectures on Architectural Design and Research Methods	5	
	AR1A066	Delft Lectures on Architectural History and Theory	5	
	AR1A080	Building Engineering Studios	10	
	AR1UA010	Urban Architecture MSc 1 Design studio	10	
MSc 2 UA		MSc 2		
Compulsory Choice				
	AR2A011	Architectural History Thesis	5	
	AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5	
25 ECTS Electives		25 ects of electives, at least 10 for an approved Design Project		
MSc 2 Elective Design Studios				
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0194	Bucky Lab A	15	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0897	Van Gezel tot Meester	20	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA020 (for A and U	Urban Archipelago	15	
students)	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR keuzevakken 2023			
	AR0049	Urban Geography	5	
AR Electives 2023				

students only)	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
	AR2AA017	Architectural Design Studio	15	
	AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
	AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
	AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
	AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
	AR2AP041	Public Building Seminar Composition and Perception	5	
	AR2AT021	Architectural Technicities Design Studio	15	
	AR2AT041	Architecture and Philosophy Lecture Seminar	5	
	AR2BO010	Borders and Territories International Design Studio	15	
	AR2CP011	MSc2 Complex Projects Design and Research Studio	15	

students)	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 and 4 UA		MSc 3 and 4 Urban Architecture		
	AR3A010	Research Plan	5	
	AR3UA100	Urban Architecture Graduation Studio	55	
Explore Lab				
MSc 3 and 4 Expl Lab		MSc 3 and 4 Explore Lab, compulsory - for A students		
	AR3A010	Research Plan	5	
	AR3EX115	Explore Lab Graduation	55	
Cross Domain City of the Future				
MSc 3 and 4 CS		MSc 3 and 4 Cross Domain City of the Future - for A students		
	AR3A010	Research Plan	5	
	AR3CS100	Graduation Studio Cross Domain City of the Future	55	
Resilient Rotterdam Graduation Studio - Veldacademie				
MSc 3 and 4 Resilient Rotterdam Studio				
	AR3A010	Research Plan	5	
	AR3RE100	Resilient Rotterdam Graduation Studio - Veldacademie	55	
variant Building Technology				
MSc 1 Building Technology				
	AR1B011	Bucky Lab Design	10	
	AR1B022	Climate Design	5	
	AR1B023	Sustainable Architectural Materials and Structures	5	
	AR1B024	Introduction to Computational Design	5	
	AR1B032	Research and Methodology	5	
MSc 2 Building Technology				
Compulsory Choice		Compulsory Choice (choose 2 courses)		
	AR0126	Bridge Design	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0134	Technoledge Façade Design	5	
	AR0137	Technoledge Health and Comfort	5	
	AR0138	Technoledge Design Informatics	5	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0145	Circular Product Design	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
Compulsory Choice		Compulsory Choice (choose 1 project)		
	AR0139	MEGA	15	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
Free Electives (5 EC)				
AR keuzevakken 2023		AR Electives 2023		
students only)	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	

AR0113	Tools of the Architect	5	
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
AR0117	Didactic coaching skills for architecture and the built environment	5	
AR0118	Experiments in Drawing Theory	5	
AR0119	Figures	5	
AR0121	Analytical Models	5	
AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
AR0126	Bridge Design	5	
AR0131	Entrepreneurship in Architecture and the Built Environment	5	
AR0132	Zero-Energy Design	5	
AR0133	Technoledge Glass Structures	5	
AR0136	Making	5	
AR0139	MEGA	15	
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
AR0142	EXTREME technology	15	
AR0143	Sustainability project - design and elaboration	15	
AR0144	BK-Launch studio	15	
AR0145	Circular Product Design	5	
AR0149	ON SITE: Landscape architectonic explorations	15	
AR0167	Architecture and Urban Design	15	
AR0168	People, Movement and Public Space	5	
AR0169	Materialisation: The Future Envelope	5	
AR0171	Geo-design for a Circular Economy in Urban Region	5	
AR0172	Globalisation - Research on the Urban Impact	5	
AR0173	Globalisation Free Choice	10	
AR0175	Campus Utopias	5	
AR0177	The Why Factory MSc2 Design Studio	15	
AR0179	Value Capturing	5	
AR0185	Research Methods 3	5	
AR0187	Transition Landscapes	5	
AR0194	Bucky Lab A	15	
AR0195	Urbanism Elective Studio	10	
AR0196	Urbanism Elective Seminar	5	
AR0202	Computational Intelligence for Integrated Design	5	
AR0203	Eco-friendly Material Choices	5	
AR0215	Form & Inspiration	5	
AR0216	Towards an inclusive living environment	15	
AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
AR0227	Infrastructure and Environment Design	10	
AR0228	Infrastructure and Environment Method Module	5	
AR0570	Worlding Theories and Concepts	5	
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
AR0771	Beyond 3D Computer Visualisation	6	
AR0796	Ornamatics	5	
AR0805	An Archeology of Digital Design	5	
AR0815	Idiosyncratic Infrastructures II	5	
AR0825	Building Stories: The Heteronomy of Urban Design	5	
AR0835	Social Sustainability in Human Habitats	5	
AR0897	Van Gezel tot Meester	20	
AR2AA010	Architectural Research and Design Seminar	5	
AR2AA015	Architectural Design Studio	15	
AR2AA016	Architectural Design Studio	15	
AR2AA017	Architectural Design Studio	15	
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
AR2AP041	Public Building Seminar Composition and Perception	5	
AR2AT021	Architectural Techniques Design Studio	15	
AR2AT041	Architecture and Philosophy Lecture Seminar	5	
AR2BO010	Borders and Territories International Design Studio	15	
AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
AR2DC010	Architectural Design Crossovers Studio	15	
AR2FO010	The Delta Shelter	15	
AR2FST010	Studio 'High-Rise Culture'	15	
AR2HA011	Building Green: Past, Present, Future	5	
AR2HA020 (for A and U	Urban Archipelago	15	
AR2MET011	Designing with Others	15	
AR2MM001	Building Ideologies Freedom	15	
AR2UA010	The Living City	5	
AR2UA020	Urban Architecture MSc2 design studio	15	
AR8003TU	Legal and Governance	5	
ARB106	The Berlage Sessions	3	

students)

	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 Building Technology				
	AR3B025	Building Technology Graduation Studio	15	
Compulsory Choice (choose 1 project)				
	AR3B012	CORE	15	
	AR3B015	User-centred Sustainability Studio	15	
MSc 4 Building Technology				
	AR4B025	Building Technology Graduation Studio	30	
variant Management in the Built Environment				
MSc 1 Management in the Built Environment				
	AR1MBE015	Research Methods 1	5	
	AR1MBE020	Design and Construction Management	10	
	AR1MBE025	Building Economics	5	
	AR1MBE030	Real Estate Management	10	
MSc 2 Management in the Built Environment				
	AR2MBE011	Building Law	5	
	AR2MBE015	Redesign of Complex Projects	10	
	AR2MBE021	Building Information Management	5	
	AR2MBE025	Urban and Infrastructure (Re)development Game	10	
MSc 3 Management in the Built Environment				
	AR3MBE010	Research Methods 2	5	
	AR3MBE100	MSc 3 Graduation Laboratory Management in the Built Environment	10	
Free Electives 15 ECTS				
AR keuzevakken 2023				
students only)	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE)	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	
	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	

AR0196	Urbanism Elective Seminar	5	
AR0215	Form & Inspiration	5	
AR0216	Towards an inclusive living environment	15	
AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
AR0227	Infrastructure and Environment Design	10	
AR0228	Infrastructure and Environment Method Module	5	
AR0570	Worlding Theories and Concepts	5	
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
AR0771	Beyond 3D Computer Visualisation	6	
AR0796	Ornamatics	5	
AR0897	Van Gezel tot Meester	20	
AR2AA010	Architectural Research and Design Seminar	5	
AR2AA015	Architectural Design Studio	15	
AR2AA016	Architectural Design Studio	15	
AR2AA017	Architectural Design Studio	15	
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
AR2AP041	Public Building Seminar Composition and Perception	5	
AR2AT021	Architectural Technicities Design Studio	15	
AR2AT041	Architecture and Philosophy Lecture Seminar	5	
AR2BO010	Borders and Territories International Design Studio	15	
AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
AR2DC010	Architectural Design Crossovers Studio	15	
AR2FO010	The Delta Shelter	15	
AR2FST010	Studio 'High-Rise Culture'	15	
AR2HA011	Building Green: Past, Present, Future	5	
AR2MET011	Designing with Others	15	
AR2UA010	The Living City	5	
AR2UA020	Urban Architecture MSc2 design studio	15	
AR8003TU	Legal and Governance	5	
ARB106	The Berlage Sessions	3	
ARB206	The Berlage Sessions	3	
GEO1000	Python Programming for Geomatics	5	
GEO1002	Geographical Information Systems (GIS) and Cartography	5	
GEO5014	Geomatics as support for energy applications	5	
GEO5015	Modelling wind and dispersion in urban environments	5	
GEO5018	Ethics for the Data-driven City	5	
MSc 4 Management in the Built Environment			
AR4R010	MSc 4 Graduation Laboratory Management in the Built Environment	30	
Cross Domain City of the Future			
MSc 3 Cross Domain City of the Future	MSc 3 Cross Domain City of the Future - for MBE students		
Compulsory for MBE students			
Compulsory			
AR3CS021	Seminar Cross Domain City of the Future	5	
AR3CS081	Graduation Cross Domain City of the Future	20	
Free electives/Research Methods 2	Free electives/Research Methods 2 (5 EC)		
MSc 4 Cross Domain City of the Future	MSc 4 Cross Domain City of the Future - for MBE students		
For MBE Students			
AR4CS030	Cross Domain City of the Future	30	
variant Landscape Architecture			
MSc 1 Landscape Architecture			
AR1LA011	Architecture and Landscape: Design Studio	10	
AR1LA021	Architecture and Landscape: Theory, Method and Critical Thinking	5	
AR1LA051	Dutch Landscape: Design Studio	10	
AR1LA061	Dutch Landscape: Theory, Method and Critical Thinking	5	
MSc 2 Landscape Architecture			
AR2LA011	Urban Landscape: Design Studio	10	
AR2LA021	Urban Landscape: Theory, Method and Critical Thinking	5	
15 EC of electives			
15 EC of electives, at least 10 EC for a design project. See AR Electives 2022			
AR keuzevakken 2023			
AR0049	Urban Geography	5	
AR0051	Online Digital Portfolio	6	
AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15	
AR0095	Social Inequality in the City, Diversity and Design	5	
AR0097	Climate proof sustainable renovation: energy use, envir impact, health	5	

	and comfort, life-cycle cost		
AR0106	Architectural Ethnography	5	
AR0107	Housing Studies: An open intersectional Archive	5	
AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
AR0109	City of Innovations Project	5	
AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
AR0113	Tools of the Architect	5	
AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
AR0117	Didactic coaching skills for architecture and the built environment	5	
AR0118	Experiments in Drawing Theory	5	
AR0119	Figures	5	
AR0121	Analytical Models	5	
AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
AR0126	Bridge Design	5	
AR0131	Entrepreneurship in Architecture and the Built Environment	5	
AR0132	Zero-Energy Design	5	
AR0133	Technoledge Glass Structures	5	
AR0136	Making	5	
AR0139	MEGA	15	
AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
AR0142	EXTREME technology	15	
AR0143	Sustainability project - design and elaboration	15	
AR0144	BK-Launch studio	15	
AR0145	Circular Product Design	5	
AR0149	ON SITE: Landscape architectonic explorations	15	
AR0167	Architecture and Urban Design	15	
AR0168	People, Movement and Public Space	5	
AR0169	Materialisation: The Future Envelope	5	
AR0171	Geo-design for a Circular Economy in Urban Region	5	
AR0172	Globalisation - Research on the Urban Impact	5	
AR0173	Globalisation Free Choice	10	
AR0175	Campus Utopias	5	
AR0177	The Why Factory MSc2 Design Studio	15	
AR0179	Value Capturing	5	
AR0185	Research Methods 3	5	
AR0187	Transition Landscapes	5	
AR0194	Bucky Lab A	15	
AR0195	Urbanism Elective Studio	10	
AR0196	Urbanism Elective Seminar	5	
AR0202	Computational Intelligence for Integrated Design	5	
AR0203	Eco-friendly Material Choices	5	
AR0215	Form & Inspiration	5	
AR0216	Towards an inclusive living environment	15	
AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
AR0227	Infrastructure and Environment Design	10	
AR0228	Infrastructure and Environment Method Module	5	
AR0570	Worlding Theories and Concepts	5	
AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
AR0771	Beyond 3D Computer Visualisation	6	
AR0796	Ornamatics	5	
AR0805	An Archeology of Digital Design	5	
AR0815	Idiosyncratic Infrastructures II	5	
AR0825	Building Stories: The Heteronomy of Urban Design	5	
AR0835	Social Sustainability in Human Habitats	5	
AR0897	Van Gezel tot Meester	20	
AR2AA010	Architectural Research and Design Seminar	5	
AR2AA015	Architectural Design Studio	15	
AR2AA016	Architectural Design Studio	15	
AR2AA017	Architectural Design Studio	15	
AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15	
AR2AI011	Interiors Buildings Cities MSc2 Design Project	15	
AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15	
AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5	
AR2AP041	Public Building Seminar Composition and Perception	5	
AR2AT021	Architectural Technicities Design Studio	15	
AR2AT041	Architecture and Philosophy Lecture Seminar	5	
AR2BO010	Borders and Territories International Design Studio	15	
AR2CP011	MSc2 Complex Projects Design and Research Studio	15	
AR2DC010	Architectural Design Crossovers Studio	15	
AR2FO010	The Delta Shelter	15	
AR2FST010	Studio 'High-Rise Culture'	15	
AR2HA011	Building Green: Past, Present, Future	5	

students)	AR2HA020 (for A and U	Urban Archipelago		
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
students only)	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
	MSc 3 Landscape Architecture			
	AR3LA011	Landscape Architecture Analysis and Visualisation	5	
	AR3LA020	Research Methodology in Landscape Architecture	5	
	AR3LA031	Graduation Studio Landscape Architecture: Flowscales	20	
	MSc 4 Landscape Architecture			
	AR4LA010	Graduation Studio Landscape Architecture: Flowscales	30	
	variant Urbanism			
	MSc 1 Urbanism			
	AR1U090	RandD Studio: Analysis and Design of Urban Form	10	
	AR1U100	RandD Studio: Designing Urban Environments	10	
	AR1U121	History and Theory of Urbanism	5	
	AR1U131	Sustainable Urban Engineering of Territory	5	
	MSc 2 Urbanism			
	AR2U086	RandD Studio: Spatial Strategies for the Global Metropolis	10	
	AR2U088	Research and Design Methodology for Urbanism	5	
	Urbanism, Free electives 15 ECTS			
	AR keuzevakken 2023			
	AR0049	Urban Geography	5	
	AR0051	Online Digital Portfolio	6	
	AR0089 (for U, LA, MBE	Design in Process, Process in Design	15	
	AR0095	Social Inequality in the City, Diversity and Design	5	
	AR0097	Climate proof sustainable renovation: energy use, enviro impact, health and comfort, life-cycle cost	5	
	AR0106	Architectural Ethnography	5	
	AR0107	Housing Studies: An open intersectional Archive	5	
	AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5	
	AR0109	City of Innovations Project	5	
	AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5	
	AR0113	Tools of the Architect	5	
	AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5	
	AR0117	Didactic coaching skills for architecture and the built environment	5	
	AR0118	Experiments in Drawing Theory	5	
	AR0119	Figures	5	
	AR0121	Analytical Models	5	
	AR0122	1:1 Interactive Architecture Prototypes Workshop	5	
	AR0126	Bridge Design	5	
	AR0131	Entrepreneurship in Architecture and the Built Environment	5	
	AR0132	Zero-Energy Design	5	
	AR0133	Technoledge Glass Structures	5	
	AR0136	Making	5	
	AR0139	MEGA	15	
	AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5	
	AR0142	EXTREME technology	15	
	AR0143	Sustainability project - design and elaboration	15	
	AR0144	BK-Launch studio	15	
	AR0145	Circular Product Design	5	
	AR0149	ON SITE: Landscape architectonic explorations	15	
	AR0167	Architecture and Urban Design	15	
	AR0168	People, Movement and Public Space	5	
	AR0169	Materialisation: The Future Envelope	5	
	AR0171	Geo-design for a Circular Economy in Urban Region	5	
	AR0172	Globalisation - Research on the Urban Impact	5	
	AR0173	Globalisation Free Choice	10	
	AR0175	Campus Utopias	5	
	AR0177	The Why Factory MSc2 Design Studio	15	
	AR0179	Value Capturing	5	
	AR0185	Research Methods 3	5	

	AR0187	Transition Landscapes	5	
	AR0194	Bucky Lab A	15	
	AR0195	Urbanism Elective Studio	10	
	AR0196	Urbanism Elective Seminar	5	
	AR0202	Computational Intelligence for Integrated Design	5	
	AR0203	Eco-friendly Material Choices	5	
	AR0215	Form & Inspiration	5	
	AR0216	Towards an inclusive living environment	15	
	AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15	
	AR0227	Infrastructure and Environment Design	10	
	AR0228	Infrastructure and Environment Method Module	5	
	AR0570	Worlding Theories and Concepts	5	
	AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15	
	AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15	
	AR0771	Beyond 3D Computer Visualisation	6	
	AR0796	Ornamatics	5	
	AR0805	An Archeology of Digital Design	5	
	AR0815	Idiosyncratic Infrastructures II	5	
	AR0825	Building Stories: The Heteronomy of Urban Design	5	
	AR0835	Social Sustainability in Human Habitats	5	
	AR0897	Van Gezel tot Meester	20	
	AR2AA010	Architectural Research and Design Seminar	5	
	AR2AA015	Architectural Design Studio	15	
	AR2AA016	Architectural Design Studio	15	
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	AR2DC010	Architectural Design Crossovers Studio	15	
	AR2FO010	The Delta Shelter	15	
	AR2FST010	Studio 'High-Rise Culture'	15	
	AR2HA011	Building Green: Past, Present, Future	5	
students)	AR2HA020 (for A and U	Urban Archipelago	15	
	AR2MET011	Designing with Others	15	
	AR2MM001	Building Ideologies Freedom	15	
	AR2UA010	The Living City	5	
	AR2UA020	Urban Architecture MSc2 design studio	15	
	AR8003TU	Legal and Governance	5	
	ARB106	The Berlage Sessions	3	
	ARB206	The Berlage Sessions	3	
	GEO1000	Python Programming for Geomatics	5	
	GEO1002	Geographical Information Systems (GIS) and Cartography	5	
	GEO5012	Land Administration	5	
	GEO5014	Geomatics as support for energy applications	5	
	GEO5015	Modelling wind and dispersion in urban environments	5	
	GEO5017	Machine Learning for the Built Environment	5	
	GEO5018	Ethics for the Data-driven City	5	
MSc 3 Urbanism				
	AR3U105	Graduation Orientation	3	
	AR3U110	Graduation Exploration	12	
	AR3U115	Graduation Lab Urbanism	15	
MSc 4 Urbanism				
	AR4U010	Graduation Lab Urbanism	30	
Cross Domain City of the Future				
MSc 3 Cross Domain City of the Future		MSc 3 Cross Domain City of the Future - for U students		
	AR3U105	Graduation Orientation	3	
	AR3U110	Graduation Exploration	12	
	AR3U115	Graduation Lab Urbanism	15	
MSc 4 Cross Domain City of the Future		MSc 4 Cross Domain City of the Future- for U students		
	AR4U010	Graduation Lab Urbanism	30	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Master AUBS

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

variant Architecture

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AD

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 Architecture & Dwelling

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1AD014	Fundamentals of Housing Design	10
Course Coordinator	Ir. P.S. van der Putt	
Course Coordinator	Ir. H.A.F. Mooij	
Instructor	Ir. H.A.F. Mooij	
Instructor	T.W. Kupers	
Instructor	Ir. P.S. van der Putt	
Responsible for assignments	Ir. P.S. van der Putt	
Contact Hours / Week x/x/x/x	Variable 4 or 8 h per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Architecture & Dwelling MSc1 studio focusses on the fundamentals of the Dutch housing design and construction practice. It deals with the themes of stacking & linking, repetition, access & circulation, dwelling typology, construction methods, climate control and sustainability. Students of the Fundamentals of Housing Design Studio design a residential complex in an urban environment in the Netherlands. The design is accompanied/preceded by research into the design assignment and the specific topics of the studio. Though topics may vary from one semester to the next, at the core of each studio lies the design of dwellings and of the dwelling environment, complemented by research and literature study. Design work is done individually, while some of the research may be performed in teams. The Fundamentals of Housing Design Studio (AR1AD014) runs in conjunction with AR1A080, run by the Building Engineering department. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects of a medium sized building on MSc 1 level; - implement the fundamentals of housing design (as defined by the course instructors) correctly, coherently and consistently, articulating the architectural appearance of a residential building; - critically reflect on his/her own design and design process.	
Education Method	Studio sessions: 96 hours (for a group of 15 students) Self-study: 184 hours	
Literature and Study Materials	Bernhard Leupen and Harald Mooij, Housing Design a Manual (Rotterdam: NAI Publishers, 2011) Frederike Schneider, Grundrissatlas Wohnungsbau (Basel: Birkhäuser, 2011) Roger Sherwood, Modern Housing Prototypes (Cambridge: Harvard University Press, 1978)	
Assessment	There are two design phases, the first in teams (of 2 or 3 students) and the second individual. Each will result in a presentation and a grade. Furthermore students perform case study analysis (in teams of 3 or 4 students) and produce a reflection paper. The final grade is the weighted average of the assignments above.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios.	
Period of Education	Semester1, Q1 & Q2	
Concept Schedule	Tuesday afternoon, Friday all day	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 AD

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebäude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicians Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicians Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages.	
Study Goals	Please contact the studio coordinator for information about the program offered in spring 2024. Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based design study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.

Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003

Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000

Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover

Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008

BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual

Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,

Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge. The course programme consists of lectures, workshops and individual mentoring.	
Study Goals	Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstool	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.). However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages. The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose. The course has 3 parts: 1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality. 2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to: 1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0. The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point. A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75. If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75. The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 Architecture and Dwelling
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Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice MSc 3 and 4

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Advanced Housing Design

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AD100	Advanced Housing Design	55
Course Coordinator	Ir. O. Klijn	
Course Coordinator	Ir. H.A.F. Mooij	
Instructor	Ir. H.A.F. Mooij	
Instructor	Ir. F.M. van Andel	
Instructor	Ir. O. Klijn	
Responsible for assignments	Ir. O. Klijn	
Contact Hours / Week x/x/x/x	Research 6 hours p/w during 16 weeks in Q1 and Q2(start Semester 1)or Q3 and Q4 (start Semester 2) Design 8 hours p/w during 24 weeks in Q2, Q3 and Q4 (Start semester1) or Q1 and Q2 (Start semester 2)	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	There is an acute shortage of affordable and accessible dwellings in the Netherlands and a collectively acknowledged need to build one million homes until 2035. The societal task of The Advanced Housing Design Studio however, is not just about quantity or the architectural design of housing, but about a holistic understanding of dwelling as a social practice and of the city as an ecology. The studio focuses on the interrelationship between inhabitants and the architectural solutions of the design project and how this relation shifts within different territorial and governmental settings in The Netherlands. The design assignment is a mid to large size residential complex or neighbourhood, combined with collective and public functions depending on the specifics of the chosen location, displaying a variety of dwelling types for contemporary living. The design must take a long-standing notion of sustainability into account and include all three registers of sustainability: social, economic and ecological. The studio adopts a multi-dimensional research approach based on comparative design analysis and literature study. From this starting point, students develop an individual take on research which can include research methods from the fields of urban and environmental studies, social, architectural and urban history and architectural and urban ethnography. In a first phase of the studio, student collectively develop an urban analysis of the chosen site and discursive frame that delineate what an urban ecosystem and an inclusive housing project can mean in the urban context of the global North and the Netherlands. In parallel, students develop their own specific angle of research and design approach based on the analysis of two to four housing projects that are exemplary for their individual research and design theme. In the second phase of the course, the design hypothesis is further developed in an iterative process between research insights and design decisions, taking the multi-scalarity of dwelling and an ecology of living together into account.	
Study Goals	Upon completion of the course, the student is able to: - design a mid to large size residential complex or neighbourhood (which responds to functional, cultural, structural, regulatory and aesthetic demands); - perform academic research (asking proper research questions, acquiring and implementing adequate research techniques and using astute modes of reporting); - establish a reciprocal relationship between research and design, and use the tools of academic research for the design process; - reflect on the architectural profession, on academic research, on his/her own position in the field of architecture, on the education she/he has received and on the ethical dimensions of architecture and academia.	
Education Method	Studio tutoring sessions, workshops, lectures.	
Assessment	Student work is assessed at P1 (ok or not ok) and at P2 (go or no-go) Only the assessment at P2 is decisive. Research and design will be assessed separately.	
Period of Education	Semester	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Global Housing

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AD105	Dwelling Graduation Studio: Global Housing	55
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	Research 6 hours p/w during 16 weeks in Q1 and Q2 Design 8 hours p/w during 24 weeks in Q2, Q3 and Q4	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Summary	This course addresses the global need for more adequate housing, focusing on alternative solutions to engage architects in design processes that take into account a critical approach to the Sustainable Development Goals. The studio aims to produce knowledge on architectural concepts, models and instruments to deal with territories facing the challenge of rapid urban growth. The studio stimulates research on design approaches that explore alternative logics of modernization for emerging urban territories. Participants will be stimulated to reconsider established modes of analysis and intervention to support the emergence of urban welfare spaces; places of everyday wellbeing, commonality, conviviality, comfort, security, health, social and environmental justice.	
Course Contents	The Global housing graduation studio explores alternative approaches to the design of affordable housing in urban conditions characterized by rapid urban growth. The studio deals with the increasing cross-cultural character of contemporary architectural practice and the disciplinary challenges associated with the perennial global housing crisis. Focusing on the current challenges in low- and middle-income countries, participants in the studio are stimulated to critically review established modes of socio-spatial analysis and housing design decision-making processes, to support the emergence of urban welfare spaces; places of everyday wellbeing, commonality, conviviality, comfort, security, health, social and environmental justice. In the first phase of this course, students develop a collective knowledge base, unpacking the complexities of the projects context, as well as the particularities of mass housing design. The fieldwork in the projects site plays a key role in this phase, allowing students to advance a design hypothesis supported by empirical evidence of vernacular social and spatial practices collected through site surveys and ethnographic research. In the second phase, the design hypothesis is further developed and elaborated to deliver a housing project that explores a multi-scalar articulation between the urban realm, the sphere of the building, and the dwelling unit, considering a critical interrelation between social, economic and environmental factors.	
Study Goals	Upon completion of the course the student should be able to: 1. Elaborate a research plan, including problem statement, research question, goals and methodology; 2. Analyse the social and spatial practices of human settlements using transdisciplinary research tools and methods; 3. Communicate research outcomes using meaningful visual outputs and academic standards; 4. Synthesise processes of transformation of the natural and built environment using cross-disciplinary research methods to relate social, economical, cultural and environmental factors; 5. Formulate a design hypothesis based on a critical assessment of the societal issues pertaining to the project's context, a stakeholder's analysis and a programme of requirements; 6. Design a project for a housing complex, exploring a critical approach to the sustainable development goals, including the definition of a wider urban approach, appropriate architectural engineering and construction systems, and adequate spatial arrangements that cater for healthy and empowering livelihoods. 7. Produce meaningful visual and physical outputs to communicate an architectural project to an audience of experts; 8. Discuss the design principles of a housing project with other stakeholders. Note: The outcome of the graduation project demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	The tutorial system is the main educational method used in this course. The tutorial sessions will be focused on integrating research and design in the studio work. Throughout the whole first semester the work for the course Global Housing Graduation Studio will comprise work developed in group and individually. The fundamental research will be developed as team work, compiling a shared knowledge base focused on the theme of the studio. The research plan, including problem statement and research question will be developed individually and support the design hypothesis. In the second semester each student will develop the project individually. The studio will have weekly tutorial sessions. Some of these sessions will include lectures and (peer-)review sessions. In the first semester, a fieldtrip excursion to the project's site will be organised, and include a collaborative workshop with local students, researchers, educators and experts.	
Literature and Study Materials	Bhan, Gautam. Notes on a Southern Urban Practice. Environment and Urbanization (31-2), 2019, 639-654. Bredenoord, Jan, Paul Van Lindert, and Peer Smets, eds. Affordable Housing in the Urban Global South: Seeking Sustainable Solutions. Abingdon, Oxon: Routledge, 2014. Caldeira, Teresa. Peripheral Urbanization: Autoconstruction, transversal logics, and politics in cities of the global south. Environment and Planning D: Society and Space 35-1, 2017, 3-20. Gameren, Dick van, Frederique van Andel, and Pierijn van der Putt, eds. Global Housing: Affordable Dwellings for Growing Cities. DASH, 12/13. Rotterdam: NAI 010 Publishers, 2015. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Marcuse, Peter and David Madden. In Defense of Housing: The Politics of the Crisis. London and New York, Verso, 2016. Mota, Nelson, and Dick van Gameren. Affordable Housing and Sustainable Development: A Tale of Two Systems. The Architectural Review, April 2016. Mota, Nelson and Yael Allweil, eds. The Architecture of Housing after the Neoliberal Turn, Footprint 24. Prinsebeek: JapSam Books, 2019. Sennett, Richard. Building and Dwelling, Ethics for the City. Penguin Books, 2018.	

	Urban, Florian. Tower and Slab: Histories of Global Mass Housing. Oxon and New York: Routledge, 2013. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018. More specific literature and Study Materials will be made known 1 week prior to the start of the course in Brightspace. Assessment In the course of the graduation process two obligatory progress reviews (P1 and P3) and three formal assessments (P2, P4 and P5) take place. The P1 and the P2 are part of the Master 3 program and P3, P4 and P5 take place within the Master 4. Evaluation 1 (P1) - Compulsory progress review Goal: Assess whether the students working method and progress guarantee he / she will be able to meet the requirements for the P2 in time. Evaluation 2 (P2) - Formal assessment Goal: Completion of Master 3; assessment students admission to Master 4; the base for passing the P2 should be that the belief is that the student can graduate in the se-mester following the P2 period with a satisfactory result. Evaluation 3 (P3) - Compulsory progress review Goal: Survey whether the students working method and progress guarantee he or she will be able to meet the requirements for the P4 in time. Evaluation 4 (P4) - Formal assessment Goal: Assessment whether content of academic fields and presentation meet the requirements to admit the student to the final public presentation (P5). Evaluation 5 (P5) - Public final presentation Goal: Public final presentation and assessment graduation project. Students will receive marks at the P5 for the following aspects; - Architecture design - Building Technology - Research - Presentation - Final Mark. The assessment of Design and Research during the different evaluation moments will be based on the EMMA rubric and focus on the following criteria: -Coherence: internal consistency, integration, essence, concept -Significance: ethical, socio-cultural and/or scientific relevance, value, meaning -Elaboration: extensiveness, degree of detail of all aspects -Correctness: accuracy, efficacy, and evidence-based -Innovativeness: personal interpretation, creativity, new, unexpected, unique situation -Knowledge and know-how: effective study and use, processing of precedents and principles -Exploration: openness, discovering and investigation, analysis and testing -Reflection: careful consideration, evaluation, effects, comparing and positioning -Presentation: clarity, intelligibility, reflection and being engaged by it as a listener or a reader. Remarks The participants in this graduation studio will be encouraged to join a field trip to the site of the assignment during Q2. The cost of the field trip is approximately 1.000,00 - 1.500,00 (return flight and accommodation for 2 weeks). Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip. During the field trip excursion, participants in this studio will join seminars with local housing experts and workshops with local students and faculty. Period of Education Autumn Semester (P1, P2) and Spring Semester (P3, P4, P5) Minimum number of participants 12 Maximum number of participants 18 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Designing for Care

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AD110	Dwelling Graduation Studio: Designing for Care in an Inclusive Environment	55
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Instructor	Dr.ir. B.M. Jurgenhake	
Instructor	E. Miedema	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	Research 6 hours p/w during 16 weeks in Q1 and Q2 Design 8 hours p/w during 24 weeks in Q2, Q3 and Q4	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	We are, now more than ever, aware of the role of our everyday (built) environment on our health and quality of life. It is clear that health (besides illness) has never been discussed as much as in the last few years. Moreover we are facing an ageing society which has certain implications for our living environment. The theme of residential and neighborhood care will become very important as our neighborhoods are not equipped for an aging society. In addition, prevention and health promotion also appear to be crucial. But how do we design for quality of life for everyone? Students will think about new concepts in which everybody may live, and which suit the (care needing) people as well as others. They will discuss health promotive architecture and how to make health- and care friendly living environments. The design will be dedicated to the specific topic of health and care in our living environment. Students may focus on new care home concepts like homes for elderly, for children or (young) adults with mental health issues, or short-term care needs (patient hotel). Or they may focus on other health/care related building typologies that are part of the network for care, such as primary care center, hospice, cancer support center or healthy neighborhoods. The project can be a design as part of a transformation with new additions, or a design of a new building. The design location(s) will be in The Netherlands and may be within one chosen city. An important part of the graduation studio is the ethnographic (human centered) research, the understanding, preparation and practice of it. This means that the user is central in the research part. Each year we try to closely work together with the target group themselves. Getting to know the target group by making contact with them is extremely important to draw conclusions for the design. Only after this, the design proposal starts. Next to this participatory contacts we collaborate with housing associations or other involved parties. Interdisciplinary research (a combination of anthropological and architectural research with observation, participation and/or interviews) will be the base of the research work and complemented by literature. Some of the research may be performed in teams. Research will be done in our studio and will feed the design.	
Study Goals	Upon completion of the course the student should be able to: 1. Evaluate the results of the research and the analytical study to formulate a critical reflection on the design assignment; 2. Translate research results to design 3. Formulate and apply strategies for a healthy and inclusive living environment, meaning a community in which all groups of our society can live according their demands 4. Compose and present a problem statement; 5. Formulate a design hypothesis; 6. Identify appropriate building techniques and construction systems to be employed in the design proposal; 7. Produce meaningful visual and physical outputs to communicate the project to an audience of experts; 8. Discuss the design principles of a housing project with other stakeholders. The graduation report demonstrates the students ability to employ moral sensibility, observation methods and analysis, creativity, judgment, decision and argumentation skills regarding architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which interdisciplinary and intercultural issues were addressed in the graduation project.	
Education Method	In the first half of the semester studio work is devoted to the research part, partly done in small groups with individual additions, and the first individual design concept. In the second part of the semester, after the P1 presentation, emphasis will shift to the design. Research conclusions will now be translated to design guidelines. Concepts will be tested, among others by making models and by discussions with the tutors. The P2 presentation concludes the MSc 3 which involves the completed research report inclusive the conclusions of the research, a master plan on neighborhood level, and a preliminary design concept of the building.	
Assessment	Examination is done with oral presentations and written paper, research report and a graduation report. The assessment of the work developed by the students will be determined by the following criteria: 1. Personal Development: Ability to self-reflect on the personal motivations to work on the assignment, capacity to elaborate a realistic planning, and demonstration of the awareness to evaluate his/her own strengths and weaknesses. 2. Research and Analysis: Ability to identify and interpret relevant information out of the research and of source material used to support the development of the design assignment. 3. Architectural Project: Ability to experiment, test and project concepts, processes, forms and materials demonstrating technical competence. 4. Materialization and Technology: Ability to articulate knowledge on the architecture discipline with other building technology, materials and construction processes. 5. Communication: Ability to create meaningful oral, written and visual communication, prepared using appropriate conventions and media. 6. Process: Demonstration of a critical attitude to the design assignment and ability to work collaboratively and independently within the main standards of the architecture profession. 7. Interdisciplinary Research: Ability to work with research methods, common in the field of anthropological and social research, like observation and interviews and combine them with the more visual methods of architectural research. Results: Study Plan with Problem Statement, Goal and Method description; Research Report, made individually or in small groups; Site assignment analysis report (group- or individual work, depending on the choice of the site); Design Proposal, including design program; All elements are integrated into an individual Graduation Report.	
Exam Hours	The examination of this course will be done in several moments: P1 examination will be done after the first quarter. It is an oral examination. P2 examination will be done at the end of the second quarter within the obligated weeks (see graduation calendar) P3 examination will be done in the third quarter. It is an oral examination. P4 examination will be done in the fourth quarter and the Msc4 semester within the obligated weeks (see graduation calendar).	

	P5 examination will be done at the end of the Msc4 semester within the obligated weeks (see graduation calendar). Retake and repair moments fall within the same academic semester, they must be workable and study able, and not cause study delays. A retake is an entirely new examination or assignment. A repair is a revision or addition to the existing assignment (with the same topic).
Period of Education	Fall semester 2023/24
Concept Schedule	A concept schedule will be distributed one week before the course starts.
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

FSA

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 FSA

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1FO010	Form, Structure and Aesthetics	10
Responsible Instructor	P.A. Koorstra	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	S.M. Osinga	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	BSc	
Course Contents	The central question in the studio is how aesthetical values can be achieved as a result of the integrated design process. Taking a stand in the discussion evoked by this question will help to understand the role of the architect in a rapidly changing environment. In the debate about sustainability we see a consensus about the need for integration of sustainable solutions in architecture. Doing research by design, inspired by the available sustainable solutions seems to be the best preparation for the near future. Students will be guided in experimenting with specific technical solutions with a focus on aesthetics. The main topics of the Studio: The expression of structure and detail in architecture The aesthetical design consequences of sustainable solutions in architecture We are looking for a formal expression, directly related to a clear guiding theme. The structure or construction and the detail are part of this expression. Sometimes they can even become the guiding theme. At least the desired expression and technical solutions will have a direct link to the proportions and materialisation. How can we use our intuition to integrate this into the design? The Form, Structure & Aesthetics Studio focusses on designing a culturally oriented building in the Netherlands. The design is accompanied and preceded by research based on analysis. Design work is done individually, while some of the research will be performed in teams.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects from a medium sized building on MSC 1 level. use and construct physical models, hand drawings and digital models to represent and research specific design themes. understand and practice the concept of the integrated architectural design. document and reflect on the design process and outcome.	
Education Method	Design Studio format with weekly assistances starting week 1. Workshops and/or seminars will be part of the studio. Analysis are made in groups, design individually.	
Assessment	Assessment based on process, analysis, documentation and final presentation. The final assessment is as a presentation of visual representations of the project and a project dossier. The dossier will combine a documentation of the design process and a written reflection. The final grade will be rounded to half or whole points. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios. The studio is not available for MSc 2 students.	
Period of Education	Semester 1, Q 1 & 2	
Concept Schedule	Week 1.1 ethics workshop monday morning (Mandatory) Week 1.1 Start Studio Week 1.8 Mid-term Presentation Week 2.8 Final Presentation	
Leerstoel	Form Studies	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 FSA

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Course Contents Continuation Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

Remarks	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Period of Education	The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Concept Schedule	The maximum marking period is 15 working days. Quarter 4.
Minimum number of participants	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Maximum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Course evaluation	For this course of the Master track of Urbanism the maximum number of participants is 30.
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process.
	Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade.
	Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting.	
Assessment	The assignment will in principle be completed in groups of 3. Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
Special Information	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0). Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolitanization of European and regional systems: Wust s, et al 2007Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location The developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on collegera. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.). However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages. The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose. The course has 3 parts: 1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality. 2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to: 1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0. The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point. A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75. If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75. The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 Form, Structure and Aesthetics

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3TA001	Technologies and Aesthetics	55
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. C.N. van Leest	
Instructor	Ir. K.B. Mulder	
Instructor	G. Coumans	
Instructor	V.L. de Vries	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	...	
Expected prior knowledge	Completed MSc1 and MSc2 Architecture Design Studio and MSc1 Building Engineering Studio	
Course Contents	Climate changes and increasing demands on material use and re-use forces the Architectural designer to constant technical innovations and re thinking of the expression of Architecture. The aim of the studio is to explore the relation between the technical design innovations and the specific Architectural form language / Aesthetics that comes along with and belongs to these developments. This will culminate in an architectural design of a small to medium size building where architectural design and techniques are fully integrated. The studio will offer each year three topics, based on technical innovations and/or climate related issues, from which the students can choose. The studio will focus on locations in Northern Europe where social or climate related problems are relevant or suited for the studio topics. The design assignment will focus on the development of the Architectural expression and integration of the topics of the studio. The specifics of the chosen location, social, economic and ecological, shall be taken into account. The studio adopts a multi-dimensional research approach based on comparative design analysis, technical analysis and literature study. From this starting point, students develop an individual take on research which can include research methods from the fields of Architecture, Arts and Building technology. In a first phase of the studio, student collectively develop an analysis of the chosen Technological topic and Aesthetic approach. In parallel, students develop their own specific angle of research and design approach based on the analysis of two to four relevant projects that are exemplary for their individual research and design theme. In the second phase of the course, the design hypothesis is further developed in an iterative process between research insights and design decisions.	
Study Goals	The student is able to: - present a coherent, significant, elaborated, correct and innovative design on main issues and on aspects to a level fit for the architectural practice. - show an attained understanding of the design process with regard to building technology and innovations, human sciences, environmental aspects and sustainability. - develop aesthetical, technical and spatial innovations and to integrate the specific expression of structure and new materials into the Architectural design.	
Education Method	Research by Design will be the leading factor during the design process. Scale models, 1:1 prototypes, material research and researching and testing the spatial conditions and consequences will be an essential part of the studio. Assistances will take place in a design studio format.	
Assessment	Assessment will be based on the Emma Assessment tool for graduation studios. Assessment will take place at P1 (research progress), P2 (research outcomes and design proposal, go / no go, P3 (design progress), P4 (design outcome, go / no go) and P5 (Public Presentation and defense)	
Special Information	This graduation studio is a cooperation between the group of Formstudies from the department of Architecture and the Chair Design of Construction from the department of Architectural Engineering & Technology. The studio is only open for students of the Track of Architecture.	
Period of Education	The studio starts in Q1, final presentation / examination in Q4. The studio is offered only once a year.	
Concept Schedule	Final to be determined. Concept: week 1.1 till 1.6 Research plan; week 1.7 till 2.8 Research / Design; week 3.1 till 4.10 Design	
Leerstoel	Formstudies (Architecture) & Design of Construction (AE&T)	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

ADC

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc1 ADC

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1DC010	Architectural Design Crossovers Studio	10
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	Ir. S. Steenbruggen	
Instructor	Ir. C.M. Calis	
Instructor	Ir. A.M.R. van der Meij	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours/week (1.1-1.8 & 2.1-2.8)	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Architectural Design Crossovers MSc1 studio is structured as an introduction to the study programs ambitions within the larger framework of a multi-disciplinary approach. The project sites and scopes within the studio intersect not only disciplinary boundaries but also provide a temporal and material widening of the scope of the architectural project. In this regard, the MS1 design studio allows the students to define the core aspects of the "universal archive" to be explored in a multi-disciplinary approach. During the design process, students will explore new architectural concepts and building types and they will build up a narrative supporting the architectural design proposal. We consider the built environment the accumulation of not only social, economic, cultural and political layers but also an intersection of environmental sources, material and immaterial flows. Shaped through intensive material and immaterial flows, these sites help the students define the "crossover" themes to be explored in the collective and individual design assignments. In this respect, the studio will focus on London, where "universal archive" will serve as a programmatic framework to explore explore new spatial configurations and building types. Combined with the on-site survey data, the collective research serves the students with critical standpoints for their individual design work. An educational excursion (3-5 days) with an on-site workshop is part of the studio program. The expenses for the excursion should be around ±250. The corresponding information is to be communicated at the first meeting of the semester.	
Study Goals	Upon completion of the MSc1 studio the students are able to: - Analyze, evaluate and pursue a range of technical, programmatic, theoretical, historical and professional implications toward the final design proposal. - Integrate theoretical knowledge and practical skills into their design process. - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field. - Design and present a coherent, significant, elaborated, correct and innovative proposal for a medium sized building on MSc1-level.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Presentation and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio to be presented as part of the final review. The assessment criteria for the individual work are: - the position formulated by the student addressing the studio theme; - the consistency of the design process, the outcome and awareness of it; - the detail and quality of the architectural project combining the research and the contextual data; - the fitness (relevance and significance) of the project with respect to the assignment; - the elaboration level of the project throughout the respective scales addressed; - the coherence and quality of the architectural position/argument, presentation, and the products. Mid-term review (Week 1.7/1.8) Workshop (Week 2.3/2.4) Final review (Week 2.8) The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Remarks	The Architecture Design Studios and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering tutors collaborate closely.	
Period of Education	Q1, Q2	
Concept Schedule	Wednesday morning & afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 ADC

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
Special Information	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0). Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicians Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicians Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicians, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the basis a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/edxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolitanization of European and regional systems: Wust s, et al 2007Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 ects experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 ects studio setting, while AR0196 offers a 5 ects lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions		3
Course Coordinator	S.E. Frausto		
Instructor	S.E. Frausto		
Instructor	J.P. Benavides Lavalle		
Education Period	1 2		
Start Education	1		
Exam Period	none		
Course Language	English		
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.		
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.		
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.		
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.		
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.		
Enrolment / Application	For more information, contact Salomon Frausto.		
Special Information	The maximum marking period is 10 work days.		
Period of Education	Semester		
Concept Schedule	Friday		
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).		

ARB206	The Berlage Sessions		3
Course Coordinator	S.E. Frausto		
Instructor	S.E. Frausto		
Instructor	J.P. Benavides Lavalle		
Education Period	3 4		
Start Education	3		
Exam Period	none		
Course Language	English		
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.		
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.		
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.		
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.		
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.		
Enrolment / Application	For more information, contact Salomon Frausto.		
Special Information	The maximum marking period is 10 work days.		
Period of Education	Semester		
Concept Schedule	Friday afternoon		
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).		

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
Course Contents	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
Study Goals	The course has 3 parts:	
	1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality.	
	2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Education Method Literature and Study Materials	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Assessment	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Period of Education Course evaluation	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75 .	
Period of Education Course evaluation	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
	Quarter 1 For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 ADC

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3DC100	Architectural Design Crossovers Graduation Studio	55
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	Dr.ir. R. Cavallo	
Instructor	Ir. S. Steenbruggen	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Instructor	Ir. A.M.R. van der Meij	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (1.1-1.8 & 2.1-2.5) 8 hours/week (2.6-2.10, 3.1-3.10, 4.1-4.10)	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Architectural Design Crossovers graduation studio perceives the heterogeneity as the pervasive urban condition in the contemporary European cities, which cannot be addressed with the conventional disciplinary division in the design field. Therefore, the studio starts with issue-based lectures and seminars to reach beyond the problem oriented design solutions. In the studio, we consider the architectural project not only a synthesis of (im)material requirements of build-ing practices but also place it at the focus of an interdisciplinary framework with an emphasis on multi-modal design research. In the studio and the seminars, the students develop an in-depth analysis of the context at different levels. Working on the different site conditions (networked/layered) the students are asked to challenge the disciplinary conventions, methodologies and approaches in a wider social, environmental or cultural context. Embracing the inherent complexities and conflicts, the studio embraces a careful curation of different modes of design representation. From spatial/morphological mapping practices to time-based analysis of urban transects, the students adopt appropriate design media pertaining to the scope and definition of their project. The subjects of study in the studio will start with the synthetic analysis of a complex phenomenon to investigate the site as a networked/layered entity. Next to design and seminar research, the studio will be complemented by master classes involving experts from related (design) fields in which the themes, practices, experiences and positions closely related to the subject matter will be openly debated with the contribution of the students. In Spring semester (MSc4) the studio elaborates on the design concepts and schemes developed during the Fall semester (MSc3) aiming at the final resolution of the graduation project that complies with the graduation and assessment criteria defined by the Board of Examiners. More detailed information (i.e. literature, precedents, schedule) for the course will be provided in the course syllabus. For questions or more information please contact the course coordinator.	
Study Goals	By completing the course the students are able to: - Define their design position addressing pressing issues specific to the project site - Conduct in-depth research and analysis of the context, design brief and spatial program - Analyse, assess and develop the technical and programmatic requirements relevant to the design position - Use techniques and instruments for corresponding their design proposal - Synthesise analytical findings into relevant spatial/architectural questions - Demonstrate self awareness and insight about their design process and position with regard to historical, theoretical, cultural contexts - Synthesise analytical design findings into a comprehensive research report - Develop individually a design proposal integrating aesthetic, technical and functional requirements of the proposed program - Present the process and the project integrally to the specialists within related fields and to a non-specialist audience - Communicate and defend complex design ideas through verbal, visual and written media to both specialist and non-specialist audiences - Integrate the relevant technical and material aspects into architectural design considering building techniques and technology - Reflect on the the design and building practices, methods and technologies in a broader social, environmental and cultural context - Demonstrate awareness of material and energy flows pertaining to the specific solutions in the proposal - Present the technical aspects of the design process and the final result in a clear and systematic way in the form of drawings, words, texts and schemes - Defend complex design position and solutions through verbal, visual and written media publicly	
Education Method	In coordination with the Research Seminar (4 hours/week 1.1-2.5)) Research Plan lectures and tutorials (AR3AP010) in weeks 1.1-1.8 Issue-based pre-design research Site survey and related fieldwork Collective and individual research report Presentations and reviews Independent design & self-study Until the mid-term review (P1), the students will work in groups that will be complemented with individual exercises as part of the seminar and tutorial classes.	
Course Relations	Research Plan (AR3010)	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Reader	To be provided at the start of the semester	
Assessment	- Design Examination (See below P1-P5) - Writing Assignment (Research Plan & Research Paper)	
	(P1) Interim mid-term evaluation (week 1.9/1.10)	

	(P2) Go-No go assessment (weeks 2.9/2.10) (P3) Progress review (week 3.8/3.10) (P4) Go-No go assessment (week 4.4/4.5) (P5) Public final presentation and diploma ceremony (week 4.9/4.10) P2 is a Go-No go examination that determines the students ability to to proceed to the final part of the studio. P4 is a Go-No go examination that determines the student's ability to proceed to the public graduation presentation and ceremony (P5). All the criteria for the evaluation are explained in the graduation manual and will determine the final mark obtained for the entire graduation studio.
Period of Education	Q1, Q2, Q3, Q4
Concept Schedule	Tuesday afternoon (seminar weeks 1.1-1.9 & 2.1-2.6) Friday morning & afternoon (studio)
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AE

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 AE

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1AE011	EXTREME architecture	10
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	A. Snijders	
Instructor	E.W.M. Hehenkamp	
Instructor	Ir.arch. G. Koskamp	
Instructor	Ir. R. Schroën	
Instructor	Ir. M. van Driel	
Practical Coordinator	Ir. R. Schroën	
Responsible for assignments	A. Snijders	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in an extreme situation, with respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on the integration of aesthetic and technology. "Die Architektur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebäude müssen etwas leisten" Stefan Behnisch. In the end, the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 1 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical and architectural solutions: the student is able to apply, reflect and transform principles concerning climate, construction, structure and architecture.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. If products are missing and the project is of sufficient quality, the teacher can offer the student a repair. For the repair, the student has to produce the missing products within two weeks after the original presentation. The products have to be uploaded to Brightspace, there will not be a new presentation. The result of the repair can be a V (=completed) or F (=fail).	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios.	
Period of Education	Semester 1, Q1 & 2	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc2 AE

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Course Contents Continuation Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
Special Information	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0). Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

Remarks	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Period of Education	The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Concept Schedule	The maximum marking period is 15 working days. Quarter 4.
Minimum number of participants	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Maximum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Course evaluation	For this course of the Master track of Urbanism the maximum number of participants is 30.
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment. Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: - Public perspective; - Private / co-creation perspective; - Civil society perspective; - Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: - The skill of writing papers is presented and discussed; - Assignments are done to support the writing of a paper; - The elements of a paper (questions, methods, theory) are discussed; - Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic land explained - Theory is consciously used to frame the work, explained and logical- - Reflection into the topic of research in relation to the place of work - Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolitanization of European and regional systems: Wust s, et al 2007Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is be made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: Formulate a conceptual or theoretical framework. Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. Identify and discuss the tension between public and private development in infrastructures and environments. Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical Reflection into the topic of research in relation to the place of work Late delivery Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habrakens theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,	
	- spatial data modelling (vector and raster spatial models),	
	- geo-data manipulation (editing, digitizing, importing, converting, etc.),	
	- overview of spatial analysis operations,	
	- production of interpretable output (e.g. maps),	
	- fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures),	
	- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),	
	- basic algorithms for raster (encoding, quad trees, map algebra),	
	- networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;	
	5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems	
	- Slides of the lectures (available on Brightspace)	
	- Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied:	
	1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.	
	AND	
	2. The grade of the final exam is ≥ 5.75 .	
	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 AE

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AE100	Architectural Engineering Graduation Studio	55
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Ir. M.J. Smit	
Course Coordinator	A. Snijders	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Climate change, resource depletion, urbanization, pandemics and economic instability face architects with new challenges; the need for carbon neutral buildings and nature inclusive neighborhoods, the lack of affordable housing, changing requirements for the existing building stock and infrastructure, etc. How can our living environment be improved whilst dealing with these challenges? What are suitable technical innovations and how can they be integrated within the architectural design? And how can transitional design and engineering strategies be organized in a socially responsible and equitable way? Specific focus areas of aE Studio are amongst others circularity, climate design, urban ecology, digital / robotic manufacturing, product design, material research, building physics, structural mechanics and computational modelling. Architectural Engineering encourages students to explore their role as architects in facing today's challenges. Under the guidance of a team of enthusiastic design tutors and together with various stakeholders, students search for innovative technical solutions for diverse problems in various contexts in both the Netherlands and abroad (i.e. Indonesia and the Caribbean). The three main research by design domains promoted by Architectural Engineering are Make, Flow and Stock. Each domain requires a different approach and relates to very diverse research and design questions. As for example: - o How to develop new building systems and methods for reusable and renewable materials? - o How to design demountable buildings and reusable building components for a circular economy? - o How to integrate digital manufacturing techniques in the design process of buildings? - o How to integrate energy production and smart management in the built environment? - o How to (re)design the existing building stock in a sustainable way? - o How to responsibly integrate infrastructure in the built environment? - o How to guarantee nature inclusiveness within the architectural design? - o How to integrate water, waste- and food flows in architectural design? - o How to design healthy living conditions (f.e. dwellings) in an increasingly polluted environment?	
Study Goals	AE graduating students achieve the following study goals: - o Students are able to set up an accurate project definition (problem definition, objectives, research questions, methods, planning, etc.). - o Students are able to incorporate a relevant technical research topic in their overall design question. - o Students are able to incorporate specific technical knowledge in their architectural design. - o Students develop skills in architectural design satisfying aesthetic as well as technical and functional requirements. - o Students develop insights in and knowledge of the design process with regard to methods for research and design - o Students are able to clearly and coherently present the project through the use of adequate means. - o Students are able to reflect upon their research and design process and the related products. - o Students are able to position themselves in the field of architecture and understand which role they can have within society as an architect. During the master 3 & 4 the complexity of the architectural design increases, leading to an optimal level required for architectural practice. The graduation report (combination of graduation products) demonstrates the students' ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding architectural ethics and his/her future role as architect. The graduation products (research plan - AR3A010, research paper, reflection paper, design output, etc) should not only contain an elaboration regarding the graduation projects' societal and disciplinary relevance, but also have to address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	aE Studio is characterized by the strong connection between technical research and architectural design. Within the MSc3 students will be guided by an architecture (first tutor) and a research tutor (third tutor), whom they will meet every week to discuss the progress of their projects. A diverse network of internal and external experts guide our students as research tutors. Within the MSc4 a building technology tutor will be added to the team. Within the Architectural Engineering Graduation Studio students define their individual graduation assignment, related to different selected contexts, societal challenges and technical topics. Based on various thematic assignments aE Studio organizes collective table meetings and presentations to discuss and reflect on each other's projects.	
Assessment	Within the MSc3 examination takes place through P1 (midterm) and P2 (final) presentations. Deliverables at the P2 are a graduation plan (for details see the Graduation Manual), research plan (part of AR3A010 course) and the aE Studio Research paper, which should at least have a grade 6 at the P2 for a Go towards the MSc4. After successfully finishing the MSc3 studio, the student will enroll in the MSc4 studio devoted to the elaboration of the design project. Within the MSc4 examination takes place through P3-, P4- and P5-presentations and a written Reflection Paper (see Graduation Manual).	
Period of Education	MSc3 + MSc4	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

A&PB

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 Architecture and Public Building

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1AP012	Public Building Design Studio	10
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Instructor	Ir. S.M. Witteman	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	2	
Course Language	English	
Course Contents	The MSc.1 Studio provides an introduction to the design of public buildings in the twenty-first century. It explores the design of hybrid buildings and sustainable public architecture and aims to address fundamental questions including: - What constitutes public architecture in the twenty-first century? - What role does public architecture play in global cities? - How can architecture promote publicness, inclusion and diversity? The Studio examines the significance of public architecture in the context of the broader social and cultural milieus. It encourages design processes and proposals that imagine innovative and diverse models of symbiosis among design intent, construction, users, and stakeholders. In addition, it encourages the exploration of hybridity to replace the limitations of mono-functional buildings and neighborhoods. Each year, the Studio is set in a select European city, exploring the unique site conditions as a foundation for the design of a medium-sized hybrid public building (approximately 3.000-5.000 square meters of floor area). The resulting design should act as a catalyst for new urban dynamics, featuring areas for leisure, contemporary media and cultures, education, and innovation. By designing a building that serves the diverse purposes and needs of the given community, the Studio promotes vibrant and sustainable urban environments. The MSc.1 Studio operates in close collaboration with the Building Engineering Studio (BES) to promote new sustainable designs, materials, and construction. The BES instructors take responsibility for the technical, structural, and performative aspects of the design exercise, implementing a collaborative educational agenda focused on sustainability, energy efficiency, designing for disassembly, and adaptability to climate change. The BES instructors prioritize specific materials, emphasize CO2-neutral and net-zero energy buildings, and reflect a circular economy. By working together, the Studios design instructors and the BES aim to develop building designs that are both innovative and environmentally responsible.	
Study Goals	Upon completion of the MSc.1 Studio, the students will be able to: - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects from a medium sized building on MSC 1 level; - develop a design position that considers specific urban conditions and reflects on a coherent design strategy; - relate the assigned functional program to a coherent architectural concept and to construction as well; - grasp the complex and heterogeneous potential of the public realm through multiplicity, density, and hybridization; - introduce properties that make architectural interventions less singular in function, but more productive, more transformative and resilient; - ensure a design outcome where aesthetic and technical aspects are explicitly integrated and mutually related; - design an innovative architecture that can improve adaptability to climate change and productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, and materials.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Architectural Design Studio (AR1AP012) and the Building Engineering Studio (AR1A080) are fully integrated to ensure the comprehensive development of each proposal. By combining these two areas of study, students will be equipped to translate their design concepts into practical, buildable solutions. Each of these Studios runs parallel activities with the same objectives, including: - lectures - public discussions - field investigations / site-visits - formal and material design-oriented assignments in drawing and modelling, preliminary to the main project - readings and writings During the first quarter, students work in teams to maximize the potential of the assignment at an urban scale. After the midterm review, students will develop individual designs.	
Course Relations	AR1A080 Building Engineering Studio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Written statements and analyses. The final design proposal is individual and Architecture and Building Technology will provide respective grades. The assessment process consists of weekly in-class discussions of progress and the midterm and final reviews. The final grade reflects:	

	Critical analysis and the urban context (25%) Design quality of the final proposal (55%) Participation, collegiality, commitment (20%)
Remarks	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio. The Architectural Design Studio (AR1AP012) and the Building Engineering Studio (AR1A080) are integrated and form a coherent whole.
Period of Education	Fall Semester (Q1-Q2)
Concept Schedule	Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 AP

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssoen and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolitanization of European and regional systems: Wust s, et al 2007Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 ects experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 ects studio setting, while AR0196 offers a 5 ects lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge. The course programme consists of lectures, workshops and individual mentoring.	
Study Goals	Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays: global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs: corporate or commercial, as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions		3
Course Coordinator	S.E. Frausto		
Instructor	S.E. Frausto		
Instructor	J.P. Benavides Lavalle		
Education Period	1 2		
Start Education	1		
Exam Period	none		
Course Language	English		
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.		
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.		
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.		
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.		
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.		
Enrolment / Application	For more information, contact Salomon Frausto.		
Special Information	The maximum marking period is 10 work days.		
Period of Education	Semester		
Concept Schedule	Friday		
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).		

ARB206	The Berlage Sessions		3
Course Coordinator	S.E. Frausto		
Instructor	S.E. Frausto		
Instructor	J.P. Benavides Lavalle		
Education Period	3 4		
Start Education	3		
Exam Period	none		
Course Language	English		
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.		
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.		
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.		
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.		
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.		
Enrolment / Application	For more information, contact Salomon Frausto.		
Special Information	The maximum marking period is 10 work days.		
Period of Education	Semester		
Concept Schedule	Friday afternoon		
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).		

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.). However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages. The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose. The course has 3 parts: 1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality. 2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to: 1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0. The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point. A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75. If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75. The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 Architecture and Public Building

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AP100	Public Building Graduation Studio	55
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Instructor	S. Lee	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. H.J. Bultstra	
Instructor	S. Corbo	
Responsible for assignments	Ir. P.A.M. Kuitenbrouwer	
Contact Hours / Week x/x/x/x	X/X/X/X	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Graduation Studio offered by the Public Building Group aims to produce future-proof architectures that are sustainable and inclusive. It investigates public-ness in its multiple, ever-changing configurations by testing the possibilities of design thinking in a world where the definition of what an architect is and does, ceaselessly shifts. Each year, the Graduation Studio focuses on a specific topic in a specific city. Architectural designs will be made for culture, health, education, work and leisure influenced by contemporary contexts and conditions. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows the built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Building Technology will be treated as an inseparable part of this process. The Studio opts for a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Throughout the MSc3 and MSc4 periods, Research by Design will occupy the core of students individual explorations. Between concept and design lies a creative process augmented by research, with design and research as intertwined and often indistinguishable acts. Design and research simultaneously mean creating design loops, variants, reiterations, intersections, impressions.	
Study Goals	At the end of the academic year, students will be able to present a coherent, significant, elaborated, correct and innovative design of a (hybrid) public building, complex or space - on main issues and on aspects to a level fit for the architectural practice. This, showing an attained understanding of the design process with regard to architectural history and architectural theory, art, building technology, human sciences, environmental aspects and sustainability, digitalization, climate control and the materialization of buildings and interiors. In developing their design proposals, students will be introduced to data-driven processes and advanced forms of architectural visualization and representation. Conventional media will be paired with AI, GIS mapping, augmented / virtual reality, digital fabrication.	
Education Method	As a response to ever-changing political and societal needs, public buildings evolve and require new spatial solutions. Research by Design allows to investigate and envision such solutions. Through a series of individual and collective assignments, students will learn to reflect on the needs of contemporary societies, to develop architectural positions and to translate their concepts into buildable spatial solutions. The Graduation Studio is organized across literature reviews, tutorials, site visits, and pre-design analysis, leading to schematic design. The Graduation Studio takes two semesters of full-time study. Semester 1 consists of Research by Design that merge Project Design, Theory and Delineation Research. They relate to analysis and collection of data, pre-design, site studies, and schematic design. As a part of Project Design, tutorials on Building Technology will be included. Building on the preceding semester, Semester 2 includes design development, materialization, and building technologies, emphasizing sustainability. Theory and Delineation Research A constitutive and integrated component of the Graduation Studio, Theory and Delineation Research is a territory of experimentation: it consists of themes, discourses, typologies, that encourage coherent design thinking and reflection. Theory and Delineation [TD] aims to facilitate the Project Design [PD] work in various ways and to enrich the outcome of the design process. TD examines a variety of literature in science, history, architecture, and related disciplines in combination with hands-on practice. TD also provides the backbone of the required MSc 3 course Research Plan (AR3A010), leading to the completion of the Graduation Plan required for the P2 at the end of the fall semester. The TD sequence provides a more precise and relevant focus on the rubric Research by Design. As part of the Studios work, each student should maintain the specificity of his/her own individual motives, interests, and approaches. TD has been therefore conceived as a journey that will gradually introduce the students to a variety of techniques and media, both analog and digital. Such media are not only intended as representational or explanatory devices. On the contrary, coherently with the nature of a Research by Design approach, emphasis will be placed on the generative character of such media their ability to act as vectors for new and unforeseen scenarios. TD addresses various topics from ideation to materialization in visionary and canonical projects. While in the first weeks of the program conventional media will be explored sketches, drawings, models, in the second block students will be introduced to more advanced media in order to develop their conceptual and spatial narratives: 3D modeling, diagramming, data visualization, scripting.	
Course Relations	AR3A010 Research Plan	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will take place during the Design Examinations, in the form of: 1. Oral presentation; 2. Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; 3. Writing assignments and analyses: Research-by-Design Journal. Criteria are:	

	a. Ability to identify a relevant topic vis-à-vis the design assignment; b. Ability to research the topic both from a present-day point of view as from a historical perspective; c. Ability to identify and interpret relevant information or source material and use it to support the development of the design assignment; d. Ability to experiment, test and project concepts, processes, forms and materials demonstrating technical competence; e. Ability to self-reflect on the personal motivations to work on the assignment, capacity to elaborate a realistic planning, and demonstration of the awareness to evaluate his/her own strengths and weaknesses; f. Ability to create meaningful oral, written and visual communication, prepared using appropriate conventions and media; g. Demonstration of a critical attitude to the design assignment and ability to work collaboratively and independently within the main standards of the architecture profession. For the final grade compound, we refer to the Graduation Manual, Master of Science Architecture, Urbanism and Building Sciences, Academic Year 2023-2024. The final grade is the average or may deviate from the average depending on the extent to which the whole does (or not) exceed the sum of its parts, or due other exceptional qualities of the work.
Special Information	The Public Building MSc Graduation Studio will require one excursion outside of the Netherlands.
Period of Education	Two semesters (Q1-Q2-Q3-Q4)
Concept Schedule	Studio tutorials on Thursday Global setup in phases: Phase 1 (in semester 1) up to P1 The foundation for the individual project. In small groups the overall framework of the assignment is analysed through site-research and a site visit, and together a site model is produced. Through individual assignments focused on a research-by-design approach, students will progressively build a thematic and conceptual framework that will drive their design explorations. Individually, students work towards (a) Spatial Diagram(s) that explain(s) the relations between program, site specific conditions, technical possibilities and so on, within the individual design framework. The work is conducted and guided in Project Design, Theory Research, Delineation Research, and Building Technology Research to be thoroughly compiled in the Research by Design Journal. Result (individual): P1 Presentation and Research by Design Journal Phase 2 (in semester 1) up to P2 The architectural concept design The Concept Design for the individual project is made and elaborated upon. The concept design shows the architectural potential and the ambition of the project within the boundaries of the individual design framework. It offers possibilities that can be assessed and shows the direction in which the design can be developed. The Concept Design is a result of, or fed by Research by Design. Students produce their Graduation Plan with the design objectives for Msc4. Result (individual): P2 presentation of Concept Design, Research by Design Journal, Graduation Plan Phase 3 (in semester 2) up to P3 From Concept Design to Preliminary Design Phase 4 (in semester 2) up to P4 & P5 Final Design Awarded Diploma
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

CP

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 CP

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1CP011	Complex Projects Design Studio	10
Course Coordinator	M. Triggianese	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Instructor	Ir. H.A. van Bennekom	
Instructor	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1 and ending in week 2.10	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	MSc 2	
Expected prior knowledge	BSc completed	
Course Contents	As an introduction to Complex Projects, the MSc1 design studio called 'Dutch Change' focuses on the Netherlands ground. By getting familiar with the Dutch environment, the MSc students will learn to understand the complex system behind buildings and establish a connection between building design and social changes. During the design process, students will explore new architectural concepts and building types and they will build up a narrative supporting the architectural design proposal. By identifying and making decisions on the balance between what changes and what remains in Dutch architecture, students will be challenged to explore both the tradition of Dutch planning and building culture. In the end, students will do an architectural design project in a Dutch city. The assignment starts from a broader understanding of the physical context, its past and future planning, environmental issues and climate adaptation on the one hand and an in-depth understanding of the building typological developments on the other hand. In fall 2023, the 'Dutch Change' studio focuses on the architecture of buildings for science and learning in the context of the Dutch Delta. Site visits to exemplary educational buildings in Europe are included in the program.	
Study Goals	Upon completion of the MSc1 Complex Projects studio trajectory: - The student will be able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects from a medium-sized building on MSc 1 level. - The student will be able to be aware of the connection between history and urban context and use architectural instruments to respond to new challenges. - The students will be able to work both with a group and independently, to coordinate different opinion and formalize an efficient design process.	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology, and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relationship between buildings, spaces and societys needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, the materialization of buildings, comfort, and climate control.	
Education Method	Tutorial once a week. Tutorials in group, supported by lectures, field trip, self study and group work. Students will be working both in groups and individually or in pairs. The individual assignment is a medium-sized building. Building Engineering Studio is integrated.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Books	Recommended Literature: Wagenaar, Cor (2011) Town Planning in the Netherlands since 1800: Responses to Enlightenment. Ideas and Geopolitical Realities, Rotterdam: 010 Publishers Barbieri, U., van Duin, L. (2003) Hundred Years Of Dutch Architecture, A: Trends, Highlights, SUN NAI Publishers. Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to the start of the studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation.	

Remarks	Repair
	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Period of Education	The Architecture Design Studio (AR1CP011) and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely.
Concept Schedule	Quarter 1-2 (fall semester)
Minimum number of participants	Tuesdays morning
Maximum number of participants	12
Course evaluation	40
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 CP

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebäude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
Special Information	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0). Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment. Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: - Public perspective; - Private / co-creation perspective; - Civil society perspective; - Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: - The skill of writing papers is presented and discussed; - Assignments are done to support the writing of a paper; - The elements of a paper (questions, methods, theory) are discussed; - Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic land explained - Theory is consciously used to frame the work, explained and logical- - Reflection into the topic of research in relation to the place of work - Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssoon and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based design study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
Special Information	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0). Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolitanization of European and regional systems: Wust s, et al 2007Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location The developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: Formulate a conceptual or theoretical framework. Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. Identify and discuss the tension between public and private development in infrastructures and environments. Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical Reflection into the topic of research in relation to the place of work Late delivery Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicians Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicians Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.). However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages. The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose. The course has 3 parts: 1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality. 2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to: 1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0. The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point. A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75. If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75. The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3	
	4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 CP

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3CP100	Complex Projects Graduation Studio	55
Course Coordinator	H. Smidihen	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Dr. O. Caso	
Instructor	M. Triggianese	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	H. Smidihen	
Responsible for assignments	Ir. J.M. van Zalingen	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	In the graduation studio, all elements of the building and design will be seamlessly integrated into a clear and coherent project that touches on all the design challenges, from analysis and problem statement, through building concept and urban implementation, all the way to materialisation and critical details. The assignment for the MSc3 CP Graduation Studio will be to develop a design brief: a clear design assignment based on the narrative generated by the research of globally relevant topics, and analyses of given urban contexts. During the MSc4, the design brief will be answered by an architectural proposal which will be developed into a clear and comprehensive architectural design to scale 1:5. The architectural design proposal needs to have a strong relation with the research and urban strategy made during MSc3.	
Study Goals	The complex projects graduation programme consists of two connected semesters: MSc3 & MSc4. Successfully completed courses from previous semesters, (MSc1 & MSc2) are a precondition for the successful start of the graduation. In the first semesters, MSc3, the students focus on the research, design investigation, and the thorough understanding of all the elements that are rendering the given context. The final goal of the semester is to transform personal fascination and research conclusions into a clear design task. The design task should be formulated as a design brief, in which design assignment, program, site and ambition of the project are clearly stated. Upon completion of the first semester, students will be required to provide an answer to the previously defined design brief in a form of a design proposal. During the development of the design, students need to keep a strong link to the previous research, as well as to the group work and to the overall site vision. The graduation ends with the final presentation of the design proposal, linked together in a group strategy. Graduated students should develop the skills in architectural design to satisfy both aesthetic and functional requirements. Additionally, skills related to understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects will be developed. Together with the training on building technology, students will be able to incorporate an understanding of the design process with regards to structural design, materialisation of buildings, comfort and climate control. The final graduation thesis demonstrates the students ability to employ analysis, creativity, judgment, decision and argumentation skills regarding architectural ethics and the students future role as an architect. The individual graduation thesis should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	The main methodology in Complex Projects is to develop each students organised and structured approach to the design process. Students are confronted with the very rigorously defined design schedule, and at the same time rather complex and demanding design assignment. By following the given structure, students will learn how to deal with complex design assignments, how to process and organise a large amount of data, how to meet deadlines and prioritise workflow. Work process, consistency & quality of weekly development will have influence on the final result. Students will be encouraged to record progress and development of the work and use it in the final narrative as well as in the argumentation of the design. Product based design development is a design method being used in Complex Projects. By having strictly defined deliverables, within the dense schedule, students are encouraged to constantly produce whilst developing a design in parallel. In this way, the design progress and the final products are much more integrated, but even more importantly deliverables are used as a mean to develop, explore and test the design. The goal is to continually refine the methods of representing, presenting and communicating the urban and architectural work. The student should explore methods of presentation, storytelling and design documentation. Group work, collaboration and collective deliverables are strongly promoted in Complex Projects. Through practising group work, students learn how to discuss, how to divide work and how to make compromises. Development of communication skills, constant usage of different communication mediums (sketches, models, presentations, reports...) is becoming more and more important in the architectural profession. Furthermore, constant communication, updates and sharing of information are becoming a medium to develop and progress design. One of the central group deliverables is a collective research book - a medium to develop a research part of the graduation. This book will be continuously updated throughout the year and will show the creative and analytic process of the group. Next to the research, students will be encouraged to investigate spatial aspects of the design questions through making large urban models. Those models will be used throughout the whole graduation to facilitate discussions, group presentations and design ideas.	
Assessment	The assessment of the studio will be based on the individual performance of each student combined with involvement and result of the group work. The students performance will be determined by the quality of his or her work, their commitment, effort and improvement over the entire course of the semester. The final evaluation will be based on the performance throughout the whole year as well as the final result. During the first MSc3 semester of graduation, there will be in total four presentation moments. The first presentation (P0.5) is an informal group presentation, where initial findings regarding graduation assignments are shared, presented and discussed. The second assessment (P1.0) is a formal presentation where students have to present their individual graduation plan and their research topic. The third presentation (P1.5) is informal, which is focusing on group work and development of the links between individual work of students, connecting them into the group strategy. The first semester ends with P2.0, a formal presentation which is go-no go review where the students continuation to MSc4 is decided upon. The evaluation of the result will be based on the performance of the student throughout the semester, the collaborative group work and the final presentation. In the final presentation, the design assignment needs to be presented in the form of a design brief, with a clearly formulated design question and a determined program, size and location for the project. After completion of the first semester, students will be required to continue developing their project further through the Msc4, and focus on the structure, materialization, comfort and climate control of the building. During MSc4, there will also be four presentation moments. The first presentation (P2.5) has an informal character and focuses mostly on presentation of the urban strategy and concept of the design. The second presentation (P3) is when students are asked to present preliminary design of their proposals. In this moment functional and programmatic aspects of the design should be developed and frozen. P4 is the main presentation of MSc4, where students receive another go-no go. In P4 a complete design should be presented, including	

Period of Education	materialization, structure, climate and façade design. After P4, students have several weeks to polish the design, create visuals and models for final presentation, P5. Complex Projects graduation consists of the two semesters. First semester, MSc3 is focusing on the research, analyses of the context, and the search for the thesis topic. Second semester, MSc4 is more dealing with the design, and providing design answers to created tasks in MSc3. Both semesters are divided in four phases, each of them lasting for five weeks. At the end of each phase the group pin-up, and individual presentation are organized. Complex Projects graduation starts in both the fall and spring semester.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MAI

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 MAI

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1MET015	Situating Architecture	10
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. P.H.M. Jennen	
Instructor	J.A. Mejia Hernandez	
Instructor	A. Stanii	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Required for	...	
Expected prior knowledge	Completed BSc	
Course Contents	Well situated buildings are evidence of architectural knowledge and skills. In a series of thematic modules you will analyze and imagine different (e.g. material, perceptual, economic, narrative) layers of a given context through concrete design tasks, leading to the individual production of a complete and well situated architectural project. Through intense design practice, you will train in the use of analytical and projective methods required to develop architectural designs that are sensibly inscribed within a particular context, understood as an entanglement (Lat. con + texere = interwoven) of natural and cultural variables. The studio will be integrated with the AR1A080 Building Engineering Studio.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze a given context using selected instruments and methods of architectural analysis and imagination. - Design a coherent, significant, elaborated, correct and innovative architectural project - on mainline and on aspects for a medium sized building on MSC 1 level.* - Develop a complete technical strategy that is integral to your project and is coherent with your analysis of the context. - Evaluate the ethical and aesthetical consequences of your design, with particular attention to the way in which it is situated in its context. * Standard learning objective required for all MSc1 design studios.	
Education Method	A first part of the course develops a series of thematic modules, which present you with different tasks that require the analysis of a particular context and the design of an architectural project. Special emphasis is put on the instruments and methods used in both the analytical and the design processes. In the second part of the studio you will develop one of your designs into a complete architectural project for a medium sized building, including the constructive method required to build it. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to situate your project within its context.	
Literature and Study Materials	...	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow you to decide which of your designs to develop further. The two final outcomes (project and evaluation) are assessed and graded individually, using the aspects and criteria determined in the EMMA feedback and assessment tool, adjusted for a MSc1 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	semester 1, Q1 & 2	
Concept Schedule	...	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 MET

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment. Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: - Public perspective; - Private / co-creation perspective; - Civil society perspective; - Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: - The skill of writing papers is presented and discussed; - Assignments are done to support the writing of a paper; - The elements of a paper (questions, methods, theory) are discussed; - Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic land explained - Theory is consciously used to frame the work, explained and logical- - Reflection into the topic of research in relation to the place of work - Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Course Contents Continuation Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Knoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolitanization of European and regional systems: Wust s, et al 2007Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicians Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicians Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicians, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions		3
Course Coordinator	S.E. Frausto		
Instructor	S.E. Frausto		
Instructor	J.P. Benavides Lavalle		
Education Period	1 2		
Start Education	1		
Exam Period	none		
Course Language	English		
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.		
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.		
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.		
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.		
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.		
Enrolment / Application	For more information, contact Salomon Frausto.		
Special Information	The maximum marking period is 10 work days.		
Period of Education	Semester		
Concept Schedule	Friday		
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).		

ARB206	The Berlage Sessions		3
Course Coordinator	S.E. Frausto		
Instructor	S.E. Frausto		
Instructor	J.P. Benavides Lavalle		
Education Period	3 4		
Start Education	3		
Exam Period	none		
Course Language	English		
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.		
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.		
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.		
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.		
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.		
Enrolment / Application	For more information, contact Salomon Frausto.		
Special Information	The maximum marking period is 10 work days.		
Period of Education	Semester		
Concept Schedule	Friday afternoon		
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).		

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,	
	- spatial data modelling (vector and raster spatial models),	
	- geo-data manipulation (editing, digitizing, importing, converting, etc.),	
	- overview of spatial analysis operations,	
	- production of interpretable output (e.g. maps),	
	- fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures),	
	- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),	
	- basic algorithms for raster (encoding, quad trees, map algebra),	
	- networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;	
	5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems	
	- Slides of the lectures (available on Brightspace)	
	- Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied:	
	1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.	
	AND	
	2. The grade of the final exam is ≥ 5.75 .	
	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3	
	4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent	
	GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 MA

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3MET105	A Matter of Scale	55
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Prof.dr.ir. K.M. Havik	
Instructor	Ir. W.W.L.M. Wilms Floet	
Instructor	Ir. P.H.M. Jennen	
Instructor	J.A. Mejia Hernandez	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	Contact Hours: 8 hours per week	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	- you can design a complete architectural project, including its technical aspects, independently. - you can communicate your architectural ideas via multiple representations, including (but not limited to) technical drawings and scale models.	
Course Contents	The quality of life in European cities is inseparable from their human scale. Through meticulous and insightful analysis and imaginative design practice you will develop a comprehensive architectural project for a European city, with particular attention to the ethical and aesthetical consequences of scaling architectural design accurately in relation to human proportions. In partnership with a local academic institution you will develop thorough analyses of (a) selected aspects of the context, and of (b) one or more buildings that contain valuable knowledge about design and construction in that context. Analytical findings will serve as basis for the formulation of a design proposal, which will be developed into a comprehensive architectural project, with special emphasis on the conditions of exploration and innovation defined by the EMMA feedback and assessment tool (see assessment, below). The conclusions of the entire process are presented as an evaluation of the instruments and methods used to achieve a human scale for your project, and its expected urban impact.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze specific layers from a chosen context using selected instruments and methods of architectural analysis and imagination. - Formulate a clear and concise design proposal for the development of a complete architectural project for that context, with special attention to its scale. - Design a comprehensive (formal, technical, functional, meaningful) architectural project as development of that proposal, expressed in a complete, consistent, and elegant set of architectural representations, prepared for an audience of experts and non-experts. - Evaluate the innovative urban impact of the resulting architectural design in ethical and aesthetical terms, with special focus on all design decisions concerned with architectural scale.	
Education Method	The studio is divided into five parts. As part of a small group, you will first develop in-depth analyses of a key aspect of the context, and of a building that contains valuable disciplinary knowledge to design and build in that context. To collect empirical evidence and carry out the necessary fieldwork you are required to participate in a studio excursion. The results of all analyses will be edited and printed into a booklet produced by the studio as a whole. In the second part of the studio you will start working individually (and continue doing so onwards), to formulate a clear and concise design proposal consisting of (a) the selection of a concrete site or situation for your project, (b) a justification for its chosen use and purpose, and (c) a first attempt to determine its overall form. In the third and fourth part of the studio all formal, functional, and technological aspects of your project are developed thoroughly, through different projective methods and multiple representations, and with special attention to scale. The fifth and final part of the course focuses on communicating the complete and final outcome of your work, including an evaluation of the instruments and methods used to design and scale your project, and its expected urban impact.	
Assessment	The analyses (P1) and designs (P3) produced in the first and third parts of the studio are subject to group and individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your project proposal (P2) and final designs (P4), produced in parts two and four of the studio, are assessed and graded with go/no-go decisions. Your final presentation (P5), at the end of the fifth and last part of the course, is assessed and graded with a final mark for the whole course. All reviews and assessments will be based on the aspects and criteria determined in the EMMA feedback and assessment tool. Related to the research carried out by the chair of Methods of Analysis and Imagination, particular attention is given to the criteria exploration and innovativeness, as defined in the tools glossary of terms, and in relation to the different aspects that are evaluated in your research and design.	
Period of Education	One academic year.	
Concept Schedule	Tuesday, Friday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

HA

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 HA

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1AH010	Heritage and Architecture Design Studio: Architectonic Design	10
Course Coordinator	Ir. W. Willers	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Responsible for assignments	Ir. W. Willers	
Contact Hours / Week x/x/x/x	4-8 hours per week starting from week 1.1 and ending in week 2.8.	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The object of this Heritage & Architecture studio is the architectural design for the re-use of a building or building-ensemble to meet requirements of contemporary and future use. A transformation framework will be made by the interpretation of the analysis of the urban context, the building and the program requirements. Various aspects of designing in existing built structures are investigated by studying reference projects and literature. By working on different scale-levels a coherent design will be made. At atelier meetings different aspects like relation existing new, urban context, functionality, spatial quality, technical aspects, material aspects will be discussed. The current debate of transformation and intervention with topics like authenticity, sustainability, layers of history, and so on is very present during this course on every single scale. Different presentations will help students to develop their presentation skills. There is a strong relation between the H&A Architectural design studio and the Building Engineering studio.	
Study Goals	Upon completion of the Master 1 design project the student is able to: - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects from a medium sized building on MSC 1 level. - produce an urban, architectural and building technical analysis, interpret cultural values on urban, architectural and technical scale. - formulates a transformation framework and starting points for a design. - interpret a project brief. - translate the transformation framework to an intervention strategy. - explore and select a scenario combining architectural and technical qualities, cultural values and program requirements. - use reference projects throughout the design process. - create a design in architectural heritage buildings concerned with preservation and renewal in existing architecture and values related to architectural history and current questions on sustainability and socio-cultural aspects such as socio-cultural, ethical, historical, philosophical, economical aspects. - elaborate elaborate of your position in designing with heritage.	
Education Method	Design coaching, 4- 8 hours per studio during education weeks, total 96 hours	
Assessment	Assessment will be based on the Emma assessment tool for MSc1 studios. Detailed information will be published in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The visit to the building, the subject of the assignment, is mandatory.	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios. The visit to the building, the subject of the assignment, is mandatory.	
Period of Education	Semester 1, Q1 & 2	
Minimum number of participants	the minimum number of participants is 12	
Maximum number of participants	the maximum number of participants is 60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 AH (nw)

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
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Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

Remarks	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Period of Education	The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Concept Schedule	The maximum marking period is 15 working days. Quarter 4.
Minimum number of participants	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Maximum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Course evaluation	For this course of the Master track of Urbanism the maximum number of participants is 30.
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssoen and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Course Contents Continuation Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevel	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 ects experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 ects studio setting, while AR0196 offers a 5 ects lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education).	
Period of Education	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Concept Schedule	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Minimum number of participants	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship 12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,	
	- spatial data modelling (vector and raster spatial models),	
	- geo-data manipulation (editing, digitizing, importing, converting, etc.),	
	- overview of spatial analysis operations,	
	- production of interpretable output (e.g. maps),	
	- fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures),	
	- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),	
	- basic algorithms for raster (encoding, quad trees, map algebra),	
	- networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;	
	5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems	
	- Slides of the lectures (available on Brightspace)	
	- Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied:	
	1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.	
	AND	
	2. The grade of the final exam is ≥ 5.75 .	
	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 HA

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc3 Adapting 20th century Heritage

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AH105	Graduation Studio Adapting 20th century Heritage	55
Course Coordinator	Ir. L.G.K. Spoormans	
Instructor	A. de Ridder	
Instructor	Ir. W. de Jonge	
Instructor	W.J. Quist	
Instructor	Ir. J. Roos	
Instructor	Ir. L.G.K. Spoormans	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Instructor	Dr. M.T.A. van Thoor	
Instructor	F. Marulo	
Instructor	Prof.dr. A.R. Roders	
Instructor	Prof.dr.ing. U. Pottgiesser	
Responsible for assignments	Ir. L.G.K. Spoormans	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The AR3AH105 Graduation Studio focusses on Adapting 20C Heritage. What is the essence of this young heritage and what is its status? How can we adapt these buildings and ensembles to make them sustainable and future proof without losing their values? The section of Heritage & Architecture holds three chairs and related research fields: Heritage & Design, Heritage & Technology and Heritage & Values. These are represented in the research and design approach of the H&A graduation studio, the curriculum and the tutors involved. The studio focusses on selected themes regarding built heritage, such as vacancy, preservation of monuments, new heritage, shared heritage or heritage communities. Research and design of the studio relates to H&A research projects and expertise of H&A staff. Students of Heritage & Architecture have to position themselves as architects in the debate on the architectural and technical characteristics of built heritage, its tangible and intangible values and design strategies for conservation and adaptive re-use. The articulation and understanding of architectural and technical values is of great importance when design decisions have to be made regarding what to conserve, what to adapt. In MSc3, students conduct research on the studio theme in a larger context and on the specific characteristics of selected cases. They develop a research question, methodology and final proposal for their individual research project. Methods can be literature study, comparative case study, mapping, archival research, field work and survey, interviews, research by design etc. Students observe, explore, identify and prioritise urban, architectural and technical values of the existing building and site in order to formulate starting points for a meaningful redesign. The research forms a theoretical base for the design project and results in an individual academically substantiated report or journal article. The individual work and the collective research by the studio contribute to the body of knowledge for the graduation studio. Based on both the thematic research and the mapping of a specific building or area that serves as a case, students develop a design brief and a design method. They explore a range of scenario's combining heritage values and program requirements, and study references on concept, strategy, materialisation, detailing, program etc. By a creative process, logical argumentation and evidence-based choices, students create a concept design that shows coherence and correctness. In MSc4, students individually develop an elaborated design based on the research, design brief and concept design of MSc3. The academic framework is applied and theoretical and practical references are studied. Strategies of conservation, intervention and transformation are explored and translated into an architectural and technical design. Students individually create a design that shows coherence and correctness, and a meaningful translation of an intervention strategy. Furthermore, they will position the project within a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts. In the graduation project the societal and disciplinary position and its relevance in relation to design ethics and intercultural issues are discussed. After reflective exploration and consideration, students present a detailed design project that addresses all important aspects of architectural design, technology and cultural value. During the course, students practice to present their design proposals both graphically and orally, and demonstrate logical argumentation and evidence-based choices.	
Study Goals	MSc3 The student: - is able to develop a full research proposal within the context of the studio - is able to conduct, synthesise and present the research using substantiated methods - is able to explain architectural and technical values, as well as their place in historical, societal, economic and environmental context - can critically reflect on research results and translate these into opportunities, obligations and dilemmas, and can prioritize and incorporate the key values into meaningful preliminary design - develop and present a design brief, method, scenarios and concept design that is relevant for contemporary academic and social society, - individually draw conclusions and present these in an academically substantiated and comprehensive report or paper as well as through a verbal presentation, MSc4 The student: - is able to position the project and its heritage aspects in a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts, - is able to apply professional knowledge and design tools related to architecture, building technology and values, - is able to argue and reflect on the design product and process in relation to current architectural discourse, - is able to demonstrate and employ moral sensibility, analysis, creativity, judgement, decision-making and argumentation skills regarding architectural ethics and his/her future role as architect, - is able to communicate complex design ideas at an advanced level through verbal presentations, visual and written media.	
Education Method	Lectures, workshops, atelier counselling	
Literature and Study Materials	Literature and study materials will be available prior to the start of the course in Brightspace.	
Assessment	Written report/ article, paper/ digital/ oral presentation	

Period of Education	Academic year
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc3 Revitalising Heritage

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AH115	Graduation Studio Revitalising Heritage	55
Course Coordinator	Ir. L.G.K. Spoormans	
Instructor	A. de Ridder	
Instructor	Ir. W. de Jonge	
Instructor	W.J. Quist	
Instructor	Ir. J. Roos	
Instructor	Ir. L.G.K. Spoormans	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Instructor	Dr. M.T.A. van Thoor	
Instructor	F. Marulo	
Instructor	Prof.dr. A.R. Roders	
Instructor	Prof.dr.ing. U. Pottgiesser	
Responsible for assignments	Ir. L.G.K. Spoormans	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3AH115 Graduation Studio focusses on Revitalising Heritage. How to bring new life to abandoned or dilapidated buildings or areas? What combination of architectural interventions and functional program leads to preservation by revitalisation? What is the capacity for change? The section of Heritage & Architecture holds three chairs and related research fields: Heritage & Design, Heritage & Technology and Heritage & Values. These are represented in the research and design approach of the H&A graduation studio, the curriculum and the tutors involved. The studio focusses on selected themes regarding built heritage, such as vacancy, preservation of monuments, new heritage, shared heritage or heritage communities. Research and design of the studio relates to H&A research projects and expertise of H&A staff. Students of Heritage & Architecture have to position themselves as architects in the debate on the architectural and technical characteristics of built heritage, its tangible and intangible values and design strategies for conservation and adaptive re-use. The articulation and understanding of architectural and technical values is of great importance when design decisions have to be made regarding what to conserve, what to adapt. In MSc3, students conduct research on the studio theme in a larger context and on the specific characteristics of selected cases. They develop a research question, methodology and final proposal for their individual research project. Methods can be literature study, comparative case study, mapping, archival research, field work and survey, interviews, research by design etc. Students observe, explore, identify and prioritise urban, architectural and technical values of the existing building and site in order to formulate starting points for a meaningful redesign. The research forms a theoretical base for the design project and results in an individual academically substantiated report or journal article. The individual work and the collective research by the studio contribute to the body of knowledge for the graduation studio. Based on both the thematic research and the mapping of a specific building or area that serves as a case, students develop a design brief and a design method. They explore a range of scenario's combining heritage values and program requirements, and study references on concept, strategy, materialisation, detailing, program etc. By a creative process, logical argumentation and evidence-based choices, students create a concept design that shows coherence and correctness. In MSc4, students individually develop an elaborated design based on the research, design brief and concept design of MSc3. The academic framework is applied and theoretical and practical references are studied. Strategies of conservation, intervention and transformation are explored and translated into an architectural and technical design. Students individually create a design that shows coherence and correctness, and a meaningful translation of an intervention strategy. Furthermore, they will position the project within a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts. In the graduation project the societal and disciplinary position and its relevance in relation to design ethics and intercultural issues are discussed. After reflective exploration and consideration, students present a detailed design project that addresses all important aspects of architectural design, technology and cultural value. During the course, students practice to present their design proposals both graphically and orally, and demonstrate logical argumentation and evidence-based choices.	
Study Goals	MSc3 The student: - is able to develop a full research proposal within the context of the studio - is able to conduct, synthesise and present the research using substantiated methods - is able to explain architectural and technical values, as well as their place in historical, societal, economic and environmental context - can critically reflect on research results and translate these into opportunities, obligations and dilemmas, and can prioritize and incorporate the key values into meaningful preliminary design - develop and present a design brief, method, scenarios and concept design that is relevant for contemporary academic and social society, - individually draw conclusions and present these in an academically substantiated and comprehensive report or paper as well as through a verbal presentation, MSc4 The student: - is able to position the project and its heritage aspects in a broader perspective of socio-cultural, historical, philosophical, economic and environmental contexts, - is able to apply professional knowledge and design tools related to architecture, building technology and values, - is able to argue and reflect on the design product and process in relation to current architectural discourse, - is able to demonstrate and employ moral sensibility, analysis, creativity, judgement, decision-making and argumentation skills regarding architectural ethics and his/her future role as architect, - is able to communicate complex design ideas at an advanced level through verbal presentations, visual and written media.	
Education Method	Lectures, workshops, atelier counselling	
Literature and Study Materials	Literature and study materials will be available prior to the start of the course in Brightspace.	
Assessment	Written report/ article, paper/ digital/ oral presentation	

Period of Education	Academic year
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Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Interiors Buildings Cities

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 AI

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1AI013	Interiors Buildings Cities MSc1 Design Project	10
Course Coordinator	Ir. S. Pietsch	
Instructor	Ir. L.M.M. de Wit	
Instructor	Ir. S. Pietsch	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	Variable 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, Interiors Buildings Cities considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. It engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. House The House of the MSc1 project is neither the representative monument of collective public life, nor is it the private house of an individual. Instead it suggests and considers those buildings that stand between; intimate figures, which both inform and are informed by the physical and social fabric of their immediate locality. Such buildings are carefully wrought and spatially rich, both generous and adaptable, yet they also embody a degree of typicality and a measure of the material culture of their place. Their interiors host the rooms and spaces that structure and draw a community into consciousness of and in dialogue with itself. Through a process of iterative drawing and large-scale modelling, supported by lectures and workshops, students will resolve the design and technical integration of such a building and its principal interior spaces, considering it as a figure set within the immediate context a carefully documented urban setting. The MSc1 Design Project is paired with the Building Engineering Studio (AR1A080) coordinated by the Department of Architectural Engineering & Technology.	
Study Goals	Upon completion of the Master 1 studio trajectory the student: - is able to present a coherent, significant, elaborated, correct and innovative design, on mainline and on aspects, from a medium sized building on MSC 1 level. - is able to demonstrate skills and knowledge to incorporate an understanding of the design process with regard to structural design, materialisation of buildings and interiors, comfort and climate design - is able to demonstrate skills and knowledge to incorporate an understanding of the design process with regards to the appropriateness of the architectural intervention in its context and its elaboration from urban setting to interior. In addition to the study goals, a specific description of the aims of the studio will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject. A characteristic working method of Interiors Buildings Cities is the making of models of varying scales appropriate to matters under consideration, in which ideas about the design project, from form to space and other relationships are tested and materialised.	
Literature and Study Materials	To be announced upon beginning of the course	
Assessment	The design studio concerns the development of an architectural project on all scale levels, from its urban setting to its materiality and elaboration of its details. The project will be assessed during an intermediate, pre-final and final presentation on: - the position that is formulated with regard to the brief and its context - the appropriateness of the intervention with respect to the assignment - aesthetic and technical / functional qualities - the elaboration throughout the respective scales - the integration of the disciplines included - the quality of the presentation, the products and the argument. - the consistency and coherence and development of the students work during his / her process The products to be assessed include the design proposal represented through drawings, models and text; the project journal and the portfolio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios.	
Period of Education	Autumn semester, first and second quarter	
Leerstoel	Interiors Buildings Cities	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 AI

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habrakens theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

Remarks	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Period of Education	The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Concept Schedule	The maximum marking period is 15 working days. Quarter 4.
Minimum number of participants	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Maximum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Course evaluation	For this course of the Master track of Urbanism the maximum number of participants is 30.
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based design study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level. Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge. The course programme consists of lectures, workshops and individual mentoring.	
Study Goals	Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicians Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicians Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,	
	- spatial data modelling (vector and raster spatial models),	
	- geo-data manipulation (editing, digitizing, importing, converting, etc.),	
	- overview of spatial analysis operations,	
	- production of interpretable output (e.g. maps),	
	- fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures),	
	- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),	
	- basic algorithms for raster (encoding, quad trees, map algebra),	
	- networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;	
	5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems	
	- Slides of the lectures (available on Brightspace)	
	- Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied:	
	1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.	
	AND	
	2. The grade of the final exam is ≥ 5.75 .	
	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 AI

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3AI100	Interiors Buildings Cities Graduation Project	55
Course Coordinator	M. Pimlott	
Instructor	M. Pimlott	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Instructor	Prof. D.J. Rosbottom	
Instructor	S. De Vocht	
Instructor	Dr. A.R. Thomas	
Responsible for assignments	M. Pimlott	
Co-responsible for assignments	Dr. J.S. Zeinstra	
Co-responsible for assignments	Dr. A.R. Thomas	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which people within or affected by the scenes made by architecture can be situated in relation to place and conditions, to time and the historical moment, and to social and material culture. Thinking beyond individual students and courses, Interiors Buildings Cities considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. Interiors Buildings Cities engages in common questions concerning the city and its territories, public space, the public interior, questions of interiority, and relations within the social and physical fabric of the city. It does so with the well-being of the entire environment shared by people and other living beings as a central concern. Palace The graduation studio theme, Palace, does not concern literal palaces, but those complex public buildings that have used the palace as a formative motif, and play a significant role in the fabric of the city. These accommodate the public's image of itself and the sense of collective and individual agencies within society. The theme responds to the history and position of such representative public buildings in the urbanised environment, and acknowledges that the figure, idea and role of the Palace have been constantly reinterpreted over time. It questions what these are in the present, and the ways through which the Palace might engage with both the contemporary city and its citizens in the present and future. To aid the study of the nature of the project and its conditions, collective research is undertaken into precedents of related exemplary projects over relevant historical periods, and in workshops and seminars aimed towards understanding their relevance to the project and participants' architectural knowledge. Further studies are made directly in relation to the conditions surrounding the project itself, from the urban environment to attendant social, cultural and political circumstances that have bearing upon them. Beyond working in existing physical and cultural contexts, we frequently work with existing buildings, their histories, meanings and materials. In the design of a large-scale project, largely the focus of our studies, we are concerned not only in the making of a building for the representations of the city and its publics, but how, and of what, it is made: the acts of building, the use and re-use of material, the sustainability of strategies of building, of the origin of materials, their extraction, processing, transport and associated labour, and the levels of responsibility of the architect, are all central considerations. We do this in association with the Department of Architectural Engineering and Technology from the outset, and regularly thereafter, so as to generate deeper questions and understanding of how to build intelligently and responsibly now, in the hope that this might be a model for future practices.	
Study Goals	Upon completion of a trajectory in the Masters programme from MSc1 through MSc4, the student: - Has developed the skills in architectural design satisfying aesthetic, relational, functional and technical requirements. During the trajectory, the complexity of the architectural design increases, leading to a level fit for architectural practice and a training period for professional accreditation as architect. - During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to research and reading in urban and architectural history, relevant theory, cultural practices, human sciences and technology. - Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects and sustainable practices. - During the Master 1, 2, 3 & 4 process, skills are acquired to incorporate insights into and knowledge of the process attained with regard to methods of investigation and architectural design. - Together with the training with regard to aspects of building technology, during the Master 1, 2, 3 & 4 process, skills are acquired to incorporate an understanding of the design process in response to truly sustainable principles, affecting approaches to building design, structural design, materialisation of buildings and their interiors, human comfort, natural diversity, and climate design. The Graduation Report, in the form of a Project Book and Project Journal, demonstrates the student's ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual Book and Journal not only contains an elaboration regarding the Graduation Projects' societal and disciplinary relevance, but addresses design ethics and the way in which intercultural issues have been addressed within the graduation project itself.	
Education Method	The object of the course is the formulation of a design project for an urban or civic institution on a representative site. This is achieved over a series of stages, ranging from discrete thematic studies to analyses of relevant models, integrated with project proposals with an increasingly significant level of definition. This includes: 1. Workshops in analysis of local urban contexts, characteristics particular to the public interior, programme, precedents and use, conducted by the design mentors. 2. Research seminars on literature, theory and history specific to the theme of the programme within the design studio, related to both the Research Plan, and research questions integrated into the development of the project, followed throughout the year through a project journal, conducted by the research mentors. 3. Research and design supervision in studio, in groups and for individuals in relation to precedents and models, in which comparative historical analysis plays a central part, and studies of local conditions, contexts and the specific conditions of both	

	site and building, conducted by the design mentors. 4. Preliminary design development, guided in workshops in MSc3 and individual tutorials towards detailed design development in MSc4, with studio presentations and critique, conducted by the design mentors. A characteristic working method of the Chair are specific design briefs on aspects of the greater project, often resolved through the making of physical models of varying scales appropriate to matters under consideration, in which ideas about the design project, from form to space and other relationships are tested and materialised.
Literature and Study Materials	A core list of reading or other material is determined for each design programme by the design and research mentors, ranging from literature and essays on cultural matters to urban and architectural theory, and critical analysis of architecture. This is published in the Studio Manual issued to participants at the beginning of the first semester. Additional reading material pertaining to issues of design and research are set forward in the Studio Manual, specific to the programme of the studio.
Assessment	The project will be assessed during intermediate and final presentations on: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; - the organisation, composition, presence and 'appearance' and the potential for functional technical qualities; - the potential for elaboration throughout the various scales; - the potential for integration of the disciplines, from aesthetic to technological; - the quality of the presentation, the argument and the product; - the consistency, coherence and development of the students work during the process of design, as demonstrated through prepared project journal, project portfolio and project book. Intelligent conventional architectural drawing and accessible models are particularly important.
Enrolment / Application	Students can enrol for this course during the pre-enrolment period for entry in first or third quarters. The studio beginning in the third quarter is based on individually generated design and research proposals, and so follows an individually-oriented research and design programme, with the same products as the course beginning in the first quarter. Enrolment for the studio beginning in the first quarter follows the convention of registering interest in the pre-enrolment period. For the studio beginning in the third quarter, students are invited to make design and research proposals before the pre-enrolment period. Those proposals which are suitable for development within the studio are enrolled through the appropriate department.
Special Information	The marking periods are directly related to the examination periods: the Go/ No Go examination (P2), and the final examination (P5), as set by the university and the department of architecture.
Period of Education	Two semesters of twenty weeks, beginning either in the first or third quarters, with a Go/No Go examination (P2) at the end of the first two quarters.
Leerstoel	Interiors Buildings Cities
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

The Why Factory

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 TWF

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A081	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Education Period	2 3	
Start Education	2 3	
Exam Period	none	
Course Language	English	
Course Contents	The main topic of the Building Engineering Studio (BES) is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. The aim of materialization - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that he or she has knowledge and understanding of: - the building engineering concept in relation to the architecture - the interaction between an architectural concept and a building engineering concept - development and elaboration in a sustainable way of technical aspects: construction, structure and climate. Thus the student must present reasoned solutions and demonstrate skill in incorporating the sustainable, technical building design effectively in the design process as a whole.	
Education Method	Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent of the theme and scale of technical elaboration. method of the studio. On the other hand the presentation products have to show the content formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualization of the concept, the elaboration of the relevant sustainable and technical aspects and a reflection on the final outcome. The information regarding presentation and assessment is more specific formulated in the course manual for the particular studio.	
Special Information	For questions please contact Ferry Adema (F.Adema@tudelft.nl). The Building Engineering Studios are taught during the 2nd quarter. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter.	
Period of Education	2nd Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1TWF011	The Why Factory MSc1 Design Studio	10
Course Coordinator	J. Arpa Fernandez	
Course Coordinator	Prof.ir. W.G.M. Maas	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Co-responsible for assignments	A.B.O. Ravon	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 1.1 and ending in week 1.10	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc1 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc1 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!) MSc1 Design Studio at the Why Factory starts in week 1.1 and ends in week 1.10. After the design studio, students follow their work in the Building Technology Seminar.	
Study Goals	The student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects from a medium sized building on MSC 1 level. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC1 level. Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 Design Studio: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 72 Number of self study hours: 208 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model	

	City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program:In the applications program model cities both are tested in real cities. The different models become counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Remarks	Students who choose the MSc 1 studio of The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. This design studio is not integrated with Building Engineering Studios.
Period of Education	From week 1.1 thru week 1.10 in the Fall semester
Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 TWF (nw)

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Course Contents Continuation Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Course Contents Continuation Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebäude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: Formulate a conceptual or theoretical framework. Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. Identify and discuss the tension between public and private development in infrastructures and environments. Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicians Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicians Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habrakens theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,	
	- spatial data modelling (vector and raster spatial models),	
	- geo-data manipulation (editing, digitizing, importing, converting, etc.),	
	- overview of spatial analysis operations,	
	- production of interpretable output (e.g. maps),	
	- fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures),	
	- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),	
	- basic algorithms for raster (encoding, quad trees, map algebra),	
	- networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;	
	5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems	
	- Slides of the lectures (available on Brightspace)	
	- Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied:	
	1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.	
	AND	
	2. The grade of the final exam is ≥ 5.75 .	
	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3	
	4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

BO

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 BO

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1BO010	Borders and Territories Design Studio	10
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week starting from week 1.1 and ending in week 2.10.	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Into the deep: abyssal frontiers [Introduction] The educational program of Borders & Territories (B&T) focuses on the role of architecture in relation to socio-political developments within contemporary society. The B&T MSc 1 Studio is a laboratory for the investigation of complex territorial, spatial, ecological, technical, and architectural practices, this semester through formulating a project for a deep sea mining institute. [Focus] Next in line in the long-standing history of extraction practices transforming the (sub)surfaces of our planet could be deep sea mining; the controversial other side of the coin of green technologies. The promise of dredging up some of the trillions polymetallic nodules containing manganese, nickel, cobalt and copper (ingredients for making the electric cars, solar panels and wind turbines), rare earths and other possible riches from the deep, announces a new dawn for the planet. Saving the planet by destroying more of it on a scale hitherto unfathomed. What is the role of architecture for future institutions and organizations, operating on a truly (blue) planetary scale, which attempt to commercialise, to regulate, to contest or to educate about the exploitation of the mineral wealth of the ocean floor, involving various geographies, nations, private commercial bodies, legal frameworks, and (robotised) infrastructures. [procedure] Precise acts of surveying and mapping are an integral part of the design process. Drawings and model-making mediate between investigation and design, both as systematization of concepts and relations and as a mode of representation. During the first quarter, a series of exercises will familiarize the students with the approach of the B&T studio. [Program] Student projects will dwell on aspects of the emerging deep sea mining industrial complex by designing a part or parts that, although separate, is/are relevant to its chain of operation, in the form of a building: the Deep Sea Mining Institute. In a brave new world, the ocean as periphery or transit space for global economic exchange becomes a vast centre, where control rooms operating remote dredgers, onshore staging grounds and mineral processing facilities, and pricing mechanisms compete with marine biology, fair distribution of concessions in and proceeds from international waters, etc. At the crossroads of geopolitical anomalies, a unique programmatic and site condition, as well as complex legal, technical and ecological requirements, the contradictions of deep sea mining need to be given form, taken into account on- and off-shore ecological cost and the complex legal and technical requirements. [Site] For the 2023/2024 fall semester, a series of sites, with both deep sea mining futures on the horizon and relevant histories of extraction, will be under consideration, including: Nauru, Okinawa (Japan), and Namibia. Each student will produce an innovative design for a medium sized building. B&Ts MSc_1 Design Studio is integrated with BES.	
Study Goals	- The student is able to develop a coherent, significant, elaborated, correct and innovative design for a medium sized building on MSc1 level - The student is able to communicate key ideas and design skills and express theoretical knowledge. - The student is able to establish an independent critical reflection and position regarding the design process and architectural representation. - The student is able to integrate the research and analysis into design process and concepts.	
Education Method	Group work (site analysis). Pin-up collective presentations. Individual Consultation. Independent design & self-study. Workshops and seminars.	
Assessment	Attendance & participation in collective presentation. Individual presentations & evaluations: midterm and final presentation. (Refer to the calendar for specific weeks). Assessment Scheme Final design (70 %) Midterm presentation (10 %) Participation (attendance, initiative, in-class discussion) (10 %) Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely.	
Period of Education	Semester 1, Q1 & 2	
Concept Schedule	Friday	
Minimum number of participants	12	

Maximum number of participants	36
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Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 BO

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.

Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003

Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000

Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover

Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008

BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual

Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,

Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based design study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location The developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicians Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicians Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicians, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.). However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages. The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose. The course has 3 parts: 1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality. 2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to: 1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0. The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point. A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75. If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75. The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 BO

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3BO100	Borders and Territories Graduation Studio	55
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	O.R.G. Rommens	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 2.1 and ending in week 2.10. 8 hours per week starting from week 3.1 and ending in week 4.10.	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The graduation studio of B&T investigates contemporary border conditions within the larger urban and territorial scale. The studio focuses on sites where new spatial conditions have emerged that are teeming with new meanings and unexpected potential but are hardly analysed within contemporary architectural discourse. Nowadays, it seems no longer possible to investigate space without considering global political developments, nor to ignore ecological anxieties, pressured coexistences or even economically driven migratory movements. B&Ts graduate studio intends to map these emerging spatial complexities, focusing on the fundamental changes regarding radical spatial differentiation, biodiversity, atmospheric and soil conditions, increased technological spatial control and economic asymmetries. The spatial patchworks of superposed political and economic processes and juxtaposed social and spatial practices will constitute the (almost) solid basis for alternative architectural interventions proposing radical counterstrategies to the current and no longer sustainable status quo. B&T considers the current developments in the border-territories related to the New Silk Road particularly intriguing testing ground(s) for speculative approaches, as the spatial transformations in these areas between the far East, Africa and Europe are resulting in a fascinating array of spatial changes, where different regimes of spatial planning, (inter-) continental infrastructural projects, political (dis-) agreements, border tensions, global capitalism, extra-state utopias and the like, are re-formatting the contemporary territorial and urban landscapes. B&Ts 2023/24 graduate studio will explore the current state of the New Silk Road by globally cross-sectioning these territories. Central to the investigation and intrinsic to the design elaboration is the production of specific mappings addressing: Border and Migration, Infrastructure and Congestion, Soil and Fault Lines, Water and Aquifers, and Contaminated Ecologies. Possible sites of investigation for B&Ts 2023/24 graduate studio: Shanghai, Chongqing, Xi'An, Tashkent, Tbilisi, Khorgas-Horgos, Mogadishu, Liège, Piraeus.	
Study Goals	Develop effective tools and techniques for implementing a design position. Analyze, evaluate and pursue a range of technical, programmatic, theoretical, historical and professional implications toward the final design proposal. Integrate and express theoretical knowledge and practical skills into design process. Develop a definitive project on her/his own that qualifies functional, spatial and aesthetic qualities through relevant research and preparation. Integrate building technology in adequate ways into the architectural design. Synthesize requirements of building technology and the architectural design. The graduation report demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	(first semester) Pre-design research. Site visit and field investigation. Formal exercises in drawings and models. Material exercises. Presentation and critique. The studio work will include and be supplemented by seminar session, informal/intermediate reviews, and a two-weeks workshop. During the first half of the semester, until the midterm review (P1), the students will work in groups. (second semester) Individual Consultation. Independent design & self-study.	
Assessment	(first semester) *P1 (Week 1.10) PDF/PPT presentation. Project synopsis, problem statement, concepts and analytical mappings. Requirements specified by the faculty rules & regulations for graduation. *P2 (Week 2.9) PDF/PPT presentation. Urban analysis/mappings (incl. methods and results) and design project proposal. The definition of the design project consists of site, site analysis, program, program analysis, site model (1:500) and a sketch design. Requirements specified by the faculty rules & regulations for graduation. The P1 & P2 weeks may be subject to change. Consult the graduation regulations for the submission requirements. (second semester) Presentations & evaluations: P3, P4 and P5. Refer to the graduation calendar for specific weeks. Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or	

	repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).
Period of Education	Refer to the graduation calendar.
Concept Schedule	First semester: Tuesday, and Thursday Second semester: Thursday
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Urban Architecture

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 UA

AR1A061	Delft Lectures on Architectural Design and Research Methods	5
Course Coordinator	Dr.ir. R.A. Gorny	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Ir. E.I. Ronner	
Instructor	Dr.ir. R.A. Gorny	
Instructor	M.F. Berkers	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	6	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A compulsory course for all students starting their Master education in Architecture at TU Delft, the Lecture Series on Architectural Design and Research Methods highlights current disciplinary issues against the background of the larger societal conditions that have an inevitable impact on the architectural practice. The course comprises a series of highly interactive talks with lecturers (professors and researchers of the Delft Faculty of Architecture, and guest lecturers), who will be addressing key contemporary positions in architectural discourse and investigate historical models and theoretical arguments in relation to contemporary discourses in research and design.	
Study Goals	The fundamental aim of the Lecture series is to foster an academic attitude based on an inquisitive approach to the built environment. Building, and the critical transformation of the built environment, is a complex field of practices that involve a multiplicity of various kinds of different forms of knowledges. After following the lecture series, students - have gained appropriate knowledge of the larger historical development of the discipline of architecture in relation to the main theoretical concepts and methods deployed of architecture and technology, their application in specific cases as presented in the lecture series addressing current issues of architectural practice and culture. - can recognize and critically reflect on different research- or design-methodological approaches in the discipline of architecture; including the larger context of the manifold relations between architecture, the city and society and the relations between design concepts, building production and materialization. - can systematically describe each their specificities and limits in understanding the built environment, and exemplify relations between specific both traditional and emerging tools and methods and for both research and design production. - can intellectually position their own approach within and toward these systems of knowledge.	
Education Method	The Lecture Series consist of weekly lectures, accompanied by interactive seminars with smaller groups of students. Generally, the lectures start with a thematic introduction, after which a weekly-differing 'architectural position' is discussed.	
Literature and Study Materials	The compulsory literature for the course is T. Avermaete, K. Havik, and H. Teerds (eds.), Architectural Positions: Architecture, Modernity and the Public Sphere, (Amsterdam: SUN Press, 2009).	
Assessment	The course is graded on the basis of a written assignment, developed from the work in the seminar. Grades will be announced within 15 working days after each assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the seminars.	
Remarks	This course is a preparation course for the graduation year.	
Period of Education	Semester 1, Quarter 1	
Concept Schedule	Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A066	Delft Lectures on Architectural History and Theory	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciú	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 2.1 and ending in week 2.8	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Required for	This course is a preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Course Contents	(Re)thinking the Global History of Architecture: History and historiography as Agents of Change Architecture, understood as the process of projecting, constructing, and maintaining buildings and structures embodies values of the past; diverse actors have used the built environment to achieve political, economic, social, or cultural goals. Over time, the role of architecture has changed extensively. From political demonstration to entrepreneurial intervention, from ideological statement to ecological practice. The memory of these earlier practices is inscribed in the built environment that surrounds us; it is also carried by historians and heritage specialists who choose to study one period over another, who focus on a specific locality, people or relationships. This course explores the ways in which architectural practice has evolved to respond to historical paradigms, responding to and shaping global and local societal conditions and frameworks. It also studies the ways in which historical investigation has shaped the course of architecture. Through a series of lectures, readings and discussion-based seminars, this course positions architectural practice in global histories and historiographies of architecture. The lectures will provide select examples from historical architecture through thematic lenses, exploring also the ways in which these histories are constructed and narrated. The course consists of a lecture series (3ECTS) and associated seminars (2 ECTS).	
Study Goals	After this course students will be able to: - recall key moments and themes in the history and theory of architecture in the context of political, economic, societal and global change; -analyze and discuss historiographical texts and presentations - develop a critical understanding of architects practice and tools through time and space; - pursue research on historical practices and buildings discussed in class, or, identifies historical examples not presented in the course; - evaluate existing research in the history and historiography of architecture; - formulate a research question and first initial idea on a personal research topic for the thesis.	
Education Method	Lectures, Readings, Discussions in tutor groups, Self study for individual research	
Literature and Study Materials	To be determined - the readings will be available on Brightspace	
Assessment	writing assignments: 1-Four written responses to readings and lectures written by a group of four students. Grading will be based on demonstrated capacity to understand, analyze, contextualize, and discuss architectural history, historiography and theory. 2-A short proposal for a history or theory thesis, written individually. The proposal is graded on a Pass/Fail basis. A rubric with the criteria for grading is available on the course Brightspace page. Repair options in case of an insufficient grade: -Group Reading Responses The group reading responses receive a group mark. But please be aware that the tutors have the possibility to give one group member a higher or lower grade, based on feedback on their performance from their fellow group members. In case a whole group fails the GRRs: the group will be given the chance to repair through an additional group assignment, deadline in week 3.2. The grade will be the average of the additional assignments and the regular assignments. In case an individual student fails the GRRs (by not attending, or not actively participating in the group work, for instance): there will be the possibility for a repair through an additional individual assignment. Deadline in week 3.2 -Proposal In case of a Fail for the proposal there is the possibility to repair this, with deadline in week 3.2: if you can improve your proposal to a sufficient level, based on the feedback from your tutor, the Fail can be changed into a Pass.	
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their preferred tutor via the course Brightspace page. This is on a first-come-first-serve basis.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is a MANDATORY preparation course for the thesis that will be written during the MSc2 (AR2A011 or AR2AT031).	
Period of Education	Semester 1, Quarter 2	
Concept Schedule	This course will be taught on Thursdays	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1A080	Building Engineering Studios	10
Course Coordinator	Ir. F. Adema	
Course Coordinator	M. Parravicini	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The Building Engineering Studios (BES) are strongly connected to the Architectural Studios MSc 1. The theme or method set in the Architectural Studio is guiding in the Building Engineering Studio as well. The main topic of the Building Engineering Studio is the sustainable design of the technical aspects (construction, climate and structure) in relation to the architectural aspects of the design. Under the larger umbrella of sustainability the BES studio intend to emphasize the circularity aspects of the design. Students are therefore required to analyze in groups an existing building in terms of circularity and to eventually to assess their own project under the same terms. The aim of materialisation - the process of integrating sustainable and technical features - is to develop the initial concept into an actual physical building design, in which the quality of the initial concept is reinforced and enriched through interaction with all relevant physical considerations. Physical and sustainable considerations can provide a valuable source of architectural inspiration.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSc 1 level. For the MSc 1 building engineering studios this means: The student's final design, as presented, must show that they have knowledge and understanding of: the interaction between an architectural concept and a building engineering concept development and elaboration in a sustainable way of technical aspects: construction, structure and climate, including circularity strategies and solutions. Thus the student must present reasoned solutions and demonstrate skills in incorporating the sustainable technical building design effectively in the design process as a whole.	
Education Method	Because of the strong relation between the Building Engineering and Architectural Studios, the educational method is set in cooperation between the two and therefore differs per studio. Basically, in all Building Engineering Studios several exploratory design studies and the development and elaboration of the technical building design are at the core of the project.	
Assessment	The assessment of the technical building design project will be based on different presentation means. On the one hand the presentation is dependent on the theme and method of the studio. On the other hand the presentation products have to show the content as formulated in study goal and course content. The presentation of the technical building design includes a poster presentation, the exploratory design studies, the visualisation of the concept, the elaboration of the relevant sustainable and technical aspects in detail and a reflection on the final outcome. The students are also required to prepare in groups and to deliver an analysis of an existing building in terms of circularity, and to reflect on circularity within their own project. The information regarding presentation and assessment is more specifically formulated in the course manual per each particular studio.	
Special Information	For questions please contact Mauro Parravicini (M.Parravicini@tudelft.nl). The Architecture Design Studio and Building Engineering Studios are integrated and taught during the 1st and 2nd quarter.	
Period of Education	1st Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1UA010	Urban Architecture MSc 1 Design studio	10
Course Coordinator	Ir. M.G.J. van Gelderen	
Instructor	Ir. M.G.J. van Gelderen	
Contact Hours / Week x/x/x/x	Variable, 4 till 8 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Msc1: What to keep? At the chair of Urban Architecture, we study medium-size urban sites that are inaccessible for the regular tools of urbanism, and where expectations of urban design can only be fulfilled by architecture. Adding a building thus means (re)designing an environment. The Msc1 studio opposes tabula rasa and addresses questions prior to the new design, concerning the existing structures on the site: whether to demolish (and what exactly) and how to re-use. The motives for preservation are not necessarily architectural value, but also rest-value of written-off material for social, non-profit uses and, especially, ecological value of used material in itself and in economizing the use of new material. A particular urban site must be restructured with the design of a new building, whilst maintaining selected parts of an existing building. Students are challenged to re-interpret as-found constellations, to re-integrate them into a new design concept and to enjoy contingencies. We want to make the best possible use of resources observed near the site, both socially, physically and technically. Other resources are buildings nearby and far-off, realized or only imagined, as an archive of human efforts to accommodate life. We favour programs that communicate their social role to the public realm sensitively and intervene in daily life, in urban areas where the right to the city, as Henri Lefebvre has put it, is in need of strengthening. The course is in close collaboration with Building Technology.	
Study Goals	- The student is able to present a coherent, significant, elaborated, correct and innovative design in general and on specific aspects from a medium-sizes building on MSC 1 level. - The student is able to creatively design meaningful relationships between the existing and the new. - The student develops awareness of the responsible uses of building materials	
Education Method	Excursion, both to the site and to relevant architectural projects Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 working days.	
Remarks	The Architecture Design Studio and Building Engineering Studios (AR1A080) are integrated and taught during the 1st and 2nd quarter. Both studios form one coherent whole and architecture and building engineering teachers will collaborate closely. Only students who choose the MSc 1 studio of Complex Projects or The Why Factory will follow the Architecture Design Studio in the 1st quarter and Building Engineering Studios in the 2nd quarter. These two design studios are not integrated with Building Engineering Studios.	
Period of Education	Semester 1, Q1 & 2	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 UA

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR2A011	Architectural History Thesis	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	Dr. R.J. Rutte	
Instructor	mr.dr. E. Korthals Altes	
Instructor	Dr. M.T.A. van Thoor	
Instructor	Dr. D.C. Baciù	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Drs. C.A. van Wijk	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.5	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The student: -Has completed the Q2 precursor course: Delft Lectures on Architectural History and Theory (AR1A066), in which a proposal for the thesis is prepared under the guidance of a tutor. - Has developed appropriate academic writing skills. For TU Delft BSc graduates, a finished AC3 paper should have provided them with skills in planning and developing a research project, critical and responsible use of sources, and logical argumentation. These skills will be applied and expanded during this course. - Demonstrates a general historical understanding of the architecture profession and the role of the architect in society. - Can apply broad knowledge of the history and theory of architecture and related art forms and the humanities, as well as of the social and cultural developments relevant to architectural design. Required Language skills: to successfully finish this course, the student must have appropriate English language skills. If in doubt, the student should consult the OpenSourceware made available through the following links: https://learn.saylor.org/course/view.php?id=42 https://learn.saylor.org/course/view.php?id=43 (These links lead to the English courses offered for free to all by the online Saylor Academy.) Please Note: Any issues regarding research skills or language capacities will have to be addressed before the start of this course, and will require serious commitment by the student. The language courses are extensive and the student will not be able to combine them with the normal thesis workload during the semester.	
Course Contents	The history thesis is a required independent research project in the Master 2. The choice of a topic and development of a proposal for the thesis are part of the precursor course AR1A066, in Q2. The history thesis may deal with architecture, urbanism, the visual arts, design and photography, film or literature. It provides students the opportunity to hone their research skills on a historical topic. If the focus is on architecture, the research can also be of a typological kind, for example on a particular type of building, preferably not through the centuries but concentrating on a particular period or aspect. If urbanism is the subject matter, the themes may vary from the regional to the neighborhood scale, design and decision making processes, the role of politics, theories (ranging from functionalism to morphological approaches, from programmatic aspects to ideas about the creative classes and gentrification). It may also be a topographical / territorial topic, where appropriate in combination with other aspects. Finally it can regard also the investigation of an abstract topic: rhythm, scale, theory of proportions, ornamentation, eclecticism and monumentality, etc. in which an historical point of view is dominant. Using mixed methods from archival research and oral history to close reading of visual and textual analysis students critically examine their topic, producing a substantial research paper based on a clear historical perspective. This analytical and conceptual experience forms an important complement to the design-based education of the master in architecture. Writing a history thesis offers students a unique opportunity to pursue a research on a specific topic and requires students to work independently. Building on historical knowledge and research skills gained in introductory and advanced courses, students focus on primary materials and pursue an original question. They develop a complex argument and grapple with multiple data sets and interpretations. Collective and individual meetings with tutors provide a framework for the production of an original, well written paper of about 6000-9000 words. Students need to be familiar with library catalogues and search engines. The papers are required to demonstrate superior and consistent understanding of scientific writing (i.e. footnotes, bibliography, front and back matter).	
Study Goals	Learning objectives After completion of the course the student: - Exhibits in depth knowledge regarding a specific field of study within architecture, urbanism, art, and or media, in relation to the socioeconomic and cultural context. - Is able to plan and develop a scientific research project. - Is able to develop a critical and logical argumentation from a scientific research question based on primary sources (text/images/artifacts), and present this in clear, coherent and correct written English, supported with images. - Is able to evaluate, interpret and make proper reference to available sources. - Is able to build on existing knowledge and develop new knowledge.	
Education Method	Students meet with the tutor during weekly group or individual meetings in the first five weeks of Q3. However, the majority of the time (5 EC = 140 hours in total) is spent on independent study, researching, writing and editing of the thesis.	
Literature and Study Materials	Course material on research and writing is available on the course Brightspace page.	
Assessment	The thesis paper is an individual assignment, and students receive a grade for their final thesis paper. A rubric with the criteria for grading is available on the course Brightspace page. The course structure has weekly assignments. These are not graded, but students receive feedback from the tutor to improve their work, building it up towards their final paper. This is also a way to check planning and progress. A month before the final hand in date, students submit a first draft for feedback. The final paper is checked for plagiarism with Ouriginal. Incorrect use of sources (plagiarism) is not tolerated and will be brought before the Board of Examiners. In case of failing there is the option of a repair; the deadline may be determined between the tutor and the student, but should be the first day of the next academic year at the latest. This repair, if passed, will be graded with no more than a 6. If they want a higher grade, students must enroll again for a retake of AR2A011 in Q3 of the next academic year. If the project is promising the student is asked to finish the project next Q3, if possible with same tutor; but if the student did not do a sufficient amount of work a full restart of the project in next Q3 will be necessary. Please note that in order to meet the requirements for the admission to the P2 examination, students need to have a passing grade	

	for their thesis course. That means that it is not possible to start with MSc 4 in February without having the MSc 2 thesis (history/theory) completed.
Enrolment / Application	Enrollment for this course, as for all courses, is through the BIS system. Once students have enrolled and the course is about to start, participants will be required to enroll to the group of their tutor from the precursor course (AR1A066) via the course Brightspace page.
Period of Education	3rd Quarter
Concept Schedule	This course will be taught on Monday afternoons
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT031	Architecture Theory Thesis Seminar - Thinking/Reading/Writing	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	4 (four) hours per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	As per MSc2 Architecture program requirements. This course is a required 'choice-course' equivalent to the History Thesis. A thesis course (5sects) is required for P2 registration.	
Expected prior knowledge	Students are expected to have a specific interest in architecture theory, philosophy and other areas, which includes previous reading and some research in these fields. Previous writing on theoretically driven topics is recommended, but not mandatory.	
Course Contents	Students participating in this course are expected to have written a "Theory Thesis Proposal" in the MSc1 Delft Lectures on Architectural Theory and History and enrolled in the MSc2 Arch. Theory Thesis in advance. The Architecture Theory Thesis Seminar 'Thinking Reading Writing' offers students the opportunity to engage the rich conceptual, philosophical and theoretical dimensions of architecture and its influence on culture through research on a topic of their own choice. The course is specifically designed to accompany our students along the exciting journey of their 'thought processes'. Through a series of lectures, group discussions, workshops and seminars, as well as self-study periods, the course helps our students to develop and practice the necessary skills in thinking, reading and writing to produce advanced forms of academic research. In this course students will learn to identify areas and topics of their interest and curiosity, and to frame them from perspectives that highlight their positions through a theoretical lens. It is a course that helps students "to feel and to think", "to identify and to frame", "to question and to problematise", and ultimately, "to articulate and to write" rough ideas and thoughts into proper academic research. As such, it is a preparation course for more advanced forms of 'research design' and academic writing in the Masters program and beyond. In our course students are encouraged to explore contemporary "matters of concern" from an architectural perspective. In this way our students dive into many exciting areas and fields of knowledge, from philosophy, theory, cultural studies, anthropology, neuroscience, psychology, ecology: a true constellation of possibilities! Thematically, the course is open to the proposal and interests of all our students: on how we speculate on architectural habits and the environment, on architecture and culture, on technologies and the future, on modes of being and existence, of models of design, aesthetics, perception and ethics, on space and time, of atmospheres and politics, and many other phenomena. Ultimately, students in our course will write an academic "thesis essay" in which they will convey the development of their thoughts and research.	
Study Goals	Upon completion of this theory course the participants will: - have a solid knowledge-base on architecture culture -its theories, methods, techniques- and its relations to other relevant disciplines - will have acquired understanding of the societal, cultural, technological, environmental and ethical dimensions and implications of conducting research on architecture, contributing to discussions concerning complex matters related to the built (and un-built) environment. - have acquired a systematic approach to academic research and practice, using appropriate theories, methods and techniques to critically investigate and analyse existing, newly proposed and self-formulated architectural ideas. - have acquired knowledge and practice on academic research and writing skills, formulating adequate questions and apply these in theoretical argumentation and the formation of discourse. - be able to critically examine and discuss existing theories, models or interpretations in the area of his or her thesis essay. - have developed an open, critical and academic attitude towards learning and the skills to continue to acquire, interpret, reflect upon, and employ new knowledge and skills independently.	
Education Method	This course is designed as a lecture-seminar course and is based on: 3 bi-weekly lectures 3 bi-weekly group seminars or thinking workshops - self-study period 1 walk-in consultation moment (prior appointment) Our education method fosters the process of research, namely, the development of specific skills and activities: reading, thinking, researching and essay writing	
Course Relations	AR1A066 (Delft Lectures on Architectural Theory and History) - required MSc1 AR2AT041 (Architecture and Philosophy) - recommended elective MSc2 AR2AT021 (Agential Materialism Design Studio) - recommended design elective MSc2 AR0570 (Worlding Theories and Concepts) - recommended elective MSc2	
Literature and Study Materials	Students are required to prepare a shortlist of references on their topic of choice. The course will provide specific reading and research venues per individual student. See course syllabus for more information.	
Prerequisites	To have accredited the following MSc1 courses: Delft Lectures on Architectural Design and Research Methods (AR1A061) Delft Lectures on Architectural History and Theory (AR1A066)	

Assessment	This course is assessed through a "Thesis Essay" (short thesis, or "werkstuk") on a topic of the student's choice. The specific characteristics of this "thesis essay" are mentioned in the course syllabus. The evaluation of the final assignment is based on the course's Rubric, available upon request. Submission of the final Thesis Essay by the stipulated deadline is a mandatory component for the accreditation of the course. Thesis essays are submitted in week 3.10, and final grades will be registered within the allowed grading and registration of the Faculty. Resist period in week 5.3 according to Faculty rules.
Enrolment / Application	Students who wish to participate in this course are kindly asked to: 1. Submit a THEORY THESIS PROPOSAL in MSC1 for the Delft Lectures in Arch. Theory and History - AR1A066 as required by the course. Please contact the coordinator of the Architecture Philosophy and Theory group (h.sohn@tudelft.nl) 2. Enrol in the Architecture Theory Thesis course during the allowed enrollment period of the Faculty. Students with known course scheduling conflicts or who are studying abroad are asked NOT to enrol in the course without contacting the coordinator in advance. Re-takers may continue working on their topics. Please contact the coordinator in advance.
Period of Education	This course is taught ONLY in QUARTER THREE weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Seminars weeks 3.7, 3.8, 3.9 & 3.10 - self-study week 3.10 - Thesis Essay due
Concept Schedule	Thursday mornings
Leerstoel	Architecture Philosophy and Theory Chair
Minimum number of participants	12
Maximum number of participants	50
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

25 ECTS Electives

Introduction 1

The Master 2 program of Architecture consists of a total of 30 credits, of which 5 credits compulsory and 25 credits free electives.

- History Thesis (AR2A011) or the Theory Thesis (AR2AT031) of 5 credits
- An approved Master 2 Architecture design project (at least 10 credits) (see list in studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576)
- Free electives as to be found in the studyguide: https://studiegids.tudelft.nl/a101_displayProgram.do?program_tree_id=21576

There are 2 possibilities for doing the Architecture Master 2 design project:

- 1 - a Master 2 Architecture design project from the 'MSc 2 design project list',
- 2 - it is also possible to participate in an (international) program of another university. For this please contact 'International Office' and Students Affairs (O&S)

The courses in this section are agreed on by the faculty Director of Education and the Master coordinator of Architecture as Architecture design projects suitable for Master 2.

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Elective Design Studios

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. Study Goals Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework. Education Method Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics Course Relations The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website. Reader A course booklet will be sent to the students before the course starts. Assessment A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. Exam Hours The end presentation will be held in week 4.10. Special Information As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting. Period of Education 4th kwarter Concept Schedule We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course. Minimum number of participants 12 Maximum number of participants 12 Course evaluation For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

Remarks	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Period of Education	The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Concept Schedule	The maximum marking period is 15 working days. Quarter 4.
Minimum number of participants	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Maximum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Course evaluation	For this course of the Master track of Urbanism the maximum number of participants is 30.
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
Special Information	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0). Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekeneken-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: - apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; - formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; - represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; - describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; - build a simplified sustainability assessment framework that consists of three indicators; - design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; - assess the impact of your EIS using your sustainability framework; - to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 ects experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 ects studio setting, while AR0196 offers a 5 ects lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge. The course programme consists of lectures, workshops and individual mentoring.	
Study Goals	Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
Course Contents	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
Study Goals	The course has 3 parts:	
	1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality.	
	2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Education Method Literature and Study Materials	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Assessment	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Period of Education Course evaluation	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75 .	
Period of Education Course evaluation	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
	Quarter 1 For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 UA

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3UA100	Urban Architecture Graduation Studio	55
Course Coordinator	Prof.ir. P.E.L.J.C. Vermeulen	
Course Coordinator	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & 2	
Course Contents	Msc3: Where to build? At the chair of Urban Architecture, we study medium-size urban sites that are inaccessible for the regular tools of urbanism, and where expectations of urban design can only be fulfilled by architecture. Adding a building thus means (re)designing an environment. This graduation studio aims at critically assessing a given situation, in order to either complete or re-direct it. The position of the building(s) relative to existing structures, its role in delineating the private and the public realm, awareness of scale and material relationships are issues that are addressed in the studio. We want to make the best possible use of resources observed near the site, both socially, physically and technically. Other resources are buildings nearby and far-off, realized or only imagined, as an archive of human efforts to accommodate life. Therefore, whether to demolish and how to re-use are considered vital questions. We favour programs that communicate their social role to the public realm sensitively and intervene in daily life, in urban areas where the right to the city, as Henri Lefebvre has put it, is in need of strengthening. The themes and design task in the msc3 studio are focussed on an elaborate contextual research, integrating the obtained knowledge of the research course (AR3UA110) the development of an urban strategy, an appropriate massing, typologies and plans, and a conscious addressing of the urban realm. The project develops over the course of two semesters. The first semester is dedicated to a collective and individual fieldwork research as well as an urban plan. The second semester focuses on the further development of an architectural project and its spatial, material and technical components. Research and design are treated as integral components that inform and mutually reinforce each other. Emphasis will be laid upon experimental research methods and notation/documentation.	
Study Goals	Upon completion of the Master 3 & 4 studio the student can: - analyse and interpret the physical and social fabric within a given urban context in relation to architectural themes by means of different experimental research methods; - formulate strategies to strengthen urban assets on a specific site, develop an appropriate programme and implement these in a design proposal for a master plan; - integrate architectural concepts derived from the research into the design of a complex architectural proposal that deals with different aspects of the research outcomes in a coherent way; - develop an architectural design that is satisfying aesthetically and answers to technical / functional requirements as well as integrates a coherent structural, material and climatic design; - acquire and incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. - acquire and incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. - acquire and incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing.	
Education Method	Excursion, both to the site and to relevant architectural projects. Group work and individual work in the studio Independent design & self-study.	
Assessment	Research & design examination. This examination is an active assessment, during and at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. All relevant studies, text(s) and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals and the general P2 and P5 criteria. The assessment criteria will be provided by the tutors at the beginning of the semester, depending on the study and tasks. A text will form part of the content to be assessed.	
Special Information	The maximum marking period is 10 working days.	
Period of Education	Semester 1&2	
Concept Schedule	Friday morning and afternoon	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Explore Lab

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 Expl Lab

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3EX115	Explore Lab Graduation	55
Course Coordinator	Ir. R.R.J. van de Pas	
Course Coordinator	Ir. E.J.G.C. van Dooren	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Research driven graduation-laboratory for exploration of fascinations in the profession of architecture. Explore Lab is an exceptional thesis laboratory for students with a unique fascination which cannot be explored in any of the regular thesis labs. This opportunity is only for highly motivated students with a specific interest. Explore Lab is a student driven studio. The first Explore Lab started in September 2005 as a student initiative. The students of the Explore Lab themselves are responsible for the program and the agenda through their graduation year. The lab coordinator and the project docents are responsible for meeting the goals that are required for examination. While engaged in their own fascinations, the explorers work together to design their own curriculum including workshops, lectures, excursions and visiting critics. Unique fascinations require unique mentor teams explorers select their mentors under the guidance of the lab coordinators. The research to be carried out will be complementary to the students graduation design project, and both will derive from the students individual fascinations. The research will produce a stand-alone product. It must deliver findings which are of use to others, students, professionals or researchers. The design is a specific project, born from the student's fascination.	
Study Goals	The study goals are dependent on and consistent with the graduation track of the individual student. For architecture students, the study goals are as follows. The student: - can carry out a substantial research project and present research results in a manner appropriate to the form of research. - has skill in architectural design satisfying both aesthetic and technical/functional requirements - has appropriate knowledge of urban and spatial planning and associated techniques - has insight into the relationship between people and architectural constructions and between architectural constructions and their environment, as well as the need to gear architectural constructions and the spaces between them to human needs and standards - has appropriate knowledge of the industries, organisations and procedures that play a role in the conversion of designs into buildings and the incorporation of plans into urban and spatial planning - has appropriate knowledge of and insight into decision-making procedures and processes The graduation report demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	Explore Lab is a student driven graduation laboratory. The educational method is therefore to be developed by the students in conversation with each other and the Explore Lab coordinators. The assumption is that studio instruction will be the primary teaching method. Students will guide their own studies and determine their own learning styles.	
Assessment	The assessment will be of both the research and design process. The student is free to determine the nature of the research and design process in discussion with their mentors, as long as the result is meeting the goals that are required for examination.	
Enrolment / Application	For enrolment please contact coordinator, Roel van de Pas (R.R.J.vandePas@tudelft.nl). Some weeks before the general date of registration for courses and projects students get an email to inform students about the Explore Lab enrolment procedure. Only ca. 25-30 students can actually participate in the Explore Lab per period. Students will be accepted on presenting a strong fascination that cannot be explored in another studio and which meets the architectural graduation requirements. It is also essential that students have finished all prior courses and projects and to have collected sufficient credits.	
Period of Education	1 year	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Cross Domain City of the Future

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 CS

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3CS100	Graduation Studio Cross Domain City of the Future	55
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours/week (1.1-1.8 & 2.1-2.5) 8 hours/week (2.6-2.10, 3.1-3.10, 4.1-4.10)	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 & MSc2	
Course Contents	This Graduation Lab is a special thesis laboratory for students that would like to develop their own (design and/ or research) fascinations in a multidisciplinary setting. Students with different backgrounds and from different TU Delft MSc tracks will work on common challenges sharing insights, approaches and methodologies proper of their own disciplines. While doing that, students work together to enrich their own graduation pathway, setting up and developing workshops, lectures, excursions and visiting critics. The students of this Graduation Lab are responsible for the program and the agenda through the thesis period. They are expected to work together as much as they possibly can, because mutual critic and collaboration is one of the important means of education in this lab. The seminar Cross Domain City of the Future (AR3CS021), shared with students coming from other MSc tracks, is included in this course - no need to enroll for that.	
Study Goals	The study goals are dependent on and consistent with the graduation track of the individual student. For MSc Architecture students, the study goals are as follows. The student: - has skill in architectural design satisfying both aesthetic and technical/functional requirements - has appropriate knowledge of urban and spatial planning and associated techniques - has insight into the relationship between people and architectural constructions and between architectural constructions and their environment, as well as the need to gear architectural constructions and the spaces between them to human needs and standards - has appropriate knowledge of the industries, organisations and procedures that play a role in the conversion of designs into buildings and the incorporation of plans into urban and spatial planning - has appropriate knowledge of and insight into decision-making procedures and processes The graduation report demonstrates the students ability to employ moral sensibility, analysis, creativity, judgment, decision and argumentation skills regarding Architectural ethics and his/her future role as architect. The individual graduation report should not only contain an elaboration regarding the Graduation Projects societal and disciplinary relevance, but has to also address design ethics and the way in which intercultural issues were addressed in the graduation project.	
Education Method	This is a student driven graduation laboratory. The educational method is therefore to be developed by the students in conversation with each other and the coordinators. The assumption is that studio instruction will be the primary teaching method. Students will guide their own studies and determine their own learning styles.	
Assessment	(P1) Interim mid-term evaluation (week 1.10) (P2) Go-No go assessment (weeks 2.9/2.10) (P3) Progress review (week 3.8/3.10) (P4) Go-No go assessment (week 4.4/4.5) (P5) Public final presentation and diploma ceremony (week 4.9/4.10) P2 is a Go-No go examination that determines the students ability to to proceed to the final part of the studio. P4 is a Go-No go examination that determines the student's ability to proceed to the public graduation presentation and ceremony (P5). The actual review weeks may be subject to change in accordance with the academic calendar. All the criteria for the evaluation are explained in the graduation manual and will determine the final mark obtained for the entire graduation studio.	
Period of Education	Fall Semester MSc3 (Q1, Q2), Spring Semester MSc4 (Q3, Q4)	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Resilient Rotterdam Graduation Studio - Veldacademie

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 and 4 Resilient Rotterdam Studio

AR3A010	Research Plan	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr.ir. R.A. Gorny	
Instructor	Dr.ir. S. Kousoulas	
Instructor	Prof.dr.ing. C.M. Hein	
Instructor	A. Stanii	
Instructor	Dr. R.J. Lee	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	The AR3A010 Research Plan course aims to help MSc3/4 students to improve their critical and analytical skills necessary to design a sound theoretical and methodological research framework through which to engage their graduation projects. The course will help students reflect on the methodologies, theories and ethics of their graduation research while supporting them to develop the necessary skills to successfully develop the research component of their individual graduation projects. Students will learn to distinguish diverse methods and approaches for research in the field of architecture. They will learn how to develop a research proposal, including a the development of a problem statement, the choice of appropriate methods and developing a frame of reference, and a reflection on the relevance of their research. They will learn how to design and formulate their research plan.	
Study Goals	Students will be able to -Develop a research plan from inception to final report -Discuss ethical questions of selected research methods -Distinguish between qualitative, quantitative and speculative research and select appropriate tools -Develop research questions and objectives to be capable of transposing their research trajectories and methods to relevant design problems.	
Education Method	The course takes place in the first semester of the graduation studio (MSc3). The active sessions will be scheduled in the first quarter, the self-study on the assignment take can continue with the studio research mentor in the second quarter, depending on the research trajectory in the studio. In weeks 1, 2, and 3 of each semester, the course will offer Plenary Lectures offered by the chairs of Methods, History, and Theory. These lectures will provide assistance in setting up a research plan, distinguishing research methods in architecture, framing the work theoretically and historically, and formulating a problem statement. In the following weeks, studio-based meetings are held in which the research approaches of each group are further explored. The sessions will help the student to develop a draft of their Research Plan in advance of the studios P1 presentation. The final Research Plan should be submitted latest 4 weeks before the P2.	
Assessment	- In week 4.5 the individual Research Plan (2000-2500 words) will be delivered to the Studio research mentor and the assigned Research Plan instructor. The assessment teams are defined based upon the research connections between the studio and the focus of the respective academic chairs of Methods, Theory and History. -The research mentor and Research Plan instructor together set the grades (50%-50%) for the research plan based upon the quality of the following aspects of the Research Plan: - Problem statement and research questions; - Definition of theoretical framework; - Methodological positioning and description of research methods; - Argument on relevance; - Bibliographical references; - Quality of writing; - Coherence and consistency of the Research Plan as a whole. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor.	
Period of Education	Semester 1, Q1 and Semester 2, Q3	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3RE100	Resilient Rotterdam Graduation Studio - Veldacademie	55
Course Coordinator	Dr.ir. M.J. van Dorst	
Responsible for assignments	Dr.ir. M.J. van Dorst	
Contact Hours / Week x/x/x/x	8 Hours per week	
Education Period	1 2 3 4	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	A resilient city is prepared for the unpredictable future, this counts for the physical environment and the social environment. The Physical city consist of the designed or (wo)men made environment and the natural environment. The social environment consist of individuals, all the social layers (formal and informal), a cultural layer, governance, etcetera. Resilience is a dynamic concept. So, for a designer of a resilient city (architect, urban and landscape architecture) there are two aspects key. First the knowledge and now-how on how your design facilitates the social environment. Secondly, the awareness of the temporal aspects, as the city of today is not the city of tomorrow. In this graduation studio, designing for a resilient Rotterdam is very much content based by the follow elements: - It is a Veldacademie studio, in which students will be working on site within a local network. Getting close to reality and the willingness to make a change in the world is dominant in the approach. - This studio is part of the TU Delft, Erasmus University and Erasmus Medical Centre convergance program titled Resilient Delta. This network will be actively used so students can get in touch with researchers, the municipality and other actors that work on (social) resilience in Rotterdam. - The knowledge developed in the studio is brought into the debate about actual city development, through publications, events and conferences in which students play an active role. The studio is design driven, yet the complexity of social-spatial reality of Rotterdam also imply an in-depth research. The research will facilitate the design and has a multi-method approach, consisting of mapping, interviews, observations. A resilient city is prepared for the unpredictable future, this counts for the physical environment and the social environment. The Physical city consist of the designed or (wo)men made environment and the natural environment. The social environment consist of individuals, all the social layers (formal and informal), a cultural layer, governance, etcetera. Resilience is a dynamic concept. So, for a designer of a resilient city (architect, urban and landscape architecture) there are two aspects key. First the knowledge and now-how on how your design facilitates the social environment. Secondly, the awareness of the temporal aspects, as the city of today is not the city of tomorrow. In this graduation studio, designing for a resilient Rotterdam is very much content based by the follow elements: - It is a Veldacademie studio, in which students will be working on site within a local network. Getting close to reality and the willingness to make a change in the world is dominant in the approach. - This studio is part of the TU Delft, Erasmus University and Erasmus Medical Centre convergance program titled Resilient Delta. This network will be actively used so students can get in touch with researchers, the municipality and other actors that work on (social) resilience in Rotterdam. - The knowledge developed in the studio is brought into the debate about actual city development, through publications, events and conferences in which students play an active role. The studio is design driven, yet the complexity of social-spatial reality of Rotterdam also imply an in-depth research. The research will facilitate the design and has a multi-method approach, consisting of mapping, interviews, observations.	
Study Goals	By the end of this course: - The student can do a social-spatial research for a complex assignment; - He or she can present to and discuss with local professionals and citizens on what spatial interventions are needed to come to a resilient city; - Gained knowledge is applied in a design in which an extensive functional and technical elaboration has been realized at all levels, from urban to detailing; - The student delivers a design that lives up to the end terms of his or her MSc tracks.	
Education Method	There will be a set of lectures by local actors, a workshop on fieldwork and interactive sessions with different teachers involved (form TU Delft and the Veldacademie). Students work in an interdisciplinary group, on site	
Assessment	Design process and product will be assessed. Design process on the way how one does inquiry in combination with designing and to what extent the student is capable to incorporate new information. The product is assessed on the relation between design products and the given context (social and physical).	
Period of Education	Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

variant Building Technology

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 Building Technology

AR1B011	Bucky Lab Design	10
Course Coordinator	Dr.ing. M. Bilow	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Dr.ing. M. Bilow	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The focus of the semester is an innovative building construction/product or facade design/product for the built environment, this may be a part of a building, a pavillion or a facade. The task is a building component or a facade product in which all the important technical and architectural aspects of a building are integrated. The course is hosted by the chair of Building Product Innovation - therefore the focus will be on architectural Product Design - this is not a general Building Construction course. The first three weeks students individually research and analyse the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. After the concept and design & engineering phase we will set up our mobile workshop and will build the concepts in a full scale prototype. The course will be finalised by a review of the prototype - possible improvements and changes will be summarised in the report as also the overall process of the semester. The final product will be presented in a public presentation.	
Study Goals	The student has a basic understanding of the field of science of Building Technology. The student is able to design a building component understanding the relation between design, society, realisation, materialisation and functioning. The student is able to test and evaluate the design on functioning and performances. The student is competent in collaborating with fellow students. The student has gained the skills to build concept and presentation models in order to help visualising their concept and ideas to a non expert audience. The student is able to visualise a complex system in an infographic.	
Education Method	We follow the methods of Design by research - as also Problem-based learning (PBL) which is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We stimulate to experiment, brainstorm and out of the box thinking. Failure is always an option in order to find the best possible solution to the problem. We strive for a high pace of idea creation in the first phase, try everything in order to be sure that the chosen concept is really the best.	
Computer Use	The students are asked to use a wide range of software on their personal computers.	
Assessment	Individual report of innovative concept and reports in team of 3-4 students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The building / making of a full scale 1:1 prototype during the building weeks. Design 80 % Model 20% Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment (or get an oral exam) within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days. First three weeks innovative concepts individually, remaining period teams of 3 to 4 students collaborating to final design. We will ask to sketch by hand till the elevator pitch / mid term presentation. We will work in the mobile Bucky Lab workshop under guidance of experts. Safety is key therefore we ask for personal safety shoes that at least suffice to a S1 level - which consist of a toe cap. For future use on building sites we highly recommend S2 level which are waterproof and therefore better suited for any kind of building site inspections.	
Period of Education	Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B022	Climate Design	5
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Dr. R.M.J. Bokel	
Instructor	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. M.J. Tenpierik	
Instructor	Dr. L.C.M. Itard	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Ir. C.J. Janssen	
Instructor	Dr. E. Brembilla	
Instructor	Dr.ir. A.C. Boerstra	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	6 hours per week in week 1.1/1.3/1.5/1.7 4 hours per week in week 1.2/1.4/1.6/1.8 3 hours in week 1.10 (exam)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Basically the Bachelor Bouwkunde is the starting point for this course and we expect students to remember (or being able to catch up) most of what was taught on building physics, building services and sustainable design in this program. We aim to make self-assessments for students available at the start or during the programme, which will direct them to study material to catch up on the basic knowledge on climate design. Please note: make sure that you are informed in advance about the expected prior knowledge. The main topics as basic knowledge are addressed below: - Knowledge of Physics and Mathematics at Dutch pre-university diploma level (VWO natuur en techniek) or equivalent: A-Level Mathematics and A-Level Physics. - Basic knowledge of Sustainable Design, Climate Design, Indoor Environment, Building Physics and Building Services: - o Sustainable design. Refer to: Jón Kristinsson, Integrated Sustainable Design, Delft/Deventer 2012. This book will (soon) be available on www.kristinssonarchitecten.nl - o indoor air quality, heating, ventilating and air-conditioning systems. Refer to: All you need to know about air conditioning the how and why of heating, ventilating and air conditioning in a nutshell by P.M. Bluyssen (Delft Academic Press) and All you need to know about indoor air A simple guide for educating yourself to improve your indoor environment by P.M. Bluyssen (Delft Academic Press). - o Daylight, thermal comfort and sound. Refer to: "Building Physics" by A.C. van der Linden (2nd edition, English), chapters 4, 5 and 10. This book will be available on https://klimapedia.nl/leerbundel/basics-of-building-physics/. - o Heat and moisture transfer, ventilation and solar radiation. Refer to: "Building Physics" by A.C. van der Linden (2nd edition, English), chapters 1, 2, 6 and 7. This book will be available on https://klimapedia.nl/leerbundel/basics-of-building-physics/.	
Course Contents	This course will give a full overview of health, comfort, building physics, HVAC systems and sustainable climate design necessary for Building Technology students. The course focusses on content knowledge primarily, linking the research-focused Research and Methodology course, the design&build-focused Bucky Lab Design course and the more specialised climate design courses offered during the MSc2 and on.	
Study Goals	The student will be able to: - describe and explain the design performance indicators for sustainable, healthy, and comfortable buildings and urban environments; - explain the various types of energy systems at building, neighbourhood and city level including their effects on a sustainable development; - optimise the energy performance and Indoor Environmental Quality (IEQ) of buildings and building designs; - make a conceptual design of HVAC systems in order to optimise energy and IEQ performances; accounting for the interactions and relations between occupant, indoor environment, energy systems and building.	
Education Method	As didactic means this course will combine lectures, demonstrations, simulations, exercises, lab tests, as well as excursions. Students work individually and sometimes in groups.	
Literature and Study Materials	Literature and Study Materials will be made known prior to the start of the course in Brightspace.	
Assessment	Written examination. One retake opportunity in the academic year for the written examination.	
Period of Education	Quarter 1 of the first semester	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B023	Sustainable Architectural Materials and Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Dr.ir. F.A. Veer	
Responsible for assignments	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	5 hours a week from week 1.1 - 1.7 3 hours week 1.9 (exam)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	Knowledge of physics and chemistry at Dutch pre-university diploma level (VWO natuur en techniek), equivalent to: - A-Level Chemistry - Core Text Fourth Edition 4th Edition - A Level Physics Fourth Edition 4th Edition Knowledge of Structural Mechanics: Statics and equilibrium of forces, centroids and moment of inertia, reaction forces of statically indeterminate beams, bending and shear force diagrams, stresses due to normal force, bending moments and shear forces, deformations of beams, buckling of columns. Refer to: "Mechanics of Materials" by J.M. Gere (9th edition, SI edition, English). Please note: make sure that you are informed in advance about the expected prior knowledge.	
Course Contents	The Sustainable Architectural Materials & Structures course equips students with the skills to develop efficient structures at conceptual stage, make appropriate materials choices and design structural elements and structural systems with these materials. This course also provides a foundation for the more advanced courses on materials and structures in subsequent parts of the BT master programme. It consists of a series of lectures and associated workshops in order to: - Cover the basic principles of practical design of typical architectural structures, with applications across a range of commonly-used materials. - Establish links between materials science and structural mechanics (taught in bachelor courses) towards the practical goal of designing safe and sustainable structural systems. - Demonstrate the importance of appropriate design decisions on materials choice and structural form at concept stage and the use of simple effective tools to generate efficient design options. - Provide the knowledge and tools required to evaluate the safety and sustainability of structural materials and systems.	
Study Goals	On successful completion of this course students should be able to: - Understanding the principles of conceptual structural design and fundamental material science. - Create viable early-design-stage options of structural forms and material choices for given realistic scenarios. - Applying simple hand calculations and simple supporting software to support building design choices on materials and structural form. - Evaluate the efficiency and appropriateness of the different options of materials choices and structural forms for building design. - Design safe and sustainable structural elements and structural systems underpinned by more detailed calculations.	
Education Method	Lectures and associated workshops.	
Literature and Study Materials	Study material will be provided on Brightspace and www.restruct-group.tudelft.nl	
Assessment	Written examination One retake opportunity in the academic year	
Period of Education	Quarter 1 of semester 1	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B024	Introduction to Computational Design	5
Course Coordinator	Dr. S. Asut	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Prof.dr.ir. I.S. Sariyildiz	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Asut	
Instructor	Dr. A. Rafiee	
Responsible for assignments	Prof.dr.ir. I.S. Sariyildiz	
Contact Hours / Week x/x/x/x	6 hours per week starting from week 2.1 and ending in 2.4. (half morning & afternoon) 6 hours per week starting from week 2.7 and ending in 2.9. (half morning & afternoon) 2 hours on week 2.10 (morning)	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	This course aims to introduce the uses of computational design methods within an integrated workflow. It is a hands-on design course in which the students develop skills by working on a design assignment. The course introduces the students to a series of practical workshops each of which focuses on a specific application such as gathering environmental data, developing spatial designs, implementing performance analyses and digital fabrication. They are then asked to integrate all of these applications within a comprehensive workflow to design, analyze, prototype and present a small scale architectural product. The course content is composed of the following topics: 1. Introduction to Computational Design (Computational Spatial Geometry, Algorithmic Design, Visual Programming). 2. Data and Information Models (Spatial Data Acquisition, Database and Information Models). 3. Performance Based Computational Design (Using algorithms for Performance Analysis, Design Iterations towards Rationalization). 4. Digital Materiality (CAD/CAM, Physical Computation).	
Study Goals	1. Understand the uses of digital technologies in the design and production of the built environment. 2. Understand spatial data acquisition techniques. 3. Use database and information models for design development and analysis. 4. Formulate algorithms to generate solutions to architectural design and building engineering problems. 5. Create visual programs for design development, analyses and fabrication.	
Education Method	Workshops: The students will work on the design assignments in groups and under the supervision of the tutors. Lectures: The students will join lectures which will be given on the campus. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (30 %) Design Examination - Final (50 %) Participation (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 1, Quarter 2	
Minimum number of participants	NA	
Maximum number of participants	NA	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1B032	Research and Methodology	5
Course Coordinator	T. Konstantinou	
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. P.G. Teeuw	
Instructor	Dr.ir. M.J. Tenpierik	
Responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	4 hours a week in week 2.1 / 2.4 / 2.7 till 2.10 8 hours a week in week 2.3 / 2.3	
Education Period	2	
Start Education	2	
Exam Period	none	
Course Language	English	
Course Contents	In this course students become acquainted with research methods applied in Building Technology, which deal with the issues of building design and research, not as competing activities but rather as complementary. Research is a way to generate new insights and new concepts for design in a systematic way, explore new realms and finding answers beyond the state of the art. It is the cornerstone on which all progress in our modern society is founded. Research methods show us the way how to conduct research in a systematic and reliable way. In this course you will learn how to execute a technical-scientific research framework for experimental research, simulations, surveys, case studies and research through design; and how to consistently and coherently report the results of such research. In the first part, the students learn about different research methods and apply some of them in practical exercises. Moreover, they learn to consistently and coherently report the results of such research. During the second half of this course, based on the Bucky Lab Design project and/or the other MSc1 courses and/or the student's own fascination, the students individually conduct a literature study on a chosen topic, resulting in a scientific (review) paper.	
Study Goals	After successfully completing this course the student should be able to: - explain scientific integrity and outline ethical issues when doing research. - execute and report on a technical-scientific research framework for experimental research, simulations, surveys, case studies and research by design; - analyse a selected topic, in a consistent, coherent and well-structured review paper, with proper referencing; - interpret the complex relationship between research and design as two aspects within building technology;	
Education Method	Lectures, practical workshops, group tutorings and self study.	
Literature and Study Materials	Literature and Study Materials will be made known prior to the start of the course in Brightspace.	
Assessment	Writing assignment: Review paper, report of the experiment. Repairs: A final mark of 6.0 or higher is needed for successfully completing this course. Each separate part should be graded at least with a 5.0. In case a partial grade is insufficient, students get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. Retakes: In case students fail the course, the course can be taken again in the next academic year.	
Period of Education	Quarter 2 of semester 1	
Concept Schedule	The First 4 weeks, before the building weeks of the Bucky Lab Design course, will be about research methods, the last 4 weeks will be the paper writing.	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Building Technology

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelaan	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelaan	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architecturally attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0134	Technoledge Façade Design	5
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Ir. A.C. Bergsma	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The course Facade Technology focuses on specific and changing design and research topics relevant to the facade industry and architectural practice and consists of lectures from TU Delft and experts from practice and several excursions. This course is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for architecture students and students of CiTG.	
Course Contents	The facade is one of the most technically challenging, complex and multidisciplinary parts of a building. The building envelope not only defines the appearance of the building and its architectural expression, but it also determines how the building technically functions in terms of comfort, energy performance, sustainability and safety. Besides these technical aspects, the envelope also plays an important role when it comes to investment costs and operating costs of a building. Because of the many aspects that have to be dealt with when designing and constructing a façade, the façade is one of the most complex and integrated parts of a building. Because so many disciplines are involved, from architecture to building physics, facade engineering, costs and construction, working in design teams and good cooperation is essential to create excellent, integrated façade concepts. The courses Technoledge Façade Design and Future Envelope focus on specific and changing design and research topics relevant to the façade industry and architectural practice and consist of lectures from TU Delft and experts from practice and several excursions. The course Technoledge Façade Design is one of the 2nd semester Technoledge elective courses of the Building Technology master program. The course is also open as a free elective course for students of Civil Engineering. The course Future Envelope is a free elective course for architecture students. General planning: The content of the course program is split into two main consecutive parts: Part A and Part B. Part A runs from week 1 to 5 (Tuesdays and Fridays). Part B runs from week 3.5 to 3.10 (Tuesdays and Fridays). Both Parts A and B consist of several lectures, excursions and a 4-5 weeks assignment. Content-wise these parts are interrelated during the course. Students that have enrolled for the course Technoledge Façade Design (AR0134, 5 ECTS) will have to go through both Parts A and B. The courses aim at knowledge application and design integration, as well as creation of awareness of the complexity of façades. In general, there will be several (thematic) lectures on weekly basis. Then, mostly in the afternoon, these lectures will be followed by studio work and/or company visits. During these afternoon studio sessions students can work on the assignments, under supervision of the experts and teachers involved. Part A focuses on a thorough façade analysis of an existing building making use of 3D and VR techniques. Part B focuses on a façade redesign of this analysed project with strong research focus. Lectures: A major part of the course consists of several lectures organized by the Faculty of Architecture, the Faculty of CiTG and several partners from industry. These partners are the VMRG, the KCG, several façade manufacturers, product manufacturers and engineering offices. Excursions: Lectures and studio work are combined with excursions to construction site and product manufacturers. For these excursions students will have to pay a contribution of max. 30 euros pp.	
Study Goals	The student is able to list and describe presented theory and knowledge on quality control, engineering and production processes of façade manufacturing, wind- and water tightness of facades, structural aspects and use of glass in facades. The student is able to analyze and explain different façade concepts, designs and details in terms of construction method, building physical and fire safety aspects, structural mechanisms, material behavior, climate aspects and quality. The student is able to make façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. The student is able to present his/her work using the appropriate drafting techniques and new 3D presentation and analysis techniques such as VR.	
Education Method	Design studio work: approx. 100 hours Lectures: approx. 2 - 4 hours per week Excursions: approx. 2 days Self study (with possibility for consults): approx. 72 hours Self study (without consults): approx. 40 hours.	
Literature and Study Materials	- literature according to course description - as part of the course several excursions will be planned. In order to finance transport by bus students have to pay a contribution of max. 30 euros pp. - literature according to course description on brightspace	
Assessment	- 1 facade analysis assignment (part A) - 2 design/research assignment (part B) Final grade is average result of part A and part B. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Exam Hours	no written exam	
Special Information	Arie Bergsma a.c.bergsma@tudelft.nl	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday morning (first 5 weeks also limited in the afternoon)	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0137	Technoledge Health and Comfort	5
Course Coordinator	Dr. R.M.J. Bokel	
Instructor	Dr.ir. M.J. Tenpierik	
Instructor	Prof.dr.ir. P.M. Bluyssen	
Instructor	Dr. E. Brembilla	
Instructor	Dr.ir. A.C. Boerstra	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	BT Master (two of the Technoledge courses are obligatory to graduate in the BT-track)	
Expected prior knowledge	AR1B022 (knowledge on Climate Design/ Indoor comfort) and AR1B032 (knowledge on doing measurements, statistics and report writing).	
Course Contents	A building is in theory designed to have the best possible indoor comfort. In practice, however, this is not always realised. This course focuses on indoor comfort and the associated physiological concepts. You will learn the cutting-edge theory on indoor comfort through lectures from experts. You will practice with these theories through measurements of the indoor comfort aspects: visual quality, indoor air quality, acoustical quality and thermal quality. You will apply this knowledge in an existing building to assess the quality of the indoor comfort and relate the indoor comfort to the design of the building and suggest improvements or execute a research project on indoor comfort.	
Study Goals	The student is able to: - explain the current theories on indoor air quality, visual quality, thermal quality and acoustical quality - analyse relevant scientific literature on indoor comfort and write a consistent and coherent scientific report on this topic - assess the indoor comfort of an existing building and suggest improvements or conduct a research project on indoor comfort.	
Education Method	Lectures Workshops using various simulation programs Measurements Literature study Discussions The first three/four weeks are individual work, the other weeks are group work.	
Computer Use	Simulation programs (Catt-acoustics, Design Builder, Phoenix (CFD) and Dialux/ Honeybee)	
Literature and Study Materials	Holistic perspective and Indoor air quality: 1. Bluyssen, P.M. (2015) All you need to know about indoor air, A simple guide for educating yourself to improve your indoor environment, Delft Academic Press. 2. Bluyssen, P.M. (2013), The Healthy Indoor Environment How to Assess Occupants Wellbeing in buildings, Earthscan, Routledge, London. Chapters 6, 8 Acoustic comfort: 1. Salter, Ch.M. (1998), Acoustics Architecture Engineering the Environment, William Stout Publishers, San Francisco. Chapters 2, 3, 4 and 6 2. Nijs, L. and Vries, D. de (2005), The Young Architects Guide to Room Acoustics, Acoustical Science and Technology 26 (2): 229-232. 3. Knowledge Base Building Physics, module A11- Speech intelligibility Thermal comfort: 1. Nicol, F., M. Humphreys and S. Roaf (2012), Adaptive thermal comfort Principles and practice, Routledge, New York. Chapters 1,2,3,4, 5 and 8 - or 2. Stanley Kurvers and Joe Leyten, Indoor Climate and Adaptive Thermal Comfort (in press, Dutch version on Klimapedia: https://klimapedia.nl/publicaties/adaptief-thermisch-comfort/) Chapters 1, 2, 4, 8, 10 - and 3. Parsons, K. (2003) Human thermal environments. Oxford: Blackwell (digitally available through TUDelft library) Chapter 2: The human heat balance equation and the thermal audit Visual comfort: 1. Baker, N., K. Steemers (2002), Daylight design of buildings A handbook for architects and engineers, Taylor & Francis Ltd, London. Chapter 10: Daylight, comfort and health	
Assessment	Written examination of the four different indoor comfort aspects (thermal comfort, acoustical comfort, indoor air quality and visual comfort). This consists of 4 individual homework assignments in the first 3/4 weeks graded pass/fail. Writing assignment (final report) supplemented with an oral presentation, group work. In case of an insufficient grade, students will have the opportunity to repair the assignment within a period of 2 weeks from the announcement of the result, with a maximum grade of 6.0. A retake is only possible in the next study year.	
Special Information	The maximum marking period is 15 work days.	
Elective	Yes	
Tags	Adventurous Fieldwork Group work Involved Modelling Small groups	
Period of Education	Quarter	
Concept Schedule	Monday (lectures and presentations) and Wednesday morning (consults/ workshops)	
Leerstoel	All chairs of the section Energy&Climate Design.	
Maximum number of participants	24	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0138	Technoledge Design Informatics	5
Course Coordinator	Dr. S. Asut	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	Ir. F.J. Gouwetor	
Instructor	Dr. S. Asut	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in 3.4. (afternoon) 6 hours per week starting from week 3.5 and ending in 3.8. (half morning & afternoon) 2 hours per week starting from week 3.9 and ending in 3.10. (afternoon)	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The students are expected to have basic skills in parametric modelling and visual programming with Rhino/Grasshopper. The students who have successfully completed the Introduction to Computational Design course (BT MSc 1) should already have the expected prior knowledge and skills.	
Course Contents	This course aims to introduce digital and robotic fabrication technologies in an integrated computational design to production workflow. It is a hands-on design and build course. The students are introduced with a series of computational design methods and digital fabrication technologies. They develop know-how and skills on these methods and technologies by practicing them in a small scale architectural design assignment. The focus of the assignment is primarily on geometry and materiality. They are asked to develop a design by using computational methods and to prototype it by using an industrial robot arm. The students of this course will have access to Lama (Laboratory for Additive Manufacturing in Architecture) to use the robot arm and other equipment here for the course activities. The course content is composed of the following topics: 1. Visual Programming in Architectural Design (Spatial Geometry in Digital Modelling, Parametric Modelling, Form Finding). 2.Design to Production through Digital Technologies (Generating Fabrication Data through Parametric Models, Organizing and Analyzing Data through Spreadsheets, Using a Scripting Language for Fabrication). 3. Fabrication with Industrial Robots (Parametric Robot Programming, Industrial Robot Programming and Simulation, Industrial Robot Operation)	
Study Goals	1. Develop an architectural design by using parametric modelling and visual programming methods. 2. Create a robot program by using parametric and offline robot programming methods. 3. Build a spatial prototype by using an industrial robot arm. 4. Create a robot simulation. 5. Use a computer script to extract, transfer and organize data.	
Education Method	Workshops: The students will work on the assignment in groups and under the supervision of the tutors. Self-paced Tutorials: The students will follow custom made online learning materials such as video tutorials and manuals. Laboratory: The students will exercise robotic fabrication at the laboratory during scheduled self-study times. Self-study: The students will study individually or in groups to work on the course exercises.	
Assessment	Design Examination - Midterm (20 %) Design Examination - Final (60 %) Participation / Individual Contribution in the Team Work (20 %) Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Semester 2, Quarter 3	
Concept Schedule	The workshops take place on Thursday afternoons. The students are also expected to follow activities in the lab in addition to the workshops. They are responsible for scheduling these activities within each team and to solve any possible scheduling conflicts that may affect the other courses they follow.	
Leerstoel	Design Informatics	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

	Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)
Period of Education	Quarter
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Free Electives (5 EC)

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based design study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course; the student is able to: - collaborate in a team with other students - work on a joint design of a specific (building) design project - integrate various aspects of sustainability into the design of the project - elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project. Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolitanization of European and regional systems: Wust s, et al 2007Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: - Formulate a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Identify and discuss the tension between public and private development in infrastructures and environments. - Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical - Reflection into the topic of research in relation to the place of work - Late delivery - Other reasons to add/deduct - Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with wording dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Technicities design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSC first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays: global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs: corporate or commercial, as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on ones understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,	
	- spatial data modelling (vector and raster spatial models),	
	- geo-data manipulation (editing, digitizing, importing, converting, etc.),	
	- overview of spatial analysis operations,	
	- production of interpretable output (e.g. maps),	
	- fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures),	
	- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),	
	- basic algorithms for raster (encoding, quad trees, map algebra),	
	- networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;	
	5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems	
	- Slides of the lectures (available on Brightspace)	
	- Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied:	
	1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.	
	AND	
	2. The grade of the final exam is ≥ 5.75 .	
	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 Building Technology

AR3B025	Building Technology Graduation Studio	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Dr.ir. T. Klein	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	1 hour per week	
Education Period	2 4	
Start Education	2 4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	MSc1 Building Technology (Bucky Lab Design and MSc1 courses), Technoledge courses, EXTREME technology or MEGA (or approved equivalent), CORE or User-centred Sustainability Studio.	
Course Contents	The Building Technology Graduation Studio continues the build-up of Bucky Lab Design and the BT MSc1 courses, the Technoledge courses, EXTREME technology or MEGA (or approved equivalent), and CORE or User-centred Sustainability Studio to the student's own graduation project in the Building Technology master track. In the graduation studio students work (preferably in groups) on their individual graduation project comprising a preparative technical-scientific study (until the P2, this first part) and design, design by research or research by design (after the P2, the second part of the graduation studio). The studio intends to be in strong coherence with themes from the research programme of the AE+T department, so students can benefit from and contribute to the research activities by staff of the AE+T department. While the graduation project's subject can relate to any of the disciplines represented by the Building Technology master track, emphasis lies on sustainability-related topics of structural design, façade & product design, climate design and design informatics. These are interlinked. The student will be assigned to two supervisors from different 'colours' of AE+T (Structural Design / Façade & Product Design / Climate Design / Design Informatics), the student projects content as well as design and research is reflected by these two supervisors. Limited consultations with other staff members might be arranged if that's desired thematically.	
Study Goals	The student: - is capable of delivering innovative contributions toward the development of sustainable structural, façade, product, climate and computational design, and to technical-scientific research in these areas; - has insight in the profession and competences of structural, façade, product, climate or computational designers and their societal and professional role and ethical responsibilities in sustainable development of the built environment; - is able to present process and result in a clear and systematic way, reflect on process and work, and argument in a scientific way.	
Education Method	Supervised self study, with possibly intermediate studio workshops	
Assessment	P1 First ideas of the graduation project, conceptual research framework, first literature study results P2 Definite research framework, literature and desktop research results, outline for the design research	
Special Information	See the current Graduation Manual AUBS for more details and for the assessment criteria. On set conditions, Building Technology students have the possibility to carry out their graduation research project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/forms. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	Quarter	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory Choice (choose 1 project)

AR3B012	CORE	15
Course Coordinator	Dr. S. Asut	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Prof.dr.ir. I.S. Sariyildiz	
Education Period	1	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	CORE (COMputational REpertoire for Architectural Design and Engineering) is a design studio course. It primarily focuses on Design Computation in an architectural and structural design context, with emphasis on building technologies. It addresses various scales within the built environment, ranging from urban to building products, by paying attention to environmental, social, cultural, and ethical aspects. The course setup involves and integrates 5 modules as; 1. Design Studio 2. Programming 3. Materials & Structures 4. Design of Construction 5. Tools, Methods, and Inspirations	
Study Goals	LO1: Integrate architectural design, structural design and computer science disciplines to develop planning and design solutions for the built environment. LO2: Apply computer programming towards developing solutions for spatial design, planning, analysis, and construction design. LO3: Analyze an urban design and planning strategy to address societal needs and sustainability demands in the given context. LO4: Design an architectural project with construction solutions, addressing the needs on different scales ranging from urban to product levels. LO5: Create a structural design by analyzing and optimizing the material and structural performances.	
Education Method	The studio provides: 1. Reviews, debates, and feedback on design development. 2. Lectures, workshops, and consultations on computer programming. 3. Lectures, workshops, and consultations on structural design. 4. Consultations on construction design. 5. Lectures and debates on various tools and methods, as well as inspirational topics.	
Assessment	The students work in teams to design and submit an architectural project which addresses the 5 tracks of the studio. The course includes formative (ungraded) and summative (graded) assessments. Formative assessments provide feedback on the work in progress. They help the students prepare for the final assessment. They are performed during: 1. Design Studio meetings (as feedback on design development and the design of construction) 2. Proposal presentation (as feedback on the proposed concept project) 3. Mid-term presentation (as feedback on the comprehensive project, prior to final presentation) 4. Programming workshops (as feedback on programming skills) 5. Structural design workshops (as feedback on topics related to structural design and materials) The only summative assessment is performed on the final presentation and the submission at the end of the course. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	

AR3B015	User-centred Sustainability Studio	15
Course Coordinator	Dr. C.L. Martin	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Dr. C.L. Martin	
Contact Hours / Week x/x/x/x	Approximately: 6 hours per week in week 1.1 / 1.2 / 1.5 till 1.9 40 hours a week in week 1.3 / 1.4 (or similar in quarter 3, week 3.1 and so on)	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	The successful completion of MSc1 Bucky Lab, BT MSc1 courses and MSc2 EXTREME technology or MEGA (or approved equivalent) is required before entry onto the User-centred Sustainability Studio can be approved. In addition, Building Technology students are strongly advised to have completed two elective courses (2 x 5 ECTS) of Technoledge in the first quarter of the MSc2.	
Parts	3 Key elements of the 15 ects course: 1. Briefing: Group site analysis, background research & readings. 2. Intervention: Onsite workshop. 2-week intensive collaborative group studio. 3. Elaboration: Group finish, detail realization & report submission.	
Course Contents	After having focused on technology in the Bucky Lab and on integrated design in EXTREME / MEGA, the aims and objectives of the User-centred Sustainability Studio are as follows: Aims: To meet societal challenges and inequalities in the urban built environment by promoting well-being, health and carbon neutral lifestyles in an era of unprecedented global uncertainty. By engaging with diverse city stakeholders, real urban contexts, global trends and future scenarios, within student groups develop design solutions that are greater than the sum of their technological parts. Ensuring that societal value is revealed at various city scales. Objectives: To collaborate effectively in trans-disciplinary groups of students and city stakeholders. To analyse the urban context and local characteristics (climatic, historic, socio-cultural and technical) of a innovative intervention assignment and to communicate the process in a suitable/effective format i.e. text, diagrams, maps, images and film etc... To develop a position in relation to future urban conditions and sustainability. To technologically respond to local circumstances with contextually appropriate interventions / engagement initiatives.	
Course Contents Continuation	Elaborate description of what students will learn (attainment): 1. Takes account of the temporal and the social context. Students to be competent in acquiring local knowledge of context, daily rhythms and rituals, lifestyle, aspirations and be competent in working alongside local inhabitants to reveal the needs of communities. The student has a critical attitude in what it means to innovate sustainably. 2. Is competent in co-operating and communicating. Students to be competent in co-operating with local stakeholder and other group members, gaining trust and feedback. Students to be competent in bridging the communication and knowledge gap between local city stakeholders and the scientific community. 3. Possesses basic intellectual skills. Students to be competent in contributing to discussions on how sustainable strategies are multifaceted and are needed at every scale. 4. Is competent in conducting research. Students to be competent in methodological research into what, why and for whom you are designing in an era of climate change. 5. Is competent in designing and/or innovating. 6. Has a scientific approach. Students to be competent in using appropriate theories, methods and/or (modelling) techniques to critically investigate and analyse existing, newly proposed and self-formulated projects.	
Study Goals	The student: - is able to design a coherent, significant, elaborated, contextually responsive and innovative urban intervention - on MSc 3 level - is able to develop a group proposition in relation to the local characteristics, sustainable futures and development, societal and ethical context. - is able to underpin the design decisions by scientific research at MSc3 level - Is able to collaborate effectively in trans-disciplinary groups of students and city stakeholders in order to communicate coherently often complex and integrated propositions to peers and local stakeholders.	
Education Method	Educational Method The User-centred Sustainability Studio aim is to meet societal & environmental challenges of urban neighbourhoods through contextual responsiveness, meaningful engagement & technological knowledge. Students are to collaborate effectively in groups to communicate sustainable innovative interventions to an audience of participating experts and local stakeholders. Studies begin by analysing the local urban context (climatic, historic, socio-cultural and technical) and global theory relating to sustainable city development. Group work is then to be elaborated within those groups to emphasize a position relating to future urban conditions, form and sustainability. The onsite workshop (Intervention) can in some cases be based abroad, as a consequence students are expected to financially contribute to this excursion. In the acquisition of self-funding a level of pro-activeness is required (amounting to around 500 euro maximum).	
Course Relations	The User-centred Sustainability Studio Studio forms an integrated part of the Building Technology master track and is aligned with previously completed consecutive courses (MSc1 Bucky Lab, MT MSc1 courses, MSc2 Technoledge and MSc2 EXTREME technology/MEGA) and with the MSc3/4 Building Technology Graduation Studio that immediately follows.	
Assessment	Design examination: In regards to the learning goals, site data is to be group collated, then group elaborated to form a design argument / city	

	engagement initiative. This process facilitating the differentiation of grades. Grades will be based on the following: 1. Quality and extent of the local analysis (in a preparative presentation with text, maps and images). 2. Quality of design investigation, design iterations, communication, final product/initiative and critical reflection. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5 students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. In case students fail (following the repair) the course can be taken in the next semester.
Special Information	The maximum marking period is 15 work days.
Period of Education	The course lasts for one full quarter (9 weeks). The combination of the User-centred Sustainability Studio work with other course material is strongly discouraged.
Minimum number of participants	No (However, in cases where it is deemed that insufficient student numbers are available an alternative programme may be used).
Maximum number of participants	40 (first semester); no (second semester).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 4 Building Technology

AR4B025	Building Technology Graduation Studio	30
Course Coordinator	Dr. M. Overend	
Course Coordinator	Prof.dr.ing. U. Knaack	
Course Coordinator	Dr.ir. T. Klein	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Course Coordinator	Ir. P.G. Teeuw	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	1 hour per week	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR3B025	
Course Contents	The Building Technology Graduation Studio continues the build-up of Bucky Lab Design and the BT MSc1 courses, the Technoledge courses, EXTREME technology or MEGA (or approved equivalent), and CORE or User-centred Sustainability Studio to the student's own graduation project in the Building Technology master track. In the graduation studio students work (in groups) on their individual graduation project comprising a preparative technical-scientific study (until the P2) and design, design by research or research by design (after the P2, this second part of the graduation studio). The studio intends to be in strong coherence with themes from the research programme of the AE+T department, so students can benefit from and contribute to the research activities by staff of the AE+T department. While the graduation project's subject can relate to any of the disciplines represented by the Building Technology master track, emphasis lies on sustainability-related topics of structural design, façade & product design, climate design and design informatics. These are interlinked. The student will be assigned to two supervisors from different 'colours' of AE+T (Structural Design / Façade & Product Design / Climate Design / Design Informatics), the student projects content as well as design and research is reflected by these two supervisors. Limited consultations with other staff members might be arranged if that's desired thematically.	
Study Goals	The student: - is capable of delivering innovative contributions toward the development of sustainable structural, façade, product, climate and computational design, and to technical-scientific research in these areas; - has insight in the profession and competences of structural, façade, product, climate or computational designers and their societal and professional role and ethical responsibilities in sustainable development of the built environment; - is able to present process and result in a clear and systematic way, reflect on process and work, and argument in a scientific way.	
Education Method	Supervised self study	
Assessment	P3 First design, design by research or research by design results, conceptual thesis report, plan for the remaining graduation timespan, draft reflection P4 Final design, design by research or research by design results, draft final thesis report P5 Final presentation of the design, design by research or research by design, final thesis report	
Special Information	See the current Graduation Manual AUBS for more details and for the assessment criteria. On set conditions, Building Technology students have the possibility to carry out their graduation research project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found at https://www.tudelft.nl/en/student/a-be-student-portal/practical-affairs/forms. The agreements can be signed at the secretariat of Education and Student Affairs.	
Period of Education	Semester	
Minimum number of participants	Not applicable.	
Maximum number of participants	Not applicable.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

variant Management in the Built Environment

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 Management in the Built Environment
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AR1MBE015	Research Methods 1	5
Course Coordinator	K. Qian	
Course Coordinator	Dr. J.S.C.M. Hoekstra	
Instructor	Dr. J.S.C.M. Hoekstra	
Instructor	Y. Chen	
Instructor	K. Qian	
Instructor	Prof.dr. P.W.C. Chan	
Responsible for assignments	Dr. J.S.C.M. Hoekstra	
Contact Hours / Week x/x/x/x	On average 4 to 6 hours per week, starting in week 1.1. and ending in week 1.10	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Research Methods 2 (AR3MBE010) and Research Methods 3 (AR0185)	
Expected prior knowledge	Bachelor in Architecture and the Built Environment or equivalent.	
Course Contents	In this course, students acquire methodological knowledge and practice their research design skills through designing their own research proposal on a relevant MBE topic. In this proposal, there needs to be a methodological fit between the research design components such as the research question(s), logic(s) of inquiry, theoretical framework, and methods of data collection. A data management plan and a reflection on research ethics is also included in the research proposal.	
Study Goals	After successfully completing the course, the student 1. is able to identify and compose research design components in the field of MBE; 2. is able to explain the concept of methodological fit; 3. is able to compose a basic research design in the field of MBE; 4. is able to identify typical ethical issues in research and plan a proper response; and 5. is able to plan research data management in terms of organization, storage, preservation, and sharing of collected data.	
Education Method	Lectures on the different components of a research design Interactive Workshops on various elements of the research process Working on individual assignments, under supervision of a tutor.	
Literature and Study Materials	To be announced on Brightspace. In previous years, the following book was used: Blaikie, N. and Priest, J. (2019) Designing Social research, The Logic of Anticipation, 3rd edition, Polity Press, Croydon, UK	
Assessment	Written assignments (100% of the final grade). If the final grade is lower than 5,5, there is a repair possibility in Q2.	
Tags	Ethics Research Methods	
Period of Education	1	
Minimum number of participants	25	
Maximum number of participants	120	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1MBE020	Design and Construction Management	10
Course Coordinator	Ing. R.J.G. van Warmerdam	
Course Coordinator	Dr. J.L. Heintz	
Instructor	Prof.dr. P.W.C. Chan	
Instructor	Dr. D.M. Hall	
Responsible for assignments	Dr. J.L. Heintz	
Contact Hours / Week x/x/x/x	35 hours per quarter / 5 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	Master Management in the built environment.	
Expected prior knowledge	Bachelor Architecture or equivalent	
Summary	Contemporary buildings require the participation of many different design professions, suppliers of materials and systems, construction companies working together to provide both better service and better product quality in the face of increasingly complex and demanding clients and society. Steering a project through this complex web of relationships and requirements has become a profession in its own right - Design and Construction Management. This course covers the knowledge and application of project management techniques, general management skills in the context of project initiation, architectural design, tendering and construction. It covers both the systemic side of construction management and the social economic context. Working in small groups, students will study a wide range of readings on these subjects and create a summary representing the knowledge content of the course. This knowledge will be contextualised through case-based exercises in which students will have to develop management responses to typical situations in the design and construction processes. A combination of peer evaluation, reports and a final exam will ensure that each student receives individual recognition for their work.	
Course Contents	The Design and Construction Management course will cover the range of knowledge required for overseeing these processes, including: - design and construction process and planning including the parties involved and their roles, responsibilities, positions, capabilities, and tasks; - project management concepts, methods, performance, and position; - project organisation on the demand side and the supply side; - quality management in design and construction; - procurement, tendering, and contracting; - Management, leadership styles, and other skills required to understand and deal with relational and contextual challenges.	
Study Goals	The student: - has insight into the real estate life cycle, the different interests of stakeholders, and the risks they bare in the building process and their management; - has knowledge of project management concepts and methods commonly used in design and construction; - has the ability to evaluate designs in terms of architectural quality, to seek a felicitous balance between cost and quality, and to defend such choices; - has knowledge of building project organisation forms and their relative (dis)advantages; - has knowledge of the types and flows of communication and information common to building project management; - has insight into the project management issues raised by the need to apply sustainability measures in both design and construction processes and products; - has insight and ability to motivate and lead in project management settings; - has the ability to critically reflect on the application of project management concepts in design and construction projects.	
Education Method	Students work in a matrix organisation of study and workshop groups. Study groups The study groups will function to amplify the students' ability to read and study a large range of literature relevant to the aims of the course. Each week, the students will be assigned a series of texts; each student must take a share of the texts and prepare a summary. The students then share their summaries and discuss the texts with each other in the study group. The students will then participate in the construction of a combined summary in which students will consolidate their summaries on each of the assigned texts. Workshops Workshops can take the form of role-playing or other exercises carried out by the group. The purpose of the workshop is to study the task and to determine the content of the deliverable. Cases The course will be structured around one or more cases upon which the assigned tasks will be based. Exam A closed book and open-question exam on the reading material, particularly the summary constructed by the students.	
Literature and Study Materials	Principal text: - Winch, G. (2010) Managing Construction Projects. An information processing approach. Blackwell Science. Oxford. - Additional texts: scientific publications and other texts made available via Brightspace.	
Assessment	Evaluation will be based on: - task performance: 30% - summary peer evaluations: 10% - peer evaluation of task group performance: 10% - exam: 50% The first three parts form one half of the final grade and the exam is the other half. The student passes if the average of the two grades is 6.0. A partial grade is at least 5.0.	
Permitted Materials during Tests	None.	
Special Information	The maximum marking period is 10 working days.	
Remarks	None	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1MBE025	Building Economics	5
Course Coordinator	Ir. E.H.M. Geurts	
Course Coordinator	Ir. H.W. de Wolff	
Responsible for assignments	Ir. E.H.M. Geurts	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Expected prior knowledge	The course requires some understanding of working with MS Excel. Extra instruction materials are provided, if prior experience is not in place. But students should expect to have a higher workload if no prior experience with MS Excel.	
Course Contents	This course is titled Building Economics (5ects) and deals with relevant financial and economic concepts that are essential for understanding the built environment. The courses focusses on real estate markets and includes: real estate financial modelling, investment decisions in real estate, the relationship between real estate markets and the macro-economy, urban economic models and patterns and the relation of real estate markets and financialisation. The course objectives are focused on the development of both knowledge and (financial modelling and debating) skill development and takes the format of a blended learning course design whereby online and on-campus are combined.	
Study Goals	At the end of this course students are able to: - relate real estate market characteristics to observed market dynamics; - apply the basic principles of urban economics in real estate market analysis; - assess the financial and economic impact of real estate development following from investment analysis; - apply basic formulas and procedures for converting future cash flows and risk patterns to present value; - evaluate the financial feasibility of (public and private) real estate investment decisions; - debate the role of finance (and in particular financialisation) in real estate and the built environment.	
Education Method	This course consists of a combination of: - Lectures - (Group) assignments - Workshops - Self-paced video tutorials combined with assignments The course is designed in a blended format whereby education is delivered online and on-campus. In particular the financial modelling component is self-guided and self-paced.	
Literature and Study Materials	Literature and Study Materials will be made known approximately 2 weeks prior to the start of the course in Brightspace.	
Assessment	Three assessment methods are employed that together determine the total weighted average grade of the course: - Written examination [weight 50%] - retake next quarter - Computer test of calculation assignments/ [weight 35%] - retake next quarter - Financialisation (group) assignment [weight 15%] - repair option in case of insufficient.	
Period of Education	Second quarter of semester 1 Master Management in the Built Environment	
Concept Schedule	The course contact hours are scheduled on 2 days a week. Note that self-study does follow a scheduled roster of submission deadlines of the assignments. The course therefore needs to be actively followed during the quarter and studies cannot be postponed until the exam week as deadlines need to be met throughout the 10 week period. Please note that there are exams in week 6 and 10 and group work in between. Retakes take place in Quarter 3.	
Minimum number of participants	15	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1MBE030	Real Estate Management	10
Course Coordinator	Dr.ing. G.A. van Bortel	
Course Coordinator	Ir. H.J.M. Vande Putte	
Responsible for assignments	Ir. H.J.M. Vande Putte	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	Real estate management is the ongoing process of aligning the built environment and the needs of users. This alignment happens at all scales of the built environment, for all types of users, and for all real estate aspects such as location, cost, function, time and quality. Real estate management is a particular type of portfolio or asset management and addresses operating the portfolio and assets as well as adapting them. This course studies real estate management from a systemic point of view and focuses mainly on the strategic management level.	
Course Contents	The course first introduces the basics of real estate management through an overview of the past, current and future real estate delivery models and their potential to fulfil user needs and meet the socio-economic and ecological challenges. The studied models range from self-provision to acquisition via cooperation, markets and state initiatives. The models are situated within the main management, social, economic and organisational theories. The course then zooms in on how two particular types of users, namely organisations and households, are currently provided with the real estate they need. Parallel to this theory immersion, students engage in an assignment case wherein they first assess the performance of the current real estate provision for these two types of users and thereafter develop a strategy to improve this provision in the future. The first type of users that is studied in detail organisations ranges from small to large organisations across all industries and services, and from the private to the public sector. The real estate management that addresses organisations is called Corporate Real Estate (CRE) management. CRE managers provide the built assets that organisations employ to accommodate their business. They aim at maximising the stake- and shareholder value of their organisation. CRE management should be well-distinguished from the approach of a real estate investor or developer. The main performance indicators of the CRE manager are the safety, reliability, adequateness, sufficiency and efficiency of the provided accommodation, not real estate profit. This course studies how CRE strategies and operations respond to an organisations expectations and how they can continue to do so in the future. The second type of users that is studied in more detail in the course are households. These range from individuals to families, communities and organised groups, from the low- and middle-income to the high-income, at the local, the national and global levels. The real estate management that addresses households is commonly called Housing management. Acquiring the right home is a crucial aspect of a households aspirations, albeit not simply a matter of individual choice as the housing market is an imperfect market. Especially in larger cities housing production and pricing mechanisms limit the possibilities of (groups of) users to attain their goals, and hinder social cohesion and urban competitiveness. As a consequence the delivery of housing in developed economies became highly institutionalised, and different types of government interventions were introduced. Housing managers are in need of improving the environmental sustainability of the housing stock, which constitutes the largest investment challenge in the built environment. New housing delivery models are emerging to better realise the right to housing for all in the future.	
Study Goals	1. Understand the key delivery models of real estate, for the wide variety of users and types of real estate, and in more depth how CRE and Housing provision can be aligned with the expectations of organisations and households respectively. 2. Apply the above knowledge through the development and execution of a performance analysis of real estate portfolios of different kinds and scale levels, and through the evidence-based design of a future strategy for these portfolios 3. Critically evaluate these delivery models from the point of view of the key challenges for the future: their implications for the end-users, society and eco-system. 4. Analyse current theories on their specificity and create building blocks for a generic real estate management theory.	
Education Method	- Project-based learning, where theory and assignment run parallel from the first week, and theory topics are chosen and presented (as much as possible) to support the case assignment. - Short and weekly exams, to help students synchronise theory acquisition and application. - Project assignment, whereby a strategy is developed for a real-life corporate real estate management OR housing management portfolio, supported by the real-life portfolio owner, developed partly in groups of 3 or 4 students and partly individually, executed and tutored in weekly studio sessions. - The writing of an essay to explore the formation of a generic real estate management theory. - Learning activities comprise lectures, expert talks, site visits to companies and projects, seminars, studio work with feedback, peer assessments, and tutorials to support the writing of the essay. Note: the one day excursion in week 1 is a compulsory part of the program of which the (limited) costs need to be covered by the student.	
Literature and Study Materials	The literature and study materials will be announced on Brightspace. They consist of: - Reader material for the theory part - Reader material for the assignment part - Slides and literature from the lectures that support both.	
Assessment	The assessment consists of three parts: 1. Theory exam, open book (individual work), consisting of a series of small and weekly written assessments, of which the 5 best of 7 are taken into account: 50% of the grade 2. Project assignment report (partly group work and partly individual): 40% of the grade 3. Essay on the formation of a generic real estate management theory (individual work): 10% of the grade To pass the course, the student must obtain a grade of 5 or higher for each part and a grade of 6 or higher for the weighted average of the three parts. Retakes: In case of a retake of one or several parts during the same academic year, the grades of 5 or higher will be kept and combined with the grades of the retaken parts. The retake of the theory exam is a one session exam of 3 hours, open book, and takes place in week 3.10. Retake of the project assignment report or the essay comes in the form of a reparation without tutor support. The maximum grade for the reparation is 6/10. The deadline is the end of week 3.5.	
Period of Education	Semester 1, quarter 2 (weeks 2.1 à 2.10)	
Concept Schedule	The course contact hours are scheduled on 2 days a week. Exams and assignment deadlines are spread over all weeks of the quarter. The course therefore needs to be actively followed during the entire quarter and studies cannot be postponed until the exam week.	

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Management in the Built Environment

AR2MBE011	Building Law	5
Course Coordinator	mr. F.A.M. Hobma	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	2 hours per week starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Required for	MSc track Management in the Built Environment. This course cannot be chosen as an elective by students Construction Management and Engineering.	
Expected prior knowledge	Bachelor of Science	
Course Contents	Building Law covers topics which are essential for understanding of the built environment. Every building project has certain needs: (a) capacity. This means that the building initiator needs designs, engineering, building materials and labor. The building initiator will use the legal instrument of contracts to arrange this. In this context, rules for procurement are relevant. These rules require that sometimes competition between suppliers of goods and services is required. (b) Land. This means that the building initiator has to buy or lease land on which the project is realised. Contracts, pre-emption right and expropriation may be used to get access to land. (c) Permission. The building initiator needs permits and plan approvals before the work can commence. Permit procedures make sure that the building project is tested against relevant norms and criteria. On top of this, the building initiator needs to make sure that conflicts are avoided as much as possible. The fields that are taught in this course are: Building contract law, Procurement law, Property law, Planning law, Environmental law and Public private partnerships. It includes private law and public law, from the building level to the level of urban development.	
Study Goals	Students are able to: 1. contrast the different building contract models. 2. evaluate the strengths and weaknesses of different contract models. 3. describe the essential rules regarding procurement. 4. address legal aspects of real estate development projects from the European and Dutch Planning and Environmental legislation perspective. 5. discuss the different forms of cooperation between public and private sectors regarding buildings and urban development.	
Education Method	Lectures, assignments and groups discussions	
Literature and Study Materials	- M.A.B. Chao-Duivis et.al., A Practical Guide to Dutch Building Contracts - F.A.M. Hobma and P. Jong, An Instrumental Approach to Planning and Development Law in the Netherlands - Uniform Administrative Conditions for the Execution of Works and Technical Services	
Assessment	Two assessment methods are employed that together determine the total weighted average grade of the course: Written theory exam [weight 70%] Law assignments [weight 30%] There is no minimum grade for the assignments. Therefore there is no retake for the assignments.	
Permitted Materials during Tests	Open book exam	
Period of Education	The third quarter of the academic year	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MBE015	Redesign of Complex Projects	10
Course Coordinator	Dr. T. Wang	
Course Coordinator	Dr. H.T. Remoy	
Responsible for assignments	Dr. H.T. Remoy	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	In "Re-design" the focus is on applying knowledge of Management in the Built Environment on analysing and redesigning complex (redevelopment) projects. This is achieved through case studies and using varying perspectives to describe the cases. The perspectives of the client and the user receive attention in particular. For the client, project delivery, procurement and tendering, and contracting parties are essential. For the user, (end-)user preferences, market opportunities, interaction with the supplier(s), and the relation between costs and quality are essential. Drawing lessons across cases from different perspectives enables to apply these in a proposal for a similar project.	
Study Goals	The student is able to: -describe a complex (re/development) project in a case study approach using a relevant theoretical model. -position a project in its context on different scales and phases: from building to markets, from initiative to operation. -describe different perspectives and place oneself in the perspectives with special attention for the perspective of client(s) and (end-)user(s). -find and select historical cases for comparison. -transfer lessons from one case to another by means of systematical comparison and explain expected benefits. -develop a project proposal for a complex (re/development) project: including a brief, actions, transactions and deadlines, including choices about the delivery model, procurement strategy and contracts to be used. -report case study results, analyses, advice and proposals in both scientific as in professional form.	
Education Method	Students work half the course individually and half in groups. The contact hours comprise a series of core lectures and tutoring sessions.	
Literature and Study Materials	Literature and Study Materials will be available 1 week prior to the start of the course in Brightspace.	
Assessment	Writing assignments: Assignment 1: case analysis (group work, 25% of the total grade); Assignment 2: event case analysis (individual work, 15% of the total grade); Assignment 3: redesign proposal (group work, 35% of the total grade); and Assignment 4: procurement proposal (individual work, 25% of the total grade). For each assignment, a repair is allowed when an assignment is graded below 6. A repair is never graded above 6. Deadlines will be shared on BS before the start of the course.	
Period of Education	First quarter semester 2.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MBE021	Building Information Management	5
Course Coordinator	Dr.ir. A. Koutamanis	
Responsible for assignments	Dr.ir. A. Koutamanis	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	Course AR2MBE021 focuses on building information management: how data and information are produced in AEEO (architecture, engineering, construction and operation of buildings), and how they can contribute to management in the built environment. The course follows the structure of the open textbook Building information representation and management, available at https://buildinginformationmanagement.pressbooks.com/ and https://textbooks.open.tudelft.nl/index.php/textbooks/catalog/book/47.	
Study Goals	The main objective of the course is to dismiss both the superficial attachment to software solutions and opaque prescriptive rules or guidelines, and provide instead a foundation for understanding the structure and potential of digital information processing: Definitions for data and information What representations are and how they organize information Current approaches to building representation entail and their utilization The relations between process and information management Decision making in relation to process and information management The connections between information management and building representations With this foundation students not only develop critical insights into current practices but also acquire practical skills for implementing better solutions with existing and future technologies. The development of these skills and insights is facilitated by weekly exercises, in which students learn to manage information in AEEO processes, including from the perspective of various actors and stakeholders who provide or require information.	
Education Method	Each week in the course focuses on a specific subject, which corresponds to one or more chapters in the textbook. Treatment of the subject takes place in three steps: lectures, weekly assignments and follow-up assignments. 1. Lectures Every week, there is a lecture in the hall indicated on the TUD timetables. Participation is mandatory and contributes to the final grade for the course. Students should be prepared to present their assignments and discuss results (see below). 2. Weekly assignments Following each lecture, students are given weekly assignments (announced on Brightspace) that constitute part of the final output for the course and are therefore mandatory. Through these assignments and the personal and plenary feedback they receive, students prepare for the final assignment. 3. Follow-up assignments Prior to each lecture, a brief assignment is announced on Brightspace, as a follow-up of the weekly assignments. Through this, students prepare for the lecture, including by evaluating their performance and formulating questions that are discussed in the afternoon lecture. Participation is again mandatory and contributes to the final grade.	
Assessment	Evaluation of student performance in the course is based (a) on the final assignment and (b) on the learning process of each student, as indicated by the weekly assignments and student participation to the weekly sessions. Both components carry an equal weight. The weekly assignments are not graded; they receive qualitative feedback intended to guide and encourage. Improvements on the basis of this feedback are expected in the following assignments, as well as in the final one. The final assignment grade is compared to the overall involvement and performance of a student throughout the course: 1. If there is evidence of progress or stability, the grade of the final assignment becomes the grade for the whole course. 2. If the final assignment is inferior to the students performance in the previous weeks, the course grade adjusted upwards form the final assignment grade, so as to take into account the learning process. 3. If the final assignment repeats mistakes and omissions that have been pointed out in earlier feedback, the course grade stands. All assignments are evaluated with respect to the following criteria: Completeness: all necessary parts are present Consistency: all parts are specified in the same way Coherence: the parts fit together in a meaningful description of a building or process For each assignment, the criteria are specified in detail on Brightspace. If the course grade is lower than six (6), students are allowed a single repair by submitting an improved version of their final assignment, on the basis of feedback they have received on BS. The repair is graded. In exceptional cases (e.g. when a student has been unable to follow the course through illness), an alternative final assignment is agreed with the student.	
Period of Education	Q4	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MBE025	Urban and Infrastructure (Re)development Game	10
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 Management in the Built Environment
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AR3MBE010	Research Methods 2	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	Research Methods 1 AR1MBE015	
Course Contents	The mission of this course is to support students starting their graduation research by teaching different methodological concepts, research and problem solving methods that can be applied by MBE students in their final year project (master thesis). Research Methods 2 (RM2) is designed to match the natural rhythm in which most graduates develop their research proposal. The course is divided in 2 parts. Part one is a common part that helps students to get started with their research proposal. This part runs in quarter 1 and quarter 3 in which students develop their research ideas into their P1 report. At the end of their P1, students should be able to choose a method. Ideally, students choose a particular research method on the basis of the research proposal, the formulated research question and general approach in a particular field. For the second part, students have to choose one of the three methods specialisations (options A, B or C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). Part two runs in parallel to the P2 report in quarter 2 and 4 in which students develop their research protocol. It is possible for students to learn about more than one method. For them there is the possibility to participate in the elective Research Methods 3 course (RM3) [code:AR0185]. In RM3, students can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. In terms of content, RM3 is an extended version of RM2-part II. Part I Getting started with your research protocol Part I runs in parallel with the MBE Graduation Laboratory and helps students to get started with their research proposal stating the problem, formulating the aim and research questions. In several workshops students develop their first ideas into a research proposal. Towards the P2, a student will develop their research proposal into research protocol. The latter includes a methods section. Part II Methods specialisations Students are required to choose one of the following 3 methods specialisations. Ideally, students choose a particular research method on the basis of the research proposal, the formulated research question and general approach in a particular field. A . Case study methods (CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria. Finally, a small case study will be conducted as an exercise. As part of this case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the RM3 version, students will conduct an additional observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. B. Applied statistical methods (ASM) The aim of this specialization is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. There will be several statistical approaches available. In the RM2 course only basic procedures will be mastered. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. The final assignment is the only assignment that will be assessed. a In the RM3 version, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will for instance use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialization starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development	
Course Contents Continuation		

	management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analyzed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalization. In RM2 students will study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. In RM3 students additionally built a preference model for the same case in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).
Study Goals	Part I Getting started with your research protocol The student: - is able to write a research proposal; - is able to develop a searchplan for relevant scientific literature; - is able to abstract data from the retrieved studies in a summary table; - is able to write an introduction to a research proposal based on the CARS model of Swales. Part II-A Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analysing and reflecting on in-depth interviews; - is able to perform a basic qualitative analysis in Atlas TI. Part II-B Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. Part II-C Operations research methods The student: - is able to characterise different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analysing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Part I - See Brightspace for a variety of recommended sources. Part II Depending on your specialisation, the student needs to study the following literature. 1. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. 2. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 3. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	Part I and II are both mandatory. If you do not carry out or fail one (or more) of the two parts, you will not receive a final mark for the course. Part I is marked with a pass or fail. The evaluation will be based on a research poster. Part II A, B and C will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the final assignment. B. Statistical methods - The mark will be based on the evaluation of a final assignment that is taken at the end of this part. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on a mathematical model. The grade that you receive for Part II will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Part I runs in quarter 1 and quarter 3 in which students develop their research ideas into their P1 report. Part two runs in parallel

Minimum number of participants	to the P2 report in quarter 2 and 4 in which students develop their research protocol. At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR3MBE100	MSc 3 Graduation Laboratory Management in the Built Environment	10
Course Coordinator	Dr.ir. A. Koutamanis	
Responsible for assignments	Dr.ir. A. Koutamanis	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	AR3MBE100 covers the first part of the graduation process, from entry in the MBE Graduation Lab up to and including the P2. Students passing the P2 proceed automatically to the second part of the graduation process, course AR4R010, where they complete their thesis and graduate from the Master of Science programme. IN AR3MBE100 each student selects a topic and supervisors for their thesis research, and develops a full research proposal for their activities after the P2.	
Study Goals	The purpose of AR3MBE100 is that each student develops a promising, substantiated and feasible research proposal that can be implemented in AR4R010. To this end, it is strongly recommended that all exploratory and preparatory work is completed before the P2: literature and market research, acquaintance with available cases, agreements with external organizations (including internships) etc.	
Education Method	Introductory lectures and workshops, followed by individual or group supervision.	
Assessment	- P1: problem statement - P2: full proposal	
Period of Education	Q1 and Q2 or Q3 and Q4	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Free Electives 15 ECTS

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment. Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: - Public perspective; - Private / co-creation perspective; - Civil society perspective; - Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: - The skill of writing papers is presented and discussed; - Assignments are done to support the writing of a paper; - The elements of a paper (questions, methods, theory) are discussed; - Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: - Layout and detailing of the paper - Research methods and reporting skills (research questions, use of sources, clear conclusion) - Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate - Research goal and relevance made clear - Research question that is adding to a general body of knowledge e.g. is interesting for a larger community - Research structure is clear, logic and explained - Theory is consciously used to frame the work, explained and logical- - Reflection into the topic of research in relation to the place of work - Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.

Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003

Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000

Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover

Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008

BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual

Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,

Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: Formulate a conceptual or theoretical framework. Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. Identify and discuss the tension between public and private development in infrastructures and environments. Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical Reflection into the topic of research in relation to the place of work Late delivery Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 ects) Semester 2, Q4 = design project and internship BSc ON project(15 ects)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSc2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSc2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSc2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.). However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages. The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose. The course has 3 parts: 1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality. 2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to: 1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0. The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point. A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75. If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75. The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbidata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 4 Management in the Built Environment

AR4R010	MSc 4 Graduation Laboratory Management in the Built Environment	30
Course Coordinator	Prof.dr.ir. V.H. Gruis	
Course Coordinator	Dr.ir. A. Koutamanis	
Responsible for assignments	Dr.ir. A. Koutamanis	
Contact Hours / Week x/x/x/x	25 hours per semester	
Education Period	None (Self Study)	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Required for	Final graduation	
Expected prior knowledge	Master 1, 2 and 3 of the MBE track, Master Architecture, Urbanism and Building Sciences.	
Parts	P3, P4 and P5	
Course Contents	AR4R010 is the second and final part of the graduation process in the Master track MBE. In it, students work towards the completion of their graduation thesis under supervision by the mentor team appointed in AR3MBE100.	
Study Goals	Based on the P2 results in AR3R010 and under supervision by their mentors, students individually conduct research towards the completion of their graduation project, in a way that demonstrates their ability to meet the final learning goals of the MBE Master track, of the Faculty of Architecture & the Built Environment and of Delft University of Technology.	
Education Method	Individual research towards completion of a Master thesis, under supervision by two or three mentors.	
Literature and Study Materials	See Brightspace	
Assessment	- P3: halfway through the semester and before submitting the P4 request, each student delivers an interim presentation and a draft graduation report. - P4: at the end of the semester, each student delivers a presentation of the completed project and a draft final graduation report. - P5: having passed the P4 assessment, each student finalises their final graduation report and defends it in the final examination session.	
Exam Hours	- P3: scheduled by each student individually - P4 and P5: scheduled centrally by the Faculty of Architecture & the Built Environment, upon request by the student	
Special Information	The maximum marking period is 10 work days. MBE students have the option to carry out their graduation research at an appropriate external organisation. Students who wish to do so are required to sign a standard internship agreement, as stipulated at http://studenten.tudelft.nl/en/students/faculty-specific/architecture/forms/ .	
Period of Education	Semester	
Concept Schedule	See Brightspace and https://www.tudelft.nl/en/student/faculties/a-be-student-portal/education/academic-graduation-calendar	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Cross Domain City of the Future

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 Cross Domain City of the Future

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory for MBE students

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Compulsory

AR3CS021	Seminar Cross Domain City of the Future	5
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. T.A. Daamen	
Course Coordinator	Dr.ir. J.H. Baggen	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	4 hours/week (1.1-1.8 & 2.1-2.5)	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	Given the framework of the project, the aim of the seminar is to: - analyse and understand which factors, tendencies, (design research) approaches, technological innovations, development and governance models will be playing a role in the future developments of our cities. - study relevant examples and/or best practices around the world and compare them with the city-study areas of the graduation studio. - collect data into research books to be shared among the participating students. - argument and develop the investigations (research positions) into a written article. Multiple aspects / disciplinary perspectives regarding the (research design) project will be presented by the involved researchers in interactive sessions. A.o., following topics will be discussed: transportation, connectivity, new mobility concepts, housing challenges, climate change mitigation, cultural planning, health and private, public and semi-public spaces, circular economy, politics, resiliency, adaptation strategies. The seminar challenges students to develop critical and comparative investigations focusing on specific insights and positions to be determined within the (research) framework of the studio.	
Study Goals	- Become aware of different (design) research methods in order to inform own graduation process. - Acquire necessary research skills in order to set up and develop a research article - Explore different inquiry methodologies and research approaches. - Work in a collaborative way within a multidisciplinary group and probe different constraints to define own (design) research approach.	
Education Method	Workshops/tutorials given by the involved tutors and/or invited guests/experts	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	- Writing Assignment - Analytical Assignment Next to the participation in the seminar, the assessment of the course will enclose a writing assignment: Option 1, a position paper Option 2, a report of the thematic seminar sessions related to a personal research standpoint. The assessment criteria will be: -Is the paper or report coherent/concise, regarding structure and clearness of style? -Is the language adequate? -How did the paper or report follow the review comments? Problem field and research question: -Has a research question / standpoint been clearly defined? -Has the question / standpoint been developed beyond its initial formulation? -Does the paper / report acknowledge a state of the art, regarding this question? Societal and scientific relevance -Is the paper /report relevant to the studio theme City of the Future from a societal point of view? -Is the paper / report relevant to the studio theme City of the Future from a scientific point of view? -Has a position been taken, in relation to this state of the art? Theoretical and operational frameworks -How is the paper / report connected with the graduation topic? -Are the sources pertinent, and well used and is the use of reference and own words balanced? -Does the paper / report follow a clear approach / methodology and shows scientific quality? On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Period of Education	1-2	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3CS081	Graduation Cross Domain City of the Future	20
Responsible Instructor	Dr.ir. R. Cavallo	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	This Graduation Lab is a special thesis laboratory for students that would like to develop their own (design and/ or research) fascinations in a multidisciplinary setting. Students with different backgrounds and from different TU Delft MSc tracks will work on common challenges sharing insights, approaches and methodologies proper of their own disciplines. While doing that, students work together to enrich their own graduation pathway, setting up and developing workshops, lectures, excursions and visiting critics. The students of this Graduation Lab are responsible for the program and the agenda through the thesis period. They are expected to work together as much as they possibly can, because mutual critic and collaboration is one of the important means of education in this lab. Students of the MBE track complete the first two assessments (P1 and P2) by selecting a research subject and mentor team; conducting literature and market research; developing a problem statement, objectives and goals, and an approach to solving the problem and reaching their goals. This course is including a student driven workshop module. Students are asked to develop a collective educational program addressing themes relevant for all participants. These activities may take the form of or include workshops, lecture series, visiting critics, excursions, charettes, etc.	
Study Goals	Problem analysis The student has knowledge and understanding of research approaches and methods for translating a subject with scientific and societal relevance into a problem analysis, problem statement, research objectives and research questions in a critical and grounded manner. Literature review The student is familiar with fundamental and recent literature in the area of MBE, and is able to conduct a comprehensive, in-depth literature review that retrieves literature relevant to their graduation project, through which they can substantiate research hypotheses and approaches. Synthesis The student has a creative, innovative and investigative approach to solving the selected problems. The student is capable of identifying relevant knowledge in their own and related areas, acquired in part through the literature review, and systematically utilising it for the definition of a coherent theoretical framework and conceptual models for their research. Methodology The student is capable of selecting appropriate research methods in a transparent and substantiated manner and of applying these in a scientifically and ethically responsible manner. Acceptability and relevance The student is capable of evaluating their own process, products and performance in relation to current scientific and professional knowledge. The student is able to formulate clear conclusions and recommendations for further research and application, and through these demonstrate that their work meets the standards of scientific research and contributes to the solution of societal problems. Time management The student is aware of the requirements for interim and end products, has sufficient time management skills to make a realistic estimate of activities and the amount of time needed for each of them and, on the basis of these, can produce a reliable working plan. Reporting and communication The student is capable of producing informative written reports, suitable to a scientific and professional audience, that provide a structured, coherent, consistent, precise and insightful account of their research process and products. The student is capable of delivering oral presentations of their work in an informative and engaging manner and at an appropriate scientific level, using valid arguments in discussing their subject. The student is open to constructive criticism and is willing to learn from feedback and comments.	
Education Method	This is a student driven graduation laboratory. The educational method is therefore to be developed by the students in conversation with each other and the coordinators. The assumption is that studio instruction will be the primary teaching method. Students will guide their own studies and determine their own learning styles.	
Assessment	- P1: halfway through the course, each student presents the progress of their work in the form of a preliminary report (draft research proposal and plan for the thesis) - P2: at the end of the course, each student presents a final research proposal and plan for the completion of their thesis	
Period of Education	Fall Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Free electives/Research Methods 2

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 4 Cross Domain City of the Future

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

For MBE Students

AR4CS030	Cross Domain City of the Future	30
Course Coordinator	Dr.ir. T.A. Daamen	
Responsible for assignments	Dr.ir. T.A. Daamen	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This Graduation Lab is a special thesis laboratory for students that would like to develop their own (design and/ or research) fascinations in a multidisciplinary setting. Students with different backgrounds and from different TU Delft MSc tracks will work on common challenges sharing insights, approaches and methodologies proper of their own disciplines. While doing that, students work together to enrich their own graduation pathway, setting up and developing workshops, lectures, excursions and visiting critics. The students of this Graduation Lab are responsible for the program and the agenda through the thesis period. They are expected to work together as much as they possibly can, because mutual critic and collaboration is one of the important means of education in this lab.	
Study Goals	Based on the P2 results and supervised by the mentor team, students individually conduct research towards completion of their graduation project, in a way that demonstrates their ability to meet the final learning goals of the MBE Master track, as well those of the Faculty of Architecture & the Built Environment and of Delft University of Technology.	
Education Method	Individual research towards completion of a Master thesis, under supervision by two mentors.	
Assessment	- P3: halfway through the semester each student delivers an interim presentation and a draft graduation report. - P4: at the end of the semester each student delivers a presentation of the completed project and a draft final graduation report. - P5: having passed the P4 exam, each student finalises their final graduation report and defends it in the final examination session.	
Period of Education	Spring Semester	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

variant Landscape Architecture

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 Landscape Architecture

AR1LA011	Architecture and Landscape: Design Studio	10
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	F.T. Toni	
Instructor	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	F.T. Toni	
Responsible for assignments	F.T. Toni	
Contact Hours / Week x/x/x/x	72 hours	
Education Period	1	
Start Education	1	
Exam Period	1 5	
Course Language	English	
Summary	The design studio Architecture and Landscape develops knowledge, skills and stances in landscape architectural research and design via an assignment for a garden-building ensemble in a semi-natural environment. With this assignment the studio takes an in-depth look at physical landscape space and human experience; delves into site and context from a (garden) design perspective; elaborates elementary landscapes dynamics; and develops skills in design composition and design elaboration at the domain scale, with an emphasis on planting design.	
Course Contents	The studio focusses on the garden (domain) scale of landscape architecture, as the most condensed form of (the ontological complex) landscape. The design project takes an in-depth look at physical landscape space, notions of context and human experience, landscapes dynamics and change, and planting design. The site has delineated boundaries, a specific scope of transformation and clearly defined ownership and operation. Integrating site and context (geographical, historical, cultural etc.) into the design forms a central motif for the assignment. Knowledge, skills and stances are developed in interpretations of a given landscape, working with landscape processes, conceptual design thinking, elaborating a landscape architectural (domain) composition, articulating the relationship between concept, composition and design details, planting design and ethical dimensions of landscape architecture.	
Study Goals	By the end of this course, the student should be able to: - identify, interpret and interrelate landscape qualities from a phenomenological perspective (human perception); - identify, interpret and work with landscape processes: previous stages of a specific site and its topography; relationships between environmental factors and resulting vegetation and topography over time; - develop active reciprocities between phenomenological landscape analysis and conceptual design thinking; - elaborate a landscape architectural composition, relating objective and subjective experiences of a site and expressing aspects of time and movement as well as the sensory qualities of the landscape; - create a coherent and clear relationship between concept, composition and detail; - elaborate a planting scheme into a proposal which expresses the various spatial, sensorial, temporal and narrative dimensions of the design; - demonstrate a variety of presentation and communication techniques, expressing human experiences (scale, materiality), time and movement; - reflect on his/her own design, involving ethical considerations as an expression of theoretical perspectives on landscape, landscape architecture and the relation between humans and nature.	
Education Method	Studio work Workshops Instruction sessions Excursions Fieldwork	
Course Relations	The quarter Architecture and Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking. Four perspectives on the analysis and design of (urban) landscapes form the backbone of the master track Landscape Architecture: landscape perception, landscape as palimpsest, landscape as scale continuum and landscape as ecological, economic and social process.	
Literature and Study Materials	- Steenbergen, C.M. and Reh, W., 'Architecture and Landscape - the Design Experiment of the Great European Gardens and Landscapes', Basel, Boston, Berlin, Birkhauser, 2003. - Dee, C. (2001), Form and fabric in landscape architecture; a visual introduction. Londen, New York: Spon Press. - Bell, S. (2004), Elements of visual design in the landscape. New York: Spon Press. - Vroom, M. (2006). Lexicon of Garden and Landscape Architecture. Basel, Boston & Berlin: Birkhäuser.	
Assessment	2D drawings and supporting visuals Physical/digital 3D-model Oral presentations Written reflection	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1LA021	Architecture and Landscape: Theory, Method and Critical Thinking	5
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. S.I. de Wit	
Instructor	F.T. Toni	
Responsible for assignments	Dr.ir. S.I. de Wit	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.7.	
Education Period	1	
Start Education	1	
Exam Period	1 5	
Course Language	English	
Summary	The Theory, Method and Critical Thinking course Architecture and Landscape introduces landscape architectural compositional principles and their application and focuses on the architectural form of plants and planting schemes over time. The course presents tools, theories, methods and techniques related to the landscape architecture discipline.	
Course Contents	In this course different knowledge fields are introduced, revolving around landscape architectural composition. Phenomenology, environmental psychology, compositional theory and planting knowledge will feed the students insight in landscape architectural composition. Seminal landscape architectonical objects will be presented and discussed, with the emphasis on the discovery and study of the complex compositional rules with which a design is built up and the relationship between composition and perception. Theories, concepts and design aspects are brought into a wider scope addressing analytical and compositional techniques and typological research. Furthermore, the role of planting will be explored and discussed in terms of space and time dynamics. The emphasis is on basic knowledge of assortment and its application in landscape design in order to define and organize space into imaginative compositions that evolve through time. Crucial is the understanding of the relationship of planting with spaces and objects, with their natural surroundings (habitat) and their evolution in time. Field trips and literature study will provide the material for studying and documenting examples of several species and their arrangements in order to discover their spatial and environmental characteristics.	
Study Goals	By the end of this course, you should be able to: - identify and interpret the historical compositional repertoire of landscape architecture and its role in current practice; - identify, interpret and discuss the formal qualities of plants and their spatial and perceptual characteristics; - demonstrate a working knowledge of environment-behaviour theories - describe, interpret, evaluate and critique landscape design in relation to composition and perception of outdoor space; - demonstrate analytical drawing and writing techniques that reveal spatial and experiential qualities of planting compositions in relation to their landscape context.	
Education Method	Workshops Lecture series Field work Discussion sessions You will work in groups.	
Course Relations	The quarter Architecture and Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking. Four perspectives on the analysis and design of (urban) landscapes form the backbone of the master track Landscape Architecture: landscape perception, landscape as palimpsest, landscape as scale continuum and landscape as ecological, economic and social process.	
Literature and Study Materials	- Dooren, N. van (2018). The landscape of critique; The state of critique in landscape architecture and its future challenges. SPOOL Vol. 5 No.1 Landscape Metropolis #4, Criticising practice - practicing criticism, 13-32. doi: 10.7480/spool.2018.1.1936. (available on Brightspace) - Kaplan, R. and Kaplan, S (2005). Preference, restoration and meaningful action in the context of nearby nature. In Barlett. P.F. (ed.) Urban Places; reconnecting with nature. Cambridge: MIT press. (available on Brightspace) - Moore, G.T. (1979). Environment-Behavior Studies, in Snyder, J.C. and Catanese, A.J. (eds.) Introduction to Architecture. New York: McGraw-Hill, 46-71. (available on Brightspace) - Steenbergen, C.M. and Reh, W. (2003). Architecture and Landscape; the Design Experiment of the Great European Gardens and Landscapes. Basel, Boston, Berlin: Birkhauser. - Wohrle, R.E. and Wohrle, H.J., (2008). Designing with plants. Basel, Boston, Berlin: Birkhauser.	
Assessment	Plan critique including an analysis in drawings and words and critical interpretation with arguments based on the offered knowledge fields. When a specific part of the plan critique is missing or insufficient, leading to a grade of 5,5 or lower, you will get the opportunity to repair. A repair needs to be handed within 2 weeks after receipt of the grade, and will result in a 6. When the plan critique as a whole is insufficient you will have the chance to retake the assignment: a plan critique on a different case. This will take place during the Summer period, to be finalised before the start of the following academic year. During this period three discussion moments with your tutor may be planned.	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1LA051	Dutch Landscape: Design Studio	10
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. G.A. Verschuure-Stuip	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	72 hours	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Summary	The design studio Dutch Landscape emphasizes on exploration, understanding and modification of landscape patterns, conditions, structures and their spatial expression. The assignment asks for interventions in a rural area in which the expression of the balance between land and water plays an important role, fitting the accommodation of a range of claims (agriculture, recreation, living, nature development) of the given site.	
Course Contents	In the design course you formulate your own assignment within a defined challenge, in which different scales and perspectives come together: water issues and the social structure of a rural society and ongoing spatial claims (like farming, energy production, recreation, nature development) on the scale of a polder landscape. The aim of the design is to explore the transition of landscape as a means of adapting and giving expression to existing landscape patterns, leading to new landscape architectural (spatial and experiential) compositions in which beauty, landscape processes, functionality and inclusiveness (of people, animals and plants) coincide. You will design through scales from concept to detail, with an understanding of the area of influence, area of effect and area of control. The design is based on a landscape analysis, understanding the spatial structure and patterns, which should lead to a spatial composition for an imaginable future, allowing for and anticipating on social and ecological processes. You learn how to design through time: from past relevant periods to new conditions for a desired future, according to the assignment that is defined and developed by yourself.	
Study Goals	By the end of this course, the student should be able to: - identify and interpret the Dutch lowlands (polder) landscape, and its spatial formation, function and patterns; - understand and integrate landscape engineering techniques and ecological knowledge related to planting succession, water/land gradients and maintenance in the design; - discuss and demonstrate the relationship between landscape analysis and landscape architectural design; - explore landscape potentials by means of options and alternatives; - understand the interdependence of land use, soil conditions, hydrological systems, vegetation characteristics and planting proposals; - apply landscape architectural design tools through scales: from ditch to horizon; - create self-explanatory drawings, expressing systems, scale ranges, texture and materials of the landscape over time; - reflect on your own design, involving ethical considerations as an expression of theoretical perspectives on landscape, landscape architecture and the relation between humans and nature.	
Education Method	Studio work (working in pairs) Workshops Lectures Excursion	
Course Relations	The quarter Dutch Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking.	
Literature and Study Materials	- Bobbink, I. en Loen S. (2013) 'Water inSight' http://repository.tudelft.nl/view/ir/uuid:e1af985b-7f72-4a55-9c07-2fc0f4c7e4f1/ - Reh, W. en Steenberg C. (2007) Zee van Land, Stichting Uitgeverij Noord-Holland - various editions of Dutch and European landscape architecture, Blauwdruk, Wageningen	
Assessment	booklet and/or posters physical models oral presentation written reflection on design	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1LA061	Dutch Landscape: Theory, Method and Critical Thinking	5
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Instructor	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. G.A. Verschuure-Stuip	
Contact Hours / Week x/x/x/x	40 hours	
Education Period	2	
Start Education	2	
Exam Period	2 5	
Course Language	English	
Summary	The Theory, Method and Critical Thinking course 'Dutch Landscape' provides insight in the interrelation of the different natural and societal forces, spatial and temporal qualities, materials and components of the cultivated predominantly rural landscape.	
Course Contents	The course addresses the structures, layers, and components of the Dutch landscape and combines the following knowledge fields: historical-geography, polder history, water management, morphology, landscape ecology (e.g., gradients, plant geography), environmental philosophy (visions to landscape and nature) and social geography. Different analytical techniques: cartography, photography, layer analysis, narratives, etc. will be used to reveal and represent the identity - which is: landscape forming processes and their topographic appearance - of a specific landscape. You learn how to critically evaluate your insights into the different aspects of landscapes in a descriptive well-illustrated document that takes the shape of a landscape biography. You need to elaborate on the outcome of the analyses as input for future transformations.	
Study Goals	By the end of this course, the student should be able to: - map landscapes and landscape elements; - identify, define, organize, value and represent different landscape forming processes and their spatial manifestation; - identify, group, value and represent the spatial and visual, ecological, technical (land/water) and social conditions of a specific landscape; - document and express the integration of different fields of knowledge; - elaborate on the outcome of the analyses for future transformations; - identify, critically assess and apply different perspectives on nature and landscape.	
Education Method	Lectures Workshops Seminars Excursions	
Course Relations	The quarter Dutch Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking. Four perspectives on the analysis and design of (urban) landscapes form the backbone of the master track Landscape Architecture: landscape perception, landscape as palimpsest, landscape as scale continuum and landscape as ecological, economic and social process.	
Literature and Study Materials	- Bell, S. (1999) Landscape. Pattern, Perception and Process. Taylor & Francis. - Cronon, W. (1999) Uncommon Ground: Rethinking the Human Place in Nature. W. W. Norton & Company. - Odum, E., Barrett, G. (2005) Landscape Ecology chapter 8, in Fundamentals of Ecology, 5th edition, p.374-411. - Ven, G.P. van de (2004) Man-made Lowlands. History of water management and land reclamation in the Netherlands. Matrijs. - Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press.	
Assessment	Presentation of a landscape as a living archive, in text and visually.	
Period of Education	Quater	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Landscape Architecture

AR2LA011	Urban Landscape: Design Studio	10
Course Coordinator	Prof.ir. E.A.J. Luiten	
Instructor	Prof.ir. E.A.J. Luiten	
Instructor	Prof.dr.ing. S. Nijhuis	
Instructor	J.R.T. van der Velde	
Responsible for assignments	Prof.ir. E.A.J. Luiten	
Contact Hours / Week x/x/x/x	72 hours	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Summary	The design studio Urban Landscape explores the role of landscape design in the context of a metropolitanising region. The studio explores the relation between systems and places and the social interaction of people in the public realm. In this design studio students work on a comprehensive spatial proposal that supports the sustainability of an urbanized landscape under transformation.	
Course Contents	Urban Landscape: design studio In the course two specific dimensions of urban landscape architecture will be elaborated and integrated: the city as an earth-life landscape system and the city as a socio-spatial environment. The project outcome includes a strategy addressing the sustainable metabolism and systemic form of the urban landscape. You are to develop a design proposal for a particular urban landscape component (green-blue sites, patterns, structures), addressing socio-spatial and physical aspects and criteria. A well-defined and thoroughly elaborated landscape architectural intervention that takes position in the balance between privatized and public space) is to be developed. You have to illustrate how the proposal contributes to large-scale and long-term urban sustainability goals and how social behaviour influences the design and responds to the result.	
Study Goals	By the end of this course, you should be able to: - identify and interpret an urban territory as an earth-life landscape system and as a social-spatial environment, using established analytical methods; - have a working competency in research-by-design techniques investigating transformations of the landscape; - discover and define (diagnose) a problematique in an extensive and complex urban territory with respect to urban ecology, urban forestry and urban agriculture - is able to formulate and develop a landscape architectural assignment in an urban context; - design on various levels of scale and size, relating the whole and the component parts in a mutual and coherent relationship; - resolve and elaborate an urban landscape design (intervention) in spatial, material and technical terms, involving water management, soil mechanics, hardscapes, planting, ecology and micro-climate aspects; - demonstrate a variety of presentation and communication techniques; - reflect on your own design, involving ethical considerations as an expression of theoretical perspectives on landscape, landscape architecture and the relation between man and nature.	
Education Method	Studio work Workshop Lectures Excursion Seminar	
Course Relations	The quarter Urban Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking.	
Literature and Study Materials	- Steenbergen, C.M., Reh, W. and Pouderoijen, M. (2011). Metropolitan Landscape Architecture: urban parks and landscapes. Bussum THOTH Publisher. - Sim, D. (2019). Soft City. Washington: Island press - Gehrels, H. et al, (2016). Designing green and blue infrastructure to support healthy urban living. https://www.researchgate.net/publication/308165682 - Henderson, K., Lock, K. & Ellis, H. (2017). The art of building a Garden City, Designing communities for the 21st Century, Newcastle upon Tyne: RIBA - McHarg, I.L. (1969). Design with nature. NewYork: John Wiley and Sons, inc.	
Assessment	Posters or booklet Oral presentation Physical (and digital) 3D-model Written reflection on design	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2LA021	Urban Landscape: Theory, Method and Critical Thinking	5
Course Coordinator	Dr. L. Cipriani	
Course Coordinator	Prof.ir. E.A.J. Luiten	
Instructor	D. Piccinini	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	40 hours	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Summary	The Urban Landscape: Theory, Method, and Critical Thinking course is dedicated to the complex relationship between city and landscape; it introduces and explains open spaces and the basic concepts of urban green-blue elements and structures as the backbone for urban development and regeneration on various scales.	
Course Contents	Historical periods, cultural and political contexts, geographical features, and critical projects have influenced the complex relationship between city and landscape, culture, and nature. The spectrum of theories, concepts, approaches, and projects on this topic forms the backdrop for this course. The lectures provide different lenses covering historical and theoretical aspects focusing on contemporary concepts and approaches on the landscape as the medium and model to guide and transform urban development.	
Study Goals	By the end of this course, you should be able to: - identify and elaborate on theories, concepts, and approaches to urban landscapes/landscape urbanism; - identify and elaborate on spatial design and its social and ecological influences and ethical implications in the context of urban landscape/landscape urbanism; - position landscape architecture concerning other spatial design disciplines like visual arts, architecture, civil engineering, ecology, and urbanism; - logically and critically reflect on a selected discourse verbally, interactively and in writing, based on main academic literature, applying the basic rules of academic writing.	
Education Method	Lecture series Workshop Seminar	
Course Relations	The quarter Urban Landscape consists of two courses: 10 ECTS Design Studio and the 5 ECTS Theory, Method and Critical thinking.	
Literature and Study Materials	Study literature will be delivered to students in class.	
Assessment	Research paper	
Period of Education	Quarter	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

15 EC of electives

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssoen and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The the focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.

Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003

Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000

Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover

Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008

BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual

Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,

Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: Formulate a conceptual or theoretical framework. Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. Identify and discuss the tension between public and private development in infrastructures and environments. Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical Reflection into the topic of research in relation to the place of work Late delivery Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeeds publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II	
	The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSc2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.). However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages. The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose. The course has 3 parts: 1. Introduction to GIS - fundamentals of Geodesy, Coordinate Reference Systems, and map projections, - spatial data modelling (vector and raster spatial models), - geo-data manipulation (editing, digitizing, importing, converting, etc.), - overview of spatial analysis operations, - production of interpretable output (e.g. maps), - fundamentals of data quality. 2. Algorithms and data structures for GIS - data structures for vector and raster data (including topological data structures), - basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.), - basic algorithms for raster (encoding, quad trees, map algebra), - networks and related algorithms such as shortest-path. 3. Applications of GIS to real-world problems Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to: 1) Explain what a GIS is and what real-world problems it can help solve; 2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS; 3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources; 4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved; 5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed; 6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems - Slides of the lectures (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0. The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point. A student passes the course only if the following two conditions are satisfied: 1. The average grade of both graded assignments is ≥ 5.75. If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it. AND 2. The grade of the final exam is ≥ 5.75. The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 Landscape Architecture

AR3LA011	Landscape Architecture Analysis and Visualisation	5
Course Coordinator	Dr. L. Cipriani	
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	Dr. D. Cannatella	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Summary	The course aims to provide students with a platform where (1) they can get to know digital and analog tools as vehicles for research and design in the field of landscape architecture and (2) enable students to experiment with alternative new approaches to thinking and communicating about spatial design issues. Workshops will be organized according to the four spatial dimensions: maps (2D), sections and models (3D), and time (4D).	
Course Contents	To represent the landscape means to understand, interpret, and transform it. It refers to a process rather than a completed product. Representation entails cartographic, and model explorations, activities that construct and visually communicate spatial and temporal knowledge and design. Representation helps landscape architects to acquire new or latent information, which is the basis for generating spatial knowledge through time. Furthermore, it is a creative act that helps the process of communication of design possibilities. This set of workshops intends to focus on representation emphasizing the four dimensions of space that are all particularly relevant for landscape architects: maps, sections, models, and time. Workshops will include learning analog and digital tools to enable landscape designers to use different approaches to think and communicate about spatial design issues. The course consists of three separate workshops where you get to know landscape architecture analysis and visualization methods in a practical setting: 1. Thinking with Maps: GIS in Landscape Architecture 2. Thinking with Sections and Models 3. Drawing Time	
Study Goals	By the end of this course, you should be able to: - apply and present different relevant design-related analysis and visualization methods and techniques in landscape architecture; - identify the potentials and limitations of the relevant design-related analysis and visualization methods and techniques in landscape architecture; - select and use suitable design-related analysis and visualization methods and techniques in a particular research and design context (e.g., graduation project).	
Education Method	Three thematic workshops with an introductory lecture followed by a short assignment focusing on applying a particular analysis and visualization method or technique. The hands-on assignments are elaborated in groups of at least two students and guided by an expert in the field.	
Literature and Study Materials	Provided during the workshops.	
Assessment	Students can pass or fail. This decision will be based on the students presence, active participation in the workshops, and outcome. After each workshop, the student must hand in a report that presents the outcome and consists of the material produced in the workshop.	
Period of Education	Semester	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3LA020	Research Methodology in Landscape Architecture	5
Course Coordinator	Dr.ir. B. Cattoor	
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. G. Bracken	
Instructor	Dr.ir. B. Cattoor	
Responsible for assignments	Dr.ir. B. Cattoor	
Contact Hours / Week x/x/x/x	45 hours per quarter	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 2 3 4	
Course Language	English	
Summary	The course provides you with a framework of academic knowledge and skills for understanding, designing and conducting design research in landscape architecture. To this end, the course presents you with an overview of the most prevailing theoretical themes and perspectives in landscape architecture, and of the discipline's proven research strategies, methods, tools and techniques. Additionally, the course familiarizes you with the academic, formal and structural requirements of a landscape architectural research design in the context of the elaboration of your graduation plan.	
Course Contents	Research skills are not only crucial to landscape architecture researchers working in an academic setting, but are also becoming increasingly important to landscape architectural professionals as the complexity of contexts and challenges landscape architects are tackling is increasing at fast pace. This course provides you with (a) an in-depth understanding of the most prevailing theoretical perspectives and themes in the field of landscape architecture and landscape architecture research, (b) an overview of the disciplines most prevailing design research methods, strategies, tools and techniques, (c) important theoretical clues and a practical aptitude for developing a critical academic attitude towards design research in landscape architecture. The course furthermore teaches you (d) how to construct a research design that meets the disciplines academic, formal and structural requirements. The course is organized in three parts: 1. Theory in landscape architecture - introduction to theory in landscape architecture; - landscape architecture as a design process; - form, meaning and experience in landscape architecture; - society, language and the representation of landscape; - ecological design and the aesthetics of sustainability; - landscape as an integrator of site, place and region. 2. Research strategies, methods, tools and techniques in landscape architecture - introduction to research strategies and related methods, tools and techniques: description, modelling, experimentation, classification, interpretation, evaluation & diagnosis, engaged action, design projection, logical systems. 3. Composing a research proposal - building blocks of a research proposal, and their interrelation: theoretical framework, problem statement, research question, research strategy, methods, tools and techniques, relevance of the proposed research; - academic, formal and structural requirements of a research proposal.	
Study Goals	By the end of this course, students should be able to: - identify landscape architecture as an academic design discipline with its own theories, methods and techniques; - critically discuss the following, major theoretical themes in landscape architecture theory: (i) Landscape architecture as a design process, (ii) Form, meaning and experience in landscape architecture, (iii) Society, language and the representation of landscape, (iv) Ecological design and the aesthetics of sustainability, (v) Landscape as an integrator of site, place and region; - compare different research strategies in landscape architecture and their related methods, tools and techniques, among others: description, modelling, experimentation, classification, interpretation, evaluation & diagnosis, engaged action, design projection and logical systems; - critically reflect on different landscape architectural methods of analysis and design - build up a coherent, feasible and effective research plan for their graduation project, complying with the academic, formal and structural requirements of a scientific research proposal.	
Education Method	The course consists of lectures and seminars. The seminars follow the flipped classroom model, where you will be asked to prepare for the seminar sessions in forehand, by means of writing and presentation assignments.	
Literature and Study Materials	Groat, L. and Wang, D. (2002) Architectural Research Methods. John Wiley and Sons. Swaffield, S. (Ed.) (2002) Theory in Landscape Architecture: A Reader. University of Pennsylvania Press. Swaffield S. and Deming M.E. (2011) Research strategies in landscape architecture: mapping the terrain. Journal of Landscape Architecture, 6(1), 34-45. - and others depending on your individual research topic	
Assessment	Presentations and written assignments.	
Period of Education	Quarter	
Course evaluation	For the course evaluations see: http://kwaltiteitszorg.bk.tudelft.nl/	

AR3LA031	Graduation Studio Landscape Architecture: Flowscales	20
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	Dr.ir. G.A. Verschuure-Stuip	
Instructor	Prof.ir. E.A.J. Luiten	
Instructor	Prof.dr.ing. S. Nijhuis	
Instructor	J.R.T. van der Velde	
Instructor	Dr.ir. N.M.J.D. Tillie	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Dr.ir. I. Bobbink	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The Flowscales studio focusses on specific context-related design projects, in which knowledge from spatial design, ecology, civil engineering and social sciences is synthesized into a coherent multi-scale landscape design proposal. Within the Flowscales studio different labs provide specific research themes. Within the lab you cooperate with your lab-mates on research and define your own project. Spatial, societal and environmental issues are explored through the application of design research and research-by-design.	
Course Contents	Flowscales interprets and elaborates landscapes as infrastructures and infrastructures as landscapes. Social, cultural and technological developments of our society demand a fundamental review of the planning and design of the landscape, in particular in relation to environmental issues. Urbanization, ecological crisis and climate change are international challenges; while the technical challenges may be considerable, the spatial and cultural challenges are by far the largest. Therefore, a renewed understanding of space-time condition of landscape and its capacity to adapt, transform and develop further offers promising opportunities to find new solutions to these problems. In this changing context the Flowscale graduation studio focusses on strategy development and design exploration of landscape compositions and systems. The emphasis is on the design of new topographies to create conditions for spatial development by employing landscape-based approaches, ecosystem services and landscape architecture design principles. Here landscape design is considered as a synthesizing activity that explores the dynamic between structure and process in various natural, cultural and urban settings, taking the specific qualities of the place (genius loci) as a starting point. By combining theoretical and functional knowledge, and by combining several scales and disciplines in a stimulating learning environment, the studio aims to prepare you for world-wide challenges related to our (urban) landscape.	
Study Goals	The thesis should prove that you are able to: - select and use suitable design-related research strategies and techniques in a particular context; - determine and design landscape architecture strategies and spatial interventions at multiple scales, which meet aesthetic, technical, ecological, functional and ethical requirements as proof of their academic knowledge; - develop and assess a landscape design assignment from an environmentally ethic perspective, focusing on moral responsibility towards the natural world. - integrate knowledge from other design disciplines and scientific fields; - write a report of an academic standard including method, theory, research and design; - can substantiate research and design decisions based on ethical and verifiable arguments; - present the work by combining oral, written and graphical media.	
Education Method	- studio meetings Flowscales - lab meetings - lectures - seminars - workshops - writing - fieldwork	
Course Relations	The accompanying courses Research Methodology and Landscape Architecture Analysis and Visualisation will help to develop your skills and attitude in theory, methods and techniques in becoming a landscape architect.	
Literature and Study Materials	Please check the semester guide.	
Assessment	In this process of graduation visual thinking and communication are considered to be crucial. Drawings, mappings and models are used to reveal and create relationships, explore and elaborate landscape systems (in terms of geometry, quantity, velocity, force, trajectory) and for critical reflection. - products of design research and research-by-design (as defined in semester book) - written and illustrated report - oral presentation	
Period of Education	Semester + MSC4	
Leerstoel	Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 4 Landscape Architecture

AR4LA010	Graduation Studio Landscape Architecture: Flowscales	30
Course Coordinator	Prof.ir. E.A.J. Luiten	
Course Coordinator	Dr.ir. I. Bobbink	
Instructor	Dr.ir. S.I. de Wit	
Instructor	Dr.ir. G.A. Verschuure-Stuip	
Instructor	Prof.ir. E.A.J. Luiten	
Instructor	Prof.dr.ing. S. Nijhuis	
Instructor	J.R.T. van der Velde	
Instructor	Dr.ir. N.M.J.D. Tillie	
Instructor	Dr. L. Cipriani	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	135 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The Flowscales studio focusses on specific context-related design projects, in which knowledge from spatial design, ecology, civil engineering and social sciences is synthesised into a coherent multi-scale design proposal. Within the Flowscales studio different labs provide specific research themes. Within the lab you cooperate with your lab-mates on research and define your own project. Spatial, societal and environmental issues are explored through the application of design research and research-by-design.	
Course Contents	Flowscales is about designing landscapes as infrastructures and infrastructures as landscapes. Social, cultural and technological developments of our society are demanding a fundamental review of the planning and design of its landscapes, in particular in relation to environmental issues and sustainability. Urbanization, ecological crisis and climate change are international challenges; while the technical challenges may be considerable, the spatial and cultural challenges are by far the largest. Therefore, a renewed understanding of space-time condition of landscape and its potential for change offers promising opportunities to find new solutions to these problems. In this changing context the graduation studio focusses on strategy development and design exploration of landscape compositions and systems. The emphasis is on the design of new topographies to create conditions for spatial development by employing landscape-based approaches, ecosystem services and landscape architecture design principles. Here landscape design is considered as synthesising activity that explores the dynamic between structure and process in various natural, cultural and urban settings, taking the specifics of the place (genius loci) as a starting point. By combining theoretical and functional knowledge, and by combining several scales and disciplines in a stimulating learning environment, the studio aims to prepare you for world-wide challenges related to our (urban)landscape.	
Study Goals	The thesis should prove that you are able to: - select and use suitable design-related research strategies and techniques in a particular context; - determine and design landscape architecture strategies and spatial interventions at multiple scales, which meet aesthetic, technical, ecological, functional and ethical requirements as proof of their academic knowledge; - develop and assess a landscape design assignment from an environmental ethic perspective, focusing on moral responsibility towards the natural world. - integrate knowledge from other design disciplines and scientific fields; - write a report of an academic standard including method, theory, research and design; - substantiate research and design decisions based on ethical arguments; - present the work by combining oral, written and graphical media.	
Education Method	- studio meetings Flowscales - lab meetings - lectures - seminars - workshops - writing - fieldwork	
Course Relations	The accompanying courses Research Methodology and Landscape Architecture Analysis and Visualisation will help to develop your skills and attitude in theory, methods and techniques in becoming a landscape architect.	
Assessment	- Products of Design Research and Research-by-Design (as defined in semester book) - Written report - Oral presentation	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Leerstoel	section of Landscape Architecture	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

variant Urbanism

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 1 Urbanism

AR1U090	RandD Studio: Analysis and Design of Urban Form	10
Course Coordinator	B. Hausleitner	
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	10 hours per week starting from week 1 and ending in week 9.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Summary	Dealing and designing with the growing complexity of a city requires a good understanding of the basic structure and key urban elements and the ways they work. Low Lands, the quarter 1 design studio of the Urbanism master track aims to develop students' critical reading, understanding and projecting of urban structure, form and functioning at different scales. Through a sequence of analysis and design exercises, students get familiar with the urban design language and the urban landscape's key structural components and fabrics. The studio approaches design with site sensitivity by emphasising the cultural values and environmental qualities of the specific place. Around the globe, Low Lands are seen historically as places with the highest degree of urbanisation. Their highly dynamic and fragile landscapes and high level of exposure to the effects of climate change question historically developed urban centres and the idea of the urban project. The studio takes the condition of the Western part of the Netherlands exemplary for such Low Lands to rethink forms and places of urbanisation. The Research and Design Studio puts the following lenses central: the landscape and site; the urban typo-morphology; the relationship and dynamics among the main elements of the urban landscape. These perspectives on the design of the urban landscape will be introduced and discussed in the lectures and workshops on History and Theory of Urbanism (AR1U121). The content of the lectures is crucial input for the work in the Studio. By making analytical and conceptual maps, drawings and models, by comparing, combining, analysing, concluding and projecting, students will learn: to identify the main layers of the urban and their relational integration; to relate and design with multiple scales and layers; to understand the relation between urban composition and performance, and the fluidity of this relation; to experiment with various urban concepts; to use design to uncover potentials and challenges in a project; to design for dealing with uncertainty via a re-composition of the main material layers; to (re)conceptualise the idea of the city. Students will build a body of knowledge by searching, selecting and using different sources and reflecting on their work in a theoretical framework of Urban Design. Students will be able to point out which urban elements and structures had, have and could have an influential role in shaping the Low Lands. Students will discover the structural qualities of the Low Lands, its permanence and fluidity, and the ruptures and discontinuities. During the entire studio, they will use drawings, models and other representational techniques to communicate their findings and explore and develop their projection of the future of the Low Lands, including long-term developments and short-term interventions.	
Course Contents	Content of Q1: Analysis and Design of City Form: Low Lands. Design Studio; Workshops; Lectures. All information is presented in the Quarter Guide available on Bright Space. Students will analyse and compare the structure, form, and space of three different Dutch places in this studio. These are related to the three main physical conditions in the Low Lands: coastal, riverine, and reclaimed land. Students work in groups of three-four with one tutor on one of the Low Land conditions. Each student will work individually on an atlas and the projection of one of the three places in the Low Lands. The atlas is based upon the diachronic and synchronic mapping of the functional and material layers across different scales from different perspectives. The projection synthesises in a designly way future forms of inhabitation. Lectures, debates, and workshops support this quarter's individual research and design assignment. The complementary workshops enhance skills in critical thinking, representation techniques, and the formation of narratives and introduce law guiding spatial plans. The students' different products (atlas, projection, exhibition) require different working attitudes. The different forms of education activate the learning process and advance the level of knowledge, skills and academic attitude.	
Study Goals	At the end of this course, the student is able to: - to recognize and understand the structural and spatial components of a city. - to test and show the effects of structural interventions on multiple scales and layers. - to use analysis and experiment as a design tool. - to point out which urban elements and structures had, have and could have an influential role in the urban (re)development. - to structure and shape a narrative on the actual condition and future potential of a city. - to (re-)conceptualize and shape the idea of the city in models and drawings. - to build a body of knowledge by searching, selecting and using a wide variety of sources and by reflecting their work to a theoretical framework in Urban Design.	
Education Method	Studio, lectures, workshops and instructions. Combination of mainly individual but also partly group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course.	

	Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 1.
Concept Schedule	The studio sessions are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. In addition to these studio sessions lectures, excursions, workshops and seminars will be scheduled on other day-parts tuned with the schedule of the parallel course AR1U121. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1U100	RandD Studio: Designing Urban Environments	10
Course Coordinator	V. Muñoz Sanz	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	M.M.E. van Esch	
Responsible for assignments	V. Muñoz Sanz	
Contact Hours / Week x/x/x/x	15 hours per week starting from week 4 and ending in week 9 (NB. also activities scheduled in week 1, 2 and 3).	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Summary	The object of the R&D studio is the design of sustainable urban environments. The focus is on the scale of the urban project, exploring the agency of urban design in addressing urgent societal and environmental challenges. Students work in groups and individually in the context of a particular environmental challenge faced by a specific urban area in the Netherlands. The goal is to improve socio-ecological inclusivity of the urban environment by developing an urban design project that embraces the understanding of the urban landscape as a complex and dynamic system. Thus special attention is given to the relation between design and engineering (parallel course AR1U131), and the application of knowledge gained during the first quarter.	
Course Contents	The Department of Urbanism has set as its mission to advance, share and apply knowledge on how to adapt the built environment to societal and environmental changes and how to apply design, planning and engineering interventions to better satisfy human needs, and to achieve fairer, healthier and more resilient urban landscapes (The Delft Approach to Urbanism, 2021). Against this backdrop, the object of this R&D studio is the design of sustainable urban environments, exploring the potential of spatial design to act as a catalyst for socio-ecological inclusivity. The ultimate goal is to improve the socio-ecological performance of a given urban environment towards a fairer, healthier and more resilient future. For that, students should embrace the understanding of the urban landscape as a complex and dynamic system. That implies considering a variety of issues which shape and impact the built environment across multiple scales and temporalities. These range from urban climate adaptation and mitigation, landscape systems, energy, typomorphology, parcel structure, density, programme, access, and ownership, to situated knowledge, architectural qualities and sensorial experiences. If the first quarter focused on analyzing and designing structures, forms and systems at the city scale, this studio combines the visionary with an ambition to create facts on the ground. Therefore, while insisting on the importance of trans-scalar relations, the focus is on the scale of the urban project. That is, students should question the agency of urban design in addressing urgent societal and environmental challenges through concrete ideas and projects on the ground. Specifically, students work in groups and individually in the context of a particular challenge faced by a specific urban area in the Netherlands. Overall, this quarter puts the stress on: 1. Connecting research and design towards a comprehensive understanding of socio-ecological inclusive urban design; 2. The integration and collaboration across disciplines, specifically design and engineering; 3. Skills in spatial thinking and communication, emphasizing parallel processes and iterative feedback loops between analysis, design, and presentation.	
Study Goals	At the end of this course, the student is able to explain: - sustainable development as a complex multidimensional-multiscale process consisting of environmental and liveability dimensions on street-, neighbourhood- and city-scale. - human dimensions in urban design related to urban design thinking, environmental psychology and urban complexity. - the characteristics of natural and human systems on both the neighbourhood and public space level. - contemporary urban transformation practices in a Dutch context. The student is able to analyse: - the urban environment from an urban landscape, typo-morphological and open space perspective. - the environmental and liveability dimensions on the street and neighbourhood scale, in the context of the city. - the synergy between natural and human systems on the neighbourhood and public space level. - the overall socio-spatial-environmental performance of the urban environment in terms of liveability. The student is able to design: - a strategy/vision on neighbourhood scale, in the context of the city. - a public space as a translation of the strategy/vision, by combining design and engineering, including natural and human systems on the neighbourhood and street-level. - context specific sustainability criteria to evaluate the overall performance of the strategy/vision and public space design in terms of liveability. The student is able to evaluate: - the strategy/vision and public space design based on the overall goals and criteria that were set by the student in the beginning. The student is able to demonstrate: - skills in writing, drawing and calculating as a means to generate design ideas and concepts. - skills in reporting and presenting to communicate to a variety of audiences. - skills in collaborating.	
Education Method	Studio, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Special Information	(External) students can only enrol for this course when also enrolling for the AR1U131 course, since the AR1U100 and the AR1U131 course are intertwined.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 2.	
Concept Schedule	The studio sessions are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and	

	Friday afternoons. In addition to these studio sessions lectures, excursions, workshops and seminars will be scheduled on other day-parts tuned with the schedule of the parallel course AR1U131. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR1U121	History and Theory of Urbanism	5
Course Coordinator	B. Hausleitner	
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Dr.ir. G. Bracken	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Dr.ir. G. Bracken	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Summary	The course gives an overview of the 'body of knowledge' of Urbanism. First, it explores the idea of the 'urban' and its major related concepts. Next, it analyses and determines the conditions of their emergence within a broader cultural context. Finally, it traces how these concepts have changed through time to enhance our present understanding of urbanised landscapes, cities, and their transformation. The programme combines urban history and theory by focusing on the historical evolution of the 'urban' as a spatial, material, socio-economic, political, cultural, and ecological project. Students will learn how to bridge the gap between theory and design and between past and present by relating critical thinking to concrete urban plans and designs.	
Course Contents	The course provides a lecture series on the theory and history of urban design and urban planning. The aim of the course is: - to foster awareness of the disciplines intrinsically historical nature, the evolution of which is motivated by concepts, interpretations, and interactions with social, political, cultural, economic, technological, geographical, and environmental processes. - to teach students how to assess the historical background of urban plans and designs from different historical periods. - to enable students to identify and explain the urban as form, structure, function and process, the relation between space, society, landscape, and site, and its composition, elements and material layers. - to provide students with the ability to evaluate the theoretical concepts underlying urban interventions, the evolution of the discipline in terms of the goals it sets itself and their - alleged - impact. - to enhance academic writing skills and to enable students to write about their discipline, urban phenomena, design and planning proposals, and theories.	
Study Goals	At the end of this course, the student: - Has knowledge of the theoretical fundaments of the discipline of Urbanism. - Is able to position historic urban designs and theories, and to relate them to the specific societal context of a historic period. - Is able to take a critical position concerning different design and planning experiments and theories of Urbanism, based upon knowledge and arguments. - Is able to position his/her own opinions, designs and plans in relation to the history and theory of Urbanism.	
Education Method	Lectures, self study (readings) and debates. Combination of mainly individual but also group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1.	
Concept Schedule	The studio sessions of AR1U090 are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. The AR1U121 course schedule is tuned with the AR1U090 course, thus not scheduled on Tuesdays and Fridays. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR1U131	Sustainable Urban Engineering of Territory	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	15 hours per week starting from week 1 and ending in week 4.	
Education Period	2	
Start Education	2	
Exam Period	2	
Course Language	English	
Summary	This quarter focusses on the making or remaking of the city and this course more specially on the city as technical construction: the natural and man-made conditions of the urban landscape, in order to integrate all interest in an urban plan. The act of integration and design requires systemic knowledge on a wide range of subjects. In this course the city is seen as a technical construction, with the central question being how to reconnect this construction to the natural systems of the territory? Therefore, this course provides vital input for the parallel studio of AR1U100.	
Course Contents	In the master track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable urban development. This quarter focusses on the making or remaking of the city and this course more specially on the city as technical construction: the natural and man-made conditions of the urban landscape, in order to integrate all interest in an urban plan. The act of integration and design requires systemic knowledge on a wide range of subjects. In the course the city is seen as a technical construction, with the central question being how to reconnect this construction to the natural systems of the territory? In new to build areas, or green fields, there is a lot of freedom to engineer or not to engineer the natural system. One of the greatest urban challenges these days is urban renewal. This is a complex enterprise in itself because you deal with the existing urban use and fabric. On top of that we need to deal with trends like climate change and the energy transition. One dimension that is reintroduced in the urban project is the subsurface system, which plays a crucial role in water management, ecology and energy supply. The subsurface system was for a long time excluded from the urban planning and design process, because it was considered a technical aspect that was dealt with by civil engineers, not as part of the urban design. However, the subsurface sets the conditions with highest impact: it is more costly and takes more time to change a cable system then it is to build a building or construct a road. Especially the idea that the natural system has already been altered for urban use, and thus lost, is preventing innovative solutions that deal with climate change and the energy transition in urban renewal. In order to incorporate natural and technical conditions in urban plans, a better cooperation with civil engineers and a better understanding of what they do not know is crucial for urban designers. This is what the course aims at. To bring order in all the technical information we make use of the System Exploration Environment and Subsurface (SEES) that uses six functional layers with different dynamics, professional domains and knowledge fields: people, metabolism, occupation, public space, infrastructure and subsurface. The subsurface layer is ordered in (for the urban planner and designer) recognizable themes: water, energy, civil constructions and subsoil. Within the themes there is a logical order in the qualities of the subsurface (www.ruimtexmilieu.nl). SEES connects the subsurface information with the urban surface in order to inspire and set clear boundaries for the development of the urban surface. It is used for analysing potential problems, chances, demands and supports a creative interaction early in the process of urban planning.	
Study Goals	At the end of this course, students are able to: - Explain what for them is sustainable development and how technology is integrated. - Evaluate the skills of an urban designer in relation to sustainability. - Explain the city as a hybrid system of natural and human characteristics both the surface and subsurface level. - Analyse the synergy between natural and human systems on the surface and subsurface level. - Identify new approaches in which the conditions and chances that are given by the natural system are re-introduced. - Demonstrate and design spatial principles or an urban ensemble with the desired performances, by combining design and engineering, including natural and human systems on the surface and subsurface level to develop ethical standards considering sustainable development.	
Education Method	Interactive lectures and intensive workshops the first three weeks of the quarter full time. Mainly group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The assessment of the assignments is done using the rubric that is available in the quarter guide on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student in the white week between Q2 and Q3, thus in the week between academic weeks 2.10 and 3.1, or the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 2.	
Concept Schedule	The AR1U131 course is 15 days scheduled in the first 4 weeks of Q2. The AR1U131 course schedule is tuned with the AR1U100 course. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 2 Urbanism

AR2U086	RandD Studio: Spatial Strategies for the Global Metropolis	10
Course Coordinator	N. Katsikis	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	V.E. Balz	
Responsible for assignments	V.E. Balz	
Contact Hours / Week x/x/x/x	10 hours per week starting from week 1 and ending in week 9.	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Regional design is the core assignment of the R&D studio in this quarter. Students work in groups on regional analysis and design for a self-defined goal in the framework of sustainable development, focusing on a metropolitan region in the Netherlands. The rationale behind this assignment is the fact that the way global economic powers influence social, cultural and environmental development is best sensible at the regional level. Such global influence results in the inability to fully control spatial development. In this sense, regional design is seen as a tool for steering development in the right direction, which -as the exploration of plausible futures- promotes and debates solutions to problems in a given context. It is a reflection on prevailing spatial conditions, political agendas and planning regimes, meant to improve good (democratic) decision-making and to inform long-term strategic planning approaches to desirable spatial change.	
Course Contents	This quarter emphasizes on (1) a comprehensive, evidence-informed understanding of regional spatial structures and development trends, (2) an understanding of interrelations among design, planning and politics and (3) communication skills that are required in collaborative decisionmaking. During the R&D studio 'Spatial Strategies for the Global Metropolis' students use this knowledge and skills to conduct a regional design. The design process knows two products, notably (1) a spatial vision and (2) a development strategy. Products are interrelated. The vision represents a desirable spatial future; it serves as a guiding normative principle for the development strategy that sets out a path towards spatial change, by means of spatial interventions that are ordered over a time and associated with capacities of actors in development. The R&D studio 'Spatial Strategies for the Global Metropolis' is the core activity of this quarter. Students conduct a regional design in groups of 4-5 students. The thematic exercises of Spatial Development Strategies (SDS) are an integral part of the studio. Knowledge on regional design and planning approaches will be provided during lectures and applied during workshops. SDS assists in and steers studio work. The series Capita Selecta also adds to the studio. It provides students with knowledge about spatial planning and discusses relations between planning, governance and design on these grounds. Parallel to the R&D studio runs the course 'Research & Design Methodology for Urbanism'. The course focuses on a theoretical understanding of design, planning and research. Students learn to position their work in a theoretical debate and write a report on these grounds.	
Study Goals	At the end of this course, the student is able to: - Understand the basic roles and instruments of strategic spatial planning in delivering public good, spatial quality and equality and sustainable regional spatial development. - Understand the complexity, multiscalarity and uncertainty of regional spatial development; can consider the limitations that these conditions set to regional planning and design. - Understand and critically reflect on roles and impacts of regional design in/on inclusive planning decision-making. - Formulate and argue for a comprehensive regional vision, drawing on commonly shared values and norms, evident regional spatial development and appropriate planning principles. - Convert a vision into a regional development strategy that is relevant and feasible in a given institutional context and robust in respect to uncertainties of long-term regional development; can estimate a fair distribution of costs and benefits among stakeholders involved. - Justify a vision and development strategy conceptually, making use of theoretical notions and an understanding of how theory and practice interact. - Use communication media that are effective in collaborative decision-making (visualize design proposals clearly, consistently and persuasively, using images and text); can engage in critical debate. - Explain the ethical issues involved in the activity of planning and designing for people.	
Education Method	Studio, lectures, workshops and instructions. Mainly group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student in the the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Special Information	(External) students can only enroll for this course when also enrolling for the AR2U088 course, since the AR2U086 and the AR2U088 course are intertwined. (External) students enrolling for this course should have developed design skills.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 3.	
Concept Schedule	The studio sessions of AR2U086 are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. The AR2U088 course schedule is tuned with the AR2U086 course, thus not scheduled on Tuesdays and Fridays. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2U088	Research and Design Methodology for Urbanism	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. M.M. Dabrowski	
Course Coordinator	R.C. Rocco de Campos Pereira	
Responsible for assignments	R.C. Rocco de Campos Pereira	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Required for	The EC of this course have to be obtained in order to be allowed to enrol for the P2 presentation of the MSc3.	
Summary	This course enables you to do academic research that will support and fundament your work in the parallel AR2U86 studio and in the Master 3 and 4 graduation project.	
Course Contents	The component Methodology for Urbanism runs parallel to the studio. It is one of the core elements of the semester. It enables you to do academic research that will support and fundament your work in the studio. This is different to the studio because here you will concentrate on traditional forms of academic research, which you will connect to less traditional forms of research, like research by design. This connection between traditional and non-traditional (design-based) forms of research is one of the characteristics of education and research in the Department of Urbanism of the TU Delft. The methodology component will help you*: EXPLAIN what a theoretical framework is. BUILD a theoretical framework that will sustain your research and design in Q3. IDENTIFY a community of authors and practitioners who write about the core ideas of your theoretical framework. WRITE an academic report, in which you will describe what are the main questions you will seek to answer in Q3 and the best methods to answer them. EXPLAIN the values connected to and the ethical issues involved in the activity of planning and designing for people. Being able to formulate your own problem statement, research questions and methodology is one of the goals of the Urbanism Master. The theoretical framework is the foundation on which the whole research and design are based. There is nothing as practical as a good theory. This is because a theory is a knowledge framework, around which you can build your own ideas, be inventive and innovative. Understanding what theories, ideas and practices exist will help you be even more innovative and groundbreaking (because you will not be reinventing the wheel). * Those are our learning objectives summarized. The guiding concepts underlying this course are: Urbanism is a trans-disciplinary field of study and practice and there are different logics of enquiry involved belonging to the human sciences, to the physical sciences and to design. These logics of enquiry conceive questions and methods differently. It is necessary to clarify these different logics of enquiry, their different questions and methods, and how they can work together, in order to be able to do research in Urbanism. The model of knowledge-building used in this course is communicative/ inter-subjective. We assume that all knowledge is constructed inter-subjectively. Knowledge needs to be communicated in order to be validated, tested, and integrated in existing knowledge. Hence the emphasis on communication. There are different ways to achieve knowledge and students and teachers need to discuss and clarify which ones are valid, relevant, ethical and effective for Urbanism. For instance, there are different ways to do research in design-based practice: how to connect design research with other (more academic) ways of doing research? Pedagogical goals: The course aims to promote: Acquisition of knowledge on basic concepts of philosophy of science. Development of critical and analytical skills. Development of argumentation skills. Clarity in presentation and communication of design and research. Excellence in writing and communication skills.	
Study Goals	At the end of this course, the student is able to: * EXPLAIN what a theoretical framework is. * BUILD a theoretical framework that will sustain your research and design in Q3. * IDENTIFY a community of authors and practitioners who write about the core ideas of your theoretical framework. * WRITE an academic report, in which you will describe what are the main questions you will seek to answer in Q3 and the best methods to answer them. * EXPLAIN the values connected to and the ethical issues involved in the activity of planning and designing for people, thus: explain what ethics in Urbanism is, explain some of the ethical challenges connected to the practice of urbanism, describe different frameworks to understand ethical issues and articulate his or her position towards specific ethical issues connected to Q3.	
Education Method	Workshops and lectures. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Combination of written and oral examination (integrated in the AR2U086 design examination). A rubric will be used for grading. The rubric will be available in the quarter guide or on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student in the the first week of summer, thus week 5.1. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Special Information	(External) students can only enroll for this course when also enrolling for the AR2U086 course, since the AR2U086 and the AR2U088 course are intertwined.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 3.	
Concept Schedule	The studio sessions of AR2U086 are scheduled in two shifts: students will be enrolled in either Tuesday and Friday mornings or Tuesday and Friday afternoons. The AR2U088 course schedule is tuned with the AR2U086 course, thus not scheduled on Tuesdays and Fridays.	

Minimum number of participants	The actual schedule will be available via Bright Space > My Timetable. For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Urbanism, Free electives 15 ECTS

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

AR keuzevakken 2023

AR0049	Urban Geography	5
Course Coordinator	R. Ordonhas Viseu Cardoso	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	5 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Course Contents	As a student of the built environment or its related disciplines, you most probably have an interest in cities. Your research concerns and design experiences are mainly focused on urban contexts, and your future outputs, either as academic or practitioner, can contribute to shape and influence those contexts. Therefore, a wide-ranging discussion about the many aspects, scales and complexities of urban space should be at the forefront of built environment education. The major questions facing cities today are far from settled, and heated debates continue about what is a city after all, how it can be defined in times of reconfiguration at higher spatial scales, how it can be managed and improved in face of growing environmental challenges and social conflicts, and how its economy can reap the benefits of urban agglomeration and interaction in a way that works for citizens, firms and organisations. This multiple and integrative perspective on cities is best captured by the broad lens of Urban Geography. This course provides an introduction to the massive problems and promises of contemporary cities, allowing you to frame your work in larger spatial, social, economic and political urban contexts. You will gain valuable insights about the main contemporary debates in urban geography and learn to recognize, criticize and gather evidence about processes of urban transformation to become a more grounded and relevant agent of change. In the course, we will discuss and reflect upon the key themes and thinkers addressing cities as a research concern. Through a variety of active learning means, such as roundtable discussions, practical assignments, a fieldtrip and empirical research, you will be introduced to the long tradition of urban geography, following a thematic approach that sees cities as the outcome as well as a generator of political ideas, economic processes, flows of people, things and information, and cultural and technological changes. We will also consider the different definitions and boundaries of the urban and discuss the main trends influencing the future of cities in policy-making, research and the media.	
Study Goals	At the end of this course, the student is able to: - Summarise the key concepts of urbanisation, urban economy, urban ecology, urban networks and governance, and discuss their role in urban development. - Apply these concepts critically to existing contexts and identify their implications for urban spaces through illustrative examples. - Analyse a real-life urban setting, using the relevant concepts of Urban Geography to support an empirical research question. - Design a research approach able to explore current and future urban transformations in a case study and develop evidence-based arguments to support his/her approach.	
Education Method	The course includes interactive lectures, short home assignments, roundtable discussions, defining a group research question, and conducting empirical research on a case study city. Interactive lectures A team of engaging lecturers will introduce the five key themes of the course as well as the associated scientific and policy debates. These lectures are broken down into discussion points where intervention by the students is encouraged and the concepts are discussed and contested. The lectures are an indispensable starting point for the course, as the students must apply the concepts learned in their home assignments and the empirical research. Besides the theoretical and conceptual lectures on the weekly topics, there will be short informative lectures providing background information on the case study city aligned with the five themes. Home assignments Each week, there will be a short home assignment aimed at consolidating the topics introduced through illustrative examples and a short reflection. These exercises should be critical, focused and able to contribute to a roundtable discussion. Roundtable discussions The instructors will moderate discussions on the weekly topics and associated scientific and policy debates. The roundtable discussions allow every student to intervene and confront their insights with the work of other students, thinking about different viewpoints and respective implications. Research preparation During the first five weeks of the course, the students should start formulating their research questions, based on their own interests, the topics and concepts covered, and the additional discussions about the case study city. With moderation by the instructors, these individual questions will be clustered according to their similarity and compatibility to form working groups to conduct the empirical study. The first task of the group is to reassess their individual approaches and formulate a group research question that serves as the point of departure for the research. You will receive instructions about developing a research question, method and plan. Empirical research The remaining weeks of the course are dedicated to conduct an empirical research study on the case study city, based on one or several of the topics covered in the course and departing from the group research question. A fieldtrip is planned for students to make observations, discuss with local actors, and visit different locations related to key urban geography debates. The outcome of this research is an academic paper (about 5,000 words) which should include the problem statement, theoretical grounding, literature review, methodology, results and discussion. You will receive a detailed assignment sheet with the instructions, desired outcomes and assessment rubrics. In week 4.10, the groups present their results to their colleagues, instructors and guests in the form of a knowledge clip. Throughout the duration of the study, the instructors are available to provide feedback to all the groups in the scheduled time slots.	
Literature and Study Materials	Recommended readings about the weekly topics will be available on the specific Bright Space page. The readings are not mandatory but we expect that you read most of them so that you can reflect more on the current debates and support better your active participation in the interactive lectures and the assignments.	
Prerequisites	Bachelor in Architecture, Urbanism and Building Sciences TU Delft or a Bachelor in a related field. The course is particularly geared towards students following the Master Architecture, Urbanism & Building Sciences and fits with all tracks.	
Assessment	A combination of assessment strategies will be applied. The main elements graded at the first stage of the course are active and informed participation in the class discussions (10%) and five brief weekly assignments (5 x 8% = 40%). At the second stage, the elements of assessment are the final piece of group research (40%) and the production of the knowledge clip (10%). A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course. The grade for the group items is given to the group, but also counts as individual grade. In case of free-riding by individual students, the coordinators can deviate from this procedure and differentiate grades within a single group.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled twice a week in the afternoon. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0051	Online Digital Portfolio	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Course Coordinator	H.P. Kiksen	
Instructor	H.P. Kiksen	
Contact Hours / Week x/x/x/x	46 hours per semester	
Education Period	1 2 3 4	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Summary	The student is taught to design and construct a presentation portfolio. This portfolio is presented on-line by means of a web browser. The portfolio contains a relevant summary of both the education parts followed by the student during the Bachelor and Master 1 and any extra-curricular activities.	
Course Contents	At the end of the course, the student will enter the labour market. To present him- or herself the student commonly uses a portfolio folder, showing the designs made in university. It is important that this portfolio is up to date and shows a selection of the best designs. By making this portfolio suitable for digital media, the students can easily present themselves internationally. During this course the student is taught in which way such a digital portfolio can be composed. During the course, the following aspects will be addressed: - research on the student's own goals for the portfolio, - research on existing portfolio for references, - practice with communicating a message, - building webpages with HTML, CSS and jQuery, - take different devices and screen sizes into account, - graphic design, with emphasis on composition, - use of colour, fonts and other means to keep the attention of the viewer.	
Study Goals	The student: - can make a portfolio for a certain goal and bring this inline with the requirements for addressing the targeted user(s), - can clearly formulate the goal of the portfolio and the requirements for addressing the targeted user(s), - can name desired/undesired behavior/style of online portfolios/websites, - can make use of HTML, CSS and JavaScript, - takes different screen sizes into consideration while making the online portfolio.	
Education Method	Lectures: 10 hours Workshop: 36 hours Self study: 122 hours	
Literature and Study Materials	- The brightspace section for this course	
Assessment	- Written plan of approach in which the intended portfolio and requirements are described, - The actual portfolio (ie website), - Written report about research and design, ending with a reflection on the learning process.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Wednesday	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0089 (for U, LA, MBE students only)	Design in Process, Process in Design	15
Course Coordinator Course Coordinator Responsible for assignments Contact Hours / Week Education Period Start Education Exam Period Course Language Course Contents	Ir. K.P.M. Aalbers Dr. F.L. Hooimeijer Dr. F.L. Hooimeijer 4 hours per week starting from week 1 and ending in week 9 plus 32 internship hours per week starting from week 1 and ending in week 9. 4 4 4 English This course is only meant for students from the MSc AUBS tracks of Urbanism, Landscape Architecture, and Management in the Built Environment.	
	Current developments such as climate change, energy transitions, technological development and the changing role of governments mean that a new approach to urban development is required. The complexity of urban issues means that in urban area developments it is increasingly important to include agility and flexibility in realising plans, collaboration between different commissioning parties and a cross-disciplinary approach involving all the separate fields of expertise. How can the changes occurring in society effectively be given direction in this context? After a century of government initiatives, this century is now needing to find a new balance of tasks between government, market and civil society. According to economist Mazzucato, private money follows public money. This is certainly true if we look at the large-scale urban challenges. It is important that government authorities make clear choices and establish policies that express and steer social developments in specified ways, that they invest and bear the risks in this, and that they safeguard continuity. Identifying and outlining a long-term perspective contributes to minimising uncertainties and helps to attract investors. In terms of social relevancy, the role of thinking through issues by the act of design so that process and design lead conjointly to realisation is a promising prospect. The basic premise for the course is to study the role of planning and design in the complexity of current urban challenges. The focus is on the roles that actors have within urban development, how information is shared and knowledge is exchanged, and how this is reflected in the products the designs, master plans and zoning plans that take shape in the urban development process. Students perform theoretical and empirical research during an internship and will produce a journal paper in which they reflect academically on a certain topic or process that they encounter in their work. Their experience and ideas are shared during three (compulsory) workshops in which the research questions, methods and output is discussed and peer reviewed. Students have to provide the internship for themselves, without the internship the course cannot be done. The workshops are compulsory; without participation the student will not be graded.	
Study Goals	At the end of this course, the student is able to: - explain the roles and professional capacities involved in area development; - evaluate the current trends in area development; - analyse the products (master plans, urban design, zoning plans) and processes of area development; - identify new approaches in information and knowledge exchange in area development; - reflect on - and interpret observations in practice and make them part of academic analyses; - demonstrate scientific approaches in developing research questions, methods and the writing of a journal paper according to scientific rules.	
Education Method	Internship (4 days per week): The internship takes minimal 9 weeks in Q4 (can be extended over summer) and can be done in the role of: Public perspective; Private / co-creation perspective; Civil society perspective; Design office (mediator, co-creation, consultancy). We can help the student to find an internship, but the student is responsible for final place of work and arrangements. There needs to be clear agreements with the place of work on the fact that next to the internship there is also other education and that for the course there are compulsory workshops. For the internship a TU Delft format contract is compulsory (maybe the office also has its own contract). It needs to be signed by the director of education. The director of education will only sign one contract per student, therefore make sure to include an extension of the internship in the contract if that is what you aim for! Intensive workshops/paper writing (1 day per week): The course aims at reflecting back from practice into academia, this is done by discussion and research, resulting in an academic paper. The discussions and mentoring of the research are done in three intensive sessions in which: The skill of writing papers is presented and discussed; Assignments are done to support the writing of a paper; The elements of a paper (questions, methods, theory) are discussed; Peer review and consultations is performed. Mainly individual work but will peer review (fundamental to science).	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Written examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done in two steps. First step is to assess if the student has participated in all group sessions, and is meeting the learning goals that consider the practical work through the logbook (deliverable 1) rewards with a pass/no pass. Second step is the assessment of the paper (deliverable 2) on the base a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical- Reflection into the topic of research in relation to the place of work Late delivery	

	Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review proces and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled 1 day(part) per week. The other day(part)s are scheduled for the internship. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0095	Social Inequality in the City, Diversity and Design	5
Course Coordinator	Dr. R.J. Kleinhans	
Instructor	Dr. R.J. Kleinhans	
Instructor	Dr. C.E.L. Newton	
Responsible for assignments	Dr. R.J. Kleinhans	
Contact Hours / Week x/x/x/x	4 hours per week, starting from week 3.1 and ending in week 3.8.	
Education Period	3	
Start Education	3	
Exam Period	3 5	
Course Language	English	
Required for	This course is compulsory for certain categories of students from the LDE master specialisation "Governance of Migration and Diversity".	
Course Contents	This course takes a socio-spatial view on social inequality, diversity and migration, and how these phenomena impact on neighbourhoods and cities, but also on individual people. The focus is on the interchange between the built environment, population composition and residential behaviour. On a theoretical and empirical level, the course will address various forms of neighbourhood change over time, in particular socio-spatial segregation and urban regeneration, and their implications for neighbourhoods and individuals. Moving on to policy and design, the course addresses various urban policies and strategies addressing social inequality with an explicit place-based diversity component (such as social mix / tenure mix). Special attention will be devoted to the roles of relevant actors in these contexts. In doing so, the course tackles the question to what extent urban policies and strategies can contribute to absorbing negatively perceived outcomes of diversity and strengthen its perceived benefits. Apart from unravelling relevant theories, the course will provide examples of recent policies and/or strategies that have attempted to address city- and neighbourhood-based issues around migration and diversity. The acquired knowledge will be applied to a real life case study. In a Design Game, interdisciplinary groups of students will create strategic design proposals for a neighbourhood which suffers from social inequality and negative implications of its diversity, and for which clever solutions must be created. The design element focuses on strategies and socio-physical interventions rather than architectural/urban design in a narrow sense. The Design Game will be conducted in co-operation with the local authorities, welfare workers, housing associations and other stakeholders connected to the chosen site.	
Study Goals	After completing the course, students can: - Explain various forms of long-term neighbourhood change and their implications for neighbourhoods and individual residents, both from a theoretical and empirical perspective; - Identify the underlying mechanisms in urban policies and urban design which aim to create positive benefits from diversity and migration, or combat their perceived negative implications; - Evaluate the socio-spatial implications of various forms of social inequality and related policies from a multi-level and multi-actor perspective; - Design solutions (in physical, social, cultural and economic sense) for diversity-related problems in neighbourhoods, taking into account the roles and resources of relevant stakeholders.	
Education Method	The course starts with several interactive lectures and group discussions. In a large part of the course, a Design Game will be conducted for a neighbourhood which suffers from negative implications of social inequality and diversity. Groups of students will conduct place-based action research and design strategic interventions that smartly connect to the neighbourhoods strengths, weaknesses, and opportunities, taking into account the (local) context. This will be supported by using materials and other information provided by key persons from housing associations, the local authorities and other stakeholders. Feedback by instructors (both written and face-to face) will be offered with every weekly assignment that builds up to the final deliverable. Active participation and attendance in the Design Game sessions (one per week) is compulsory. Repeated absence without notifying the course co-ordinator in advance will result in deduction of points for the individual student (not the group).	
Literature and Study Materials	Will be provided as PDFs or URLs on Brightspace.	
Reader	For the Design Game, a separate reader will be provided on Brightspace.	
Assessment	Both instructor and peer feedback will be provided on a weekly basis on students Design Game assignments and the final report. The summative assessment is based on two elements: 1. The Design Game report (60% of final grade); 2. A written exam, based on the lecture, literature and other acquired insights (open questions: 40% of final grade). Each assessment element should be passed with a minimum grade of 5.5 before the final grade will be determined. If students fail to achieve the pass mark of 5.5, they have an opportunity to resubmit an improved version of the Design Game report within four weeks from the last work day of quarter 3. In this case, the maximum score will be sufficient, meaning that the students passes this course.	
Special Information	The maximum marking period is 10 work days.	
Elective	Yes	
Period of Education	Quarter 3	
Concept Schedule	Every Thursday morning in Q3	
Maximum number of participants	This course is also offered as a module in the LDE master specialisation 'Governance of Migration and Diversity'. Therefore, a maximum number of 20 students from the Faculty of Architecture can enroll in this course. Enrollment through BIS is done on a first-come-first-serve basis.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0097	Climate proof sustainable renovation: energy use, envir impact, health and comfort, life-cycle cost	5
Course Coordinator	Dr. A. Meijer	
Instructor	Dr.ir. A. Straub	
Instructor	Dr. L.C.M. Itard	
Instructor	Dr. A. Meijer	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Course Contents	Housing renovation processes are challenging social-technical systems. In this course, students gain knowledge of sustainable renovation from different perspectives (social, technical, financial, environmental) and apply gained insights and knowledge in a feasibility study for a housing renovation project. The interests of all stakeholders have to be merged to result in a successful renovation.	
Study Goals	Understand the roles of existing and new stakeholders in integrated renovation processes. Insight in life-cycle cost analysis (LCCA) and financial case, energy demand and energy calculations on building and neighbourhood scale, environmental impact and environmental life cycle assessment (LCA) of buildings and renovation. Insight in health and comfort of residents. Implementation and integration of gained knowledge in a feasibility study of a renovation project.	
Education Method	Challenge-based didactical approach: the assignment is a tender game of a feasibility study for a housing renovation project. Lectures cover several perspectives of integrated climate proof adaptations of the built environment.	
Literature and Study Materials	All literature will be handed in class.	
Assessment	Tender Game 50% of grade and Written Exam 50% of grade.	
Exam Hours	3	
Permitted Materials during Tests	Calculator	
Elective	Yes	
Period of Education	10 weeks	
Concept Schedule	Monday- and Tuesday morning	
Minimum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0106	Architectural Ethnography	5
Course Coordinator	N.J. Amorim Mota	
Instructor	Ir. P.S. van der Putt	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Instructor	Dr. A. Campos Uribe	
Responsible for assignments	N.J. Amorim Mota	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course stimulates students to research the correlation between environmental design and human behaviour from a cross-cultural perspective, acknowledging their positionality. The course explores a transdisciplinary approach, architectural anthropology, and explores a combination of research methods that include, but are not limited to visual ethnography, spatial analysis, and participant observation as key research methods to investigate the relations between humans, non-humans and the environment. In the first phase of the course, the participants will develop field work, literature review and data analysis to produce a research report and a poster. In the second phase, the students organized in groups will prepare a synthesis of the data collection, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. The final phase of the course is a personal reflection on the challenges, opportunities and relevance of using ethnographic methods for architectural research.	
Study Goals	By the end of this course students will be able to: 1. Identify elements of the correlation between environmental design and human behaviour from a cross-cultural perspective; 2. Analyse and synthesise the relations between humans, non-humans and the environment using a combination of architectural and ethnographic methods; 3. Elaborate a research report with a synthesis of the field work, literature review and data analysis, using adequate written and visual media; 4. Use story-telling as a medium to elaborate a visual narrative based on the results of the fieldwork, data collection and analysis; 5. Elaborate a critical synthesis of the research outputs, using architectural drawings as a preferred medium to enable meaningful communication between different audiences. 6. Formulate a critical reflection on the research methods, analytical process and criteria for the preparation of the research outputs, acknowledging the researchers' positionality.	
Education Method	The course Architectural Ethnography comprises group assignments and individual work. The main educational methods used in this course are lectures, tutorial sessions, and peer review sessions. While the group work will be the most important component of the course, each student will individually produce a critical reflection on the challenges, opportunities and relevance of Architectural Ethnography for Architectural research, based on the methods, processes and results of the work developed for the course. The participants in the course Architectural Ethnography will investigate different neighbourhoods / communities in a Dutch city. The participants will be divided in teams aiming at conducting fieldwork, observations and other forms of data collection in a case study area. Each group will be responsible for the production of the following deliverables: a) Research Report; b) Poster; c) Narrative and Visual Synthesis of the Ethnographic Research; Additionally, each student should produce an: d) Individual critical reflection on the research process and production of outputs.	
Literature and Study Materials	The course will use the following publications as main textbook references: Lucas, R. (2020) Anthropology for Architects: Social Relations and the Built Environment. London; New York: Bloomsbury Visual Arts. Kaijima, Stalder and Iseki. (2018). Architectural Ethnography - Japanese Pavilion Venice Biennale. Tokyo: Toto Pink, S. et al. (2020b) Methods for Researching Homes, in Making Homes: Ethnography and Design. London and New York: Routledge, pp. 93126. Rose, G. (2016) Visual Methodologies: An Introduction to Researching with Visual Materials. Thousand Oaks, California: SAGE. Stender, M., Bech-Danielson, C. and Hagen, A.L. (eds) (2021) Architectural Anthropology: Exploring Lived Space. Abingdon, Oxon and New York: Routledge.	
Assessment	The evaluation methods in this course comprise a combination of formative and summative assessments. The work handed in at the end of each instructional unit will be the object for the summative assessment and will be based on qualitative aspects. The tutorial sessions, the progress review sessions and the in-class peer-to-peer learning activities are the main formal methods of formative assessment. The summative assessment will be based on the following deliverables: a) Analytical Assignment: Ethnographic Research: Research Report (Group Work) b) Practical Exercises: Poster: Synthesis of (Quantitative) Data Collection (Group Work) Ethnographic Research: Narrative and Visual Synthesis (Group Work) c) Writing Assignment: Critical Reflection (Individual Work) For each student, the final grade is determined by a weighed calculation of the results achieved in group work a) + b) and individual work c). The weight of the different deliverables will be announced 1 week prior to the start of the course in Brightspace.	

	Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring semester, 3rd quarter (weeks 3.1-3.10)
Concept Schedule	Tuesday afternoon
Minimum number of participants	12
Maximum number of participants	48
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0107	Housing Studies: An open intersectional Archive	5
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course advances critical tools for the analysis and projection of twentieth-century affordable collective housing designs across the globe, examining the relevance and applicability of intersectionality theory to housing design. Geographically, the course looks into both western and eastern traditions of housing production. Methodologically, it investigates these traditions from the critical lens of global intersectionality. The course starts with analyses of case studies, followed by the development of a critical intersectional archive, organized in the form of a collectively-curated exhibition. The course starts with a discussion of intersectionality theory and its applicability to global housing design. Students will then develop analyses of each case study, revealing the historical and environmental circumstance of the projects commissioning in relation to its morphogenetic features (the projects DNA), as follows: - Students will critically reflect on the socio-economic, political, urban and territorial contexts from which each of these projects emerges, including institutional and governance frameworks and policy-making processes; - They will investigate how each case study and its designers are embedded in the field of global housing histories and cultural production, and how this embeddedness relates to particular design features (typo-morphology, composition, structure, details, materialization, technology); - Students will focus on how a specific form, discourse and territorial situation allow for care work, maintenance and other types of labor, and how they promote or challenge notions of gender roles, from the projects conception to its afterlife. In the last phase of the course, students will develop a critical intersectional archive that will be organized in the form of a collectively-curated exhibition at the Faculty of Architecture, featuring the analyzed case studies through research-based documentation, including graphic and textual outputs.	
Study Goals	Upon completion of the course the student is able to: - Analyze and synthesize the main generative components of a housing project, as well as the societal factors that impact it and vice-versa, by using adequate textual and visual outputs; - Understand the concept of (global) intersectionality and its relation with the architecture of housing; - Assess the influence of gender, socio-racial, economic and environmental factors in the development of an affordable collective housing project in relation to its particular circumstance, using the notion of global intersectionality as a critical lens; - Compose and present critical reflection in the form of an exhibition design using adequate academic protocols and archival research; - Apply the results of an analytical study to develop a collaborative curatorial project for a research-based exhibition on affordable collective housing design; - Present a curatorial design strategy to an audience of experts and non-experts.	
Education Method	The course Global Housing Studies will be structured in three phases: Phase 1 - Concepts and Theory: This phase is based on literature review; each session will be organized on the model of flipped classroom through lectures and peer-review assignments. Phase 2 - Analysis: This phase will be based on the application of theoretical and historical research allied to morphological and typological analysis. This phase will be based on tutorial sessions. Phase 3 - Projection: This phase will be based on the development of a critical projection of the analytical outputs in the form of a curatorial project. This phase will be based on tutorial sessions and group presentations.	
Course Relations	The Global Housing Studies is related with the theme and contents developed in the MSc2 Dwelling design studio 'Global Housing' (AR2AD012).	
Literature and Study Materials	The course will use the following publications as main references: Bond, Johanna. Global Intersectionality and Contemporary Human Rights. Oxford: Oxford University Press, 2021. Fitz, Angelika and Elke Krasn, eds. Critical Care. Architecture and Urbanism for a Broken Planet. Vienna, Cambridge MA, London: AzW, MIT Press, 2019. Fraser, Nancy. Contradictions of Capital and Care. New Left Review, no. 100 (2016): 99117. Glendinning, Miles. Mass Housing, Modern Architecture and State Power a Global History. London: Bloomsbury, 2021. Grossman, Vanessa and Ciro Miguel, eds. Everyday Matters: Contemporary Approaches to Architecture. Berlin: Ruby Press, 2021. Hayden, Dolores. The Grand Domestic Revolution: A History of Feminist Designs For American Homes, Neighborhoods, and Cities. Cambridge, MA: MIT Press, 1981. _____. What Would a Non-Sexist City Be Like? Speculations on Housing, Urban Design, and Human Work. Signs, vol. 5, no. 3, Supplement. Women and the American City (1980): S170S187. Kockelkorn, Anne: Palace on Mortgage. The Collapse of a Social Housing Monument in France, in Neoliberalism on the Ground. Architecture and Transformation from the 1960s to the Present, edited by Kenny Cupers, Catharina Gabriëlssohn and Helena Mattsson, 1944. Pittsburgh: University of Pittsburgh Press, 2020. Lane, Barbara Miller, ed. Housing and Dwelling: Perspectives on Modern Domestic. Architecture. London and New York: Routledge, 2007. Roberts, Marion. Living in a Man-Made World: Gender Assumptions in Modern Housing Design. London and New York: Routledge, 1991. Staub, Alexandra. Conflicted Identities: Housing and the Politics of Cultural Representation. New York: Routledge, 2017. Wakely, Patrick. Housing in Developing Cities: Experience and Lessons. Abingdon, Oxon: Routledge, 2018.	
Assessment	The evaluation methods in the course Global Housing Studies comprise a combination of two assessments:	

	- Analytical assignment: The student performs an analysis and reports the results in the form of both a written report and graphic material towards the constitution of an intersectional archive. - Practical exercise: The students collectively curate and design an exhibition based on the documents and materials assembled and produced in the analytical assignment. - The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.
Period of Education	Spring Semester Third Quarter: Week 3.1 - week 3.10
Concept Schedule	Friday Morning
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0108	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage)	5
Course Coordinator	Prof.dr. A.R. Roders	
Instructor	Dr. I. Nevzgodin	
Instructor	Ir. A.C. de Ridder	
Responsible for assignments	Prof.dr. A.R. Roders	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	Mastermind: CRASH (Conservation, Reuse, Architecture, Sustainability and Heritage) is a hands-on course, that introduces students to critical thinking in architectural redesigns, while playing the classic game Mastermind. Students are the code-crashers. The architect(s) involved in the redesign are the code-makers. The hidden code reveals the nature and impact of the architectural redesign. Who will win, the students, the architect(s) or both? Join, and find out.	
Course Contents	Throughout the course, students are offered a series of lectures on how to best code the nature and impact of architecture redesigns, and on the different domains (Conservation, Reuse, Architecture, Sustainability and Heritage). Students work in groups of five (CRASH) on a selected case study, and are tutored in peer review sessions, to monitor and give feedback about the progress of their research. As a team, the students reveal and discuss the impacts of an architectural redesign on a building (listed or unlisted) and/or building layer (i.e., site, structure, skin, services, space plan, stuff). Individually, each student chooses his/her domain. By the mid-presentation, students should have collected all the data needed to perform the analysis. By the final presentation, students are to present their proposal for the code and justification. The code of the architect will be revealed and compared to the one defined by the students.	
Course Contents Continuation	The five domains are respectively: [C]onservation, reveals the nature and impact of the architecture redesign concerning the building physical condition / deterioration, by comparing the building condition assessments, before and after the architecture redesign. [R]euse, reveals the nature and impact of the architecture redesign concerning the buildings functionality (i.e. use, performance, behavior), by comparing the building functionality assessments, before and after the architecture redesign. [A]rchitecture, reveals the nature and impact of the architecture redesign concerning the buildings typology (e.g. style, form, proportion, geometry), by comparing the assessments of the building typology, before and after the architecture redesign. [S]ustainability, reveals the nature and impact of the architecture redesign concerning the buildings sustainability (social, economic and ecological), by comparing the building sustainability assessments, before and after the architecture redesign. [H]eritage, reveals the nature and impact of architecture redesigns on the buildings cultural / heritage significance (attributes and values), by comparing the cultural / heritage significance assessments, before and after the architecture redesign.	
Study Goals	Allowing the architect to master critical thinking in architectural redesigns, a form of reflective reasoning that evaluates facts, information and arguments, by applying a range of intellectual skills to form a clear, logical and coherent judgement on the nature and impact of architecture redesigns. Sub-goals: Develop an investigative attitude towards the nature and impact of architecture redesigns, by cross-relating the domains: Conservation, Reuse, Architecture, Sustainability and Heritage. Understand the added value of critical thinking, sometimes confirming, others contesting own opinions/general assumptions. Experience multi-disciplinary teams and shared decision-making, when comparing and integrating individual results per domain. Assess a selected domain individually, comparing before and after architectural redesign Reach consensus on a co-created assessment, making use of a pre-defined framework Produce a documentary of a building by means of text, drawings, graphs and figures, reporting the nature and impact of the architectural redesign in the respective domains, as well as, explain their interrelations. Produce chapters, documentaries and argue in discussions with team members and stakeholders, using an appropriate professional scientific language.	
Education Method	Lectures (including Presentations): 12 hours Tutorials/Presentations: 20 hours Independent study: 108 hours (78%)	
Course Relations	The content of the course is complementary to the content of the elective course MSc 2 CSI - Heritage. It is suggested to HA students to attend both electives.	
Literature and Study Materials	Mastermind booklet, Book chapters, journal articles and other lecture materials.	
Assessment	Chapter (48%), PechaKucha (17%), Lectures (23%) and Attitude (12%)	
Period of Education	Q 3	
Concept Schedule	Wednesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0109	City of Innovations Project	5
Course Coordinator	M. Triggianese	
Instructor	M. Triggianese	
Instructor	Y. Söylev	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Required for	MSc3	
Expected prior knowledge	BSc and MSc1 completed	
Course Contents	Scheduled in Q3 in the MSc2 semester of the Architecture track, the elective City of Innovations gives students the possibility to develop specific design projects with a strong exploratory approach. The elective aims to connect education to urgent issues in dialogue with stakeholders outside the Faculty of Architecture and the Built environment. As part of Complex Projects ambition, the definition of City of Innovations guides the research-based design project. Research challenges can include the relationship between mobility and public space; architectural typologies and new ways of living, working and moving; cultural change and the development of our cities. Please contact the course coordinator to know this year's case studies.	
Study Goals	Upon completion of 'City of Innovations project' elective the student will be able to: - link theories of architecture and urban design with projects and visions in the field of environmental technology, sustainability, history and architectural composition - analyse the morphology, functioning of the urban fabric and social and cultural aspects of the studied location - design both in group and independently an architectural project and urban vision with graphic skills - integrate the individual design aspects (such as on public space, building envelopes, sustainability) into the group vision in a collaborative way - formulate argumentations and positions in oral, written and graphic forms. The course includes a 'gaming session'. The students will learn about: - The complex forces that drive contemporary urban interventions (architectural and urban projects which are interdisciplinary and which involve many stakeholders) - How to simplify this complexity into a narrative, building knowledge, in order to develop scenarios/solutions - How to communicate with the clients and to design or define collaboratively the brief for the architectural and urban intervention - How to achieve a consensus among stakeholders and develop a common agenda for their site	
Study Goals continuation	Upon completion of the MSc1, 2, 3 & 4 studio trajectory the student: Has developed the skills in architectural design satisfying both aesthetic and technical/functional requirements. During the trajectory the complexity of the architectural design increases leading to a level fit for architectural practice. During this trajectory, skills are acquired to increasingly incorporate an understanding of the design process attained with regard to architectural history and architectural theory, art, technology and human sciences. Additionally, skills are acquired to incorporate an understanding of the design process attained with regard to the relation between buildings, spaces and society's needs, including environmental aspects. During MSc1, 2, 3 & 4 process skills are acquired to incorporate insights in and knowledge of the design process attained with regard to methods of investigation and designing. Together with the training with regard to aspects of building technology, during the MSc1, 2, 3 & 4 process skills are acquired to incorporate an understanding of the design process with regard to structural design, materialization of buildings, comfort and climate control.	
Education Method	The studio is organized with the method of charrette (period of intense design activity and short-term design project, usually developed in teams in workshops) Research will be done in groups of max 10 students and the design in groups of max 4 students. Workshops, lectures, tours are included in the studio program. The course includes a 'gaming session'. See above (study goals) the scope of the session. Tutorial once week (please check with the studio coordinator)	
Course Relations	The elective is part of Complex Projects MSc program. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. As part of Complex Projects ambition, the search for definition of City of innovations will guide this research-based design project.	
Books	Recommended literature: Avermaete T., Havik K., Teerds H. (Ed.) Architectural Positions on Architecture, Modernity and the Public Sphere, SUN, 2009. Reyner, R., Megastructure: Urban Features of the Recent Past, New York, Harper & Row, 1976 Sennett, R., Together: The Rituals, Pleasures, and Politics of Cooperation, New Haven: Yale University Press, 2012 Shannon, K., Smets, M; Landscape of Contemporary Infrastructure, Nai010 Uitgevers, 2010 Zeidler, E.H. Multi-use architecture in the urban context, Nostrand Reinhold, 1985 Additional literature and study material will be made known one week prior to the start of the course in Brightspace.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio. For information about the studio work, please visit City of Innovations book series (TU Delft OPEN Books): https://books.open.tudelft.nl/home/catalog/series/city-of-innovations	
Assessment	Assessment will take into consideration the research approach, dedication, commitment, effort and improvement of the team in the investigation of the context and project area, as well as the quality of design and presentation. Concrete aspects for evaluation are: research work, clarity of the problem statements, originality of the final presentation. Standard condition for repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio. The assessment will be based on the following type of examination: - Oral examination. The research and design projects are presented during the course and at the end of the course to tutors and guest critics (experts on the theme of investigation). - Design examination. Students deliver at the end of the educational period drawings (digital), analysis reports and presentations	

	as final products (analytical assignments). During the educational period the student receives feedback on the progress and how to develop the research and design process. Assessment criteria (see EMMA rubric): - Design and Research to be assessed on coherence, significance, elaboration, correctness and innovativeness both on main line and on aspects. Design & Research counts for 80% in the final grade. - Presentation: to be assessed on the degree to which it is clear, intelligible, reflective and engaging both on main line and on aspects. Presentation counts for 20% in the final grade. Besides studio program students are expected to fully engage with events and people which the case studies have to offer.
Period of Education	Quarter 3 (spring semester), 10 weeks
Concept Schedule	Wednesday morning
Used Materials	You can find the students' work of previous editions of City of Innovations Project in the following (open access) publications: https://books.open.tudelft.nl/home/catalog/series/city-of-innovations
Minimum number of participants	12
Maximum number of participants	40
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0110	Adaptive Strategies Past, Present, Future: Topics in the History of Architecture and Urban Planning	5
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	Drs. C.A. van Wijk	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This course explores prominent themes in the history of architecture and urban planning. Ways of investigation and the relationship with the current scientific and practical disciplines will be discussed and questioned. History matters, yes, but in what way?	
Study Goals	This course explores the past, present and future of architecture and urban form. Students will 1. learn about the methodologies, terminologies and practices of historical research and they will apply this knowledge to a research topic of their choice related to the subject of the course. They will 2. Learn how to connect the analysis developed during their research in a meaningful way to their design proposal. The output of the course will be presented as a group project and students will be 3. learn to improve their collaboration skills. The topics for this course will depend on the teacher.	
Education Method	Lectures, Discussions	
Literature and Study Materials	Study material on research and writing is available on the course Brightspace page.	
Assessment	Writing assignment: final paper The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	3rd Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0113	Tools of the Architect	5
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Responsible for assignments	Prof.dr.ir. K.M. Havik	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The chair of Methods of Analysis and Imagination offers an elective seminar for the study of the different tools of the architectural discipline, and their experimental use for the analysis and imagination of built environments. In this seminar you will examine and reflect upon the nature, potential, and shortcomings of drawing, modelling, and writing as as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Study Goals	Upon completion of this course you should be able to: - Describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination. - Analyze one or more aspects of a particular built environment through the experimental use of drawings, models, and writings. - Develop one or more imagined alternative futures for a particular built environment through the experimental use of drawings, models, and writings. - Evaluate the effects of using drawings, models, and writings experimentally in your analysis of, and your imagined alternative futures for a particular built environment.	
Education Method	In a first part of the course you will describe the theoretical and historical antecedents of one or more tools of architectural analysis and imagination, by reading a selection of texts and discussing them in a seminar setting. The second part of the course is divided into thematic modules, in which you will depart from those antecedents to analyze and develop alternative futures for the selected built environment through the experimental use of drawings, models, and texts. The conclusions of the entire process are presented as an evaluation of the nature, potential, and shortcomings of the tools you have studied, as means to achieve alternative and unconventional readings of architecture, as well as unforeseen representations of imagined built space.	
Assessment	Your study of the theoretical and historical antecedents of different tools of architectural analysis and imagination are subject to progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far. Your analyses and your development of possible futures for the selected built environment are assessed per thematic module, and graded using the aspects and criteria determined in the EMMA feedback and assessment tool, adapted for this seminar. Your final evaluation is graded individually, on the basis of the abovementioned criteria. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	One academic quarter, Q3	
Concept Schedule	Tuesday	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0114	Architectural Translations: Drawing, Recoding, Tectonic	5
Course Coordinator	A.S. Alkan	
Instructor	A.S. Alkan	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	2 hours/week (3.1) 4 hours/week (3.2-3.8) 6 hours in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This MSc2 elective seminar course inquires into the link between representational and tectonic codes in architecture and how those translate dialogically. Although the term (code) gained currency in digitally mediated design practices, within the seminar, different modes of coding and codification will be explored in order to draw up spatial readings leading to tectonic translations: formulations between images and objects. With the premise that there is an inherent link between the modes of representation and design, the seminar will inquire in architectural de/re/coding between images and media-constructs (drawing to tectonic). In this respect, departing from Karl Bötticher's distinction between core-form and art-form, the recent turn of materiality in architecture will be investigated with a specific focus on media discourse in architecture. The course is a hands-on seminar with a theoretical framework, in which the students explore the medial connections between conceptual and procedural aspects while working with different media guided by thematic readings on a weekly basis. The analytical phase leads to the specification of the individual thematic mini-bibliographies supporting each student's individual work. The students may follow this course independently. For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Study Goals	The course has four main objectives for the students to: - Gain theoretical literacy in architectural representation and design media - Describe major debates, methods, techniques and issues in architectural representation - Analyse design medias formative role in architectural design process - Develop/exercise project-specific media constructs/techniques	
Education Method	- Lectures, guest lectures, tutorials and presentations - Weekly readings and seminar discussions - Experiments with media-constructs, image-objects, portfolio. - The students will be completing bi-weekly reading responses and their portfolio along the seminar.	
Course Relations	For the students who will follow the MS2 design studio Intersections the seminar is particularly recommended.	
Literature and Study Materials	Literature and Study Materials will be released on Brightspace one week prior to the start of the course.	
Reader	The reader will be provided with the course syllabus.	
Assessment	- Portfolio and final presentation of individual work/media-construct (60%) - Participation/presentations at seminars, lectures & discussions (20%) - Weekly assignments and reading responses (10%) - Booklet and exhibition (collective) (10%) The portfolio compiles: - Analytical assignments - Progressive documentation of the individual work - Writing assignment (by choice) On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Enrolment / Application	a.s.alkan@tudelft.nl	
Elective	Yes	
Period of Education	Q3	
Concept Schedule	Thursday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0117	Didactic coaching skills for architecture and the built environment	5
Course Coordinator	Dr.ir. S. Zijlstra	
Course Coordinator	Dr. M.H. Arkesteijn	
Instructor	Dr. H.J.F.M. Boumeester	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Dr. M.H. Arkesteijn	
Responsible for assignments	Dr. M.H. Arkesteijn	
Contact Hours / Week x/x/x/x	8 hours per week from 1.1 to 1.5. Week 1.6 to 1.11 no more than 2 hours per week.	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	Dutch	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0118	Experiments in Drawing Theory	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Instructor	N. Sanaan Bensi	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The elective Experiments in Drawing Theory is a seminar, offered by the chair Borders&Territories, in which the speculative power of drawing towards developing a reasoned architectural design approach is both discussed and experimented with. The course consists of the development of an individual drawing experiment, with seminar discussions. During the seminar sessions, the themes experiment, drawing and theory will be introduced and contextualized as a specific field of architectural research. The incredibly profound history of architectural drawing will be investigated through a series of case studies, though these are not exclusively linked to the architecture discipline but belong to the much broader field of creative disciplines. During the seminars exercises, participants probe how specific means of representation relate to specific conceptions of space. Drawing is not only considered to be a technique, though this aspect should not be underestimated at the same time, it has a lucidity that is intrinsically connected to thought (teoria) as well. Drawing is an autonomous instrument of architectural knowledge, while it is also simultaneously simulacrum of reality and reality, memory and anticipation, subject and object. The individual assignments will consist of the production of one or a series of architectural drawings, positioning an innovative notational system and its performance. The seminar course aims to approach this complex theoretical question about the specificity and un-specificity of drawing, herein intended both as a concept and instrument of innovative architectural thinking. In this present context, the focus is directed to the challenging of the convention governing a design approach and the definition of an alternative notational system of signs, rules, and techniques preceding the idea of the architectural object.	
Study Goals	The student is able to initiate and develop a reasoned experimental architectural design approach. The student is able to express and crystallize the innovative aspects of the architectural design at the level of the architectural representation. The student is able to perform architectural design research through drawings.	
Education Method	Readings and discussions of theories regarding (architectural) drawing. Seminars and tutoring development of drawing exercises. Guest lectures and presentations.	
Assessment	Attendance and participation in the seminars, discussions and collective presentations. Weekly presentation of the individual design development. End-term submission of drawing-design and collective exhibition (the instructor will specify the paper and drawing requirements and the deadline at the start of the seminar). Assessment Scheme - Assignment (70 %) - Weekly development assignment (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation, exhibition) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Concept Schedule	Tuesdays between 13:45 and 17:45 Week 3.1-3.8, and 3.10 (exams)	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0119	Figures	5
Course Coordinator	D.H.G. Somers	
Instructor	D.H.G. Somers	
Instructor	Dr. J.S. Zeinstra	
Instructor	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Thinking beyond individual students and courses, the Chair considers its educational programme as a collective and reflective space of study and discourse: an attitude that is intended to encompass the work of both students and staff. The Chair engages in common questions concerning the public interior, questions of interiority, and their relations with the social and physical fabric of the city as a whole. Figures The Figures of this elective project refer to the constellation of formal, spatial, typological and material conditions through which architecture has been composed and physicalised across its history. This has often been expressed in terms of difference and change: as movements, styles and ideas that succeed or compete with one another. An alternative history might address what connects things: the elements that relate or repeat between architectures made in very different times and places. This course explores these architectural continuities. An ongoing research project for the chair, each year a particular concern or condition is chosen to research through a series of precedents, chosen to represent context that might encompass but go beyond the orthodoxies of Western architectural history. Each addresses the architectural interior, questions of interiority and the boundaries that define these, in relation to the wider context of the city or the landscape. Investigations will encompass not only the physical condition but also the social and cultural contexts that underpin it. Case studies are collated, represented and analysed in respect to one another, through media which might include drawings, models and descriptive texts; constructing a body of knowledge that will grow into an archive for publication and exhibition.	
Study Goals	Upon completion of the elective course the student is able: - to analyse architectural case studies through different historical, social and cultural contexts, and understand the ideas that informed them - develop a position with regards to these projects and study them within a collective research project - represent the findings in those studies through the making of models, drawings and texts, within a collectively developed format A specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The elective studio proceeds through a variety of working methods: group work, individual tutorials, internal lectures and thematic exercises specific to the studio.	
Assessment	Assessment will focus on the research work undertaken within the set theme and the specific research questions raised within it; the study that responds to those questions; the representation of that study through the making of an artefact. Products: texts with illustrations; drawings; models.	
Remarks	The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course. The maximum marking period is 10 work days.	
Period of Education	Spring semester, Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0121	Analytical Models	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	Ir. R.R.J. van de Pas	
Instructor	G. Coumans	
Responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	An explorative formal study exercise, aimed at the development of imaginative and thematic architectural analysis skills. The analysis is based on architectural precedents, making active use of a variety of conceptual, digital and physical Modelling approaches.	
Course Contents	The Analytical Models course aims at furthering the understanding of architectural composition and perception, though thematic, analytical study, making active use of design visualisation and model-making techniques. This precedent-based designerly study initiative addresses issues of the architectural composition and perception. Aspects of study include: - Development of study approaches for the benefit of precedent-based composition analysis; - Explorative visualisation of architectural concepts and identification of design phenomena; - Design-driven enquiry making use of various digital 3D modelling applications; - Design-driven enquiry making use of physical scale modelling and graphic representation. The course can be considered as an interactive learning environment and laboratory for thematic formal study. Previous themes of study have included the architectural oeuvres of Robert Venturi and Aldo van Eyck. The exercise brings together ambitions of composition research and evocative, analytical 3D modelling with the didactic opportunities of tangible, physical modelling. The aim is to stimulate the participating students to develop meaningful insights and knowledge on the level of architectural designing and to develop professional skills in the field of design visualisation and communication.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and analytical exploration of a project. - use and construct models; digital, graphical or physical models representing design issues. - formulate and defend findings and conclusions, orally and in writing. - contribute to the group process in a constructive way; carrying out specific tasks and determining the rule within the group as a whole as well as contributing towards an integral group product.	
Education Method	Free choice Master exercise (5 credits) offered by the FormStudies BK group, department of Architecture, Faculty of Architecture, TU Delft.	
Literature and Study Materials	Various applications of digital and physical modelling techniques plus graphic analysis and presentation means.	
Assessment	Assessment on the basis of process, end-result, documentation and analysis. The maximum marking period is 10 work days. If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Special Information	coordinator	
Period of Education	2e Semester, 3rd Quarter, The course is scheduled on the Wednesday morning.	
Concept Schedule	Wednesday morning	
Leerstoel	Form Studies	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0122	1:1 Interactive Architecture Prototypes Workshop	5
Course Coordinator	H.H. Bier	
Responsible for assignments	H.H. Bier	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	This course involves skill building in computational design and robotic prototyping by employing Design-to-Robotic-Production -Assembly and -Operation (D2RPA&O) methods. Focus is on small scale interventions in urban context that are robotically produced and operated. Such physically built robotic environments consist of reconfigurable, adaptive systems incorporating sensor-actuator mechanisms that enable buildings to interact with their users and surroundings in real-time.	
Course Contents	Students learn in a workshop set-up to conceptualize, design, produce and/ or operate buildings and building components by applying D2RP&O methods, which consist of parametric design, robotic fabrication and interactive operation techniques. In this context, D2RP&O is understood as a systemic approach for the design, construction and operation of buildings.	
Study Goals	Students learn to develop a coherent, elaborated, and innovative design - on mainline and on individual aspects - at MSc 2 level.	
	Specific for this course, Design-to-Robotic-Production and Operation (D2RP&O) for Interactive Architecture is taught in a workshop set-up wherein: (1) Students understand the principles and possibilities of D2RP&O and are able to incorporate D2RP&O in the design process of a small urban intervention. (2) Students develop skills in architectural design resulting from D2RP&O processes satisfying both aesthetic and technical / functional requirements. (6) Skills are acquired during the D2RP&O process to incorporate an understanding of the design process with regard to structural, environmental, and materialisation design.	
Education Method	Design research and practice are implemented in a workshop/seminar set-up by employing computationally advanced design, robotic manufacturing, and interactive operation techniques.	
Literature and Study Materials	Bier, H. and Knight, T., Digitally--driven Architecture, Footprint Issue 6, Stichting Footprint, 2010 (https://www.researchgate.net/publication/44444960_Digitally-Driven_Architecture) Bier, H. and Knight, T., Data Driven Design to Production and Operation, Footprint Issue 10, Stichting Footprint, 2014 (https://www.researchgate.net/publication/281404980_Data-driven_design_to_production_and_operation?ev=prf_pub) Bier, H. Robotic Building, TEDx Delft 2015, TEDx Delft Salon, The Future, (https://www.tedxelft.nl/2015/04/tedxelft-events-tedxelft-salon-the-future/) Bier, H., Robotic Building (http://www.roboticbuilding.eu/education/msc3-4/) Bier, H. and Mostafavi, S. Structural Optimization for Materially Informed Design to Robotic Production Processes, AJEAS, 2015 (https://www.researchgate.net/publication/286477508_Structural_Optimization_for_Materially_Informed_Design_to_Robotic_Production_Processes) Liu Cheng, A. and Bier, H., An Extended Ambient Intelligence Implementation for Enhanced Human-Space Interaction, ISARC, 2016 (https://www.researchgate.net/publication/305999106_An_Extended_Ambient_Intelligence_Implementation_for_Enhanced_Human-Space_Interaction) Bier, H., Robotic Building, Adaptive Environments Springer Book Series, 2018 (https://www.researchgate.net/publication/327338545_Robotic_Building?_sg=IX8dERr6Sd19HPExhcJvg3MiT7hYFgb9SqxWl4QJ1cH-RifcjAZgUY1J5mHqP0nqqSLnjEff5dyqoquqZmL9oMDiMbQX0Y8_JzpwWMC2.aD38bz1jL9FW5GmBVY6HvjbgxDNIIIL82JzAEx_vrVK0pkyOeYUwj_Xre6ybor4aBNjathDC2d5TbYoMWxonjQ) Bier, H. et al., Actuated and Performative Architecture: Emerging Forms of Human-Machine Interaction, Spool CpA 3, 2020 (https://journals.open.tudelft.nl/spool/issue/view/834)	
Assessment	Process and final results are evaluated by means of scaled and 1:1 virtual and/ or physical 2-4D prototypes, written reports, and oral presentations. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Quarter 3	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0126	Bridge Design	5
Course Coordinator	Dr.ir. J.E.P. Smits	
Responsible for assignments	Dr.ir. J.E.P. Smits	
Contact Hours / Week x/x/x/x	Week 3.2/3.3/3.5/3.6/3.8/3.9 - 2 hours per week = 12 hours	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Bridge Design; an integral approach through landscape, architecture and structure. The design of bridges is a fascinating field of work. Whether it is a simple crossing or an intricate steel structure; a bridge appeals to the imagination. Bridges overcome barriers, create connections and bring people together who were thus far separated. Whether a bridge is part of an urban context or a landscape setting, bridges are symbols of culture that deserve the attention of good designers. The attention for the aesthetic design of infrastructure is growing since the 90s. Bridges are no longer seen as mere functional objects. For a long time, the design of infrastructure works have been the sole domain of the engineer. Nowadays bridges, viaducts, tunnels, and even whole road designs have obtained a renewed interest from architects, landscape architects and urban planners. Yet the number of architects and landscape architects with a solid portfolio in this area is limited. Engineering companies that specializes in bridge design lack the skills to make an aesthetically pleasing design that is firmly embedded in the context and forms part of a public space of high quality. Bridge Design' is an elective in MSc2 and is meant for students in the master tracks of either Architecture, Urbanism, Landscape Architecture and Architectural Engineering + Technology. CiTG or ID students are also welcome. The course focuses on the design of bicycle bridges. The design process stretches from the integration of the design in the urban or landscape context to the architectural engineering of the design.	
Study Goals	The student gets familiarized with the multidisciplinary design process and the different disciplines involved in the design of a bridge/civil structure. The student is able to derive design criteria for the bridge/civil structure from the spatial and societal context. The student is able to structurally elaborate a conceptual design of a bridge/civil structure. The student acquires knowledge of the different disciplines involved in the design of a bridge/civil structure such as: landscape design, urban design, architectural design and structural design.	
Education Method	Lectures, Design studio, Masterclasses from renowned bridge designers,	
Assessment	Students work in small multidisciplinary groups, in which different aspects of the assignment are addressed. Oral presentation and final report. + Posters or slides with texts, drawings and images. + physical models. Assessment by the course manager and other lecturers. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Q3, Wednesdays: 3.1 Wednesday all day; MANDATORY excursion 3.2 Wednesday afternoon; tutoring 3.3 Wednesday afternoon; tutoring 3.4 Wednesday all day; MANDATORY masterclass 1 3.5 Wednesday afternoon; tutoring 3.6 Wednesday afternoon; tutoring 3.7 Wednesday all day; MANDATORY masterclass 2 3.8 Wednesday afternoon; tutoring 3.9 Wednesday afternoon; tutoring 3.10 Wednesday all day; MANDATORY final presentations	
Concept Schedule	Wednesdays Week 3.1/3.4/3.7/3.10 - 8 hours per week = 32 hours	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0131	Entrepreneurship in Architecture and the Built Environment	5
Course Coordinator	Prof.dr.ir. J.W.F. Wamelink	
Instructor	Prof.dr.ing. U. Knaack	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wamelink	
Contact Hours / Week x/x/x/x	8 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture or comparable	
Summary	This elective course is developed for students (either architecture, urbanism, technology, or management) who like to investigate the opportunities for starting a venture in the built environment, or want to develop an entrepreneurial attitude in taking on exciting challenges in the built environment. In this course students are guided in developing their ideas about a self-owned venture (design-firm, innovator, service provider, project management expert, etc.). A venture that gives them freedom for personal development, pursuing personal goals. Within the course, the student's competencies are central and joint work is being done on the development of an entrepreneurial plan, making use of an entrepreneurial mindset. The course is provided by a team of teachers with an entrepreneurial background from the faculty of architecture and the built environment complemented with teachers from the Delft Centre for entrepreneurship.	
Course Contents	The common theme of the course is the translation of a students individual idea into a viable plan to start his or her own company. The course contains the following elements: - discovering various aspects of being an entrepreneur through the analysis of a case and interviewing entrepreneurs - imaging the students personality with respect to an entrepreneurial mindset. - designing a blue-print for the self-owned company, resulting in an entrepreneurial plan and a roadmap. Using his/her knowledge, skills and network of contacts, the student will be challenged to design a blue-print for his/her company. A clear vision, a way to to differentiate the company, and proper understanding of the potential client's values are key aspects. - pitching the ideas within their own network and incorporating feedback into a final plan - reflecting on general feedback from their network and teachers In parallel, various lectures and exercises are given focusing on the theory of business models, financing, market forces and social entrepreneurship. Various appealing entrepreneurs from the sector will provide inspirational lectures.	
Study Goals	After finishing the course, the student: - has developed a broader insight into the value and meaning of entrepreneurship in architecture and the built environment - has a broader insight into his/her own personal character and drive related to starting a self-owned company - has more insight about the feasibility of starting the self-owned company, making use of his/her passion, knowledge, skills and network. - has introductory knowledge about business plans, financing and market influence - has the ability to critically reflect on his/her entrepreneurial skills and plans - has the skills to articulate the viable business proposition in a concise and convincing pitch presentation - knows what the next steps could be realising the self-owned company.	
Education Method	The course's learning activities comprise: - lectures: theory - self-study: developing entrepreneurial plan - groupwork: peer reflection, and inspiration from the inside world - guest lectures: inspiration from the outside world - tutorials: to develop the entrepreneurial plan and roadmap	
Literature and Study Materials	Reader	
Assessment	Individual report and pitch, including the final entrepreneurial plan, roadmap, and personal reflection	
Period of Education	Quarter	
Concept Schedule	Tuesday afternoon and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0132	Zero-Energy Design	5
Course Coordinator	Ir. S. Broersma	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Instructor	Dr.ir. L.J.J.H.M. Gommans	
Instructor	T. Blom	
Responsible for assignments	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Contact Hours / Week x/x/x/x	Week 3.1-3.9 4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A BSc. in Architecture is highly preferred and recommended in order to have design skills and some basic knowledge of climate design and sustainability. The MOOC Zero Energy Design is also recommended to do, especially for students that did not do their BSc. in Delft. This can also be done at the start of the course.	
Course Contents	The urgent issue of an energetically poor performing existing building stock is the subject of Zero Energy Design. The energy use of these buildings, to become comfortable for its users, comes with considerable carbon emissions; as does the electricity use of its residents. But also the use of new materials to upgrade or transform these buildings causes some initial carbon emissions. Within the assignment of ZED, an existing residential building block has to be transformed into a zero carbon building with net zero energy use and net zero carbon emissions. So, all carbon emissions, from the energy use of the building and the users, as well as the carbon emissions embedded in the material use for renovation, have to be compensated by sustainable energy production on-site. With the successive steps (of the New Stepped Strategy) of reducing the demand, re-using waste streams and producing the remaining demand with renewables, a combination of smart measures has to be defined to reach this goal. Smart energy connections with the surrounding built environment will also be considered. The focus of this course lies on a well-integrated climate design and well-balanced energy system that makes the building carbon neutral or even provides a negative carbon footprint. The building has to be transformed in an architectonically attractive way, including additional space and functions, to become future proof. With an energy potential mapping analysis of the neighbourhood and an energy performance calculation program, tools are provided to quantify and prove the final carbon and energy performance.	
Study Goals	The student is able to: - develop an integrated energy-neutral climate design - make energy calculations and optimize the energy performance of a building	
Education Method	Lectures, interactive lectures, excursion, computer exercises, writing a report, presenting. The assignment will in principle be completed in groups of 3.	
Assessment	Knowledge of the theory is tested through a report and an oral presentation. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Monday afternoon	
Minimum number of participants	10	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0133	Technoledge Glass Structures	5
Course Coordinator	Dr. M. Overend	
Course Coordinator	F. Oikonomopoulou	
Instructor	Dr.ir. F.A. Veer	
Instructor	F. Oikonomopoulou	
Instructor	T. Bristogianni	
Instructor	mr. J.D. O'Callaghan	
Instructor	Ir. H. Rozendaal	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	AR1B021 Bucky Lab Engineering - the parts: Structural Mechanics + Material Science or AR1B023 Sustainable Architectural Materials and Structures	
Course Contents	In many famous architectural works, it is the structure and materiality of the building that manifest the projects innovation. In such works, the structural system of the building becomes the primary form of expression. Ergo, the ability to understand the principles of structural engineering and design, and the motivation to experiment with new types of structures and materials are invaluable assets for both architects and structural engineers. Towards this goal, this elective aims to bridge the gap between architects and structural engineers and familiarize students with the basic and some advanced skills required to successfully design structures via the design, construction detailing and structural verification of a column-free, long-span structure made primarily of glass (including for its load-bearing structure). The use of a novel, non-conventional structural material like glass highlights the need for integrated thinking when designing a structure, having to take into account not only the strength of the material when dimensioning the individual elements, but also limitations linked to its manufacturing and processing processes and the need for redundancy in the design. Besides the structural verification, special attention is given to the assembly order of the structure, as well as on the design of the key connections to be visually discreet and most importantly, allow for the desired load transfer. The course evolves around the (structural) design assignment, which is done in groups of (max.) 4 students (3 BT students and 1 student from another track). Each year the design assignment is different: previous examples include the design of a full-glass conservatory in the Kew Gardens in UK, of a full-glass observatory for the Aurora Borealis in Iceland, of a transparent protective shelter for the temple of Hagar Qim in Malta and of a glass museum for the Khufu boat in Egypt. The students are asked to develop their own design solution and apply to it the fundamentals of long-span structures and structural glass, which are introduced through a series of lectures. The development of the design is guided during the weekly studio sessions by one-to-one consultations with the instructors. The consultations during studio sessions, as well as the workshops and lectures, tackle the following aspects of the structural design solutions: architectural and structural design, built-up of components, assembly sequence of the structure, risk analysis, design of connections, sustainability and structural verification through numerical modelling and/or hand calculations. During the course, the students give 3 presentations: one on their initial design idea, a mid-term presentation and a final presentation. Through these presentations the students can receive formative feedback from all tutors as well as from their peers on their work. The final assessment of the students work is based on their design progress, final presentation and final report.	
Study Goals	After successfully completing this course the student is able to: - design and engineer a safe and feasible structure on a basic (MSc2) level; - apply non ordinary materials, like glass, within a loadbearing structure; - solve practical challenges related to the engineering and construction of a glass structure.	
Education Method	Lectures, Structural Workshops and Studio Consultancy Sessions	
Assessment	Final presentation and final report Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	3rd quarter	
Concept Schedule	Thursday	
Maximum number of participants	28 (21 Building Technology students + 7 students from other tracks)	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0136	Making	5
Course Coordinator	Ir. H.A. van Bennekom	
Responsible for assignments	Ir. H.A. van Bennekom	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	completed BSc	
Course Contents	Making is a special course that combines research and design with practical hands-on workshops, focused on designing and making objects of concrete with specific, innovative properties and expressions. Students will obtain theoretical and practical insight in the interdependencies between research, design, testing, constructing, making and final architectural expressions. Because, after all, the choice and knowledge of the material and its practical possibilities and impossibilities, have ultimate consequences for the performance, durability and aesthetics of the built object, and is as such a crucial experience for architectural education. Through excursions and meetings, the course brings students in contact with the professional industry.	
Study Goals	GENERAL: Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal. SPECIFIC: The students achieve experience in: - experimenting and prototyping for divers casting techniques for concrete objects. - knowing and exploring issues and possibilities of innovative molding techniques and materials and the consequences for the design. - designing and developing architectural applications and details in concrete, considering added values, connections of building components, articulation, form properties, de-molding techniques, surface qualities and esthetics. - collaborating with professionals of the Dutch and international cement industry, and international research programs.	
Education Method	Weekly tutorials in studio, hands-on workshops, supported by lectures and possibly a field trip. The development and making of concrete objects requires insight in existing techniques and at the same time an understanding of societal/global trends and necessities for the built environment. Therefore, the education method used is an interdisciplinary activity that combines research techniques with design consultancies and guided practical experience. The developed proposals are based on individual and/or group research and design work, and include investigation of themes about architectural components and expression, innovations for the cement industry, trends, new geometries and materials, sustainability, circularity, durability and sustainability within the concrete industry. The existing research done in previous studios will be part of the expected prior knowledge, which we will use and continue to build on. Next to research consulting and design tutorials, the method involves practical work consisting of building molds, pouring sessions, and developing casting and de-moulding strategies. During a final presentation event with professionals, students will present their casted concrete experiments and prototypes products as well as their presentation panels. They will reflect on their experiences, considering the performance of prototypes, new processes and possibilities, and the expression.	
Books	- Beeld Schoon Beton (in Dutch only), Stichting ENCI Media (2005) - Depending on current theme, will be announced during course.	
Assessment	Tutorial once a week. Tutors and invited specialists from the cement/engineering/design industry will assess the results in line with the specific theme and set goals. Tangible results, presented in an exhibition setting, get a paramount role. Deliverables will include a collective research/design/workshop book, presentation panels and final concrete prototype models. Regarding the final presentation students will be requested to have a complete narrative to defend their proposals, based on their research and experiments, well positioned in social, technical and global awareness. Reflection on experiences, performance and processes will be taken into account, results can be published on the internet. Course Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are; research work, argument formulation, hands-on experiments, design, and presentation.	
Elective	Yes	
Period of Education	Q3 (1x/wk)	
Concept Schedule	Tuesday morning	
Leerstoel	Complex Projects	
Minimum number of participants	8	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0139	MEGA	15
Course Coordinator	Dr. M. Overend	
Course Coordinator	Dr. A. Luna Navarro	
Course Coordinator	Dr. S. Brancart	
Course Coordinator	M. Turrin	
Responsible for assignments	M. Turrin	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	MEGA is a collaborative integral multi-disciplinary design of a special big and/or tall building. This could be a multifunctional skyscraper or a multifunctional building with a large span, such as a stadium, a sports facility, a museum, or transport hub. The course targets master students in Architecture, Real Estate & Housing, Building Technology (MSc 2); welcomes students from Civil Engineering; is open to non-TU Delft students, conforming with TU Delft regulations. Students work in teams. The design team of 5 to 8 students is responsible for delivering an integrated design as a multidisciplinary team; while each student is responsible for one discipline. Disciplines involved are: architecture, structural design, climate design, façade design, design/construction management and computational design/BIM. Sustainability runs transversally across these disciplines. All disciplines work based on digital models. The design process occurs in a collaborative digital design environment, supporting the workflow across the different disciplines. The collaborative digital design requires an integrated 3D approach with BIM (Building Information Modelling) principles, parametric design, performance analysis and multi-disciplinary computational optimization/design exploration. The workshop is very realistic and closely matches the design process of large international projects in the competition phase; it is a very good preparation and experience builder for your future career. It is highly appreciated by future employers. The course is supported also by external international design/engineering offices. With them, the location of the project will be chosen and the brief of the design assignment will be developed. As examples from recent years, support was given by Arup and UNStudio, by ABT and Neutelings Riedijk Architecten, by MVRDV, etc. In past editions, firms like Techniplan, Deerns, DGMR, Esteco, and others consulted the students on specialized disciplines, with a perspective from practice. Examples of past collaborations include also Municipalities and Provinces, such as the City of Rotterdam, Almere and Den Haag, and the Province of Friesland. Disciplines: The team is organized on disciplines: -Architectural Design -Climate Design and building services -Computational Design -Façade Design -Structural Design -Management The disciplines are divided amongst the team members; each member is responsible for the contribution and integration of these aspects in the collective design. Students are encouraged to match their role in the team with the specialization they follow in the Master track. Phases: The course is structured in 3 phases: -Lectures; excursion; intensive learning -Sketch design of 2-3 options; presentation of options; choice of one option -Preliminary design of the chosen option; final presentation The first phase includes lectures by professors, external experts and architectural/engineering firms. During the excursion, the project site is visited. Intensive sessions allow studying and practicing group dynamics, collaborative work, computational design. The second phase focuses on the design of multiple options. The daily design activities are facilitated by tutors who are expert in the disciplines. Each discipline has a weekly time for individual consults. During a presentation, one design option is chosen for further development. The mid-term presentation is facilitated also by external experts. Feedback by them and tutors inform the design and decision-making. After the mid-term presentation, the design option is detailed with the team, leading to the end presentation. The end presentation is an important event with external experts assessing the designs. The design is summarised in reports about each discipline. Site: The assignment has an actual site where the building is planned. Past examples are in Amsterdam, Rotterdam, London, Brussels, Guangzhou. Learning objectives regard team work and individual specialized contributions: Collaborative design (whole team) The student will be able to: - design (with digital models) together with different disciplines (different goals and backgrounds) - design in a realistic design environment Sustainable design (whole team) The student will be able to: - identify key goals of sustainability for an interdisciplinary project - contribute as a specialist to the holistic sustainability of an interdisciplinary project Architectural Design (specialist) The architectural designer will be able to: - direct interaction between architecture/masterplan/environmental context - develop architectural design concepts based on interdisciplinary inputs	
Course Contents Continuation		

- integrate structural, façade, climate concepts into architectural design
- integrate sustainability and construction into architectural design
- develop the interdisciplinary project until preliminary design

Climate design (specialist)

The climate designer will be able to:

- develop climate and building services concepts based on interdisciplinary inputs
- evaluate different climate and building services systems in relation to architectural design
- integrate with architecture, structure, façade
- calculate climate performances to provide feedback in design decisions
- dimension the HVAC installations
- develop the interdisciplinary project until preliminary design

Computational Design (specialist)

The computational designer will be able to:

- set a collaborative digital workflow across disciplines / BIM
- set multi-disciplinary parametric design strategies/methods
- set multi-disciplinary processes for performance analysis with simulation tools
- set multi-disciplinary computational optimisation processes for design exploration
- coordinate digital interactions across disciplines in different design phases

Façade/envelope design (specialist)

The façade designer will be able to:

- develop façade/envelope concepts based on interdisciplinary inputs
- evaluate different façade/envelope systems in relation to architectural and climate design
- integrate with architecture, structure, façade, building services
- collaborate with the climate design specialist to provide feedback in design decisions based on numeric assessments
- dimension the elements of the façade/envelope
- develop the interdisciplinary project until preliminary design

Structural Design (specialist)

The structural designer will be able to:

- develop structural concepts based on interdisciplinary inputs
- evaluate different structural systems in relation to architectural design
- integrate with architecture, façade, climate design
- calculate structural performances to provide feedback in design decisions
- dimension the structural elements
- develop the interdisciplinary project until preliminary design

Management (specialist)

The manager will be able to:

- develop balance of costs and revenues for design optimisation based on interdisciplinary inputs
- develop real estate perspectives with stakeholder- and functional strategies in design and operational phase
- integrate construction methods/planning and site management and logistics
- collaborate interdisciplinary to provide feedback in design decisions based on numeric assessments
- define and coordinate objectives, tasks, deliverables in the group process

Study Goals

The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.

Specified for this course:

After successful completion of the course, the student will be able to:

- work in an interdisciplinary design process supported by digital workflows;
- understand and apply discipline-related knowledge in projects for large or tall buildings.
- develop design strategies to achieve high building performances;
- integrate numeric analysis and simulations data to address design choices.

Education Method

In this course, the education methods are:

- Lectures by professors and specialists
- Collaborative working sessions with other students
- Exposure to external architectural practice and external experts
- Consults with tutors
- Making presentation and receiving/integrating feedback

Special is the involvement of external practitioners and external experts linking this course to practice.

For this course several multidisciplinary teams of students are formed, which are each responsible for one integral design. Each student has a different role in the design team and is tutored by instructors specialized in her/his discipline. When possible, students take roles according to their specialization during the Master studies.

Apart from focussing on his/her own discipline, the aim for each team-member is to achieve the best integral design paying special attention to collaborative design, sustainable design and computational design.

Feedback is received during the mid-term and final presentation from the external experts and tutors.

Literature and Study Materials

Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content.

- Rem Koolhaas, Delirious New York: A Retroactive Manifesto for Manhattan, 1978.
- Iñaki Ábalos and Juan Herreros, Tower and Office: From Modernist Theory to Contemporary Practice, 2003
- Barnes, M., Dickson, M., (Ed.), Widespan Roof Structures, Thomas Telford, London, 2000
- Kloft, E., Eisele, J., (Ed), (2003) High-Rise Manual, Hardcover
- Ali M, Armstrong P. Overview of sustainable design factors in high-rise buildings. CTBUH 8 World Congress, Dubai. 3-5 March 2008
- BREGlobal Ltd. BREEAM International New Construction 2016. Technical Manual
- Borhani, A., Dossick, C.S., Meek, C., Kleiner, D. and Haymaker, J., 2019. Adopting Parametric Construction Analysis in Integrated Design Teams. In Advances in Informatics and Computing in Civil and Construction Engineering (pp. 351-358). Springer,
- Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods

Assessment

Presentations and Reports

Assessment is twofold:

- Group assessment for integral group design based on presentations
- Individual assessment for discipline report

The students mark is a combination of the group assessment and individual assessment.

Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0.
(A retake can be done in the next year)

Period of Education

Quarter

Course evaluation

For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0141	CSI Heritage (Conservation, Survey, Investigation of the Built Heritage)	5
Course Coordinator	Dr. B. Lubelli	
Instructor	W.J. Quist	
Instructor	Dr. B. Lubelli	
Instructor	S. Naldini	
Responsible for assignments	Dr. B. Lubelli	
Contact Hours / Week x/x/x/x	4 hours per week, starting on week 3.1 and ending on week 3.10 (excluding week 3.9)	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Summary	The course CSI Heritage aims to provide students in architecture, and especially those interested in the field of conservation, rehabilitation and re-use of heritage buildings, with a sound and practical insight in building materials and construction techniques, including their development and use during time, their properties and degradation mechanisms. The course contributes to the development of an investigative attitude towards the technical aspects of conservation and rehabilitation interventions on heritage buildings. The knowledge gained in the course will support the future architect in guiding the survey of a building, the investigation of the damage processes, the diagnosis process and the decision on the conservation and rehabilitation interventions. [C]onservation, concerns interventions aiming at preserving and rehabilitate existing buildings, taking into account not only technical aspects but also the historic value of the building and its components. [S]urvey, examines the physical condition of a building, its components and materials and forms a standalone assessment at a moment in time in order to adequately maintain and plan future interventions and use of a property, [I]nvestigation, involves the application of a broad spectrum of methods, technologies and sciences to answer those questions of interest discovered in the survey, in order to identify specific causal links between damages and their origins.	
Course Contents	The course gives students the opportunity to deal with the technical aspects of survey and investigation on heritage buildings, with the final aim of integrating them in the decision-making process on the conservation and rehabilitation interventions. The course will deal with the following subjects: Materials: history, properties and use of building materials, including both traditional (e.g. brick, natural stone, mortars) and more recent materials (concrete, glass, plastic) Construction techniques: specific use of materials and components and their development in time. Damage processes & diagnosis: survey of the state of conservation, formulation of hypothesis and validation through investigation and diagnosis of the damage process. Technology of conservation and rehabilitation interventions on heritage buildings: interventions at both the level of the materials (e.g. reintegration, protection through surface treatments) and of the building (e.g. intervention against rising damp, strengthening of the structure)	
Study Goals	At the end of the course, the student : - has appropriate knowledge of the history of building materials and construction techniques and is able to identify them correctly in a building; - is able to formulate hypotheses on the damage processes, suggest appropriate investigation methods to come to a diagnosis and understand the outcomes of research; - is capable to advice on technical aspects of conservation and rehabilitation interventions of buildings taking the historic values and the future use into account. - is able to document study results visually, in written text and verbally using appropriate technical language	
Education Method	Lectures, interactive sessions and on-site survey: 36h Independent study: 104h (74%) The course consists of lectures, interactive sessions and on-site survey and investigation. Lectures provide background knowledge to the students, enabling them to approach interactive sessions and on-site work. On-site survey and investigation of case studies ensure the application of the learned notions in practice through a hands-on approach. Throughout the entire course, students work in groups on a case study and are tutored accordingly. Students are to meet the teachers to coach them on their research, but will also coach themselves in groups on different topics. Case study options differ with respect to building materials and technologies involved, degradation patterns and mechanisms, and type of conservation and rehabilitation interventions required. Supported by instructors and different specialists, the students will carry out a survey of the building, develop an investigation plan, validate their hypothesis through on-site research, come to a diagnosis of the damage processes and give an advice concerning the interventions related to conservation and rehabilitation of the building.	
Course Relations	The content of the course is complementary to the content of the Heritage&Values elective. It is suggested to Heritage & Architecture students to attend both electives.	
Literature and Study Materials	Reader, journal articles, on-line education material, including recorded lectures, specific lecture material on the selected case studies	
Books	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Reader	Literature and study material will be made known in Brightspace one week prior to the start of the course.	
Assessment	Analytical assignment (analysis report on the selected case study). Students who received a mark 5.5 or lower (in a scale from 1 to 10) are eligible for repair. The repair consists in submitting a revised version of the analysis report. A mark will always be assigned to the repair: the maximum mark for a successful repair is 6. The date of the repair is fixed at 4 weeks (20 working days) from the submission of the original analysis report.	
Period of Education	Q3	
Concept Schedule	Wednesday afternoon	
Minimum number of participants	15	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0142	EXTREME technology	15
Course Coordinator	Ir. R. Schroën	
Course Coordinator	Prof.dr.ing. U. Knaack	
Co-responsible for assignments	Prof.dr.ing. U. Knaack	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The project is about building in a extreme situation, in respect to climate, location and function. Essence is the interaction between the extreme circumstances, the technical solutions, and the architecture. Extreme circumstances do request technical solutions which will be the starting point for the design development. The designer has to direct the 'engineer questions and answers', towards the articulation of the form which is based on integration of aesthetic and technology. "Die Architectur des 21 Jahrhunderts hat ihre Unschuld verloren, Gebaude müssen etwas leisten" Stefan Behnisch. In the end the student is able to understand technical solutions, to reflect on them, to applicate them and to transform them. And the student is able to design a coherent design result.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on Master 2 level. Specified for this course: In the end the student is able to design a healthy coherent building in extreme conditions with a focus on technical solutions: the student is able to apply, reflect and transform principles concerning climate, construction and structure.	
Education Method	In EXTREME students make an individual design project. Students attend lectures, do self study, and meet with their teachers once per week.	
Assessment	Design examination. A design examination is an active assessment, during or at the end of the educational period, with a design (drawings, models, reports, oral presentation) as a final product. During the educational period, the student receives feedback on the progress and how to develop the design and design process. Examples of end products: drawings (on paper, digital), scale models, reports, reflection, presentations. If products are missing at the final presentation the project cannot be assessed and the student fails the project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Special Information	The maximum marking period is 15 work days.	
Period of Education	Quarter	
Concept Schedule	All lectures and teaching is on Tuesdays.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0143	Sustainability project - design and elaboration	15
Course Coordinator	Prof.ir. M.F. Asselbergs	
Course Coordinator	Prof.dr.ir. A.A.J.F. van den Dobbelsteen	
Course Coordinator	Ir. P.G. Teeuw	
Contact Hours / Week x/x/x/x	Varies.	
Education Period	Different, to be announced	
Exam Period	none	
Course Language	English	
Course Contents	This course is connected to active involvement of students participating in design teams related to practice. This course deals with the architectural and technical design and elaboration. The course is not regular offered but incidental.	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level.	
	Specified for this course; the student is able to:	
	- collaborate in a team with other students	
	- work on a joint design of a specific (building) design project	
	- integrate various aspects of sustainability into the design of the project	
	- elaborate on components of the design challenge, related to architectural design, structural design and engineering, envelope design and engineering, climate design and engineering, etc.	
Education Method	Tutorials, workshops, (mid-term) presentations, reporting, exhibiting (if applicable).	
Literature and Study Materials	Literature and Study Materials will be made know prior to the start of the course.	
Assessment	Design examination. Portfolio of the design, report and oral presentations will be assessed by different criteria. Also the group attitude and pro-activity of the student will be reviewed. All depending on the specific project.	
	Repairs: In case of incomplete or insufficient work, students get the opportunity to repair the assignment within a limited (agreed) period, a successful repair will be assessed with a pass (= 6.0).	
Special Information	Enrolment for this course is not by BIS. In case the course is offered it will be announced how to enrol.	
Period of Education	Varies.	
Concept Schedule	Depends on the project (varies).	
Minimum number of participants	Varies per project.	
Maximum number of participants	Varies per project.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0144	BK-Launch studio	15
Course Coordinator	Prof.dr.ir. J.W.F. Wameling	
Instructor	Dr.ir. R.M. Rooij	
Instructor	Ir. H.A. van Bennekom	
Instructor	Prof.ir. M.F. Asselbergs	
Responsible for assignments	Prof.dr.ir. J.W.F. Wameling	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Bachelor Architecture, Urbanism & Buildings Sciences or comparable.	
Course Contents	Many of the current societal challenges have a strong relation with the built environment, such as the energy transition, the circular economy, the scarcity of raw materials, spatial resilience and justice, and recently the consequences of the COVID-19 crisis, eg urban health and indoor climate issues. Solving these complex problems requires a creative, entrepreneurial, interdisciplinary approach. Young people with an entrepreneurial attitude can make an important contribution to this. In particular, design-oriented students with great creativity and profound interdisciplinary knowledge can make impact. Innovation research shows that new market initiatives can be an important stimulus to achieve innovation in the sector. Architectural innovation, for example, often arises from a new office (often with young people). We also see many new entrants in the energy transition and circular construction designs that create surprising solutions with a completely new perspective. In the faculty many students show an intrinsic motivation to get started with these challenges, and go along with an idea in order to market their solutions as a company or concept, often together with others. In this design studio, we are looking for ground-breaking solutions for the society related problems mentioned. In this design studio, individual students or an interdisciplinary team of students will design a solution in such a way that it will be both a showcase for the outside world, and a possible start of a new venture. The project is guided by a variety of tutors from all departments of the faculty in order to emphasize the interdisciplinary character. To create this combination of design and entrepreneurship, creative, enterprising students come together in this design studio. They work on the development of their idea in the form of a design proposal and they think about how their idea has additional societal value, can create societal impact, and can be brought to the market. The idea can be a physical product, but also a strategy, service, approach or alike. Upon completion of the project, the interdisciplinary groups present themselves to an independent jury. The BK-launch studio is part of the BK-launch platform for innovation and encouragement of entrepreneurship in the faculty. After finishing the design studio successfully, students can decide to participate in the BK launch platform (see https://www.tudelft.nl/bk/samenwerken/bk-launch).	
Study Goals	Upon completion of the design studio the students can: create a coherent, significant, elaborated, correct and innovative design proposal for an architectural, urban, of building technological challenge, including a viable business plan or implementation strategy. argument why their project can have (additional) societal value and impact for solving a large(r) socio-spatial challenge. interdisciplinary collaborate with students from other disciplines via the development of a joint and integral design proposal. demonstrate an entrepreneurial attitude and mind-set and related skills, such as creativity skills, value assessment skills, and the integration of market and business constraints in the design development process. present, discuss and defend their design proposal and business plan/implementation strategy convincingly to an audience of experts from the field.	
Education Method	The course's learning activities comprise: - Tutorial in studio - Workshops - Lectures	
Assessment	Grades will be based on course participation, assignments, presentation, and the final project. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	Monday afternoon and Thursday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0145	Circular Product Design	5
Course Coordinator	Dr. O. Ioannou	
Contact Hours / Week x/x/x/x	Tuesdays from 3.1 to 3.4 is for lectures; 3.5 is for interim presentations; weeks 3.6-3.9 are dedicated to studio work and 3.10 is for final presentations. Fridays is time for self study.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	A background in design is required.	
Summary	The course focuses on Circular Building Product Design. It is one of the Technoledge elective courses of the BT Master track. The course is also open as a free elective course for students from other faculties.	
Course Contents	Building products are the basic components of the built environment. The choice of materials, subcomponents and their internal and external interfaces provide them with properties that have a great impact on the life-cycle performance of a building such as reuse, repair and remanufacturing. As such, they play a crucial role in the transition from the Linear to a Circular Built Environment. Our input to students is structured across four domains of inquiry: materials, design, manufacturing and management. Our intention is to identify key parameters, but also the complex interdependencies of the aforementioned domains. The course comprises of in-class lectures from selected guests, constant online exchange and in-situ excursions to manufacturing facilities.	
Study Goals	After successfully completing this course the student is able to: - Identify key parameters for making building products circular, - Correlate the key parameters to reason complex domain interdependencies, - Design a circular product or circular product concept by prioritizing key parameters and relations, - Communicate design artefacts and self-evaluation results by using a clear and coherent verbal and visual narrative.	
Education Method	Lectures, design studio work, blended learning, self study.	
Assessment	Analysis of benchmark products and context. Conceptualisation of product configurations and functionality. Design of a building product and its presentation in mock-up and drawings. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Tuesday mornings between 08:45-12:45 and Friday afternoons between 13:45-17:45.	
Leerstoel	Building Product Innovation	
Minimum number of participants	10	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0149	ON SITE: Landscape architectonic explorations	15
Course Coordinator	Dr.ir. N.M.J.D. Tillie	
Course Coordinator	Dr.ir. S.I. de Wit	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / 4	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Required for	students need to be master students	
Expected prior knowledge	design skills	
Summary	In the On Site elective students are invited to investigate, understand and re-design an existing urban landscape of several hectares. You are challenged to make use of unorthodox explorative methods and develop concrete proposals for improvement, if possible, physically constructed during the course. On Site offers a multidisciplinary design setting in which you interact with the users or owners of the area.	
Course Contents	In this elective course that is organised by the section of Landscape Architecture, the spatial potentials of a specific site (e.g. a university campus, a neighbourhood district, an infrastructural left-over) are the central design issue. We aim at participants with different disciplinary backgrounds. We will select one or two urban landscapes in which a large variety of current societal and spatial needs can be operationalised. Landscape interpreted as public domain, ecological resource, social space and healthy environment requires new approaches and proposals for the physical improvement of the outdoor over-all quality. Students are challenged to review their ways of spatial exploration and diagnosis and to develop substantial landscape ideas for a better place. Through fieldwork, the site will be analysed applying experimental methods and techniques, some of which are borrowed from disciplines like ecology, social sciences and visual arts. The experimental analysis depicts the subjective, dynamic and intangible characteristics of the place such as: processes, activities, memories, stories, experiences, rituals. Through sensorial perception, tracing narratives, investigating historic sources, mapping spaces, experimental photography you dis-cover the identity of the site. The final goal of the course is to develop designed proposals for landscape-based actions in the area. Potential execution of the design should be taken into account while working on the proposal. Preferably, hands-on landscape engineering and construction work is part of the course, as well as interacting with the stakeholders and the public. This course is being developed in close collaboration with area managers, users and advisors to enlarge the chances of actual adoption and implementation of the design proposals.	
Study Goals	By the end of this course, you should be able to: - to enlarge the disciplinary repertoire used for the investigation, the visualisation and the understanding of topography and for the clarification of spatial identity of a specific landscape; - to understand, internalise and apply the potential interaction between landscape architecture tools, other design disciplines and other fields of science; - to develop a concrete landscape architectural proposal for a specific site; - to elaborate a design proposal in terms of engineering, construction and maintenance.	
Education Method	studio work interactieve lectures workshops fieldwork work on site	
Assessment	drawings models films or if possible: real constructions in the public realm	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 4	
Concept Schedule	Monday	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0167	Architecture and Urban Design	15
Course Coordinator	Dr.ir. M.G.A.D. Hartevelde	
Course Coordinator	Dr.ir. R. Cavallo	
Responsible for assignments	Dr.ir. R. Cavallo	
Contact Hours / Week x/x/x/x	12 hours per week (4.1-4.5) 8 hours per week (4.6-4.8, 4.10)	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	The course is open to students of the Masters degree programmes in architecture, urbanism and landscape architecture. If you are in a different programme: please consult coordinators before enrolling and ask approval. MSc track Architecture: it is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080). Skills are acquired to incorporate an understanding of the design (process) attained with regard to architectural/urban history, theory, art and technology as well as relevant general knowledge of human sciences. Additionally, skills are acquired to incorporate an understanding of the design (process) attained with regard to the relation between buildings, public spaces and society's needs, including environmental aspects.	
Course Contents	Massive urbanisation puts pressure on public space and demands new programmes for instance, alternative gathering places such as public interior spaces and a variety of forms of collective spaces. This diversity of programme cannot be planned in advance, but interventions in the city need constantly to be grounded on sharp design approaches in order to respond adequately to the necessities of our times. In general, mobility and public life manifest themselves in various forms as carriers of urban development. Design experiments, as put forward in this course, have to show how to work with continuously changing urban conditions, how mobility transforms the city and public space can take various forms, how programs hybridise, and how new technologies can be used to keep up with the urban dynamics. Given these themes, designs also present awareness of the inclusiveness and accessibility of various systems and places, facilities and technologies. In this interdisciplinary Masters design studio, you combine these issues and present them to your peers and a team of interdisciplinary supervisors. You focus particularly on the consequences of urbanisation for the major foundations of the city of the future urban infrastructure and public space and you envision an experimental design, within a larger set of visions produced by you and your fellow students. In these designs, students and staff are interested on one hand to the urban intervention in the built environment and its effect on architecture, and at the other hand to the architectural treatment of the city and its effect on urbanism. The studio is supported by an interdisciplinary lecture series which provides an overview of vested theories and cutting edge research on people movement, urban vitality and public space. This includes seminal works by Gehl, Whyte, Jacobs, Appleyard, Lynch and research work by Cullen, Smithsons and Venturi & Scott Brown. The role of citizens and designers in shaping vibrant urban public space is explored through readings, film and active discussions with students. This is certainly not your average dry theory course. The course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - creates presentations analysing the subject on an academic level. And, the student: - understands the interrelation of architectural and urban design, to evaluate and create proposals for strategic interventions, with regard to spatial-social patterns and the culture of the city - evaluates skills in architectural and urban design to create an elaborate design proposal in typological terms related to use, ownership and meaning - creates an elaborate design proposal on the edge/overlap of both professions, satisfying formal, technical and functional requirements, including materialisation.	
Education Method	The course consists of interactive studio work and lectures. Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists create strong design propositions as critical thinkers. In class, you are encouraged to question the course material, the case, the lecturer and the general state of urban theory. Studio work includes group analyses* and individual design of a challenging case. As such, the course provides contextual insight in the problematique highlighted in the course. The case will be updated annually. It serves as test-bed for a design proposition, which stands for a more general statement in the sphere of interdisciplinary design approaches. Lectures are followed by discussion groups* that challenge you to discuss and apply the theories covered in class in your urban analyses. Small weekly homework assignments are covered in these groups. Therefore, come prepared! Your final statement is based on research and represented in an elaborated design. These will be presented at the last day of class. *) the discussion groups ideally consists of four/five members, who divide topics and peer each other.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Studio work 80% - Lectures 20% Assessment of studio work: Analyses and design, presented in drawing form with written commentary and a model. Assessment of lectures: Class participation and homework assignments together with final presentation (including 5 pages individual contribution to a collaborate report, 1 group poster (A1) and verbal presentation (Q&A) proving integration with class readings. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page.	

	Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Special Information	This course includes AR0168 - People, Movement and Public Space (so it cannot be combined with this course). The studio work includes an excursion to the site. Please, do not hesitate to inform with the course coordinators what this year's case studies is.
Remarks	The maximum grading period is 15 work days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Tuesdays. The actual schedule will be available via Bright Space > My Timetable.
Leerstoel	Urban Design Design of Public Space Architectural Crossovers
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 32.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0168	People, Movement and Public Space	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Dr.ir. M.G.A.D. Harteveld	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 10.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	Cities are alive! People move through the city to reach their work, school, go shopping, or to enjoy culture, art, or other people. But what environments accommodate the urban vitality we all crave for? What is the influence of policymakers, engineers, urban designers, and architects? This course aims to provide an overview of vested theories and cutting-edge research on people, movement, and public space. This includes seminal works and studies from different disciplines. The role of policymakers, engineers, designers, and citizens in shaping vibrant urban public space is explored through readings, film, and active discussions with students. This is certainly not your average dry theory course the course material will come alive through active discussions and the direct application of theories in analysing real urban settings.	
Study Goals	At the end of this course, the student: - knows key literature and recent research on people, movement and public space - understands main theories on people, movement and public space - applies these theories in analysing real urban settings - evaluates critically on these theories - defines recommendations and assignments on the basis of analyses - creates presentations analysing the subject on an academic level.	
Education Method	The course consists of on-demand video lectures, mandatory literature, and other material to be studied. Lectures are followed by smaller discussion groups* that challenge you to discuss and apply the theories covered in the lectures and literature in real urban analyses. Five weekly homework assignments are covered by peer-review in these discussion groups. Therefore, come prepared in advance and take critics into account! Active participation and discussions are greatly welcomed and reading the course materials is absolutely required. These are not consumer classes! Great urbanists are critical thinkers questioning the course material, the lecturer and the general state of urban theory is strongly encouraged. The class concludes with the reviewed material, a final statement based on your weekly work, defining a design assignment without elaborating the design, and a poster. This will be presented/uploaded at the last day of the course. *) the discussion groups ideally consists of four/five members, who divide topics relevant for a given location and peer each other.	
Literature and Study Materials	1 People, Movement & Public Space - Introducing today's Problem Setting and pioneering Problem Solving: Sert (1952), van Ecyk (1956), Hertzberger (1956), Mumford (1958), Gruen (1964), Breines and Dean (1974), Alexander (1978-1984), et seq. 2 Path Systems On Physical Urban Patterns and Use Patterns: Kahn (1951-53), Venturi, Scott Brown and Izenour (1968), Wurman (1970), Alexander, Ishikawa, Silverstein et al (1977), et seq. 3 Pedestrian Perspective On Street Views and People's Views: Cullen (1961), Smithsons (1961), Appleyard, Lynch and Myer (1964), Rowe and Kuoetter (1978) McCluskey (1979), Passini (1984), et seq. 4 Psychology of Place On Senses of Place and Mental Images: Debord and Jorn (1957), Lynch (1960), Steinberg (1973), Canter (1977), Relph (1976), Prak (1979), Peattie (1987), et seq. 5 Public Life On Learning Live and Understanding Public Space: Whyte (1958, 1979), Jacobs (1961), Gehl and Gehl (1968/1971), Rudofsky, (1969), Appleyard and Lintell (1972) et seq. 6 Presentation Presentation Hand-In / Uploading	
Assessment	Peer-reviewing of five weekly homework assignments within the student peer groups. Grading individual final work, which includes assessment of a booklet with (i) the five (improved) weekly assignments, and (ii) a concluding part, defining recommendations and design assignments, and the assessment of (iii) an academic poster presentation. Late assignments will not be graded. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The course an open elective for the MSc AU&BE, targeting the MSc Architecture, Urbanism, Landscape Architecture, Architectural Engineering, and Management of the Built Environment. The course is already included in the course AR0167. So, please, understand that you can't be enrolled in both. The course is mandatory for the MSc TIL Policy track, and a recommended external elective for the other MSc TIL tracks, including Design, Operations, and Engineering. In all cases, please understand, you have to be enrolled in Brightspace as well as via the registration system of the home faculty (BIS) of this course: The faculty of Architecture and the Built Environment.	

	If you're not part of one of the above programmes, you may still be welcome. Please ask the course coordinator. If you have e-mail permission to join the course, do send this to 'intekenen-bk@tudelft.nl' (E&S registration at the home faculty), and ask them if it is still possible to put you in the system. You have to be registered as such in order to be assessed in and credited for the course AR0168. The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Mondays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any (elective) course the minimum number of participants is 15.
Maximum number of participants	For this course the maximum number of participants is 100+.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0169	Materialisation: The Future Envelope	5
Course Coordinator	Ir. F.R. Schnater	
Instructor	Ir. A.C. Bergsma	
Responsible for assignments	Ir. F.R. Schnater	
Education Period	3	
Start Education	3	
Exam Period	4 5	
Course Language	English	
Course Contents	This course focuses on the technical development of the enveloping architecture. The object of the course is to challenge the student to think about future developments and innovation- or research possibilities. The course is an elective for architecture- and building technology students as well as students civil engineering. AR0169 is combined with the course AR0134, Technoledge Façade Design. BECAUSE OF THIS COMBINATION/OVERLAP IT IS NOT POSSIBLE TO DO BOTH COURSES. The course is made up out of a series of lectures, excursions and studio-work. Studio work: Under supervision of (external) experts from TU Delft and/or industry, students will work in groups (groupsize max. 4 persons) on assignments and façade analyses that are related to several façade aspects and topics. Lectures: During the course a series of lectures relevant to the design- and analysis assignments will be organized. We will invite lecturers from the TU, the industry and engineering firms. These lecturers will also be involved in the tutoring during the design- and analysis assignments. Excursions: The lectures and studio work are combined with excursions to a construction site and/or product manufacturers. For these excursions students will have to pay a nominal fee.	
Study Goals	The student - is capable of understanding technical developments and reflecting on façade designs. - is able to respond adequately to technological issues and formulate conceptual solutions to technological problems in ad hoc situations. - is able to understand and develop façade designs and concepts that are coherent, integrated and feasible in terms of building physical, structural and constructional aspects. - is able to present his/her work using the correct (3d)drafting techniques and other appropriate presentation techniques and is able to use modern visualization tools to make a compelling presentation. - shows initiative and keen interest in technical aspects of façade design	
Education Method	lectures, studio sessions, and excursions	
Assessment	writing assignment (report) analytical assignment (drawings and (virtual) models) oral examination (presentation)	
Period of Education	the maximum grading period is 15 work days.	
Concept Schedule	10 weeks in q3	
Minimum number of participants	Tuesday afternoon and Friday morning (first 5 weeks also in the afternoon)	
Maximum number of participants	4	
Course evaluation	30	
	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0171	Geo-design for a Circular Economy in Urban Region	5
Course Coordinator	A. Wandl	
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	How can we achieve a circular economy on an urban and regional scale? What are the spatial consequences of such a transition? In this course, students apply a geo-design approach to develop eco-innovative solutions for circular urban regions in collaboration with (inter)national stakeholders and researchers to tackle this task. Background: Europe's economy is not circular. About 60 per cent of the land used to meet the EU's consumption demand is located outside its territory. Transitioning towards more circularity is crucial to delivering the resource efficiency agenda established under the Europe 2020 Strategy for smart, sustainable, and inclusive growth. In the H2020 Research & Innovation Action project REPAiR (REsource Management in Peri-urban Areas: Going Beyond Urban Metabolism), we developed a cross-scale approach to developing spatial development strategies for circularity in urban regions. In this course, students will build upon methods developed by this international research project on two specific case studies, Amsterdam and Naples. The course's key aim is to develop spatial development strategies that support the development of a circular economy in European metropolitan areas. With the help of geodesign and stakeholder interaction, the students will develop eco-innovative solutions, which will be modelled and tested in the case study area. Approach: The course applies a geodesign approach to reveal the local space-specific challenges and possible strategies. Campagna (2014) defines geodesign as 'an integrated process informed by environmental sustainability appraisal, which includes project conceptualisation, analysis, projection and forecasting, diagnosis, alternative design, impact simulation and assessment, and which involves a number of technical, political and social actors in collaborative decision-making'. The students will focus on the aspects of analyses, alternative design and assessment with the following objectives: To develop an understanding of the characteristics, mechanisms, and inter-scalar dynamics of the resource management systems and the relations between waste flows, environmental and spatial quality, allocation and governance in peri-urban areas. To interpret the link between metabolic flows and urban processes by extending the assessment of urban metabolism by notions of urban drivers and urban patterns and environmental and spatial quality and co-benefits. To develop and assess place-specific eco-innovative solutions for resource management, which improve the environmental and spatial quality and the quality of life. To understand the decision-making structure and processes in the case study areas concerning different stakeholders' diverse interest and priorities. The case study areas are either a Dutch or a European Metropolitan area and will be related to the NEW European Bauhaus project NEB-STAR Theoretical Framework - Extended Urban Metabolism Urban metabolism here is a framework for modelling complex urban systems' material and energy streams as if the city was an ecosystem. This approach allows for studying the dynamics of cities (beyond 'traditional' mobility and the relationship between built/(un)cultivated environments) in relation to scarcity, carrying capacity and conservation of mass and energy [Newman et al., 2009]. UM challenges traditional urban planning, in which social, cultural, political and technical dimensions prevail over the biophysical dimension. In this course, we build on the extended UM approaches [see for example, Minx et al. 2010, Schremmer et al. 2011, Pincetl et al. 2012, Goldstein et al. 2013], in which urban subsystems with their environmental and spatial impacts are addressed more explicitly. We emphasise the notion of synergism in UM studies, focusing on the benefits that may arise from the intrinsic relationships existing within the urban metabolic system [Zhang et al. 2014].	
Study Goals	At the end of the course the student is able to: apply the geodesign framework to develop eco-innovative solutions (EIS), which support the spatial transition towards circularity; formulate the idea of an eco-innovative solution in a way that it responds to challenges formulated by policymakers; represent the - for the transition towards more circularity - most relevant environmental, social and economic subsystems and their spatial structures; describe, the economic, social and spatial process that are influenced by an EIS, using flow maps, system diagrams and systemic sections; build a simplified sustainability assessment framework that consists of three indicators; design an alternative future, anticipating the changes in physical spatial structure and resource flows based on the consequence of the application of an EIS; assess the impact of your EIS using your sustainability framework; to inform decision-makers about the positive and negative effects of your EIS and how they relate to their aims.	
Education Method	Lectures to explain key concepts and methods the students will use and can apply. Topics include Geodesign, Circular Economy, decision making, systemic design, GIS-based mapping and spatial analysis, sustainability assessment and graphic representations. Briefs and poster templates: On Bright Space, students can find a short brief of the task of each session/poster. For each poster, we have prepared an (InDesign) template, with the main questions the students should use to guide their exploration. Studio - Group work: The course was isdesigned, that the students spent one whole day per week together in a (virtual) room. This way, a knowledge-sharing atmosphere is established, and the possibility to overhear other groups' ideas and quickly demonstrate and discuss things that are relevant to more group is possible. Iterations: The course is built in a way that students iterate the development of their posters in two different ways, the first is after four sessions, there is one session to iterate the first three posters, based on the feedback and the last two days are dedicated to iterating all posters again. The second form of iteration is that the six of the posters are actually pairs, one applied to the status quo the other to a desired future . Stakeholder input: stakeholders give presentations providing their knowledge for the students. stakeholders act as external critiques at midterm as well as during the final presentation (2,8) Pressure cooker design: The one poster, one day, one question approach forces students to transition quickly from research towards design in order to have concrete and developed ideas that can be assessed.	

Literature and Study Materials	Combination of individual and group work.
Assessment	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course. The final assessment has two parts, (i) the final poster presentation in front of colleagues, teachers and stakeholders and (ii) the hand in of the final poster. Each individual session finishes with a formative assessment in the form of either a review by the course instructors or a peer-review between the students, this way the students get weekly formative assessment. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.
Remarks	The maximum marking period is 15 working days.
Period of Education	Quarter 4.
Concept Schedule	The sessions are scheduled on Wednesdays. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 24.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0172	Globalisation - Research on the Urban Impact	5
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Course is research base and is organize to explore the concepts and dimension of the metropolization process in general so to understand the particularities in an emerging economy and is based on the complex city region research group within the chair of Spatial Planning and Strategies. The research Platform where its constructed within the following research lines: Metropolitan spatial structures The metropolitan spatial structure and its forming region considering in its complexity- The evaluation of the diverse regional structures: economics competitiveness and environmental sustainability as well the search for social wellbeing How the role of planning strategy and its related practices can be improving by the better knowledge of the spatial structure and its performance? Regional Governance, planning and design The governance of metropolitan regions in the context of increasing complexity and fragmentation of spatial relationships To what extend can urban and regional planning and design methods serve as a catalyst for territorial transformation? International Planning and developing regions The focus is on comparatives studies on the way diverse form of intervention trough spatial planning and territorial management searching for the validation on diverse methodologies How are approaches and tools changing to deal with critical territorial challenges, particularly risk associated with clime change, the spatial dimension of the knowledge economy ad the networked metropolitan region? Delta Urbanism The focus is on the new approaches in design and planning of urbanized delta areas-how to balance the diverse claims and interest-balancing competing claims require the finding on new relationships to be forged between design, engineering science and governance. How can we define a new balance between planned, designed and engineered interventions in the systems of the delta on the one hand and a freedom for self-organization of natural and societal processes in the other?	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies.	
Education Method	Lectures, seminars, working groups and studio sessions. Combination of individual and group work. The approach for this course is define in 3 pre set thematic lines that determine diverse methodologies to follow: Thematic line 1: Balance development in a large and expanding metropolis Understanding the role of the city and its constitutive elements and linking its functions in a metropolitan perspective towards a more sustainable development Analysing concept of Globalization/world city model/ and urban competitiveness from metropolitan level to empowerment at the local level Analysing the Planning framework process and its direct relation to the urban space and form, within land uses plan and evaluations Thematic 2: The urban/regional structure-analysing the urban form Understanding the polycentric developing model and its correlated Fragmentation processes From the monocentric sustainable model to the functional base structure-Developing Centralities- as a base to understand the regional structure and its socio spatial effects The accessibility/ mobility dilemma within the diverse users and its divers networks Thematic 3: Strategic Sustainable planning in a metropolitan condition, How to integrate the future development with existent stagnated areas in a sustainable developing perspective, at least improving actual development levels Considering a base for more endogenous type of development within the main constitutive agglomerations that define it. All considering the diverse potentialities of transformation and management of the deltaic condition.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus design examination. The expected output is an essay and a new developing vision with concrete planning strategy within key interventions. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 32.	

Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).
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AR0173	Globalisation Free Choice	10
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. D.A. Sepulveda Carmona	
Responsible for assignments	Dr. D.A. Sepulveda Carmona	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The course is design as a research base project that investigates: The main forces on the formation of metropolitan structure, the potential arising for urban regeneration and how to construct spatial strategies for socio-spatial integration with an integrated, complex and collaborative approach. Friedmann 2007; Kratke S. (2007) The metropolization of European and regional systems: Wust s, et al 2007 Metropolization and economic crisis; www.atlas debuenosaires.gov.ar/aaba The delta landscape conditions, dynamics and constraints that shape the potential for integral and resilient development. H. Meyer (2012) The urbanization in a delta landscape: a flicke history; www.delta-alliance.nl/deltas/parana The socio-spatial conditions for development. Taking an actor relational approach in networks, we explore the need and potential for collaboration of diverse actors in a common search for development. Sepulveda& Janches (2009) exploration on socio spatial integrative strategies; Rozenblat. C (2012) http://www.unil.ch/webdav/site/iguurban/shared/Rozenblat_IGU_2012.pdf; A. Da Cunha (2012) Urban Geography in the era of globalization: The city of the future The context and location A developing countrys main metropolitan area, Buenos Aires, Argentina, in the southern periphery within environmental conflict conditions: the whole of the Matanza -Riachuelo river basin. The study case The main characteristic of the case is express as urban fragmentation, socially and spatially resulting in a dual city, express in an unbalance distribution of opportunities and urban qualities. A main driver over the last 30 years has been adaptation of regional and urban form to the new division of labour arising from forces of globalization. The Aim of the study is: To define spatial options and supportive urban programmes as conditions for sustainable integration of fragmented spatial development in the marginal areas of the Buenos Aires Delta. This in order to determine a more responsive metropolitan structure. This requires the formulation of a planning strategy.	
Study Goals	At the end of this course, the student is able to: - understand the dynamics of an urban metropolis in a developing country, including the metropolitan /urban analysis approaches at different scales, the diverse actors and their interests, and recognising the many systems (functional networks, natural systems) that define the metropolis. - understand the relations and interactions between the diverse stakeholders with divergent interests and the impact on urban development and the distribution of costs and benefits. - explore synergies between changes brought by globalization forces in the existing city, and the influence (or not) of planning tools and interventions and strategies. - explore how the performance of the environmental system can be addressed under a more integral perspective for development (water-soils). - explore through a regional design intervention the potential programmes and spatial strategies of development, by applying research by design methodology - integrate disconnected areas to the urban fabric & urban structure - understand the effect of delta conditions (via the analysis at the Matanza-Riachuelo River basin) on the potential and constraints of sustainable urban development and the planning framework. - work in groups to develop a strategy for socio-spatial integration and functional and spatial synergy of the region, including strategic projects that may help to initiate the proposed strategy.	
Education Method	Studio sessions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The schedules of the courses AR0172 and AR0173 are tuned. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 30.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0175	Campus Utopias	5
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Course Coordinator	Ir. E.H. Gramsbergen	
Instructor	Ir. E.H. Gramsbergen	
Responsible for assignments	Ir. E.H. Gramsbergen	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	In this course we will work collectively on a comparative analysis of a selection of influential modern campus designs and their emblematic buildings. One of the themes will be the tension between global and local forces. To what extent were the projects influenced by general academic ideals and architectural models? And what was the role of the specific local cultural and spatial conditions on the other hand? We will follow a typological approach with attention to five different scale levels and their interconnections: from the territory to the campus to the building to the interior and finally to the ornaments.	
Study Goals	1. Criticality: to develop a scientific approach towards selecting and handling source material such as literature and archival material 2. Competence: to develop advanced visual research techniques by making use of plan analysis and comparative analysis methods 3. Contextualization: to be able to reflect on the international socio-economic and cultural context in which modernist campus designs came into being 4. Communication: to develop adequate ways to present research findings to peers and a larger audience 5. Collaboration: to develop collaboration skills by working on a collective research theme and comparable outcomes	
Education Method	Weekly seminars, field trip t.b.d.	
Assessment	Assessment is based on both individual and group work in a 60%-40% ratio. Final products are a series on analytical drawings of a single case study accompanied with a written explanation (individually) and a group presentation on the outcomes of the comparative analysis (format to be determined during the course)	
Period of Education	Q3, wk 3.1-3.10	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0177	The Why Factory MSc2 Design Studio	15
Course Coordinator	J. Arpa Fernandez	
Instructor	A.B.O. Ravon	
Responsible for assignments	J. Arpa Fernandez	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary	The Why Factory (T?F) is a global think-tank and research institute, run by MVRDV and Delft University of Technology, and led by professor Winy Maas. It explores alternative possibilities for the development of our cities in particular and of our Planet in general, by focusing on the production of models and visualizations for the Planet of the future. Education and research at The Why Factory are combined in a research lab and platform that aims to analyze, theorize and construct future cities and a better Planet. The Why Factory investigates within the given world and produces future scenarios beyond it; from universal to specific and global to local. It proposes, constructs and envisions hypothetical societies and cities and landscapes; from science to action and vice versa. The Why Factory thus acts as a future World scenario making machinery. Moreover, we want to engage in a public debate on architecture and urbanism. The Why Factory's findings are therefore communicated to a broad public in a variety of ways, including exhibitions, publications, workshops, and panel discussions. The research at the Why Factory produces observations, hypotheses and statements in a visual and direct manner. The images produced are a combination of science and fiction, in an approach integrating systematic observations and gathering of data with speculation and imagination through spatial and architectural means. A systematic, parametric exploration of parts of the design is an integral part of the research approach. STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO. During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. For more information about our previous studios, please visit: https://thewhyfactory.com https://thewhyfactory.com/education/	
Course Contents	MSc2 offered by The Why Factory focus on exploring how the future of architecture and the city will be. The students are asked to rethink, research, reshape and enhance the image of future of architecture and urban life. Studios include highly integrated research and design meant to contribute to the development of The Why Factory's agenda. During the Why Factory MSc2 Design Studios, we invite students to research on visionary, green, fantastic, fast, self-sufficient, austere, cute, transparent, biodiverse, intimate, adaptable, free, open, emotional, surprising, natural, wonderful and common future architecture and cities (and Planets!)	
Study Goals	- Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. - Upon completion of the design studio the student is able to demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. There are six qualitative aspects expected from students at the end of their MSc1 and MSc2 Design Studios: 1. Critical Thinking: The ability to create a conceptual framework, work with studio concepts and self-reflect on work developed over the course of each semester. 2. Craft: Commitment to refining how a project is investigated and represented, including simulations, models, drawings, analysis, etc.. 3. Rigorous Investigation: Thorough and complete investigation of ideas through research, iteration of drawings and models, and rhetorical elaboration. 4. Response to feedback: Ability to respond to and incorporate feedback from studio instructors. 5. Imagination and Creativity: Spirit and originality in proposed project approach and its subsequent development. 6. Capacity to integrate in a large group and produce collective research and design. It is very important during the studio to work in large teams and be able to adapt to team-work, as an essential training for future professional life.	
Education Method	Number of studio hours: 80 Number of self study hours: 332 STUDENTS WORK IN LARGE GROUPS AND PRODUCE ONE SINGLE PRESENTATION AT THE END OF THE STUDIO During the studio, several individual interviews with the instructors will take place so as to evaluate the individual student's progress within the group. The Why Factory runs research projects, which are positioned in a classical research tripod of models, views and software; of model cities, applications and storage. The research on the Future City is undertaken through the interactive composition of three fields. It speculates on possible theoretical models in the model city program. It makes counter proposals for existing cities. It stores its knowledge through an evolutionary gaming program. Model Cities Program: Model Cities concentrates on the conceptualisation and modelling of cities, each within its own limited set of parameters that allow for maximal exploration of a specific subject in order to engage with possible futures. The Model City Program theorizes abstract cities and translates them to physical models to explore spatial qualities and quantities, potentials and limitations. T?F seeks for a refined combination of science and fiction in order to bring our dreams and desires closer to reality. Applications Program: In the applications program model cities both are tested in real cities. The different models become	

	counter proposals for existing cities. T?F collaborates with local institutions to test different hypotheses and discusses them with local governments and citizens. Software Program;How can we store all the information that derives from the model city and applications programs? Can we create a library that is not only passive but can behave actively? Maybe we can store knowledge in gigantic software, an evolutionary game, that not only collects data but also positions them and makes them visible, comparable and in the end even productive? It combines the role as a library with the one as a connector or a communicator and even generator. It becomes a city itself; an evolutionary city; a data cloud. Such a tool combines the more collective agendas with the individualistic tendencies of the current societies; a developing series of urban software is imagined.
Assessment	Oral examination and design examination: a collective research and design proposal will be presented at the end of the studio by two or three members of the group. These two or three students are just representatives of the team and present the work undertaken by everyone. Students will receive individual grades according to their performance during the studio. Instructors will monitor de individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.
Permitted Materials during Tests	On-screen presentation, printed materials and models.
Special Information	The maximum marking period is 15 work days.
Period of Education	spring semester, Q4
Concept Schedule	Tuesday and Thursday from 8.45 to 12.45 from week 4.1 to 4.10
Leerstoel	The Why Factory
Minimum number of participants	12
Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0179	Value Capturing	5
Course Coordinator	Ir. H.W. de Wolff	
Instructor	Prof.dr. W.K. Korthals Altes	
Responsible for assignments	Ir. H.W. de Wolff	
Contact Hours / Week x/x/x/x	0/4/0/0	
Education Period	2	
Start Education	2	
Exam Period	2 3	
Course Language	English	
Summary	This course deals with financial mechanisms behind urban (re)development. Urban (re)development influences land prices in an area, on the short as well as on the long term. For urban (re)development, different kinds of investment are needed. Besides investments in the construction or the renewal of real estate, also investments in public services and public space are necessary. Often, urban development takes place in a multi actor context, with different property owners, tenants, project developers, investors, the government as well as people living and working in the surrounding areas, companies, interest groups, etc. This multi actor context also changes, during the life cycle of a project. As a consequence of this, costs and profits of urban (re) development often are not distributed equally amongst the different stakeholders having an interest in the development. This might lead to a suboptimal decision process, a less sustainable outcome and sometimes to a stalemate. In which way can the rise of property values in urban (re)development be re-used within the project, to pay for the less profitable parts of the development, including the provision of merit goods like social housing and public goods like landscape elements, parks, infrastructure and parking facilities? Which strategies can be used by the government involved? How can be dealt with issues that might complicate such a strategy, like uncertainty within these projects, necessity of public accountability, protection of private ownership? How can value capturing be integrated in the management of urban areas? Which system boundaries or scale level needs to be taken into account: can value capturing help engineering metropolitan solutions?	
Course Contents	The course focuses on analysing and designing strategies for value capturing for municipalities in urban (re)development. In the course, the following topics are addressed: - strategies for value capturing in urban (re)development in multi-actor situations with private ownership to stimulate integrated development of an area, taking care that public facilities, public spaces, social housing and other less profitable parts of the development can be realised; - instruments that can be used within such a strategy for capturing the plus value and for equalizing the costs and profits amongst the different stakeholders, with a focus on legal and financial instruments, possible effects of these instruments and pitfalls. Learning from international experiences with regard to different strategies and instruments is an important element of this course. In analysing and designing strategies and instruments, effectivity, efficiency, resilience, legitimacy, accountability and transparency are important concepts.	
Study Goals	After completion of the course, the student is able to: - summarize different strategies and instruments for value capturing that can be used in (re)development projects - identify different aspects that determine the potential of these strategies as well as that might be the pitfalls - understand innovations in the field of value capturing taking into account the international context with regard to the experiences with value capturing - assess a value capturing strategy - design a proposal for (the improvement of) a value capturing strategy for urban (re)development projects, taking complex ownership and use rights and the role of the government into account.	
Education Method	Interactive lectures, in which articles and cases are discussed. Every week, a short assignment has to be prepared based on the suggested literature, that will be used in the lecture	
Course Relations	AR2MBE011 Building Law AR1MBE025 Building Economics	
Literature and Study Materials	Literature list is given with the course outline on Brightspace	
Assessment	The final grade is based on an individual assignment (paper following a specified structure) and an exam (open questions), both having equal weight. Each assessment element should be passed with a minimum grade of 5.0 before the final grade will be determined. Prerequisite for obtaining a grade is a positive evaluation of participation, which is assessed on the weekly assignments that should be handed in on time and be satisfactory	
Period of Education	Second quarter	
Concept Schedule	Tuesday morning and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0185	Research Methods 3	5
Course Coordinator	Dr.ir. J.S.J. Koolwijk	
Responsible for assignments	Dr.ir. J.S.J. Koolwijk	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	MBE Students who participated in Research Methods 2 and want to learn about more than one method should participate in this elective Research Methods 3 course (RM3). The mission of this course is to learn students research methods. Students can choose between three methods specialisations (options A, B and C). A distinction is made between methods focusing on generating insights into evidence-based solutions (A: Case study methods and B: Applied statistical methods), and problem-solving methods that try to improve decision-making (C: Operations research methods). RM3 is also advised for non-MBE students who want to deepen their knowledge about one of the research methods. Students who come from another faculty are responsible to ask their faculty if they can use this course for elective credits. MBE students who participated in the mandatory Research Methods 2 course (RM2) can only choose one of the two remaining specialisations. For instance, if a student chooses to follow Case study methods in RM2, this student is only allowed to follow Applied statistical methods or Operations research methods in RM3. Students who follow RM3 will work together with students who follow RM2. Students who participate in RM3 will do an extra assignment. Methods specialisations Students are required to choose one of the following 3 methods specialisations. A . Case study methods(CSM) This specialisation will discuss the theory of case study research taking an interpretive approach. This specialisation will go into the rationales (why), methods (how), kind of results (what), and important considerations, such as reflexivity and methodological quality criteria. Students will exercise different practice based research methods to conduct case study research, such as interviews and observations. Students analyse a qualitative research paper to learn about quality criteria for qualitative research. Finally, two small case studies will be conducted as the main assignments. As part of the first case study each student needs to perform an in-depth interview. Practising includes developing an interview protocol for in-depth interviewing, performing an in-depth interview, transcribing the interview, using software to analyse the data and reporting. In the second case study, students learn how to conduct an observational study by developing a research question, identifying variables and a population of interest, and developing an appropriate observational study including materials. Course Contents Continuation B. Applied statistical methods (ASM) The aim of this specialisation is to teach applied statistics for building sciences. Statistical methods consists of a series hands-on blended learning practices, provided as an approximately three to four weeks intensive course. This is followed by a discrete choice assignment. There will be several statistical approaches available. The students first will master basic procedures. The concept of the course is that one learns to run statistical procedures in SPSS and how to interpret the statistical output that SPSS produces. The course will be given as a series of (online) practices and is on purpose scheduled as a series of multiple practices per week. During the practices one can work on self-tests using video tutorials and the book of Andy Field. For the final SPSS assignment, students need to show competences in applying and interpreting SPSS procedures. To prepare to the SPSS assignment, students are encouraged to practice self-tests multiple times. Therefore, the self-tests are not graded. Secondly, students either receive an individual assignment or learn how to systematically collect data using Virtual Reality. In case of the latter, one will use an already programmed VR model of a Healthcare Hub in which a discrete choice experiment has been included as an illustration of a Research-through-Design approach at the VR-Zone (in the Library). The VR model was developed to obtain the input of different stakeholders in developing an evidence-based design of a healthcare hub. One then will use the (already) collected data from the discrete choice experiment and learn to use statistical software to identify what design characteristics in VR influence peoples choices and thus reflect their preferences. C. Operations research methods (ORM) This specialisation starts with an introduction to the domain of problem solving methodologies in science, both in the technological design sciences as well as in the social management sciences, including mathematical models, operations research, logical argumentation and mathematical-formal logical systems (software supported) and a critical appraisal of these methodologies. The differences and similarities between problem solving in operations research methods, focused on design research, and in empirical research methods will be explained using from the fields of real estate management and urban development management. On the basis of case studies, comparative analysis and the systems approach (system thinking and system theory) methodological difficulties concerning practical application and integration of knowledge, theories, methods and techniques will be analysed. The exercises focus on basic concepts, problem solving strategies and strategic inter-actor design methods, project set-up and operationalisation. Students will first study a case and build a linear programming model that enables them to design different alternative solutions for the problem at hand that will support decision making. Secondly students additionally built a preference model for the same case as in RM2 enabling decision makers to take into account all stakeholders, their goals, criteria, weights and preferences and select an optimal solution).	
Study Goals	A. Case studies methods The student: - understands the interpretative approach and can apply and reflect on the interpretative approach in practice-based studies; - is able to use practice-based methods to perform case studies to generate knowledge and to answer the research question; - is able to use theoretical concepts in in-depth interviewing and can apply these concepts in analyzing and reflecting on in-depth interviews.	

	B. Applied statistical methods The student: - is able to perform several basic statistical approaches in SPSS - is able to properly interpret the resulting output in SPSS - is able to indicate which analyses and syntheses fit the questions to be solved at the relevant level of scale; - is able to use and elaborate the method(s) chosen to generate knowledge and answering the research question. C. Operations research methods The student: - is able to characterize different types of management, decision making and design problems in the fields of architecture, urbanism and building science; - is able to describe the overall process of formulating, analyzing and re-structuring a management, decision making and/or design problem in a solvable way; - is able to represent and re-structure a management, decision making and/or design problem in a mathematical design and decision model and critically reflect upon it; - is able to make a critical methodological appraisal of scientific quantitative operations research studies.
Education Method	Lectures, master classes with discussions and presentations of staff and students, combined with assignments and practical exercises.
Literature and Study Materials	Depending on your specialisation, the student needs to study the following literature. A. Case study methods - Gherardi, S. & Strati, A. (2019). How to conduct a practice-based study: Problems and Methods, 2nd edition. Cheltenham, UK: Edward Elgar Publishing Ltd. ISBN: 9781788973557 (exact pages will be published on BrightSpace). - Moerman, G. (2010). Probing Behaviour in Open Interviews. PhD. VU University Amsterdam, Amsterdam, the Netherlands. ISBN: 9789086594436. Chapter 1 and Chapter 4. B. Applied statistical methods - Field, A. (2018). Discovering statistics using SPSS, 5th revised edition, Thousand Oaks, CA, USA: Sage Publications Ltd, ISBN 9781526419521 C. Operations research methods - Reader Operations Research Methods for managerial multi-actor design and decision analysis. Editor Binnekamp R., Barendse, R. Binnekamp, R.P. de Graaf, L.A. van Gunsteren P.P. van Loon, IOS Press (draft). - Dym, C.L., Little, P., Orwin, E.J., Spjut, R.E. (2014). Engineering Design, a Project-Based Introduction, 4th edition. Hoboken, NJ, USA: Wiley International, 2004. ISBN: 9781118324585
Assessment	All three specialisations will be examined by means of assignments that are specific for each part: A. Case study methods - Some minor assignments that are marked with pass or fail. - The mark will be based on the evaluation of the two main assignments. B. Applied statistical methods - The mark will be based on the evaluation of a final SPSS assignment and an individual discrete choice assignment. C. Operations research methods - The mark will be based on the evaluation of a written assignment and on two mathematical models. The grade that you receive for a specific specialisation will also be the grade you receive for this course. Your minimum mark is a 6,0. Students are allowed one resit per assignment. When you fail the resit, you need to retake the full course.
Period of Education	Research methods 3 runs in Quarter 2 and 4.
Minimum number of participants	At least 6 per part. In the event of insufficient interest or the illness of a tutor, we reserve the right to cancel a part (A, B and/or C) course.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0187	Transition Landscapes	5
Course Coordinator	Dr.ir. G.A. Verschuure-Stuip	
Course Coordinator	Dr.ir. I. Bobbink	
Responsible for assignments	Dr.ir. I. Bobbink	
Contact Hours / Week x/x/x/x	0 / 0 / 0 / X	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Summary	The course offers spatial (landscape and urban) design training in actual heritage assignments by participating in a workshop with interdisciplinary teams. You work a few days on-site provided by stakeholders and/or governmental bodies.	
Course Contents	Identity, continuity and transformation are essential notions of today's spatial design practice. In this course, you will learn how to analyse the given situation, interpret the characteristics of the transformative mechanisms and design a proposal that supports the (spatial) identity and narrative of the site. The scale of the assignment can vary from a historical garden to an (urban) landscape. Your role is to participate actively in ongoing transformation processes from a multidisciplinary angle. You will prepare design proposals to modify a heritage site in cooperation with stakeholders and governmental bodies. The results will serve as the ground for discussion within these communities. In preparation for the workshop, you study literature on theory and methods of heritage transformation and team up with students from different disciplines and debate on the theory on place-making and heritage. During the workshop you will work with experimental analysis visualisation methods and techniques on heritage representation, like sensorial perception, tracing narratives, investigating historical sources, mapping space in various ways, experimental photography, etc. The results of the design workshop will be presented to local stakeholders. By offering this course, the section of LA wants to strengthen the interaction with stakeholders and the public and work with students from different disciplines, and meet professional colleagues, teachers and researchers.	
Study Goals	By the end of this course, you should be able to: - identify, group and value the main aspects of the identity of a landscape project; - debate methods and tool on heritage transformation and place making; - use methods and tools to present visionary transformations for a larger audience (participation); - cooperate in a multidisciplinary setting.	
Education Method	lectures literature study three-to-five-day design workshop in the Netherlands	
Literature and Study Materials	Hermans, R., Kolen, J., Renes, H. (2015) Landscape Biographies. Geographical, Historical and Archaeological Perspectives on the Production and Transmission of Landscapes. Amsterdam University Press. Janssen, J (2014). Modernising Dutch heritage conservation: current progress and ongoing challenges for heritage-based planning and management; tijdschrift voor Economische en Sociale Geografie 2014, Vol. 105, No. 5, pp. 622-629.	
Assessment	Oral presentation written description of the project in the form of a booklet, exhibition, model a.o.	
Remarks	The maximum grading period is 15 work days.	
Period of Education	Quarter 4	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0194	Bucky Lab A	15
Course Coordinator	Dr.ing. M. Bilow	
Responsible for assignments	Dr.ing. M. Bilow	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The focus of the course is the development of an innovative building product. This will vary from a small building, part of a building, pavillion, component or a facade. Within the course you will learn how to imagine, design, develop and finally build an architectural related concept that shows its potential in function and aesthetical qualities. Hosted by the chair of Building Product Innovation you will learn how to design a building product that will solve a predefined or self defined problem. The first weeks, students individually research and analyse the given topic of the assignment in order to come up with an innovative concept. The remaining weeks of the semester are dedicated to a design by research process in which all the main aspects of the design, from applied mechanics, material properties to production techniques are researched ending in an integrated final design. Experimentation, sketching, small and big scale model making is part of this process. Finally we will build the prototype in our mobile Bucky Lab workshop. This course is a shorter version of the famous Bucky Lab which is the best course of the Building Technology Master Track, so expect the same fun but in a smaller package! We focus more on the design and construction and will reduce the building physics and structural engineering part for this version. Have a look on www.buckylab.nl for more informations or search for it online to see pictures and videos of the last 12 years. We really build in our mobile workshop - so every student has to wear safety shoes (S2)	
Study Goals	The student is able to design a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. Specified for this course: the student - has an understanding of the relation between design, society, realisation, materialisation and functioning. - is able to design and evaluate building components based on their function and performance.	
Education Method	Problem-based learning (PBL) is a student-centered approach in which students learn about a subject by working in groups to solve an open-ended problem. This problem is what drives the motivation and the learning. We love to experiment and thrive on the curiosity of the students.	
Assessment	Individual report of innovative concept and reports in team of three students of design by research process from concept to final design, main focus the level of integration of all the researched aspects. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday and Thursday - details may vary	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0195	Urbanism Elective Studio	10
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 10 erts experimental elective studio is about future tasks for urban planners and designers. Students work on a group project together with individual sub projects. Since topics may vary, this course is capable of always choosing current spatial subjects. Sometimes this course may be in close cohesion with the AR0196 course, sometimes these two courses have completely separate topics.	
Course Contents	The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism. Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed.	
Study Goals	At the end of this course, the student: - is competent in urban design and urban planning; - displays a scientific approach; - is competent in collaborating; - is competent in communicating (oral presentation + report including visualisations).	
Education Method	Studio sessions, lectures, workshops and instructions. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The studio sessions are scheduled on 2 different day parts per week, most likely Tuesday and Friday mornings or Tuesday and Friday afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0196	Urbanism Elective Seminar	5
Course Coordinator	Ir. K.P.M. Aalbers	
Responsible for assignments	Ir. K.P.M. Aalbers	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Summary	This 5 erts experimental elective course offers a lecture / seminar / workshop setting in which future urban tasks will be addressed. Since topics may vary, this course is capable of always choosing current spatial subjects.	
Course Contents	Sometimes this course may be in close cohesion with the AR0195 course, sometimes these two courses have completely separate topics. The courses AR0195 and AR0196 provide space during the fourth elective quarter of the master track of Urbanism for various and changing initiatives each year; AR0195 offers a 10 erts studio setting, while AR0196 offers a 5 erts lecture / seminar / workshop setting. In these two electives new didactical ideas and / or substantive ideas for the Urbanism curriculum may be tested, and - after a success - may be implemented in the obligatory programme of the master track Urbanism.	
Study Goals	Thus the content and educational methods may vary. The AR0195 is a studio set up in which design and designerly thinking related to future urban tasks will be key; The AR0196 is a lecture, seminar or workshop in which future urban tasks will be addressed. At the end of this course, the student: * is competent in conducting design and planning research; * displays a scientific approach; * is competent in collaborating; * is competent in communicating (oral presentation + report including visualisations).	
Education Method	Lectures, workshops and / or seminars. Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Oral examination plus design examination. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Students should achieve a minimum grade of 6,0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student within 5 working days. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on 1 day part per week, most likely Monday or Thursday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 50.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0202	Computational Intelligence for Integrated Design	5
Course Coordinator	M. Turrin	
Course Coordinator	Prof.dr.ir. I.S. Sariyildiz	
Instructor	M. Turrin	
Instructor	Dr. C. Andriotis	
Responsible for assignments	M. Turrin	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Computational Intelligence encompasses theory and application of computational methods, techniques and tools that have the ability to learn based on given datasets, models and tasks. It includes AI comprising machine learning, bringing together concepts from probability and statistics to programming and optimisation. It is increasingly applied in the building sector, both to help understand the current status of built environment and to make informed (design) decisions based on predicted future responses. It mines data and translates them into actionable information. It harnesses and helps understanding information to turn it into applicable knowledge. This course will focus especially on the potential of Computational Intelligence for Integral Design in architecture and engineering, intended as a process of integration across disciplines. In this course you will learn about the current state-of-the-art of Computational Intelligence applied to architectural design and engineering, and about the theory and fundamental knowledge required to understand how to critically use (and eventually develop) your own Computational Intelligence tools. Topics of optimisation, probabilistic analysis, and machine learning will be covered, from distribution fitting and sampling, to regression, neural networks, and evolutionary algorithms, among others. You will also experience a design process where you will apply such techniques to a small-scale project, developing your design process with Computational Intelligence methods and tools.	
Study Goals	After the completion of this course you will be able to: Critically understand the current state-of-the-art, the potential and limits of Computational Intelligence for architectural and engineering design; Understand the theory of and apply basic Computational Intelligence methods, techniques and tools; Create a concept design by applying Computational Intelligence methods, techniques and tools, especially towards multi-disciplinary integration.	
Education Method	The students will be acquainted with and understand the state-of-the-art through lectures and self-study. Theory and basic application of methods, techniques and tools will be introduced through lectures, practical workshops and self-study. Application in design processes will be experienced based on self-study, working sessions (with other students), consults with tutors, making presentations and receiving/integrating feedback. During the course students work in part individually and in part in small groups.	
Literature and Study Materials	Specific literature is provided at the start of the course in Brightspace. The literature below provides an indication on relevant general content. Wortmann, T., 2018. Efficient, Visual, and Interactive Architectural Design Optimization with Model-based Methods Wortmann, T., Cichocka, J. and Waibel, C., 2022. Simulation-based Optimization in Architecture and Building Engineering - Results from an International User Survey in Practice and Research. Energy and Buildings, p.111863. Ekici, B., Turkcan, O.F., Turrin, M., Sariyildiz, I.S. and Tasgetiren, M.F., 2022. Optimising High-Rise Buildings for Self-Sufficiency in Energy Consumption and Food Production Using Artificial Intelligence: Case of Europoint Complex in Rotterdam. Energies, 15(2), p.660. Pan, W., Sun, Y., Turrin, M., Louter, C. and Sariyildiz, S., 2020. Design exploration of quantitative performance and geometry typology for indoor arena based on self-organizing map and multi-layered perceptron neural network. Automation in Construction, 114, p.103163. Andriotis, C., 2019. Data driven decision making under uncertainty for intelligent life-cycle control of the built environment.	
Assessment	This course uses two types of assessment: writing assignments and design examination. Specifically, your work will be assessed by reviewing the following end products: A short essay on critical positioning and identified opportunities based on the state-of-the-art; A critical reflection on workshops content, process and outputs; A presentation and report on the process and results of the design-related project. Students should achieve a minimum grade of 6.0 to pass this course. In case a grade is 5.0 or 5.5, students will get the opportunity to repair the assignment within a period of 2 weeks from the result announcement, a successful repair will be assessed with a 6.0. (A retake can be done in the next year)	
Period of Education	Quarter	
Concept Schedule	Wednesday	
Leerstoel	Design Informatics	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0203	Eco-friendly Material Choices	5
Course Coordinator	D.P. Peck	
Course Coordinator	Dr.ir. F.A. Veer	
Responsible for assignments	Dr.ir. F.A. Veer	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	msc 1 building technology or equivalent	
Course Contents	basics of sustainable materials and eco based materials selection	
Study Goals	understand the problems of critical materials and be able to do a correct materials selection allowing for several scenarios	
Education Method	lectures and workshops	
Computer Use	laptop required with edupack software	
Literature and Study Materials	ashby materials and the environment 3rd edition	
Assessment	exam	
	One retake opportunity in the academic year for the examination.	
Permitted Materials during Tests	book and laptop	
Period of Education	3rd quarter	
Concept Schedule	Monday	
Maximum number of participants	80 80 80	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0215	Form & Inspiration	5
Course Coordinator	M.G. Vink	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	M.G. Vink	
Responsible for assignments	M.G. Vink	
Contact Hours / Week x/x/x/x	4 uur per week, week 3.1 t/m 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	Master 1	
Summary	The assignment is to develop a personal interpretation of a specific architectural aspect and to subsequently work this out to a thoroughly designed, articulately detailed and documented spatial object.	
Course Contents	The Form and Inspiration course is a 10 week MSc2 elective (5ECTS) consisting of a personal journey from inspiration to form. Students will thoroughly explore a chosen Inspiration in order to reinterpret and represent a classical architectural element (e.g. window, ceiling) which we call Form. Guided by your personal inspiration, you will actively explore the concepts documentation, expression, evocation, translation, imagination and innovation on the scale of an architectural element. A large scale model (1:10 - 1:20) forms the final product of a research process concluded in a visual artwork. The course is a process of attentive observation and physical making with the aim to arrive at a new interpretation and expression of a classical architectural element through artistic exploration, inspired by a personal fascination.	
Study Goals	Upon completion of the course the student is able to: LO1 - [analyse] a personal inspiration through an artistic process of ; a. observation (multisensory: seeing, hearing, feeling, smelling) b. documentation (collecting and representing) LO2 [apply] choose and experiment with a relevant artistic technique to document and represent observations. LO3 [create] produce an artwork that expresses/communicates the concluding, found quality of an inspiration LO4 - [create] design a single architectural element based on the study of a personal inspiration. LO5 [create] Represent (express) a designed architectural element ceiling / window in an accurate and evocative 1:10 physical scale model.	
Education Method	Tutoring talk with sketches Making is central and sketches are expressive thought products. Each week requires the production of experiments which we call sketches. You will explain what you have produced and what you learned from it and we will ask questions in short conversations. The course is an individual project with a personal process. We invite and stimulate you to share what moves you in this course. Workshops - short assignments in studio Lectures - for context (sub) Group - conversations	
Assessment	Deliverables: Sketches - Artwork - Sketches - Physical Scale Model Criteria: SKETCHES INSPIRATION Selection of observational sketches show relevant approach of the inspiration Sketches show effective and explorative study with chosen technique ARTWORK Artwork expresses relevant concluding quality, distilled from the (LO1,2) analysis in an evocative and precise way DESIGN SKETCHES Sketches show effective study with chosen technique Sketches show coherent, imaginative, innovative translation from inspiration to arch. element Sketches show functionality and meaning of architectural element MODEL Model is a relevant and correct representation of the design Model has relevant and evocative material choices Model shows technical 1:10 design choices Model has a proper composition and detail level appropriate to scale 1:10 Knock out criteria: presentation presence, complete set of deliverables, process booklet hand-in Grades are communicated within 10 working days after the presentation. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	for more information you can contact the responsible instructor or course coordinator.	
Period of Education	2e semester, 10 weeks in quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various materials for drawing, painting, collaging, photography and modelling.	
Leerstoel	Formstudies	
Minimum number of participants	12	

Maximum number of participants	24
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0216	Towards an inclusive living environment	15
Course Coordinator	Dr.ir. B.M. Jurgenhake	
Responsible for assignments	Dr.ir. B.M. Jurgenhake	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Students of the Elective Studio Towards an inclusive Living Environment design a residential, residential + mixed function or alternative project in an urban environment. The main question of this elective is to what extent can architecture provide an inclusive and healthy living environment for all. The design is accompanied by a short phase of human-centered research as start of the elective (visual anthropology with observation, participation and/or interviews) Design work is done individually or in groups of two students, the research may be performed in teams of max. three students. Each semester the design assignment may be different from the one before. It includes projects for special groups of our society (more vulnerable people like the elderly, children, people who need care...) or it focuses more on the topic of a health promotion or a specific health care environment. The design may end up in a series of small scale interventions, a design of a transformation of one or more existing buildings or the design of new building(s). Though topics may vary from one semester to the next, at the core of each studio lies the question: what does an Inclusive and Healthy Living Environment mean for the architecture? What is needed in the neighborhood/or in one specific building complex (chosen by the tutor)? We will explore the question by looking at one chosen neighborhood (or one specific building complex) as a multi-domain structure and by working on different scales. We will discuss new ideas for an inclusive living environment. Each semester we try to closely work together with the target group themselves, municipalities and/or housing associations. Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Study Goals	Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - Studio specific study goal 1: The student is able to combine different (interdisciplinary) research methods and to translate and discuss research outcomes into design. - Studio specific study goal 2: The student is able to understand the potential multiple user groups and their demands In addition to the specific focus of each design studio (track), upon completion of the design studio the student is able to: - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal. - demonstrate sufficient insight in and knowledge of the design process - position the project within a particular theoretical, historical, social or contextual framework.	
Education Method	Workshop day(s) incl. an excursion to the site and lectures as a start of the Msc2. Getting acquainted with the method of the studio; research fieldwork on location; in-depth research on location - preferably combined with a stay at the location (one or several days). Weekly tutoring of the research and the design in the design studio; possibly additional tutorial days with specialists, research presentation, midterm presentation and end presentation with visiting critics	
Course Relations	The studio is emphatically looking for a cross-over between architecture and other fields of expertise. This may be expertise in the specific target group; urban- and landscape planning; taking a look into the possibilities for a financial realization of the project. Further explanation can be found in the flyers or on our website.	
Reader	A course booklet will be sent to the students before the course starts.	
Assessment	A Research Report: a written document made by the whole group about the human centered fieldwork, done in the neighborhood. Students deliver a Draft version after 4 weeks and will get feedback to be able to finalize the product. The assessment will be supplemented with an oral presentation to explain the product directly after the fieldwork phase of the first weeks. The report has to be delivered halfway the course. A1 poster Drawings: Students make A1 posters of their design. One day before the end-presentation they have to be delivered. The end-presentation will be held in week 4.10. Process Presentations will be held throughout the semester; Exact requirements to be announced at the start of the studio. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Exam Hours	The end presentation will be held in week 4.10.	
Special Information	As we define the design assignment each year, depending on the recent topics in society, or new questions coming out of the field of Health and Care, please read our flyer and come to our information meeting.	
Period of Education	4th kwarter	
Concept Schedule	We will meet weekly on Tuesday morning at the faculty. Next to that we will have second meetings, or at the location, or online, or at the faculty. These second meetings will be announced at the beginning of the course.	
Minimum number of participants	12	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0226	MSc2 Studio Elective Urban and Infrastructure (Re)development Game	15
Course Coordinator	Dr. A. Ersoy	
Instructor	Prof.dr. P.J. Boelhouwer	
Instructor	Prof.dr. E.M. van Bueren	
Instructor	mr. F.A.M. Hobma	
Instructor	Dr. E. Louw	
Instructor	Dr.ir. M. Spaans	
Instructor	Dr.ir. S.C. van der Spek	
Instructor	Ir. H.W. de Wolff	
Instructor	Y. Chen	
Instructor	Dr.ir. S. Zijlstra	
Instructor	Dr.ir. T.A. Daamen	
Instructor	Dr.ing. G.A. van Bortel	
Instructor	Dr.ir. E.W.T.M. Heurkens	
Instructor	Dr. W.J. Verheul	
Instructor	V. Muñoz Sanz	
Instructor	K.B.J. Van den Berghe	
Instructor	Dr. H. Hou	
Instructor	Ir. E.H.M. Geurts	
Responsible for assignments	Dr. A. Ersoy	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The study focuses on skills of integration and analysis based on the knowledge acquired in the first semester. Students will play roles in project simulation and learn how to assess, analyse, research and improve operation practice in this professional field. The course aims to train students to grasp an integral approach when managing urban (re)development both at the urban area scale and at the portfolio and object scale. Through a role-playing simulation project, students will be given design assignments that drive them to (re)develop a complex urban location with both residential and non-residential elements.	
Study Goals	Understanding the changing context of the global and local environment and economic, social and cultural elements that contribute to various urban problems; understanding the context, content, players and means of implementation during the cyclic phases of urban area development; evaluating positions, objectives and means as well as strategies of involved parties in different phases; analysing the social-economical and urban context as well as the status and function the area can possibly achieve in the future; setting up functional programmes for the area in question; analysing spatial possibilities and the feasibility and financial consequences of investments; developing institutional and financial plans for different phases in order to manage and oversee the development design and implementation process, thereby effectively coordinating the input of the various actors in the project; conducting feasibility studies of the real estate portfolio strategy with involved and/or potential stakeholders and the cost-benefit analysis; working in multidisciplinary teams, negotiate and communicate with different parties, present project results and reflect the development process with an analytical report.	
Education Method	Flip learning; classroom exercises; online reading; group work	
Assessment	Essay writing; and final report	
Period of Education	Q4	
Concept Schedule	Tuesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0227	Infrastructure and Environment Design	10
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	7 hours per week starting from week 1 and ending in week 9.	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	With urgent urban challenges such as climate adaptation, energy transition, and continued urbanisation, the urgency of integrating planning and design with urban engineering increases. The implementation of new technological interventions and the utilisation of the natural system is hampered by the lack of an integrated approach incorporating urban planning and design decisions. Meanwhile, urban and economic growth increasingly competes for infrastructure and environment, affecting the success or failure of the daily operating systems of cities and thereby urban competitiveness. The challenge is to fundamentally re-think the urban landscape in light of new technologies. The question is how to renew existing cities by integrating the parameters of the natural system, as well as technological innovations directly into urban development opportunities arising from spatial planning and design. In order to stimulate and design the synergy between design and engineering this course offers the possibility for architects, urban designers and landscape architects to get well acquainted with the concepts and language of civil engineers on the subject of infrastructure and environment; at the same time the civil engineers will get acquainted with the world and language of designers. In order to create an emerging path where synergy between the disciplines makes sure that technology becomes embedded in the design process, this course offers possibilities for both urban designers and civil engineers to get well acquainted with each others discipline.	
Study Goals	Students will be able to: - Formulate their design perspective that is based in a conceptual or theoretical framework. - Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. - Analyse and design infrastructures on a regional scale and on the scale of the section. - Identify and discuss the tension between public and private development in infrastructures and environments. - Apply methods concerning the appraisal of sustainable urban environments and infrastructure. - Demonstrate in a design the connection between the natural system and technical possibilities in urban environments. - Be able to translate analyses into design and the design into a formal plan. - Perform inter-disciplinary working.	
Education Method	Lectures, self study, workshops and working groups. Combination of individual and group work. Readings in the field of knowledge brokerage, technical entrepreneurs, landscape ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Exercises in building a theoretical or conceptual framework and translating analyses into design. Interdisciplinary learning by taking class with civil engineers and policy students in which understanding can be created for each others knowledge and skills, where fences between the knowledge fields can be broken down, where contacts can be made for later in professional careers. The Urban Water Management course starts in Q3 with 8 lectures of which the compulsory ones are indicated in the schedule, the others can be viewed on colleggerama. In Q 4 there is an assignment, excursion and workshop with the urban water management students. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group project as elaboration of the workshops.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	The course results in group analysis and an individual project. The content of individual project is: 1) Use of theory to frame your research and design perspective. 2) Research and analyses of technical data/infrastructure of your site resulting in an environmental and infrastructure potential map. 3) Research and analyses of the surface of your site, resulting in a surface potential map. 4) Synthesis between 2 and 3 and together with 1 resulting in a (spatial) concept. 5) Concept translated in a performance based urban design that will be translated into a formal plan. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. There are a set of grades in which there is differentiated between assessment of 1) the individual contribution to the group results/products and 2) the quality of the groupwork itself. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before the end of Q2 (grading is done end of week 6). The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Elective	Yes	
Tags	Analysis Design Group work Research Methods Sustainability Transport & Logistics Underground Water management	
Period of Education	Quarter 4.	
Concept Schedule	The sessions are scheduled on Tuesday and Thursday mornings. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any (elective) course of the Master track of Urbanism the minimum number of participants is 15.	

Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 25.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR0228	Infrastructure and Environment Method Module	5
Course Coordinator	T. Kuzniecowa Bacchin	
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. F.L. Hooimeijer	
Responsible for assignments	Dr. F.L. Hooimeijer	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Course Contents	In the track Urbanism students learn to integrate social, cultural, economic and political perspectives with the natural and man-made conditions of an urban landscape in order to shape and plan for more sustainable development. This course is the theory and methodological support of the design course Infrastructure & Environment Design (AR0027) that focusses on the making or remaking of the natural and man-made conditions of the urban landscape. This course conceptualises and theorises the act of integration and design. It provides the necessary systemic knowledge.	
Study Goals	The course programme consists of lectures, workshops and individual mentoring. Students will be able to: Formulate a conceptual or theoretical framework. Identify and discuss the synergy between natural conditions and technological potential and possibilities in urban environments. Identify and discuss the tension between public and private development in infrastructures and environments. Perform inter-disciplinary working.	
Education Method	Self study, workshops and working groups: Readings in the field of knowledge brokerage, technical entrepreneurs, ecology, sustainability and urban theory for a better understanding and theoretical framing of the individual project. Workshops with professionals and with students of technical background to understand differences in language and concepts and learn to apply the technical information to the spatial context. Individual or group elaboration of the workshop results Combination of individual and group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page.	
Assessment	Literature review and paper. A rubric will be used for grading. The rubric will be available on the course specific Bright Space page. Each student studies the role of his/her place of internship with the focus on the role of design in the development process or the process in the design activities and will formulate their specific research question. Through theoretical, empirical and action research the students will deliver a scientific paper as result of the course. The assessment of the course is done on the basis of two results. The literature review and paper. The assessment is done on the basis of a rubric counting the following aspects: Layout and detailing of the paper Research methods and reporting skills (research questions, use of sources, clear conclusion) Academic attitude: clear argument, adequate sources, critical, transparent, references accurate and appropriate Research goal and relevance made clear Research question that is adding to a general body of knowledge e.g. is interesting for a larger community Research structure is clear, logic and explained Theory is consciously used to frame the work, explained and logical Reflection into the topic of research in relation to the place of work Late delivery Other reasons to add/deduct Published in Journal This results in a grade, when the student continues with putting their paper into a scientific journal review process and succeed publishing their grade will be improved in a 10. Students should achieve a minimum grade of 6.0 to pass this course. Grades 5,0 5,5 are eligible for Repair. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student during the summer quarter. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the original grade between 5,0 and 5,5 is registered in Osiris. In case of a final grade below 6,0 the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 work days.	
Period of Education	Quarter 2 and quarter 4	
Concept Schedule	The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For any elective course of the Master track of Urbanism the minimum number of participants is 15.	
Maximum number of participants	For this course of the Master track of Urbanism there is no maximum number of participants.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0570	Worlding Theories and Concepts	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. R.A. Gorny	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	6 hours per week, starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	A general interest in theory and philosophy is desirable, but not a prerequisite for participating in the course.	
Course Contents	Worlding and worlding practices, in their inherent pluriversality, call for a general reinterpretation of built and lived environments and related cultural and spatial practices, architecture included. With this in mind, this course focuses on architecture as a material-discursive practice and a technique of environmental design that engages in the purposeful transformation of the world. It familiarizes students with a set of key theoretical concepts from recent discourses in architecture and urban/environmental design, as well as attendant (environmental, social, cultural, philosophical, technological, economic) discursive streams to understand and engage with worlding dynamics. The general aim of the course is to establish a conceptual basis around both more-canonical and newly-emerging concepts and theories, and foster an intellectual sensibility for their critical and creative (analytical/methodic) use within architectural research and design. To demonstrate their understanding of different theories and concepts, students will write a series of short entries to a collective Glossary.	
Study Goals	Upon completion of this course, students - have gained an overview and understanding of fundamental concepts used within recent architecture-theoretical (and attendant) discourses; - can articulate both general relationships and differences between these notions and frameworks, and position more specific understandings of singular notions discussed; - can appropriately apply these notions in a nuanced and layered ways to current research and design questions; - will have developed academic writing and research skills	
Education Method	The course is taught in 8 interactive seminar-style lectures and student-led discussion based on preparatory readings.	
Course Relations	AR2AT031 (Architecture Theory Thesis required choice course) recommended MSc2-Q3 AR2AT041 (Architecture and Philosophy elective course) recommended MSc2-Q3 AR2AT021 (Architectural Techniques design studio) recommended MSc2 -Q4	
Literature and Study Materials	Course reader and reading material, source and reference lists.	
Reader	A course reader will be made available two weeks in advance of w.3.1	
Assessment	The course is assessed through active participation and, as detailed in the course syllabus, a series of short, academic reflections (1500 words in total) that form the main grading component. The final submission is due in week 3.10; the marking period is 10 working days after the submission date.	
Enrolment / Application	Enrolment during Faculty periods. No special enrolment for this course.	
Period of Education	The course is taught in Q3 only.	
Concept Schedule	Weeks 3.1-3.8 on Tuesday mornings OR Friday mornings: Lecture-seminar sessions 18 and individual tutorials/feedback sessions. Week 3.9: self-study Week 3.10: Due date assignment (check your Timetable for final schedule)	
Leerstoel	Architecture Philosophy and Theory	
Minimum number of participants	12	
Maximum number of participants	75	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0625	Climate Change as a Game: (Co)designing with Children the Landscape of the Future	15
Course Coordinator	Dr. L. Cipriani	
Responsible for assignments	Dr. L. Cipriani	
Contact Hours / Week x/x/x/x	0/0/0/4	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have completed a MSc1 design course (in Landscape Architecture, Architecture or Urban design).	
Course Contents	Today's new generations, particularly young people, will face consistent landscape changes in their lifetime. Can we create consciousness in young generations of how climate change will modify landscapes and cities? Can we involve children and university students in co-designing the landscape considering climate change? Starting from exploring different forms of outdoor games (role play, art installations, etc.), this course aims to educate and engage younger generations in climate change issues. The game becomes the tool through which students can acquire knowledge, observe the world from different perspectives and ultimately imagine and transform the future world. Games are designed experiences where players can learn through doing and being rather than absorbing information in traditional educational formats. By assuming various roles and perspectives, the educational experience of play triggers emotions that help to acquire new awareness, develop a more complex vision of the future, and finally make decisions. In the course, students from landscape architecture, architecture, and urban disciplines will explore an assigned landscape, adopting the concepts of play, design, and climate change, building a collective outdoor game for children on-site. In the first phase, students will investigate the assigned site and collect best practices on climate landscape games as "designed experiences." In the second phase, students will meet children on-site to understand their perception of climate issues and explore with them possible games. Then, students will develop games for the site context. The best game(s)/installation(s) projects will be collectively selected to be co-constructed. An education expert (Anita van Oosten) will also help select the game to be feasible for children. In the third phase, students will co-construct the selected games in the study area. Finally, students will invite the involved children to co-designing/co-playing/co-testing the landscape games on climate change. The project will also be disseminated involving multiple schools at the national level in a one-day final event. Some funding for running the course (trips and game construction) will be available. Contacts with the children and primary school are already established. The Comenius program sponsors the studio.	
Study Goals	By the end of this course, you should be able to: - tackle climate transition as a game, developing playful, emotional, artistic, and outdoor learning methodologies and co-design methods with children; - to research a specific site and collect best practices on climate change as a game in the landscape; - elaborate, build, and present a coherent, artistic, playful, and innovative design proposal for a specific landscape site and a child audience; - conduct and communicate design research throughout the scales by using photos, videos, physical models, hand and digital drawings, and landscape construction throughout the design process; - co-design, co-play, and co-test the landscape games on climate change with children and the class.	
Education Method	Lectures, design studio, fieldwork, on-site work. Students will work in large groups and will coordinate collectively throughout the process. Reviews will be on an individual, group, and class basis.	
Literature and Study Materials	Materials will be provided during the course. Invited guests will present their design works to inspire students.	
Assessment	The assessment will be based on design examination and oral examination. A final booklet will present each group's collective research, design and on-site work. It will include text, photos, drawings, models, videos, and on-site construction of the whole process (including on-site events). Group members will present the final work at the studio's end. Students will receive individual grades according to their performance during the studio. Instructors will monitor the individual progress within the group work. During the semester, several intermediate reviews will be scheduled. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Site visits and outdoor excursions in the selected site will be part of the design course. Two co-design sessions with children are foreseen: one on-site and one at the selected school. Construction phases and co-test sessions with children will also be on-site.	
Period of Education	Quarter 4 - 10 weeks starting in week 4.1.	
Concept Schedule	Wednesday morning, Friday morning	
Minimum number of participants	-	
Maximum number of participants	25 students	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0682	Heritage and Architecture Design Studio: Research and Architectural Design	15
Course Coordinator	Ir. W.L.E.C. Meijers	
Course Coordinator	Ir. W. de Jonge	
Instructor	Ir. A.C. de Ridder	
Instructor	Ir. W. Willers	
Instructor	Ir. A.W. Hermkens	
Instructor	Ir. W.L.E.C. Meijers	
Responsible for assignments	Ir. W.L.E.C. Meijers	
Contact Hours / Week x/x/x/x	12	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Heritage & Architecture defines research and design on all levels of scale: the use of materials and technology, the reuse and redesign of a building or a building complex, and the development of landscape and urban structure. For all scale levels the value of the entire context is taken into account, which is of vital importance to address the challenges and face the responsibilities of working on existing built structures. Particular attention is paid to values regarding architecture, urbanism, construction and interior, related to architectural history and current questions on sustainability. Re-designing and researching buildings of significance in cultural-historical context is the main concern of Heritage & Architecture. In this course, the research of existing built structures leads to conclusions that provide the position and interpretation for the transformation or conservation in the architectural design. The developing discussion in this studio by Learning from others, by theory and reference material is guiding for the architecture of the re-design. In small groups students research related questions to the proposed subjects for the transformation design. Students individually create a re-design that shows a meaningful translation of an intervention strategy into the spatial, functional, contextual, material and architectural-technical design. The design choices are based in an understanding in relation to cultural value.	
Study Goals	Upon completion of the design studio the student is able; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal for a cultural-historical context. - analyse relevant precedents concerning their societal context, cultural historical, technical and material aspects and aspects of use. - to understand the focus on moral sensibility, analysis, creativity and judgement skills regarding architectural ethics - to position the project within a particular theoretical, historical, social or contextual framework. to present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project. The coaching is during educational weeks. At the beginning of the course, there are specific thematic exercises and lectures that are related to and provide information on the studio subject. The design project includes group work for the research of the studio subjects next to the development the individual design based on a design strategy. The result is based on the studio research and represented in an elaborated design with an argued position in the field of Heritage and Architecture. It will be presented in the last week of the course.	
Literature and Study Materials	To be announced upon the beginning of the course and on Brightspace. It is recommended that students have studied; Kuipers and de Jonge (2017) Designing from Heritage https://books.bk.tudelft.nl/press/catalog/book/isbn.9789461868022	
Assessment	Presentations will be held during the quarter. A final presentation is at the end of the quarter. Products of drawings, texts, models and a project journal documenting the design process are presented in a visual and oral presentation. The project will be assessed on the basis of the following aspects: the position that is formulated with regard to the brief and its wider context appropriateness of the design with respect to the assignment and its translatability into a physical manifestation the coherence, elaboration and integration of the final design the quality of the presentation (visual and oral) the consistency, coherence and development of the students work during the design process The student has achieved a sufficient result on scale 1 to 10 with 6, the student has the possibility to take a repair with a mark of 5 or 5,5 and the student failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	The project takes place in the second quarter of the Spring semester.	
Concept Schedule	Wednesday during the whole quarter. In the first 3 weeks are additional studio days depending on the organisation of building visits, lectures and excursions.	
Leerstoel	Heritage & Design	
Minimum number of participants	12	
Maximum number of participants	60	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0771	Beyond 3D Computer Visualisation	6
Course Coordinator	Ir. J.J.J.G. Hoogenboom	
Instructor	P. de Ruiter	
Instructor	Ir. J.J.J.G. Hoogenboom	
Responsible for assignments	P. de Ruiter	
Contact Hours / Week x/x/x/x	46 hours per quarter	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	No prior knowledge required.	
Summary	The students create within 10 weeks a visualization of a quote derived from a book, speech or song. The visualization can be a single high resolution poster or a VR environment. They use advanced software like Maya, Mudbox, Substance Painter and the Unreal game engine in combination with the HTC Vive for the VR. In weekly sessions specific themes will be covered and the corresponding techniques will be practised.	
Course Contents	This course is called Beyond 3D. It means that the content of the course goes beyond the traditional 3D visualization and enters the realm of advanced modelling, texturing and rendering which can be found in the film and gaming industry. The topic is the visualization of a quote. The quote can be chosen from a book, a speech or from a song. The complexity of the task is to translate the essence of the quote into an image or a virtual environment communicating this essence, a process which also can be found in the visualization of an architectural design idea. The result can vary from medieval castles attacked by dragons to cityscapes floating through space and everything in between and beyond. Students who have successfully completed this course are adept at independently implementing computer applications for the effective visualization of any idea or concept.	
Study Goals	The student can: - translate a quote into a 3D representation and create a high-quality visualization, - demonstrate the effective implementation of 3D computer visualization using high-end animation software, - create complex geometric models in a 3D environment, - set up an efficient workflow and data-exchange, - explain the difference between material shading models and apply these models to his/her project.	
Education Method	Contact time: 42 hours - 7 weekly 4 hour workshops - 7 lectures of 2 hours Individual study: 94 hours	
Computer Use	Own laptop with a dedicated graphics card is mandatory.	
Literature and Study Materials	Online literature, TOI-Pedia: http://wiki.bk.tudelft.nl/toi-pedia/AR0771	
Assessment	The assessment is based on: - A poster as digital file or the virtual environment as an Unreal project. - All the related project files. - A portfolio (breakdown) of the project. The assessment takes into account the quality of the above mentioned deliverables, the process and the used techniques.	
Special Information	The maximum marking period is 10 work days.	
Remarks	This course is especially designed for students who want to expand their knowledge beyond what is needed for a traditional architectural visualization.	
Period of Education	Quarter	
Concept Schedule	Monday morning lectures, workshops on Wednesday and/or Thursday.	
Leerstoel	Design Informatics	
Minimum number of participants	10	
Maximum number of participants	45	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0796	Ornamatics	5
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	P.A. Koorstra	
Co-responsible for assignments	G. Coumans	
Contact Hours / Week x/x/x/x	4	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Summary	The Ornamatics course explores the potentials for new ways of studying, evolving and realising architectural ornaments in contemporary architectural design, using computer-aided modelling and manufacturing techniques.	
Course Contents	In recent years, ornamentation has increasingly (once again) become a compositional issue in architectural design. At the same time, computer-aided modelling protocols - in combination with new production technologies - have contributed to wholly new ways of shaping building elements. Some examples of new techniques, which have recently become very successful in building production and in architectural design education, are: 2D Laser cutting; 3D Rapid Prototyping; 2,5D and 3D Milling. Such new approaches not only create new opportunities for traditional production processes (including physical modelling), they also offer new perspectives for design and manufacturing on the level of architectural components and connections. The course combines a focus on the opportunities for new forms of ornamentation, with the active utilisation of computer aided-modelling and manufacturing techniques. The course couples the analysis of historical and contemporary aesthetic paradigms with the opportunities and evolvments of a variety of 3D digital platforms. It tries to stimulate the discussion about the role and meaning of ornament and decoration in the present, by setting the design in relation to an existing project. The issue of Ornamentation involves study on the level of historical architecture styles and production techniques, finding / analysing / categorizing of typical examples, development of a critical view on aesthetics related to building components.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design proposal for an ornament. - apply knowledge and understanding in the fields of composition, materialisation and detailing as well as the attainment of skills in the fields of (computer-aided) manufacturing and representation. - approach a design problem from a cultural and intellectual point of view and give a 400 words reflection on this.	
Education Method	design studio format and lectures	
Assessment	Assessment on the basis of process, end-result, documentation, analysis and presentation. The maximum marking period is 10 work days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	Coordinator	
Period of Education	2e semester, 10 weeks Quarter 3	
Concept Schedule	Wednesday afternoon	
Used Materials	Various modelling approaches physical as well as digital are utilised in the context of the Ornamatics course. Active use is made of the facilities of the facultys CAM-lab.	
Leerstoel	Form studies	
Minimum number of participants	12	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0805	An Archeology of Digital Design	5
Course Coordinator	Dr. T.G. Vrachliotis	
Course Coordinator	M. Mateljan	
Course Coordinator	D. Pohl	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The seminar explores the impact of the digital paradigm of the 1970s on architecture and the built environment. Students will gain a comprehensive understanding of how contemporary questions around the digital relate to their recent past. Through archival research, oral history interviews, software analysis, and image historiography, students will investigate the innovative history of digital design, from the late 1950s to today's AI Labs. The main assignment will be to design an exhibition about the digital, based on a time-line research, and oral history interviews. Students will be challenged to develop an original hypothesis that allows for a new storytelling of the digital in architecture that goes beyond parametric design and takes cultural, economic, and social changes into account. The research seminar promotes both collective group work and independent thinking, and encourages students to address two major challenges. In the first part of the semester, students will perform archival and library research in the collection of the Het Nieuwe Institute to understand how the digital entered architectural education, research and practice. This will include exploring institutional histories, curricula, research projects, and new computing machines. In the second part of the semester, students will develop a conceptual timeline of the digital, reflecting the findings of the research phase. The research will be shared with the public, by developing and curating an exhibition at BK.	
Study Goals	- to understand social, economical and technological relations of digital tools and methods - to showcase how contemporary questions around the digital relate to their recent past - to demonstrate sufficient insight in and knowledge of the digital in design - to perform archival and library research in the collection of the Het Nieuwe Institute and the TU Delft library - to record, edit, analyse and publish oral history interviews with key actors - to contextualise key actors, media, and technologies geographically and historically in a collective time-line - to demonstrate argumentation and graphic skills aiming to consolidate and strongly communicate a historical narrative about the digital - to collaborate collectively to curate and implement an exhibition of visual media, that follows a conceptual organisational logic, and that meets objectives of the class collective vision	
Education Method	Research will be conducted in thematic groups, the contribution to the exhibition is either individual or in groups of max 2 students. Findings will be presented and discussed in seminar sessions.	
Assessment	Students are assessed through Design examination and Oral examination, in a form of weekly pin-ups showing research progress, arguments and concepts, organised in specific formats, as well as on the basis of the final deliverables. The criteria for assessment will be communicated in the course Reader (syllabus). The midterm assessment will take place halfway through the course program (not graded), and the final assessment will be done at the end of the course program (graded). Final presentation consists of a collective research group booklet in A4 (each student with four pages), one individual narrative with a contribution to the exhibition (one object per student) and the timeline (one A0 per student). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Special Information	The locations of the consulting archives are mainly in Delft, Rotterdam or The Hague. Students might consider additional costs for printing, and travelling, which could be quantified around 50 euros per person, depending on location and possibilities. The seminar will take place in the weeks 3.1-3.10 on Friday morning.	
Period of Education	Quarter 3 (spring semester)	
Concept Schedule	Friday morning	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0815	Idiosyncratic Infrastructures II	5
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	Ir. F. Geerts	
Instructor	Ir. F. Geerts	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Idiosyncratic Infrastructures II The course intends to remedy a knowledge gap, by collecting, redrawing and categorising pieces of infrastructure. Analysing the specific circumstances conditioning these artefacts, investigating modes of representation specific to infrastructure, and focusing on the very "thingness" of the infrastructural artefact, will contribute to a catalogue of idiosyncratic infrastructures. Infrastructure and its component parts are dominated by standards, codes and conventions that are intended to enhance efficiency, safety and feasibility, cemented in a repository of proven knowledge that is above all normative. At the same time infrastructural objects are always grounded in complex pre-existing realities, produced by contradictory desires, and often influenced by conflicting agencies. The customised intersection of standards, codes and conventions with the specificities, resistances and opportunities of a real terrain has produced often clever, inventive, and imaginative solutions. These idiosyncratic solutions have however often remained off the radar, and do not prominently contribute to the body of knowledge of infrastructure design, mainly because of being too specific and exceptional to categorise. The course practises the inverse of integrated design striving to analytically unpack the multi-disciplinary synthesis of the highly -specialised architectural objects of infrastructure. This Borders&Territories elective takes existing infrastructure case-studies at the intersection of architecture, city and landscape, as the basis for a drawing and modelling experiment. Seminar-discussions on different representational conventions will feed the speculation towards a final exhibition/catalogue.	
Study Goals	At the end of the course a student: 1. has an advanced knowledge of key modes of representation of infrastructure in art, design, and engineering and can reflect on these in discussions, drawings and writings; 2. can reverse-engineer by means of drawing and modelling particular infrastructural artefacts; 3. is capable to interpret and reflect on non-standard cases from practice from a theoretic and design point of view; 4. can reflect on the historical and conceptual relationship between architecture and infrastructure.	
Education Method	1. Lectures within a seminar setting. 2. Seminar tutorials with student participation through class discussion and student presentations. 3. The drawing/modelling-assignment progresses week-by-week, guided by different weekly sub-themes.	
Assessment	Class participation Weekly progress End-term submission and collective exhibition The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Elective	Yes	
Period of Education	Semester2, Q3	
Concept Schedule	Tuesday afternoon	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0825	Building Stories: The Heteronomy of Urban Design	5
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	3 hours per week starting from week 3.1 and ending in week 3.10.	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	Urban design can be regarded as an autonomous discipline that develops according to its own internal logics. The reinvention of the grid as an urban planning tool, from Hippodamus in ancient Greece to Cerda in 19th century Barcelona and Bernard Tschumi in 20th century Paris, is a good example. However, this seminar will focus not on the autonomy of the discipline, but its heteronomy. We will investigate how urban design responds to external developments that take place in the social, economic, political and cultural realm. During the 20th century, architects and urban designers were faced with giving shape to both new and existing programmes while responding to a novel set of external challenges, such as growing democratization, emerging ecological concerns, increasing secularisation, the rise of neo-liberalism, etc.	
Study Goals	The goals of this course are three-fold. First, it aims to offer students a concise overview of the history of urban design in the 20th century, and demonstrate how the practice and profession of urban design are influenced by challenges that (strictly speaking) are external (or heteronomous) to the discipline. Second, it aims to familiarize students with the use of databases of architectural journals (such as the Avery Index and the RIBA Library), and also teach them how architectural journals can be used as a medium of research. Third, and finally, it aims to offer students insight into the subjective nature of historiography and familiarize them with alternative modes of architectural and urban design historiography, and of narrating history in general.	
Education Method	The course consists of two parts. The first part spans the four weeks and the second part spans five weeks. In the first part, students work in groups. Each week, they will receive a lecture by the course instructor(s). In addition, they will be given readings to present in class, and be asked to engage in research using databases of architectural journals and the collection of architectural journals held by the TUDelft library. In the second part of the course, students work individually. Emphasis in the second part is on project research (through archives, literature review, interviews, etc.) and analysis through narrative construction and graphic design. The final outcome of this course is a short story graphic novel.	
Assessment	There are 3 key assignments that need to be completed/submitted: (1) During the first part of the course the students will be given readings which they need to present orally and accompanied by slides, during the course. (2) At the end of week 4 (which is the end of the first part of this course), each group of students should have finalized an analytical assignment using the databases of architectural journals as well as the collection of journals held by the TUDelft library to identify 20 potential case-studies for further research (in the second part of the course). (3) At the end of the second (and final) part of the course, each student should submit a graphic novel short story, which is to be presented orally and publicly. All three assignments are graded. For assignments 1 and 2 a grade is given per group, whereas for assignment 3 an individual grade is given.	
Special Information	This course will take place on Tuesday mornings.	
Period of Education	Semester 2, quarter 3 This course will take place on Tuesday mornings.	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0835	Social Sustainability in Human Habitats	5
Course Coordinator	D.K. Czischke	
Responsible for assignments	D.K. Czischke	
Education Period	1 3	
Start Education	1 3	
Exam Period	none	
Course Language	English	
Course Contents	Human habitats refer to the environment in which human beings exist and interact. The term habitat (derived from the Latin habitare, to inhabit) comes from ecology, and includes many interrelated features, especially the immediate physical environment, the urban environment and the social environment. At the individual and family levels, habitat consists of peoples homes and the buildings and spaces where they go about their daily life. The cornerstone of human habitats is therefore, the home. Housing is strongly linked to social, economic and political developments on various levels individual household, housing estate, neighbourhood, urban, regional, national and even international. The basic function of housing is to provide shelter, so it directly affects peoples quality of life and wellbeing. However, housing is also an economic good; over the last decades housing has increasingly been considered as an asset to be traded in the market. This multidimensional nature of housing often creates tensions between different agendas: social, economic, environmental. Across the world, housing issues reflect local and national specificities and historical pathways. Nonetheless, globalization is creating convergence on key housing issues and challenges in large parts of the world. In this course, students will learn about the role of housing to achieve socially sustainability human habitats in different geographical contexts. Key concepts, policies and practices aiming at maintaining and improving social inclusion and wellbeing in housing and living environments today and in the future will be presented and discussed. The course takes a global lense and the concepts discussed are applied to various country contexts. It adopts a multi-dimensional understanding of the concept of social sustainability in relation to housing, which identifies at least four core dimensions: 1) the social preconditions for sustainable development (i.e., values, habits and rules); 2) the equitable distribution and consumption of housing; 3) the quality of social relations in housing and living environments; and 4) the physical conditions or livability of housing and living environments.	
Study Goals	At the end of the course, the student will be able to: 1. Explain the main problems and challenges related to social sustainability in human habitats in different parts of the world. 2. Apply key concepts and theories to explain and assess social sustainability challenges in human habitats. 3. Examine a policy or programme applied to the solution of a social sustainability problem in a given context and provide recommendations for its improvement.	
Education Method	The course is structured in three basic modules: Module 1: Challenges Overview of social sustainability challenges across the world (e.g., equitable housing distribution and consumption; social relations; liveability, etc.) Module 2: Theories A selection of theoretical approaches that can help explain the causes, effects and meaning of the above challenges in different world regions (e.g., Capabilities approach; sustainability transitions; commons; collective action; social movements; critical theory. Module 3: Interventions Policy and practice approaches to improve social sustainability (e.g., cooperative housing; Community Land Trusts; Incremental housing programmes; Social mix policy approaches). Each module will be taught through a mix of lectures by teachers from the BK Faculty and invited guests (including practitioners) and flipped classroom methodologies involving a selection of educational videos produced by teachers from BK and from the IHS at Erasmus University Rotterdam. The content of the reading and videos will be discussed with teachers and invited guests in seminar-style activities.	
Literature and Study Materials	Will be provided on Brightspace.	
Assessment	The course will be assessed through two main activities: a) Individual essay (50% of final grade): Students formulate a social sustainability problem and examine it by applying a theoretical lens of their choice. b) Group assignment (50% of final grade): In groups of three or four, students choose a case of an intervention (policy or programme) designed to tackle a social sustainability problem and critically examine its effectiveness. The list of cases is provided by their tutors. At the end of their assignment, they must provide recommendations to improve the given intervention and justify these by drawing on literature and material from the course.	
Period of Education	Q1, Q3	
Minimum number of participants	20	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR0897	Van Gezel tot Meester	20
Course Coordinator	Ir. E.J.G.C. van Dooren	
Instructor	Ir. E.J.G.C. van Dooren	
Instructor	V.L. de Vries	
Responsible for assignments	Ir. E.J.G.C. van Dooren	
Contact Hours / Week x/x/x/x	Eerste kwartaal 4 uur per week, 2e kwartaal 8 uur per week	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	Dutch	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Learning to design is a matter of doing and becoming aware what to do. Teaching designing is a matter of making the design process explicit and training meaningful actions and skills. Both are main subject in this MSC 2. The design process and the didactics of design are studied and practiced at the hand of a frame work of 5 generic elements. Basically, designing is a process of experimentation (exploring and testing), in a laboratory (sketching and modelling). The designer has to address aspects in different domains (form, material, function and context), using common known and proved knowledge, often in the form of patterns and principles (frame of reference). In the end (s)he comes up with a coherent meaningful, adequate elaborated design, addressing the specific design situation at hand.	
Study Goals	Be aware: course is in Dutch, because of the internship in the BSc first year Upon completion of the design studio the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level.	
Education Method	Specific for this course, the student is able to demonstrate sufficient insight in and knowledge of the design process demonstrate sufficient insight and knowledge of the didactics of design In a number of short design projects, the design process will be done implicitly and studied explicitly. This may lead to insight into generic design process actions and skills. In a number of seminars the design process and the didactics of design will be studied. In an internship (assistant teacher BSc first year) being a design teacher will be explored. The experiences will be discussed in the gezel meester studio.	
Assessment	*) In case of specific circumstances, the internship can be replaced by other ways to explore design education Assessment will be based on the results of the design projects and reflective studies on design (education). The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Q3 = seminars (5 erts) Semester 2, Q4 = design project and internship BSc ON project(15 erts)	
Concept Schedule	Q1 = Friday afternoon Q2 = Tuesday afternoon + Friday afternoon + internship	
Minimum number of participants	12	
Maximum number of participants	18	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA010	Architectural Research and Design Seminar	5
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is an Elective workshop/seminar under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. This Elective opens the possibility to develop and offer an unique and experimental limited design or research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods and content will vary. The final topic and content will be presented at a studio information meeting of the responsible studio and described in the syllabus before the enrollment for the spring semester starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design or research result- on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design or research with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings or a report. -collaborate and communicate by making active use of various methods to present the design or research result in all its aspects. -is able to position the result within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be published at the beginning of the course.	
Education Method	The workshop or seminar features individual and group tutorials which will be study specific to the design or research topic as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research or design work undertaken by the individual student within the set theme; the specific research or design questions raised within; the specific study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention or conclusions with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and ethical conclusions or results; - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Period of Education	Quarter 3	
Concept Schedule	Education starts week 3.1, final presentation week 3.10. No education in week 3.9	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA015	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA016	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	Quarter 4	
Concept Schedule	Tuesday	
	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	15	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AA017	Architectural Design Studio	15
Course Coordinator	P.A. Koorstra	
Responsible for assignments	P.A. Koorstra	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Summary		
Course Contents	The course is an approved Architecture Design project under supervision of the department of Architecture. The course will be executed by one of the disciplines of the track of architecture. The course makes it possible to develop and offer an unique and experimental design and research project on MSc2 level. Since every year the course will be organised by a different group the theme and methods will vary. The final topic and content will be presented at a studio information meeting and described in the syllabus before the enrollment starts.	
Study Goals	Upon completion of the course the student is able to - present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. - is able to demonstrate the appropriateness of the design with respect to the assignment - conduct design research and research-by-design by using physical and/or digital models, digital and/or hand drawings as a tool throughout the design process. -collaborate and communicate by making active use of various methods to present the design in all its aspects; the architectural composition, materialisation and integration of construction. -is able to position the design within a particular theoretical, historical, social or contextual framework Next to the general study objectives formulated by the Faculty, a specific description of the aims of the studios will be published in the Studio Manual, to be distributed at the beginning of the course.	
Education Method	The design studio features individual and group tutorials, and study specific to the design project as well as several dedicated thematic exercises, lectures and seminars that pertain to and inform the subject. There will be weekly assistances in groups as well on individual basis.	
Assessment	Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. Products: will be described in the syllabus which will be published at the beginning of the course The project will be assessed on: - the position that is formulated with regard to the brief and its context. - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation. - aesthetic and technical / functional qualities; the elaboration throughout the respective scales - the quality of the presentation, the products and the argument. - the consistency, coherence and development of the students work during his / her process If graded as 5 or 5.5 a repair option will be discussed. A repair is a revision or addition to the existing assignment. The deadline is within two weeks after the final presentation day.	
Remarks	An Excursion can be part of the course; it will be announced at the presentation of the studio before the enrollment.	
Period of Education	quarter 4	
Concept Schedule	Education starts week 4.1, final presentation week 4.10	
Leerstoel	Department of Architecture	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AD012	MSc2 Dwelling design studio 'Global Housing'	15
Course Coordinator	N.J. Amorim Mota	
Instructor	Prof.ir. D.E. van Gameren	
Instructor	N.J. Amorim Mota	
Instructor	R. Varma	
Responsible for assignments	N.J. Amorim Mota	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design studio and the Building Engineering Studios (AR1A080).	
Course Contents	This design studio challenges students to find appropriate methods for the analysis and design in cultural contexts that are not their own. Participants in the studio develop housing proposals that advance new possibilities to negotiate local cultures and techniques on the one hand, and global developments on the other. Against this cross-cultural background, students are invited to develop their own position and to find design strategies that take as key premise the development of adequate housing for regions undergoing a process of rapid urbanization. To support the development of the project, participants in this course develop spatial and situational analysis in the projects location. Using a combination of different research methods, from design analysis to architectural ethnography, students investigate local patterns of inhabitation, urban and building morphology and typology, interdependence between dwelling characteristics and lifestyles, and negotiations between individual aspirations, collective welfare, and environmental protection. The socio-spatial analysis is used to support the development of a project that critically addresses the challenging negotiation between processes and narratives of globalization and situated practices.	
Study Goals	Upon completion of the design studio the student: 1. Produce analytical outputs that account the social, morphological, typological and environmental characteristics of a specific dwelling environment. 2. Elaborate a problem statement and critical reflection on the challenges and opportunities associated with a specific urban condition. 3. Formulate a design strategy for affordable housing in relation to the particular circumstances of a specific site and/or urban condition in view of the framework of the sustainable development goals. 4. Design and develop an urban housing project based on a multi-scalar design strategy, articulating the design decisions from the scale of the dwelling unit to the neighbourhood scale. 5. Design and develop adequate dwelling types taking into account the available resources, as well as the needs, aspirations and lifestyle of an urban community. 6. Identify and explain the qualities of the proposed design in relation to a specific socio-political, economic and environmental context. 7. Identify appropriate building techniques and construction systems to be employed in the design strategy and architectural project. 8. Produce meaningful written, visual and physical outputs to communicate the design process and the project to peers and experts.	
Education Method	The course is structured in two phases, based on education methods that comprise individual initiative and self-study, weekly tutorial sessions, complemented with lectures and reviews by experts and peer-to-peer discussions within the studio. In the first phase students are invited to join an excursion to the project's site and develop a multi-layered analysis of the site's existing environmental situation, including fieldwork, desktop research, literature review, and analysis of precedents of housing design in similar conditions. During the field trip excursion, the participants in this course will be invited to participate in a one-week workshop, working in collaboration with local students, and attending lectures delivered by local researchers, educators and experts. In the second phase the students will attend tutorial sessions with the course instructors and develop a problem statement, followed by a proposal for a master plan. The masterplan plan should be based on a clear design hypothesis, which should entail a coherent narrative framing the acquired knowledge into a design proposal for the project's site. The outcome of this phase will be presented to the peers and reviewed by the course instructors. The design hypothesis, developed in groups, will be followed by a design proposal, where each student working individually will develop the architectural characterisation of a significative part of the masterplan, including plans, sections, elevations and spatial-material relations showing the qualities of the urban housing neighbourhood in relation to the site's socio-economic, cultural and environmental circumstances.	
Assessment	Throughout the duration of the design studio, there will be regular moments for formative feedback (at every weekly tutorial session), and at the end of each phase. The summative feedback will be based on the assessment of group and individual work, concerning the following phases: PHASE 1_Foundations 1.1. Fieldwork / Contextual Research (Group work) 1.2. Concepts and Analysis of Design Precedents (Individual and Group work) PHASE 2_Project 2.1. Design Hypothesis / Masterplan ((Group work) 2.2. Housing Project (Individual Work) The relative weight of each deliverable will be made known 1 week prior to the start of the course in Brightspace. Retake and repair regulations In accordance with Article 1.23 of the Teaching and Examination Regulations Bachelor, and Article 1.24 of the Teaching and Examination Regulations Masters, students are entitled to at least two opportunities to take a written examination, in the period of one academic year. Examinations refer to all variants of examinations used in our faculty, except for very special circumstances such as practicals or excursions. Repair: The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Remarks	Participating in this studio requires a field trip to the project's site for approximately two weeks in the Spring semester (mid-April/early-May). The cost of the field trip is approximately 1.000,00 - 1.500,00. Each participant in the studio should support this cost, and/or find possibilities to secure (co-) funding for the trip.	
Period of Education	The course is offered in the Spring semester, Q4	
Concept Schedule	Tuesday afternoon and Thursday afternoon	

Minimum number of participants	12
Maximum number of participants	36
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AI011	Interiors Buildings Cities MSc2 Design Project	15
Course Coordinator	Ir. S.S. Mandias	
Instructor	Ir. L.M.M. de Wit	
Instructor	D.H.G. Somers	
Instructor	Ir. S. Pietsch	
Instructor	Ir. S.S. Mandias	
Responsible for assignments	Ir. S. Pietsch	
Contact Hours / Week x/x/x/x	48 hours per week starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The Chair of Interiors Buildings Cities focuses on buildings and interiors that accommodate the different scales and gradations of public life within the city, from the street to the public interior. It addresses the ways in which these can be situated in relation to place, time and material culture. Each course in the programme refers to a particular building or interior type, acknowledging its significance in the past and exploring its capacity for adjustment, adaptation or transformation in response to the needs of contemporary society and culture. The Salon of the MSc2 project refers to the tradition of the large public room, which receives and shapes the society of people that it gathers. A society brought together not through proximity, but rather through discourse, in relation to shared interests. Originally the salon was both a cultural phenomenon and a specific space within the European aristocratic home during the 17th and 18th centuries. Mostly initiated by women (salonnières), they were social gatherings in which participants engaged in the art of conversation, dedicated to the exchange of ideas and the pursuit of knowledge. This course considers the relevance of such a notion in a contemporary setting. Students will design the structure and fabric of a contemporary space for conversation, in response to an existing building and a specific community and site. The rooms scale and elaborated interior, structure the orders and arrangements of the building in which it is set. It offers opportunities for both intimacy and publicness and, whether through its physical relationship with the outside, or as a consequence of the conversations or events that it hosts, it engages the city. Through a process of iterative drawing and large-scale physical modelling, supported by lectures, workshops and seminars, students will design the structure and fabric of such an interior, responding to an existing building and including consideration of its furnishing, relevant technical aspects, material finishes and the possibilities for its inhabitation.	
Study Goals	Upon completion of the MSc2 design project the student is able to: - analyse relevant precedents concerning their societal context, technical and material aspects and aspects of use. - develop a consistent and coherent design process, making informed and well-argued decisions, using appropriate analogue and digital tools for drawing and model making, and respond to feedback from tutors and peers. - develop, on the basis of the brief (as specified in the studio manual), the given site and the precedent research, an architectural idea for the project - On the basis of this idea, design a coherent, elaborated and integrated interior project in terms of technical aspects, material aspects and aspects of use. - present the proposal in a clear and coherent way, both orally and by using appropriate analogue and digital tools for drawing and model making.	
Education Method	The design studio features individual and group tutorials, as well as several dedicated thematic exercises, internal lectures and seminars that pertain to and inform the subject.	
Literature and Study Materials	to be announced upon beginning of the course.	
Assessment	The assessment of students work will be based on a project journal documenting the design process, and the visual and oral presentations of the precedent analysis and the design proposal. The project will be assessed on the basis of the following aspects: - the position that is formulated with regard to the brief and its wider context - appropriateness of the design with respect to the assignment and its translatability into a physical manifestation - the coherence, elaboration and integration of the final design - the quality of the presentation (visual and oral) - the consistency, coherence and development of the students work during the design process	
Special Information	The maximum marking period is 10 work days.	
Period of Education	The project takes place in the second quarter of the spring semester.	
Concept Schedule	Different days	
Leerstoel	Interiors Buildings Cities	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP010	MSc2 Public Building Design Studio Multiplicity and Identity	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Instructor	S. Corbo	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc.1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	In envisioning radical solutions for the public domain, the MSc.2 Public Building Design Studio investigates the emergence of new building typologies as a response to the urgent problems that many cities face nowadays global warming, environmental pollution, spatial segregation, invasive privatization, etc. The MSc.2 Studio focuses on solutions that support socially, culturally and ecologically sustainable communities, in order to translate the urban environments challenges into experimental architecture. Emphasis will be placed on hybrid complexes where different functions and users coexist, embedded with a wide range of spatial articulations, including living, working, leisure and culture. Via experimentation and speculation, the Studio aims to rethink types and typologies that are traditionally associated to mono-functional programs corporate or commercial as well as conventional building systems. The Studio adopts multiplicity as crucial property of architecture and its elements. Multiplicity endows built environment with more conducive, transformative, and resilient qualities: it provides conceptual tools for creating new places, buildings, and building elements; reinvents conventions, and questions the present toward the future. Each year, the MSc.2 Studio is set in a different city, exploring the unique site conditions as a foundation for innovative architectural proposals that act as a catalyst for new urban dynamics.	
Study Goals	Upon completion of the Design Studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - investigate the processes of adaptation and transformation of the given urban conditions, by constantly relating the human aspects of the changing society to the urban effects of their actions; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - focus on the qualitative aspects of multiplicity in society and design; - design an innovative architecture which can contribute to improve adaptability to climate change as well as productivity of technical solutions, materials and building physics; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Public Building Group adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. Rather than developing one single architectural proposal throughout the quarter, this Studio challenges the conventional learning curve which goes from the universal to the particular. The Studio, in fact, will explore specific and limited (in time and scope) aspects of the design process to build, around those, new spatial narratives. It is, therefore, conceived as a sequence of design exercises aimed to address a diverse range of spatial and typological issues. The Studio will be structured across: - lectures - field trips - readings, writings, and public discussion - experimental research - design tutorials - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Written statements and analyses. The design proposal is individual. The assessment process implies: - in-class participation on a weekly base with public discussion of in-between results - Midterm and Final Reviews The final grade reflects: - Critical analysis and the urban context (25%) - Design quality of the final proposal (55%) - Participation, collegiality, commitment (20%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and	

Special Information	failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	Studio tutorials on Friday
Leerstoel	Public Building
Minimum number of participants	12
Maximum number of participants	32
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2AP031	Public Building Seminar Aesthetics of Sustainable Design	5
Course Coordinator	S. Corbo	
Course Coordinator	S. Lee	
Instructor	S. Lee	
Responsible for assignments	S. Lee	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Expected prior knowledge	Software for design, graphics, data representation, and presentation Research skills English proficiency in formal presentation The course is also open to non-architecture majors.	
Course Contents	This seminar course takes place in the first eight weeks of spring semesters. The central question of the course is how sustainability-oriented design may contribute to the overall quality of design. In addition to that, the course will introduce the students to the current debate on sustainability and circularity, by offering updated literature review, as well as by presenting innovative design tools and techniques. The course investigates which architectural and aesthetic potentials that are not always visible in the basic premises of sustainable built environment can be uncovered and used. It also looks at how the theories and rationales behind sustainable-oriented design lead to certain aesthetic presumptions and assumptions. Even though we speak of "sustainability" daily, we should make it clear what it actually entails. This course aims to establish a foundation that sustainability presents aesthetic opportunities, rather than limitations, to the conception, design, and construction of buildings and places. Through case-studies, literature review, excursions, and lectures we will discuss some widely used sustainable building materials and the pertinent concepts and sciences behind them. We will explore some primary elements in sustainable design and apply them to individual small-scale design exercises.	
Study Goals	By completing the course, the students will gain concrete knowledge and applicable skills toward sustainability as an integral part of design practice: Understand the foundations of sustainable design. Understand the application of sustainable thinking in design. Learn to assess and evaluate the potentials of sustainable design. Learn how to apply what kind of sustainability concepts to a given design task. Understand the relationship between sustainability as cultural discourse and the role of design for it. Develop viable scenarios for sustainable lifestyle.	
Education Method	Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The seminar will consist of literature review, site visits, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: Lectures & discussions Literature review & design analyses Research by design Presentations & reviews The class will be divided in groups depending on the number of participating students.	
Literature and Study Materials	The course employs a variety of weekly references that deal with each weeks topic. The literature list will be announced in the specific course syllabus of the term.	
Assessment	Assessment of the students work will be based on the following: Oral presentation; Design output: drawings and images, compiled in presentation posters, physical models, material samples, etc. The final grade reflects: Class participation & discussions (20%) In-class presentations (30%) End-term design presentation & submission (50%) IMPORTANT: The first session lays out the overall framework and crucial information regarding the contents of the course. Participation in the first class is mandatory. Absence without the instructors consent prior to the first session may result in the dismissal from the course. Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Wednesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	32	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AP041	Public Building Seminar Composition and Perception	5
Course Coordinator	S. Corbo	
Course Coordinator	Ir. A.M.F. van Dam	
Instructor	Ir. A.M.F. van Dam	
Responsible for assignments	Ir. A.M.F. van Dam	
Contact Hours / Week x/x/x/x	36 hours per quarter	
Education Period	3	
Start Education	3	
Exam Period	3	
Course Language	English	
Course Contents	This seminar addresses the perception of public buildings and its role as a design vector. In our time, the hegemony of vision has been reinforced by a multitude of technological inventions and the endless multiplication and production of images. Perception can be conceived of as a process of interaction that relates to visual composition, articulating and suppressing details while focusing on identification. This process of interaction is taken as the starting point to investigate transformations in architectural composition. Technical education usually focuses on phenomena like form and program. However, to include the experience of perception will enlarge the architects repertoire substantially. For example, the way in which colour affects our perception of the architectural composition, shows us how its application can be considered as a powerful tool. The course presents and investigates methods that can be applied for site-specific design. Moreover, a more theoretical reading program supports the thematic workshops, contributing to the understanding of notions like the formal and informal in architecture and urbanism. The readings are centered on writings about perception in the context of architecture and give evidence of the great variation in the approaches and tools. Learning to understand architectural composition in the context of perception offers the potential to make architecture more productive and more resilient.	
Study Goals	By completing the course, the students will be able: - to experiment with different design methods and techniques; - to explore the specific relationship between image and language; - to work with the different constraints that define the potential for innovative design.	
Education Method	Students will engage both individual and collective assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide solutions that are both innovative and responsive. The seminar will consist of literature review, pre-design analysis, and tutorials leading to an individual/group result. The seminar runs several activities, including: - workshops - research by design - presentations - lectures - individual portfolio	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, physical models, material samples, etc.; - Writing assignments and analyses. The final grade reflects: - Text comprehension (25%) - Diversity in tools (25%) - Portfolio (25%) - Attendance and participation (25%) Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.	
Period of Education	Half semester (Q3)	
Concept Schedule	Tuesday afternoon	
Leerstoel	Public Building	
Minimum number of participants	12	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT021	Architectural Technicities Design Studio	15
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week beginning in week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	As per MSc2 Faculty requirements: It is expected that students have the knowledge from a MSC1 design studio course and the Building Engineering Studios (AR1A080). Affinity with architecture theory is desirable, but not required.	
Course Contents	In the Architectural Technicities Design Studio we examine how humans relate to and transform their environments through technology, and how these relations transform humans, technology, and the environment in return a process known as a technicity. Therefore, if different modes of urban development correspond to radically different architectural technicities, then the studio will examine how they both result in the production of different urban subjects. In simple terms, we will not examine architecture as a product of culture or ideology, but rather claim the opposite: architecture produces culture and ideology. As such, the studio aims to complement architectural and urban methodologies that are based only on typological assumptions and will necessarily open architectural discourse to philosophy, media theories and affect-oriented studies. The thematic and design assignments of our studio vary per year, but always depart from actions rather than programmatic or functional prerequisites, foregrounding the potentials of architectural, technological, and environmental agencies involved in the design process. This studio is highly experimental and hands-on regarding the material aspects of theory as practice. It welcomes students who are inclined to explore unfamiliar (yet exciting) themes, raise interesting questions and architectural problems, and experiment with ideas, concepts, and methods to make their design practice and skills more meaningful.	
Study Goals	Upon completion of the design studio the student can convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. The student will be able to: develop a transdisciplinary approach to identify and address shared problems and concerns regarding the production of contemporary urban environments. apply their own methodological innovations that may inform and trigger new ways of speculative design. adopt a transdisciplinary, cross-domain mentality and attitude based on in-depth reading of seminal literature and its design application in real-life contexts.	
Education Method	This studio is taught with the aid of a set of mini-lectures & group discussions; short study-trip/excursion; design studio sessions and studio-specific workshops.	
Course Relations	AR2AT031 (Architecture Theory Thesis Seminar)- required choice course MSc2 AR2AT041 (Architecture and Philosophy Lecture Seminar) -elective course MSC2 AR0570 (Worlding Theories and Concepts) - elective course MSc2	
Reader	A course reader will be made available for the studio (varies per semester). Please consult syllabus in Brightspace.	
Prerequisites	MSc1 Studio accredited.	
Assessment	This design studio is assessed with: midterm presentations (analysis: research, argument and conceptualization) final design project presentations studio report (multiple media are allowed)	
Enrolment / Application	Enrolment per Faculty regulations & periods. For queries contact the course coordinator.	
Special Information	Short field excursions or study trips may be programmed for this studio	
Period of Education	This course is taught only in Q4 of each academic year.	
Concept Schedule	Tuesday and Thursday morning / afternoon	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	48	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2AT041	Architecture and Philosophy Lecture Seminar	5
Course Coordinator	Dr.ir. H. Sohn	
Instructor	Dr.ir. A. Radman	
Instructor	Dr.ir. S. Kousoulas	
Responsible for assignments	Dr.ir. H. Sohn	
Contact Hours / Week x/x/x/x	12 hours contact time per week starting in week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Expected prior knowledge	An interest in philosophy and theory is desirable, but not required for this course.	
Summary	A common (mis)understanding is that Architecture is active practice, while Philosophy is passive contemplation. In this course we will correct this misunderstanding, approaching philosophy and architecture from unexpected and fresh angles. We will see how both are engaged in a dynamic process of exchange and transformation: where philosophy encourages us to think otherwise, to produce concepts and experiment with problems, architecture provides a framework and a field of operations.	
Course Contents	Students in this course will be introduced to a host of concepts through a slow, collective reading of three short texts. In these texts, we will examine current issues and concerns that may be articulated through philosophy and architectural thinking. Examples of such issues are the complex, trans-disciplinary encounters of technology, culture, the environment, architecture and so on. Students in this course will be encouraged to 'freely associate' thoughts that emerge from the reading of these texts. In this way, participants in this elective lecture seminar will engage in rich conversations and group discussions on many areas and fields of knowledge that intersect in areas that relate to architecture, understood as the design of the lived environment, present and future.	
Study Goals	Upon successful completion of this course, the student has: - acquired appropriate knowledge on philosophical and architectural thinking, and the production of related art forms, literature and media. - developed sufficient intellectual and inquisitive skills and an academic and critical attitude towards the analysis, setting and solution of complex problems; formulate adequate questions and evaluate the validity of knowledge claims. - become aware of the rootedness of ideas, designs and plans in a particular temporal, and societal context. - learned to conduct independent, ethical research.	
Education Method	This course is based on the newly developed pedagogy of 'collective reading', namely, the slow reading out loud of short texts (or segments) and their simultaneous analysis and group discussion. The course is designed as a lecture seminar: 3 bi-weekly lectures: Architectophilosophical Lab 3 bi-weekly reading seminars: Collective Readings / Reading Collective	
Course Relations	AR2AT031 (Architecture Theory Thesis Lecture Seminar) AR2AT021 (Agential Materialism Design Studio)	
Literature and Study Materials	This course has a Course Reader: a compilation of three segments or texts that will be read in the seminars. The Course Reader will be available on Brightspace in advance of the course start, but will be read during the meetings. No further or prior reading is required for this course.	
Reader	This course has a Course Reader. It changes every academic year. See Brightspace and the Course Syllabus for the current Reader.	
Assessment	This course is assessed with a specific WRITING ASSIGNMENT: Students are asked to write a set of three "Thought Pieces". "Thought Pieces" are short, open-scope texts in which the students will generate writings and other forms of creative expression : reflections of the discussions of the seminars; speculations on specific ideas, thoughts or topics; narrative or story-telling experiments; etc. Students are free to choose the modality and thematic of these thought pieces. More information on the "Thought Pieces" is available in the course syllabus. Students will decide themselves whether to submit each piece after each seminar, or compiled as a set of three, at the end of the quarter. The maximum marking period is 10 working days after the final deadline. The due date for all assignments is in week 3.10 as per academic calendar.	
Enrolment / Application	This course is taught in Q3, enrolment during Faculty periods. No special enrolment for this course.	
Elective	Yes	
Period of Education	This course is taught only in Q3	
Concept Schedule	Wednesday afternoons weeks 3.1, 3.3, 3.5 - Lectures weeks 3.2, 3.4, 3.6 - Reading Seminars weeks 3.7-3.10 - self-study week 3.10 - due date "thought pieces"	
Leerstoel	Architecture Philosophy and Theory Chair	
Minimum number of participants	12	
Maximum number of participants	50	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2BO010	Borders and Territories International Design Studio	15
Course Coordinator	Dr.ir. M.G.H. Schoonderbeek	
Course Coordinator	S. Milani	
Instructor	Ir. F. Geerts	
Instructor	Ir. M.J. de Haas	
Instructor	Dr.ir. M.G.H. Schoonderbeek	
Instructor	S. Milani	
Responsible for assignments	Dr.ir. M.G.H. Schoonderbeek	
Contact Hours / Week x/x/x/x	12 hours per week starting from week 4.1 and ending in week 4.10.	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc2 International Design Studio of Borders&Territories (B&T) will focus on the relation between architectural research and architectural design. The studio will deal with the research topics of the B&T group, which can be summarized in the following main components: (1) MEGA-MICROS, namely the relationship between the extremely large and small scale of architecture; (2) NEW GROUND, investigating the relationship between new land reclamation projects and architecture; (3) ZONES OF CONFLICT, investigating the entanglements of milieus created by conflicts of (soiled) substances. The course consists of three parallel studios focussing on different locations changing yearly, while the research structure will remain unchanged. The studios will be offered as cooperation with other universities and (when possible) kick-started by an on-site workshop. The course will also offer a series of lectures on studio-related themes.	
Study Goals	Upon completion of the design studio, the student is able to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSc2 level. Understand the relationship between architectural work and its context, as well as ways to relate (or implement) architectural research findings to architectural construct. Develop the ability to clarify a design project to others by means of images, spoken and written words. The student is able to position the project within a particular theoretical, historical, social and contextual framework.	
Education Method	Group work (research and site analysis). Excursion (TBC) Lectures and workshops. Pin-up collective presentations. Individual consultation. Independent design & self-study.	
Assessment	Studio attendance & participation. Individual presentations & evaluations. Mid-term (week 4.5) and final (week 4.10) reviews. (Specific weeks & dates of the presentations may be subject to change according to the official academic calendar of the university). Assessment Scheme - Design (70 %) - Weekly development assignment/mid-term (10 %) - Participation (attendance, initiative, in-class discussion) (10 %) - Final Exam (Clarity of presentation) (10 %) Deadlines and times for retakes or repairs, are at the discretion of examiners. The deadlines fall within the same academic year, must be workable and studyable, and not cause study delays. The deadlines are the same for all students; prior to the retake or repair, there is an opportunity for inspection and/or discussion of the results (in accordance with Art. 1.28-1.29 Teaching and Examination Regulations Masters).	
Period of Education	Quarter (Fourth quarter - Q4)	
Concept Schedule	Tuesday and Thursday	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2CP011	MSc2 Complex Projects Design and Research Studio	15
Course Coordinator	M. Triggianese	
Course Coordinator	Prof.ir. C.H.C.F. Kaan	
Instructor	Prof.ir. C.H.C.F. Kaan	
Instructor	M. Triggianese	
Instructor	H. Smidihen	
Responsible for assignments	M. Triggianese	
Contact Hours / Week x/x/x/x	10-12h per week, starting from week 4.1 and ending in week 4.10	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The MSc 2 design and research studio explores a specific theme with the aim of positioning the architectural project into a broader social, cultural, political and economic context. In the last years, students have conducted thorough research including data analysis and urban context analysis for a specific topic of global relevance. Students are asked to translate the outcomes of research into an architectural and urban design proposal tackling several different scales in parallel: network, city, building and interior. To foster imagination, both conceptual and realistic representations of design and research are welcome. Students are encouraged to present their work in a creative and original manner, from axonometric line drawings to mixed-media collages. Please contact the studio coordinator for information about the program offered in spring 2024.	
Study Goals	Upon completion of MSc2 Complex Projects design and research studio, the student is able: - to convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal on mainline, and on aspects relevant to the MSC2 level. - to position the project within a particular theoretical, historical, social or contextual framework; - to understand the fundamental design process with regard to architectural theory, art, technology and human sciences; - to demonstrate sufficient insight in and knowledge of the design process; - to develop critical thinking while approaching a complex urban scenario; reflecting upon the relationships between analysis, conceptualization, method and composition of a design proposal; - to develop technical skills regarding the architectural drawing on different scales; - to develop argumentation and graphic skills aiming to consolidate and strongly communicate a design narrative.	
Education Method	Tutorials in studio. Research will be conducted in thematic groups, design is either individual or in groups of max 2 students. The studio includes seminars with lectures in the research phase.	
Course Relations	Chair Ambition Complex Projects (CP) encourages students to explore an architecture of dialogue, one that is dialectic, inclusive and relational. It does not content itself with the notion of architecture for architects, addressing purely an elite selection of connoisseurs and making sense only within the bounds of its own field. It engages with reality to transform it from within. Architects develop designs of buildings and spaces which are only constructed if they are regarded as useful and embraced by stakeholders. Complex Projects explore how the normal can become both exceptional and useful, refrains from formal prejudice, and is implicitly sustainable. Architectural projects are fully integrated designed buildings. Integrated design requires a process that is highly complex and has a strong architectural guidance. Besides, the number of participants and stakeholders in any architectural project has increased exponentially. Not only when designing large projects, but at any scale the requirements (functional, spatial, and technical) are ambitious, often contradictory, and the expectations on meeting them are high. Design proposals must be verified and validated against these demands. In Complex Projects the objective is to engage this complexity with professional knowledge, a set of skills and critical thinking. We ask our students to be inquisitive and open minded. Complex Projects targets all scales of the architectural thinking: detail, building, city, and region. It aims to expand the knowledge on design of the built environment, nourish critical thinking and broaden the minds of future architects.	
Reader	Reader (syllabus) with the studio programme, the basic literature and the weekly schedule will be provided prior to start studio	
Assessment	Assessment The assessment in use consists of two parts: Analytical assignment for the research phase and design examination of the design phase. Example of end products: research report and presentation, design drawings, representation and presentation. The midterm assessment will take place halfway through the studio program (not graded), and the final assessment will be done at the end of the studio program (graded). The proportion of different assessments will be described in the syllabus. Evaluation Evaluations will be based on the overall performance within the studio. The students performance will be determined by the quality of his/her work, commitment, teamwork, effort and improvement over the entire course of the semester. Concrete aspects for evaluation are: research work, argument formulation, translation argument into the concept, urban plan, architectural design, presentation. Repair The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	The locations of the Complex Projects MSc2 project can be in the Netherlands or abroad. Please contact the studio coordinator to know this year's site visits. Students might consider additional costs for printing, travelling and accommodation, which could be quantified between 50 - 150 euros per person, depending on location and possibilities.	
Period of Education	Quarter 4 (spring semester)	
Concept Schedule	Tuesday afternoons and Thursday mornings	
Minimum number of participants	12	
Maximum number of participants	40	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2DC010	Architectural Design Crossovers Studio	15
Course Coordinator	A.S. Alkan	
Course Coordinator	Dr.ir. R. Cavallo	
Instructor	A.S. Alkan	
Instructor	Ir. J.A. Kuijper	
Instructor	Ir. J.P.M. van Lierop	
Responsible for assignments	A.S. Alkan	
Contact Hours / Week x/x/x/x	12 hours/week (4.1-4.5) 8 hours/week (4.6-4.8 & 4.10)	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a MSc1 Design Studio.	
Course Contents	MSC2 "Intersections" studio considers experimenting as a central axis of architectural design investigation with a multidisciplinary and intescalar approach within different geographical and territorial contexts. Sharing the same etymological origin with the words experience and expert, the term experiment defines the investigative yet formative characteristics of architectural design process. By geographical displacement, biennales and international workshops, Architectural Design Crossovers MSC2 studio will provide a central theme to be renewed every semester. The studio couples experiencing and experimenting within different geographical and territorial contexts to help the students form expertise along their research and design interests. Therefore, the studio engages in critical design practices and their theoretical and historical foundations with emphasis on process-based design inquiries. The studio guides the students to apply research-oriented critical approaches to analyse and reflect upon design actions, positions, methods and outputs which lead to site-specific interventions across spatial and temporal scales. Due to the nature of the studio, international collaboration and workshops and participation at architectural events are integral to the studio. A relatively long educational excursion (5-7 days) with on-site workshops is part of the studio program. The corresponding information is to be communicated at the introductory meetings and via Brightspace.	
Study Goals	Within / Upon completion of the MSC2 studio the students are able to: - Recognise critical design approaches from/within other related fields; - Use and develop experimental methods of investigation and synthesis; - Define critical design position within the theme of the studio; - Integrate relevant theoretical knowledge and practical skills into the design process; - Reflect on the cross-disciplinary role of architecture within the wider discourse of the design field; - Communicate and defend the architectural project through investigative and critical methods.	
Education Method	- Lectures and seminars - Pre-design research & precedent studies - Site survey and fieldwork (Excursion) - Individual and group tutorials & workshops (pre-design & preliminary design) - Interim presentations and reviews	
Literature and Study Materials	Literature and Study Materials will be made available on Brightspace one week prior to the start of the course.	
Assessment	The assessment takes into consideration not only the quality of the design work but also the process and the development of appropriate design instruments for investigative and critical design research to be undertaken by the students. The consistency in the ideation, projection and materialisation process is an integral component of the final evaluation. The collective documentation of the fieldwork, investigations and the results will be compiled in the form of a portfolio and book to be presented as part of the final exhibition. More specifically, the assessment criteria for individual work are: - the critical design position formulated by the student addressing the studio theme; - elaboration of the project throughout the respective scales addressed; - the coherence and quality of the presentation, the products and the argument. Mid-term (week 4.4-4.5) and final review (week 4.10). The actual review weeks may be subject to change in accordance with the academic calendar. On the scale of 10, the student achieves a sufficient result with the final grade of 6; has the possibility to take a repair with a grade of 5,0 or 5,5; and fails with a mark of 4,9 or below. The repair has to be completed within 2 weeks after the completion of the course.	
Elective	Yes	
Period of Education	Q4	
Concept Schedule	Tuesday morning (W4.1-4.4) and Thursday morning & afternoon (W4.1-4.10)	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FO010	The Delta Shelter	15
Course Coordinator	Ir. M.G. Vink	
Course Coordinator	G. Coumans	
Course Coordinator	P.A. Koorstra	
Instructor	P.A. Koorstra	
Instructor	G. Coumans	
Instructor	Ir. M.G. Vink	
Responsible for assignments	P.A. Koorstra	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	The assignment is to design a small project in a Delta environment; a dynamic and natural surrounding on the border of water and land. The infinity of the location and the constant changing conditions invite to research the meaning of boundaries and the integration of the landscape in the design. The experience of the specific and poetic qualities of this environment will be one of the explicit themes in this course; the contradiction between the human scale and the unrestricted landscape, the influence of wind and tide, the flora and fauna and the position of human within this often vulnerable ambience. The role, impact and contribution of architecture in such places is part of the research in this assignment. More specific the typology and manifestation of the architecture will be discussed and developed on the basis of the design proposals. The ethics and aesthetics of architecture will be discussed regarding questions as; What are the necessary conditions for architecture to give a satisfying contribution to this environment? Is it inevitable that architecture is a disturbing factor, can it only be of temporary presence, or can architecture contribute to the appreciation and preservation of these kind of environments? The project will be developed by using physical scale models, hand sketches and text during all the phases of the design process; the analysis, design and presentation. The aim of this method is to stimulate the creative process by using the physical model and drawing as a feedback and inspiration tool to develop the concept into a design.	
Study Goals	Upon completion of the course the student is able to present a coherent, significant, elaborated, correct and innovative design - on mainline and on aspects on MSC 2 level. conduct design research and research-by-design by using physical models and hand drawings as a tool throughout the design process. collaborate and communicate by making active use of various scale models to present the design in all its aspects; the architectural composition, materialisation and integration of construction. The student will be able to communicate his/her contemplations and reflect on the role and position of the architect in this assignment.	
Education Method	Lectures, seminars and design studio format. Weekly assistances in groups as well on individual basis.	
Assessment	Assesment on the basis of process, analysis, documentation and (re)presentation of the end result. A brief reflective statement of max 450 words is part of the assesment. Presentation will contain a variety of physical models, drawings, photographs and text. The products should give a clear insight in spatial design, the construction and the relation and meaning of the design towards its environment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Special Information	coordinator	
Remarks	An Excursion within the Netherlands is part of the course A site visit will be part of the studio.	
Period of Education	Q4, 10 weeks, starting in week 4.1	
Concept Schedule	Education starts week 4.1, final presentation week 4.10, week 4.9 no education	
Leerstoel	Form Studies	
Minimum number of participants	18	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2FST010	Studio 'High-Rise Culture'	15
Course Coordinator	Prof.ir. N.A. de Vries	
Course Coordinator	S. Corbo	
Course Coordinator	P.A. Koorstra	
Course Coordinator	Dr.ir. D. van den Heuvel	
Instructor	Ir. P.A.M. Kuitenbrouwer	
Instructor	Ir. P.S. van der Putt	
Instructor	Ir. O. Klijn	
Instructor	W.C. Yung	
Instructor	G. Coumans	
Responsible for assignments	Dr.ir. D. van den Heuvel	
Contact Hours / Week x/x/x/x	112 hours per quarter	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	completed MSc1	
Course Contents	The Studio High-Rise Culture - offered by the section Form, Space and Type - seeks to address the ongoing urban densification by developing new typologies that will inject our cities with vibrant urban spaces, open and accessible, diverse and future-proof. Due to issues of sustainability, the current housing crisis and changing lifestyles, there is an urgency to further densify our cities. A new wave of high-rises is being constructed, not only in the high-speed urbanizing economies of Asia and Africa, but also in the ageing cities on the European Continent. But what could be a desirable mixed-use approach to this new moment in city construction? How can, in high-rise developments, different housing typologies be combined with collective and public programs in order to have lively streets as well as the possibility to create vertical neighborhoods? What sort of city can we create with new vertical open forms, in which collective spaces that invite chance encounter, as well as generous and protective interiors of your private apartment, are combined? The section Form, Space and Type contains the chairs of Public Building, Dwelling and Form Studies. Regular tutoring in the Studio High-Rise Culture is performed by Public Building and Dwelling. Form Studies will use the physical model as a research tool to explore and discover the possibilities of explicit expression of high-rise buildings, their influence on the surrounding space and buildings. Central themes are perception, composition and material expression. The Studio is integrated by a theory seminar, in which the students will collectively read background literature and will write a position statement regarding their design project. The statement will address the current debates on high-rise typologies, city culture, diversity and inclusion, and gentrification processes. The MSc.2 Studio High-Rise Culture is curiosity-driven and combines speculative architectural design with experiments in urban living. Exploratory workshops are combined with research assignments.	
Study Goals	Upon completion of the design studio, the students will be able to: - convincingly present and discuss a coherent, significant, elaborated, correct and innovative design proposal in general, and on aspects relevant to the MSc.2 level; - perform critical comparative research that results in a clearly formulated design hypothesis; - demonstrate how urgent societal issues are addressed in the design project; - demonstrate the necessary argumentation skills to explain and reflect upon the relationships between analysis, conceptualization, method and composition of a design proposal; - represent space in its complex interpenetration of people, architectures, technologies, materials; - confront questions of flexibility, growth and hybridization when developing a comprehensive design proposal.	
Education Method	The Studio adopts a pedagogical approach known as Research by Design. This approach treats design work as a form of research in itself, in which theory and practice, analysis and speculation come together to conceive and develop architectural ideas. Research by Design presents both projection and speculation, as well as a means to prepare, describe, and explain. It enhances the creative process that occurs when concept and design merge, blurring the line between design and research. The research process in design involves loops, variants, reiterations, intersections, and impressions, which all contribute to a continuous progression. Students will engage different assignments, which will enable them to develop a critical perspective on the needs of contemporary societies and to provide architectural solutions that are both innovative and responsive. The Studio will be structured across: - lectures - readings, writings and public discussion - design tutorials / workshops - specific exercises challenging innovative thinking	
Literature and Study Materials	Literature and Study Materials will be available in the courses Brightspace one week prior to the start of the semester.	
Assessment	Assessment of the students work will be based on the following: - Oral presentation; - Design output: drawings and images, compiled in presentation posters, physical model(s), material samples, etc.; - Writing assignments and analyses. Assessment will focus on the research and design work undertaken by the individual student within the set theme; the specific research questions raised within; the specific design study that responds to those questions; the representation of that study in a physical presentation made by the student. The final grade reflects: - the position that is formulated with regard to the brief and its context; - the appropriateness of the intervention with respect to the assignment; the feasibility and translatability of the idea into a physical manifestation; - aesthetic and technical/functional qualities; the elaboration throughout the respective scales; - the quality of the presentation, the products and the argument; - the consistency, coherence and development of the students work during his/her process. Products: will be described in the syllabus which will be published at the beginning of the course.	

	Repair regulations The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the Studio.
Period of Education	Half semester (Q4)
Concept Schedule	For this interdisciplinary design studio, full dedication throughout the week is required; designated days for studio tutorials, workshops and seminars: to be fixed during AY 2023-2024.
Leerstoel	Combined Studio of the groups of Dwelling, Public Building and Form Studies
Minimum number of participants	12
Maximum number of participants	30
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

AR2HA011	Building Green: Past, Present, Future	5
Course Coordinator	Prof.dr.ing. C.M. Hein	
Instructor	J.M.K.K. Hanna	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	4-6 hours per week starting from week 3.1 and ending in week 3.8	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	This lecture/seminar course provides a historical foundation for the creation of sustainable architecture and is designed for students to investigate issues of sustainability in architectural and urban form, past and present. It posits that traditional vernacular design holds many inspirations for contemporary design and life and therefore starts with a historical analysis.	
Study Goals	Students will learn how to evaluate architecture and urban form in regard to architectural and urban, but also ecological, economic, political, cultural, social sustainability in cities over time and through space.	
Education Method	The course is a mixture of lectures, seminars and discussion sessions. It requires attendance and close reading of texts and careful analysis of buildings and practices of architectural and urban design.	
Assessment	This course requires active class participation. It aims to model scientific research through in-class research, presentation of small research themes, and in-class presentation of readings and personal research topics. Students will submit a final paper or a portfolio on the research.	
Period of Education	3 Quarter	
Concept Schedule	Wednesday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2HA020 (for A and U students)	Urban Archipelago	15
Responsible Instructor	Prof.dr.ing. C.M. Hein	
Course Coordinator	Prof.dr.ing. C.M. Hein	
Course Coordinator	J.M.K.K. Hanna	
Course Coordinator	Dr.ir. M.G.A.D. Harteveld	
Responsible for assignments	Prof.dr.ing. C.M. Hein	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	The Urban Archipelago studio explores the future of living, working and producing at a time of rising sea levels, increased migration via the seas, and socioeconomic challenges generated by the energy transformation. It posits that we need new architectural and urban forms that allow for integration of diverse populations, people and practices. Along European coasts and in port city territories the relation between sea and land is shifting. We need porous spaces to accommodate climate- change related spatial, social and cultural transformation. We need to host new spatial approaches that accommodate living with water as well as new hyper diverse populations and that provide climate justice for all. We need new public spaces that capture water-based lifestyles where newcomers, refugees and migrants encounter citizens, and where skilled employees cross the paths of service or knowledge workers. We argue that architects and urban designers need to include water awareness into their local strategies to achieve both a balanced composition of the population and regeneration of residential areas, sustainable urban design and architecture, and to facilitate blue economic growth and transshipment. Students will develop scenarios that allow for an integration of marine and maritime practices into the complex reality of a hyperdiverse population living in an urban archipelago. Designing new architectures for an urban archipelago asks architects, planners, anthropologists, economists, and historians to rethink the logic of research and design through the lens of water. In collaboration with the Department of Architecture of University of Pescara (Urbanistica 1 - course code 700M) and the Faculty of Architecture of University of Ljubljana (Studio Aman - course code AW1, AW2, AW3), the Urban Archipelago course will focus on the Adriatic Sea as a space of coexistence and conflicts. We will focus on Trieste, an emblematic coastal city that is part of the larger Adriatic system which includes cities on water such as Venice and Ravenna in Italy, or Koper in Slovenia and Pula in Croatia. Trieste today needs to rethink the relationship between land and water, between the sea and abandoned industrial heritage. New scenarios and maritime mindsets are needed to reimagine Trieste within the Mediterranean, and beyond, to better respond to the challenges posed by social, spatial and environmental transitions.	
Study Goals	Students will be able to: - understand and investigate the role of public spaces in the creation of more sustainable port-city territories, while paying tribute to their global and interconnected heritage. - connect questions of space, society, and culture, while building on a long-term analysis of port-city relationship with the aim to be used in urban and architectural design for the creation or improved ecosystems. - apply mixed-methods relevant to understand, interrelate and redesign port-cities and public spaces, like mental mapping, close reading, exploration of long-term developments, which help to understand the terms and practices for sustainable port-city territories.	
Education Method	Lectures, Seminars, Discussion, Studio design	
Assessment	Assessment will be related to the study goals and the quality of the products mentioned in the course syllabus. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	Semester 2, Quarter 4	
Concept Schedule	tuesdays and thursdays, week 4.1 till 4.10	
Minimum number of participants	12	
Maximum number of participants	36	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MET011	Designing with Others	15
Course Coordinator	A. Stanii	
Course Coordinator	Prof.dr.ir. K.M. Havik	
Instructor	J.A. Mejia Hernandez	
Instructor	Dr. A. Sioli	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Course Contents	The built environment is not produced by architects alone. Inhabitants, builders, and experts from other professions are also responsible for co-creating the built environment together with architects and engineers. Through experimental design practice you will examine and use analytical and projective instruments and methods of the architectural discipline and other related professions in collaboration with the different actors who participate in the production of architecture. The final outcome of the studio is a real-size built architecture, produced in collaboration between architects and non-architects.	
Course Contents Continuation	The three design studios offered by the chair of Methods of Analysis and Imagination form future architects with the skills and knowledge required to: (a) inscribe architectural designs within concrete sites and situations (MSc1, Situating Architecture), (b) collaborate with others productively in the development of architectural designs (MSc2, Designing with Others), and (c) scale fully developed architectural designs meticulously and insightfully in relation to human proportions (MSc3/4, A Matter of Scale). Well situated, co-created, human-scaled architectures result from the thorough and creative analysis and imagination of possible futures for particular architectural contexts. In the three courses analysis and imagination are fully integrated in practice through the proficient and explorative use of instruments and methods of the architectural discipline and other related professions.	
Study Goals	Upon completion of this course, you should be able to: - Analyze the context defined for a project brief in relation to the different actors that determine the production of architecture in that context. - Design a buildable architectural project for that context, in development of the project brief, and in collaboration with selected individuals or groups. - Build the resulting design(s) in collaboration with others. - Evaluate the ensuing architecture, and the instruments and methods employed in its design and construction, in relation to the collaborative process.	
Education Method	A first part of the course develops a series of thematic modules, that present you with different tasks that require the analysis of a particular context and the design of an architectural project. Emphasis is put on the ways in which analysis and design are carried out in collaboration with other architects and non-architects. In the second part of the studio, you will plan and build one or more co-created designs in collaboration with other architects and non-architects. The conclusions of the entire process are presented as an evaluation of the instruments and methods used to analyze, design and build with others.	
Assessment	The analyses and designs produced in the first part of the studio are subject to individual progress reviews (formative assessments, not graded), which make clear to what degree you have fulfilled the learning objectives developed so far, and allow all participants involved in the studio to decide which design(s) to build. Two final outcomes (project and construction) are assessed and graded collectively, and one (evaluation) is assessed and graded individually, using relevant aspects and criteria determined in the EMMA feedback and assessment tool, adapted for the MSc2 level. On a scale from 1 to 10 you achieve a sufficient result with a mark of 6 or more, have the possibility to take a repair with a mark of 5 or 5,5 and fail with a mark of 4,9 or less. Repair has to be completed within 2 weeks after completion of the studio.	
Period of Education	One academic quarter.	
Concept Schedule	Tuesday afternoon.	
Maximum number of participants	24	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2MM001	Building Ideologies Freedom	15
Course Coordinator	J. Gosseye	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Course Contents	What does freedom mean, for whom, and how can it be defined in terms of architecture and urban design? From the early 20th century, many architects have reflected on this question. In the 1930s, Frank Lloyd Wright, for instance, began developing Broadacre City, a utopian urban model that was to allow Americans greater freedom by giving each citizen one acre of land on which they could grow their own crops. In the 1960s and 1970s, numerous conceptual models were developed some of which were (partially) built that sought to give people greater agency in the shaping of their environment. Think, for instance, of John Habraken's theory of Open Building. From the 1980s then, under neoliberalism, which literally signifies new freedom, individual freedom and self-expression were increasingly celebrated, resulting in different kinds of architectural and urban experiments. Around the turn of the 21st century, an increasing number of architects subscribed to the belief that the best response to our contemporary liquid modernity is to design spatial frameworks in which change can occur. Andrea Branzi's Agronica is a good example of this trend. What this short list of examples demonstrates is that the consideration of freedom in architecture and urban design is highly dependent on the place and time in which it emerges, as well as on one's understanding of freedom. This studio will examine what freedom (might) mean in architecture and urban design today, and how our urban environment might change as we weigh individual freedom against civil liberties and the common good. Students will design a work-dwelling complex that offers housing for 50-150 people in an inner-city plot. The final design should be presented in the form of a short story graphic novel that follows one day in the lives of three different occupants/ inhabitants/users of the work-dwelling complex. It should clarify how the design promotes freedom as understood by these protagonists, and how they weigh the importance of (individual) freedom against civil liberties and the common good.	
Study Goals	Upon completion of the course, you will be able to: - Conduct design research and research-by-design; - Position architecture within theoretical, historical, social and contextual frameworks related to architecture and freedom, and define your own position vis-à-vis the conundrum of architecture and freedom; - Communicate design through an alternative, narrative method the graphic novel short story that presents the design in all its aspects: the architectural composition, materialisation and (to a lesser extent) the integration of construction.	
Education Method	The studio will consist of three parts. The first part, Readings and Research will consist of group work; it will comprise four weeks and will focus mainly on gaining an understanding of different interpretations of the concept of freedom. During this first part of the course, students will be given readings that focus on different aspects of freedom in architecture and the built environment. Students will be asked to present a summary of these readings orally during the course. During this first part of the course, students will also be invited to research and analyse reference projects that respond to the notion of 'architecture and freedom'. Identifying, analysing and presenting (the analysis of) these reference projects will also be group work. The second part of the studio, Storylines Protagonists will function as a hinge. It will take up only one week during which each (individual) student will be asked to develop and present the characters and storylines of your three chosen protagonists. These three chosen protagonists should be people that you know and that you interview to sharpen the brief for your own design assignment (i.e. the design of a work-dwelling complex for 50-150 people). The third and final part of the studio, Designing for Freedom, which comprises five weeks, will focus on developing the design based on the three storylines of the chosen protagonists. This third part of the course is also individual work.	
Assessment	- Group presentations of the assigned readings (first part of the course): 20% - Group presentations/submission of reference projects: 20% - Final design: 60% The final design (and deliverable) will be assessed on: - The critical position that is formulated with regard to the brief; - The appropriateness of the design intervention with respect to the assignment; - The conceptual and aesthetic quality of graphic novel, and the consistency and coherence of the narrative (short story) elaborated within it.	
Period of Education	Semester 2, quarter 4. This course will take place on Tuesdays. In some weeks, additional meeting moments might be scheduled on other days to engage in fieldwork, visit reference projects, etc.	
Maximum number of participants	12	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA010	The Living City	5
Course Coordinator	Dr. L.G.A.J. Reinders	
Instructor	Ir. E.I. Ronner	
Contact Hours / Week x/x/x/x	4 hours per week	
Education Period	3	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The MSc2 elective 'The Living City' explores the intersections between architecture and everyday life through experimental methods of notation and visualization. The course locates architecture in the flows, rhythms, cycles and seasons of nature and human life. It explores the assemblage of urban places, the stories of buildings, the lives and flows of people, and the ecology of non-human species. You learn how to do anthropological fieldwork; to read, de-code and analyze a place, to observe people, buildings and spaces, and to translate your findings into visual scripts and movies. To make architecture move!	
Study Goals	Upon completion of the course the student can 1. Organize a fieldwork study and apply different methods of notation, analysis and visual representation. 2. Relate architecture to real life and develop an understanding of the complex relations between humans, buildings and urban spaces. 3. Logically explain and reflect upon the relations between fieldwork, analysis and visual script/movie. 4. Work efficiently and constructively in a collaboration with other students.	
Education Method	Excursion to the site/city. Group work and individual work in the studio Independent design and self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. The maximum marking period is 10 working days. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR2UA020	Urban Architecture MSc2 design studio	15
Course Coordinator	Ir. E.I. Ronner	
Course Coordinator	Drs.ir. E.P.N. Schreurs	
Instructor	Ir. E.I. Ronner	
Responsible for assignments	Drs.ir. E.P.N. Schreurs	
Contact Hours / Week x/x/x/x	12 hours per week	
Education Period	4	
Start Education	4	
Exam Period	none	
Course Language	English	
Expected prior knowledge	It is expected that students have the knowledge from a master 1 design course and the Building Engineering Studios (AR1A080).	
Course Contents	Msc2 material culture Material culture pertains to the physical objects, resources and spaces that people make and use to define their culture. In the context of climate change and the current carbon footprint impact of the building industry, the profession needs a radical and fundamental shift in its building cultures. Architects can use their capacities as 'signifiers' to challenge the industry's building conventions and develop material points of view that offer new solutions and trigger appropriate design motives. While theories of new materialism suggest that the affordances of materials or what they enable should be the starting point for future design, material culture theories add a notion of cultural continuity. The studio will merge both ideas in an assignment that will develop new material attitudes and products from the study of existing examples. This will be done by creating and (as far as possible) implementing elements and details that work from an enhanced understanding of material properties and their cultural values and a meaningful integration of old and new. The precise brief and materials with which the studio will work are still under development, but aims at refreshing conceived ideas while making your hands dirty.	
Study Goals	Upon completion of the course, students can: 1. Analyse existing examples of material applications, reflect on their their potential and architectural motives, and make them applicable to current design challenges. 2. Integrate architectural ideas from theories of new materialism and material culture into an argued position that applies to the design proposal. 3. Produce an elaborated design proposal that treats the different aspects of the assignment in a coherent way and presents the work in with a critical attitude.	
Education Method	Excursion to relevant architectural projects and production places. Group work and individual work in the studio Independent design & self-study	
Assessment	All relevant studies and their presentation are assessed at the end of the semester. Assessment is in accordance with the study goals. Marking will take place within max 10 working days after the assessment. The student has achieved a sufficient result on scale 1 to 10 with 6, has the possibility to take a repair with a mark of 5 or 5,5 and failed with 4,9 or minor. Repair has to be completed within 2 weeks after completion of the course.	
Period of Education	Semester	
Concept Schedule	Tuesday- and Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR8003TU	Legal and Governance	5
Course Coordinator	mr. F.A.M. Hobma	
Instructor	mr.dr. P. Jong	
Responsible for assignments	mr. F.A.M. Hobma	
Contact Hours / Week x/x/x/x	4 hours per week starting from week 1.1. and ending in week 1.8.	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Required for	This course is part of MSc Construction Management and Engineering	
Expected prior knowledge	Bachelor of Science	
Course Contents	This course is about legal & governance aspects of the construction process, especially the institutional legal & framework settings concerning the major players and the main interests concerned. Legal aspects mainly rest in limitations and possibilities created by the legal system, as a context for interactions within and across the public and the private sector boundaries, during the complete life-cycle of the construction process. The relevant legal and institutional frameworks and their underpinnings will be explained and analysed and put into the perspective of 'real-life' problems. Insight into legal reasoning, as a specific methodology, will be practised through the use of legal literature and jurisprudence. The main course topics will deal with both public and private law. The place and position of the future graduate in the construction process will serve as a guideline in the selection of these topics.	
Study Goals	After successfully completing this course, the student is able to: - recognize important aspects of the institutional framework of the construction process and the governance modes that develop within this framework; - recognize relevant legal aspects (within different phases) of the construction process and analyze these in the context of public and private institutional frameworks; - estimate, analyse the weight and complexity of these aspects; - develop ideas to deal with these aspects and provide solutions for problems or conflicts resulting from these aspects in accordance with the fundamental positions of the public and private sectors;	
Education Method	(Video)lectures; assignments; self study without guidance	
Literature and Study Materials	- E.M. Bruggeman et. al. , A Practical Guide to Dutch Building Contracts. Published by: IBR, The Hague. - Hobma, F.A.M., Jong, P. An Instrumental Approach to Planning and Development Law in the Netherlands. Published by: IBR, The Hague. - Journal articles.	
Assessment	This course will be assessed by means of a written exam (70%) and assignments (30%). There is no minimum grade for the assignments. Therefore they cannot be retaken.	
Remarks	This course can not be taken as an elective by students from MSc track Management in the Built Environment (Master Architecture, Urbanism and Building Sciences)	
Period of Education	One quarter	
Concept Schedule	Thursday morning	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB106	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	1 2	
Start Education	1	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

ARB206	The Berlage Sessions	3
Course Coordinator	S.E. Frausto	
Instructor	S.E. Frausto	
Instructor	J.P. Benavides Lavalle	
Education Period	3 4	
Start Education	3	
Exam Period	none	
Course Language	English	
Course Contents	The course is a thematic lecture series focusing on scholarly research and critical approaches to architecture discourse. The aim is to provoke dialogue and debate, engaging students in linking architectural thinking and practice to the history of ideas and changing social and cultural conditions. Invited speakers consist of international emerging and leading practitioners, scholars, and experts. The course is organized by the Berlage Center for Advanced Studies in Architecture and Urban Design. Each semester's theme relates to the education and research programs of the Department of Architecture.	
Study Goals	The main objective is to bring students in contact with leading and emerging practitioners, scholars, and experts. Focusing on scholarly research and critical approaches to contemporary architecture and urban design, the course provokes dialogue and debate. Students are encouraged to link architectural thinking and practice to the history of ideas and changing sociocultural conditions.	
Education Method	The course consists of lectures, followed by discussion between students and the lecturer, in a seminar-like setting.	
Prerequisites	While this course is a post-master's-level course, it is open to select MSc. students as an AR elective course.	
Assessment	Students are required to complete a report, essay, or written equivalent project. In accordance with Article 1.24 of the Berlage Post-Master Teaching and Examination Regulations students are entitled to at least two opportunities to take a written examination, in the period of one academic year.	
Enrolment / Application	For more information, contact Salomon Frausto.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Semester	
Concept Schedule	Friday afternoon	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1000	Python Programming for Geomatics	5
Course Coordinator	Dr.ir. B.M. Meijers	
Responsible for assignments	Dr.ir. B.M. Meijers	
Contact Hours / Week x/x/x/x	6 hours per week (2h lecture / 4h practical)	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	It is assumed that students know how to use command line tools, how to work with files and directories, and know what are the parts of a computer and how they work together.	
Course Contents	This course gives an introduction to the Python programming language and focuses on applications for Geomatics in its assignments. Introduction to programming with (mainly) Python and (some) C++, topics that will be covered: - Interactive mode of the Python interpreter and writing scripts. - Basics of the Python programming language: variables; data types: integer, float, boolean, string; expressions; assignment statements. - Control constructs: if-statements, loop constructions (for, while, range). - Input and output. - Advanced data types: list, tuple, dictionary. - Functions, modules, recursion. - Objects, classes. - Using a development environment. - Main differences between interpreted (e.g. Python) and compiled (e.g. C++) languages.	
Study Goals	After following this course, the student should be able to: 1. explain and use the basic elements of a programming language; 2. describe and give examples of some Object Oriented programming features; 3. translate a (simple) Geomatics related problem into an algorithm; 4. construct a correctly functioning program used in the Geomatics domain; 5. understand the difference between an interpreted and compiled language and explain when to use one or the other.	
Education Method	Lectures, programming assignments, self-study.	
Literature and Study Materials	Think Python: How to think like a computer scientist, available from: http://greenteapress.com	
Assessment	Laboratory assignments (60%) and exam (40%). To pass the course both assignments (with their individual weights) and exam have to be graded sufficient to pass the course (weighted average of part ≥ 5.75). For assignments that are submitted late, points will be removed. A repair option is offered (after the regular exam) for one of the programming assignments, in case the course can be passed by executing this repair option (note, the repaired assignment can get a maximum grade of 6.0). The repair option should be executed within 4 weeks after the regular exam. For the exam, the regular resit exam is available.	
Special Information	The maximum marking period is 10 work days.	
Period of Education	Quarter	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO1002	Geographical Information Systems (GIS) and Cartography	5
Course Coordinator	Dr. G. Agugiaro	
Instructor	A. Petrovi	
Instructor	Dr. D. Cannatella	
Instructor	Dr. G. Agugiaro	
Responsible for assignments	Dr. G. Agugiaro	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	1	
Start Education	1	
Exam Period	1 2	
Course Language	English	
Expected prior knowledge	This is a GIS introductory course, so no previous specific skills are required. It open to students with heterogeneous backgrounds (architecture, urban planning, civil engineering, etc.).	
	However, prior knowledge of scripting/programming in at least one language (e.g. Python) will be of help. For this reason, attending course "GEO1000 - Python Programming for Geomatics" in parallel is encouraged (also considering the following courses within Geomatics MSc).	
Course Contents	The course provides an overview of Geographical Information Systems (GIS) and digital Cartography, and of how GIS can be used in practice to solve real-world problems. The course also provides students with theoretical background knowledge of concepts, data types and GIS-related typical processes and algorithms of GIS packages.	
	The course has both a theoretical and a practical part in which students do exercises to get hands-on experience with GIS packages. The open-source software QGIS and GRASS GIS/SAGA GIS packages and FME by Safe Software are used for this purpose.	
	The course has 3 parts:	
	1. Introduction to GIS	
	- fundamentals of Geodesy, Coordinate Reference Systems, and map projections,	
	- spatial data modelling (vector and raster spatial models),	
	- geo-data manipulation (editing, digitizing, importing, converting, etc.),	
	- overview of spatial analysis operations,	
	- production of interpretable output (e.g. maps),	
	- fundamentals of data quality.	
	2. Algorithms and data structures for GIS	
	- data structures for vector and raster data (including topological data structures),	
	- basic algorithms for vector (point-in-polygon, Boolean operations, intersection, area, etc.),	
	- basic algorithms for raster (encoding, quad trees, map algebra),	
	- networks and related algorithms such as shortest-path.	
	3. Applications of GIS to real-world problems	
	Real-world problems related to the built environment (e.g. urban spatial analyses, estimation of energy demand for buildings, determination of noise impact due to construction of infrastructure, etc.) will be solved with the help of GIS packages.	
Study Goals	After the course Introduction to GIS and digital cartography the student will be able to:	
	1) Explain what a GIS is and what real-world problems it can help solve;	
	2) Describe the quality aspects of geo-datasets and compare the two conceptualisations of space (field versus objects), and how these are modelled in a GIS;	
	3) Use a GIS to visualise, convert and analyse geographical datasets coming from different sources;	
	4) List the main spatial data structures and algorithms used in GIS, compare and discuss them in terms of applicability depending on the problem to be solved;	
	5) Explain and analyse what the basic spatial operations are and consist of, and how they are performed;	
	6) Generalise the GIS knowledge to solve more complex spatial problems by integrating existing tools and developing tailored solutions/workflows.	
Education Method	Lectures: 26 hours; Labs (supervised individual and group hands-on exercises): 20 hours; Self-study: 94 hours	
Literature and Study Materials	- Book: Principles of Geographical Information Systems	
	- Slides of the lectures (available on Brightspace)	
	- Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	Written exams (1 mid-term quiz + 1 final exam), 2 graded assignments (practicals with a GIS package). The exam, the quiz and the assignment are graded each between 1.0 and 10.0.	
	The final grade is the average sum using these weights: 5% Quiz, 20% Assignment 1, 25% Assignment 2, 50% Final written exam. The final grade is rounded to the nearest half-point.	
	A student passes the course only if the following two conditions are satisfied:	
	1. The average grade of both graded assignments is ≥ 5.75 . If a student does not pass, a supplementary retake assignment will be handed out after the end of the course with 3 weeks time to submit it.	
	AND	
	2. The grade of the final exam is ≥ 5.75 .	
	The grades of the assignments are valid for 1 academic year, i.e. till (but not including) the beginning of the course in the following academic year.	
Period of Education	Quarter 1	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5012	Land Administration	5
Course Coordinator	Prof.dr.ir. P.J.M. van Oosterom	
Course Coordinator	mr.dr. H.D. Ploeger	
Instructor	Prof.dr.ir. P.J.M. van Oosterom	
Instructor	mr.dr. H.D. Ploeger	
Responsible for assignments	Prof.dr.ir. P.J.M. van Oosterom	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This course gives an introduction to the field of land administration. Proper Land Administration (LA) is considered to be a condition for sustainable economic development in a country. Land Administration Systems (LAS) serve different purposes, such as legal security (of ownership), fair valuation/taxation, land use planning, etc. In some countries there are multiple centuries of LAS track record (via Public Registers, Cadastral Maps, etc.). In other countries LAS is still not available, as implementation is non-trivial due to finding the right legal, organizational and technical solutions for the different components. The role of Geomatics (survey, spatial data management, updating and map editing, data dissemination) in LAS is very significant. Often in a single country multiple organizations (ministries, agencies or other authorities) are involved in land administration. The world of land administration is getting more international and information is used beyond country boundaries; e.g. INSPIRE cadastral parcels. This implies that land administration standards are crucial for interoperability. The key concepts (person/party, right, restriction, spatial unit/parcel, boundary, legal and spatial source documents) are described in a conceptual information model (UML domain model). The (international) standards are based on established practices and are a good starting point for renewal or initial implementation of LAS. The Land Administration Domain Model (LADM, ISO 19152) will therefore be discussed in detail. Due to automation and digitalization, LAS implementations have changed a lot over the last couple of decades. Also because of ever evolving societal needs, and increasing technological possibilities, the field is continuously faced with new challenges, such as 3D Cadastral registration (to name just one example). In this course students will learn about the organizational, legal and technical aspects of land administration. Based on a system approach, i.e. a study of a system with emphasis on the relations between its elements and the common goal, they will be able to perform an assessment of a specific LAS and to make proposals for development and further improvement.	
Study Goals	After the course the student is able to: - Describe and explain the underlying legal principles, institutional arrangements, and technology of Land Administration Systems (LAS); - Assess, based on a systems approach, the practical value of a specific (national) LAS and possibly propose improvements; - Interpret the data from a LAS in a specific case, and give a reasoned opinion about the meaning and value of this information on rights, restrictions and responsibilities in this specific case; - Explain the increasing importance of novel developments such as 3D Land Administration, international standards (ISO19152, the Land Administration Domain Model), integration of land registration, valuation and spatial plan information, and support of sustainable development.	
Education Method	- 14 Interactive lectures (28 hours) - 14 Blended learning and reading assignments (42 hours) - 4 Assignments (each 8 hours finish/report, in total 32 hours) - Self-study and exam preparation (35 hours)	
Assessment	20% Four assignments (no retake possible), 80% Written exam (3 hours).	
Period of Education	Quarter 3.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5014	Geomatics as support for energy applications	5
Course Coordinator	Dr. G. Aguiaro	
Instructor	mr. C.A. León Sánchez	
Responsible for assignments	Dr. G. Aguiaro	
Contact Hours / Week x/x/x/x	6 hours per week, from (first week of Q1) till (last week of Q1)	
Education Period	1	
Start Education	1	
Exam Period	1	
Course Language	English	
Expected prior knowledge	GEO1004 (3D Modelling of the built environment) or equivalent GEO1006 (Geo-database management systems) or equivalent	
Course Contents	Semantic 3D city models play globally an increasing role as hubs of harmonised and integrated information that include both spatial and non-spatial data. For certain applications, the geo-spatial dimension plays a fundamental role. In bottom-up approaches, for example, several data regarding buildings are used and processed to further compute meaningful characteristics and KPIs (Key Performance Indicators) about the building stock. Such data can be then further aggregated up to the district or city level and so on. Depending on the specific application, the spatial dimension can play an important role. A simple list of examples could be: the computation of the volume of complex buildings, the computation of shared walls between adjacent buildings, the estimation of the solar irradiance on the roof surfaces or on the building façades. In such cases, 3D data is needed, as the usual 2D data that may not suffice. The course will focus on the use of 3D city models, based on the international standard CityGML, as support for energy-related applications in the framework of the energy transition. A non-exhaustive list of possible applications is: - Bottom-up approaches for estimation of energy performance of buildings - Coupling of 3D city models with specific simulation tools - Assessment of photovoltaic potential at urban scale - Integration with supply networks (e.g. gas, district heating, etc.) - Data modelling, definition and testing of (energy-related) data standards. The course has both a theoretical and a practical part. Every year, a specific topic will be selected and treated during the course. Every year, depending on the selected topic, the necessary theoretical background will be provided during lectures.	
Study Goals	The overall objectives of the course are: 1) Understand the main concepts of the international standard CityGML and its extension mechanisms, e.g. via ADEs (Application Domain Extensions) 2) Understand the requirements in terms of data to develop energy-related applications at urban scale based on semantic 3D city models 3) Depending on the selected topic, (re)use or implement algorithms that allow to solve a specific problem 4) Depending on the selected topic, couple existing simulation software tools with a semantic 3D city models by defining and implementing bi-directional data interfaces. After the course the student will be able to: 1) Understand the fundamental requirements for urban energy modelling 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model based on CityGML 3) Use (and, if needed, adapt) software tools to generate, store and visualise 3D city models 4) Depending on the specific application, implement the required procedures or, alternatively, define a proper interface between the 3D city model and the simulation tool 5) Apply the acquired knowledge to set up and run a proper simulation environment to solve a specific problem 6) Gather and analyse the simulation results, and possibly make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours. Students are encouraged to work in groups during the laboratories.	
Literature and Study Materials	- Slides of the lectures (available on Brightspace) - Hand-outs (available on Brightspace) - Additional selected book chapters or scientific articles (available on Brightspace)	
Assessment	2 graded assignments, one final oral exam. The assignments and the oral exam are graded each between 1 and 10. The final grade is the averaged using these weights: 30% Assignment 1, 35% Assignment 2, 35% final oral exam. The final grade is rounded to the nearest half-point (i.e. ..., 6.5, 7.0, ..., etc.). It must be ≥ 5.75 to pass the course. Following conditions apply: - The average grade of both Assignments must be ≥ 5.5 to pass. If not, a supplementary resit assignment will be handed out after the end of the course with 3 weeks time to submit it (i.e. approximately in the second half of November with deadline in December). - The grade of the final oral exam must be ≥ 5.75 to pass. If not, there is a resit exam, also oral, generally at the end of Q2 (i.e. in January) - The grades of the assignments are valid for 1 year, i.e. till (but not including) the beginning of the course in the following academic year. - If a student passes the final oral exam, it is still possible to take the resit. For the final grade, the higher of the two grades is used.	
Period of Education	Quarter 1 of the second year of MSc Geomatics	
Concept Schedule	Each lecture/lab session takes place during a two-hour time slot, three times per week. There are, on average, 2 lectures on theory and 1 lecture with hands-on practicals per week.	
Minimum number of participants	15	
Maximum number of participants	30	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5015	Modelling wind and dispersion in urban environments	5
Course Coordinator	C. Garcia Sanchez	
Responsible for assignments	C. Garcia Sanchez	
Education Period	4	
Start Education	4	
Exam Period	4	
Course Language	English	
Expected prior knowledge	Knowledge of one programming language is required (e.g. Java, Python, etc.). Prior knowledge of linux and calculus would be an advantage. In any case, at the beginning of the course, the average knowledge of the students will be evaluated in order to adjust the level of the course accordingly.	
Course Contents	The course focuses on the modelling of winds and dispersion around 3D city models. The goal is to further the students experience in geomatics knowledge by learning tools with direct application to real urban scenarios. The course covers the necessary fundamentals of fluid dynamics and computational fluid dynamics methodologies to perform simulations in urban environments. The simulations will cover wind predictions and dispersion of pollutants, which can play an essential role when designing and/or improving urban areas to assess and ensure urban sustainability, liveability, energy efficiency and comfort.	
Study Goals	After the course the student is able to: 1) Understand the fundamental requirements for wind and dispersion simulations; 2) Perform data requirement analysis for the modelled phenomenon starting from (but not limited to) a semantic 3D city model; 3) Apply the acquired knowledge to set up and run a correct simulation environment to analyze wind and dispersion in an urban area; 4) Gather and analyse the simulation results, and make them available for further applications.	
Education Method	Lectures and laboratories: 6h/week; Self-study: 92 hours	
Reader	Slides of the lectures (available on Brightspace); Handouts (available on Brightspace); Additional selected book chapters or scientific articles on specific topics (available on Brightspace).	
Assessment	The assessment consists of 2 hands-on assignments: one individual assignment (25% of the overall course grade) and one group assignment (75% of the overall course grade). Retakes are possible for independent assignments.	
Period of Education	Quarter	
Concept Schedule	Tuesday, Wednesday and Thursday	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

GEO5017	Machine Learning for the Built Environment	5
Course Coordinator	Dr. L. Nan	
Course Coordinator	Ir. S. Du	
Course Coordinator	N. Ibrahimli	
Responsible for assignments	Ir. S. Du	
Contact Hours / Week x/x/x/x	6 hours per week	
Education Period	3	
Start Education	3	
Exam Period	3 4	
Course Language	English	
Course Contents	This is an introductory course for machine learning to equip students with the basic knowledge and skills for further study and research of machine learning. It introduces the theory/methods of well-established machine learning and state-of-the-art deep learning techniques for processing geospatial data (e.g., point clouds). The students will also gain hands-on experiences by applying commonly used machine learning techniques to solve practical problems through a series of lab exercises and assignments. The topics of the course include: - Introduction to machine learning [-] Applications of machine learning [-] The scope of machine learning -) Regression vs classification -) Supervised learning vs unsupervised learning [-] Limits and dangers of machine learning - Clustering [-] K-means [-] Hierarchical [-] Density-based - Linear regression [-] Closed-form solution [-] Solution via optimization [-] Gradient descent - Classification [-] K-nearest neighbors [-] Bayesian classification [-] Logistic regression [-] Support vector machine (SVM) -) Maximum margin classification -) Soft-margin SVM [-] Decision trees and random forest - Neural networks [-] Multi-layer perception [-] Backpropagation - Deep learning (focusing on CNN) [-] Convolution [-] CNN architecture	
Study Goals	After the course, the students will be able to: - explain the impact, limits, and dangers of machine learning; give use cases of machine learning for the built environment; - explain the main concepts in machine learning (e.g., regression, classification, unsupervised learning, supervised learning, overfitting, training, validation, cross-validation, learning curve); - explain the principles of well-established unsupervised and supervised machine learning techniques (e.g., clustering, linear regression, Bayesian classification, logistic regression, SVM, decision tree, random forest, and neural networks); - preprocess data (e.g., labelling, feature design, feature selection, train-test split) for applying machine learning techniques; - select and apply the appropriate machine learning method for a specific geospatial data processing task (e.g., object classification); - evaluate the performance of machine learning models.	
Education Method	Lectures, reading materials, assignments, lab exercises, and (optionally) student presentation.	
Course Relations	This course is closely related to other Geomatics courses in geospatial data acquisition, data management, data visualization, data analysis, and 3D modeling.	
Literature and Study Materials	- Christopher Bishop. Pattern recognition and machine learning. Springer. 2006. - Kevin Murphy. Machine Learning: A Probabilistic Perspective. MIT Press. 2012 - Ian Goodfellow, Yoshua Bengio, and Aaron Courville. Deep Learning. MIT Press. 2016. - Lecture notes (will be distributed during the course).	
Practical Guide	In the assignments and lab exercises, students will experiment with the machine learning techniques introduced in the lectures using popular python frameworks and tools (e.g., Jupyter notebook, scikit-learn, NumPy, Matplotlib, PyTorch). - Each assignment will be announced when the related lectures are delivered; - Be creative with experiments; try different scenarios and discuss the pros and cons; discuss the effect of parameters (if possible). - For group assignments, discussions between groups are highly encouraged; - The report should include a short description of 'who did what' and a brief reflection on how the feedback received from others improves the work.	
Prerequisites	- Linear algebra - Calculus (a good understanding of partial derivative and gradient is necessary) - Basic probability or statistics - Python programming language	
Assessment	The assessment of this course consists of two group assignments and the final exam. The final grade is based on the evaluation of both the assignments and the final exam, i.e., - Group assignments (40%). All assignments have equal weight in the final grade. It is possible to resubmit your work after incorporating the feedback/suggestions received from the teachers. However, the evaluation of an assignment is mainly based on the first submission. Students who have improved their work may receive a slightly higher grade depending on the significance of the improvement (but no more than 5%). Assignments submitted after the deadline: the grade will be deducted with 10% per	

	day late, and not accepted after 3 days late. - Final exam (60%). The final exam consists of multiple-choice questions and open questions. Example questions will be given two weeks before the exam. Grading is done on a 10-point scale, and the final marks are rounded off to the nearest full and half figures. To pass the course, all assignments and the final exam must be graded greater or equal to 5.5. A total of 5.75 or above is required to pass the course. Repair and retake options: any component of the assessment graded 5.5 or lower is eligible for one more opportunity to retake (for the exam) or repair (for assignments). A repair is a revision or addition to an existing assignment, and a successful repair will only be assessed with a 6.0. A retake is an entirely new examination.
Elective	Yes
Tags	Artificial intelligence
Period of Education	Q3
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

GEO5018	Ethics for the Data-driven City	5
Course Coordinator	Dr. S. Calzati	
Course Coordinator	mr.dr. H.D. Ploeger	
Responsible for assignments	mr.dr. H.D. Ploeger	
Contact Hours / Week x/x/x/x	4 hours per week, from week 4.1 to 4.8	
Education Period	4	
Start Education	4	
Exam Period	4 5	
Course Language	English	
Course Contents	This course discusses the main principles of data ethics and the relevance of applied ethics in the domain of the data-driven city to answer the main question of how to safeguard the human dimension in a data-driven world. Eight billion people and their devices produce unimaginable amounts of data, seven days a week, 24 hours a day. Data on all aspects of human existence, that promise, in combination with increasingly powerful computers and the application of AI, mankind the opportunity to influence all aspects of life. This will be especially the case in the urban setting. The concept of the 'smart city' assumes that the flows of data that the city and its residents produce will lead to better (after all smart) solutions for designers, industry and, at the end, citizens. For example, the smart city will be a better planned city, smart mobility will allow for efficient and safe use of space and time, and a smart home will be able to cater optimally the daily needs of the user, based on the personal profile as determined on the basis of the available data. However, these technology driven perspectives leave the question open how we want the life in Urbiadata, the data-driven city of 2050, to look like? The choices we make today determine our world of tomorrow. Taking this perspective it is clear that the development of our data-driven future cannot be just a question of engineering and the economic models supporting the solutions. Should all possible development and use of data-driven solutions (e.g. by architects and urban planners) be permissible in the light of the consequences? How about values as autonomy, privacy, transparency, consent and fairness? Although the posed question of the interpretation of the data-driven city appears at first sight a question of policy, which could lead (or not) to the creation of a normative (legal) framework setting standards via enforceable rules, a crucial discussion should precede this: what are the ethical concepts and principles that guide us in determining what solutions are right or wrong? Data ethics are needed to be able to answer the crucial question for people living and working in Urbiadata: how to safeguard the human dimension in a data-driven world?	
Study Goals	After this course, the student is able: - To explain the relevance of applied ethics in relation to the design and governance of the data driven city in general - To identify concrete ethical issues within the field of the use of data in a concrete situation - To identify and apply the relevant ethical principles for this concrete situation. - To create a framework for the application of technology in the data-driven city and to formulate a substantiated standpoint regarding the admissibility of this technology.	
Education Method	12 Interactive lectures and workshops. Part of the workshops is a short presentation, one by each student, about a (scientific) paper to be chosen from the reading material included in the syllabus. All students are expected to participate in the discussion. (24 hours) Self-study, Individual assignments (60 hours) Design of the artefact and writing of the report containing the explanation and reflection. The artefact must provide an answer on the central question of this course: how to design ethical principles for the data-driven city and how to design a data-driven city based on those principles (56 hours)	
Assessment	The student will design an artefact and submit a written report (providing explanation of the artefact and containing a reflection). The course is concluded with an oral exam. The artefact can be a video (min 5 minutes, max 10 minutes), an animation (min. 5 minutes, max. 10 minutes) or a model. The final grade is constructed as follows: artefact: 40%, written report: 40%, oral exam: 20%. Students need to score at least a 5.75 grade for both the artefact, written report and oral exam, and need to give one presentation on an assigned topic in order to pass the course. Only for the written report a repair option is available. A successful repair will result in a maximum grade of 6.0 for the repaired written report. The repair should be submitted within 4 weeks after the regular exam.	
Period of Education	Quarter 4	
Maximum number of participants	20	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 Urbanism

AR3U105	Graduation Orientation	3
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	8 hours per week in week 1 and 2, and 4 hours per week starting in week 3 ending in week 9.	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	Within this Graduation Orientation course, students will connect their graduation project research to the main research questions of the (cross-cutting) themes of the Urbanism research programme.	
Course Contents	Within the Graduation Orientation course, students are asked to develop a first, preliminary proposal for their graduation project and to propose their studio of preference and their 1st and 2nd mentor of preference. Students will be connected to the Urbanism graduation studios based on their individual proposal. Students will be asked to critically reflect on the variety of urbanism research themes and to position their intended graduation project within.	
Study Goals	At the end of this course, the student is able to: develop a first proposal for his/her graduation project. develop an overview of the different research themes in the department of Urbanism, and is able to state his/her own position within. provide clear arguments of the connection between his/her own graduation topic, the Urbanism research program, and the preferred graduation studio(s). communicate his/her first proposal for his/her graduation project in an academic way. demonstrate skills of researching and academic writing.	
Education Method	Seminars and interactive peer-feedback (and expert-reflection) sessions. Combination of mainly individual but also partially group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment. A rubric will be used for grading. The rubric will be available in the year guide on the course specific Bright Space page. The assignment, developing a first proposal for his/her graduation project, will be assessed in two ways. The course coordinator team will assess the assignment according to the format (downloadable from Brightspace), based on the following criteria: Critical reflection skills on the positioning within the Urbanism research portfolio Academic writing and referencing Scientific and societal relevance Reflection on ethical issues The proposed studio coordinator or main mentor will assess the assignment based on the following criteria: Clarity of motivation, problem (field) statement, aim of study, project approach In depth analysis and underpinning of the project/thesis proposal Appropriateness (of argumentation) for the proposed main mentor and studio As this assignment is a start to build towards P1/P2, the grading will be done by a Fail, Repair, or Pass. Students should achieve a minimum grade of Pass to pass this course. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before P2. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3 (intensive first two weeks).	
Concept Schedule	The sessions are scheduled on Mondays and Thursdays / Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3U110	Graduation Exploration	12
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	12 hours per week in week 3 and 4 and week 6 and 7.	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course consists of 2 intensives/workshops. In the Graduation Orientation course students define their graduation topic, find a studio and 1st mentor. The studio choice defines which studio intensive the student will join. In this studio intensive the essential topics for the specific studio are addressed. Together with their 1st mentor and studio coordinator students select which other intensive is most fit for their graduation project. These intensives focus on a specific skill, technique, approach or method. Students used the gained knowledge of the intensives to sharpen their graduation project.	
Study Goals	At the end of this course, the student: - is able to analyse the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. - is able to evaluate the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. - is able to reflect on the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. - is able to use the content of the intensives to create and polish his/her graduation project. - shows social awareness and professional responsibility.	
Education Method	Intensives/workshops of 2 weeks each. Combination of individual and group work	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus for some intensives/workshops design examination. A rubric will be used for grading. The rubric will be available in the year guide on the course specific Bright Space page. Each intensive results in an assignment. The combination of and reflection on these assignments results in a final assignment which supports the more precise definition of the graduation project. The assignments together result in 1 final grade. Students should achieve a minimum grade of 6,0 to pass this course. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before P2. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3.	
Concept Schedule	The sessions might be scheduled on all day parts of the week. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3U115	Graduation Lab Urbanism	15
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Hartevelde	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	B. Hausleitner	
Instructor	T. Kuzniecowa Bacchin	
Instructor	Dr. C. Furlan	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	MSc1 & 2 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the first part of the graduation laboratory; students (individually) develop their graduation project.	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the first part of the graduation laboratory; students (individually) develop their graduation project. After getting to know the main Urbanism research topics and research questions, the student works in content-driven studios with specialized researchers. The studios provide activities for the students, such as lectures, peer review sessions, master classes dedicated to improve students skills and lectures, symposiums and workshops organised by students themselves according to their own interests.	
Study Goals	At the end of this course, the student: - is able to describe and map the problem field of his graduation work on the basis of a motive, fascination, or question (Problem field). - is able to define a relevant field of graduation objectives, concerning research questions and design tasks (Field of graduation objectives). - is able to define an approach, with specific methods, techniques and design instruments for the graduation work (design and research), based on the results of the Master 3 AR3U013 course, which suits the objectives, the design task and the research questions. (Approach). - is able to present a consistent and adequately constructed theoretical framework for the graduation topic, based on the results of the Master 3 course AR3U023 (Theoretical framework). - is able to define and describe the project location and design task, together with an urban analysis, in line with the formulated problem field (Design and research location). - is able to define the in-between and end products appropriate for the aim of the graduation project (In-between and end products). - is able to put forward arguments on how the graduation work will provide a substantial contribution to society and science (Relevance). - is able to present a first concept or hypothesis, in which a first solution or direction for the design task or the main question is embedded (Concept);. - is able to provide the agreed time frame with the formulated in-between and end products (Planning).	
Education Method	See the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the master track of Urbanism. In this quarter, and according to their topic of interest, students will be assigned to work in studios where they will work closely with researchers specialized in their specific topic. Students will choose their first mentor from this studio, and a second mentor from a different section.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated two times: at the first Evaluation: Compulsory Progress Review (P1) at the second Evaluation: Formal Assessment (P2) A rubric (EMMA) will be used for grading. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities.	
Special Information	On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.	
Remarks	The maximum marking period is 15 working days. The final product of this course is the still growing thesis report, an integral product that the student has to deliver at the end of Master 3, 1 week prior to the P2 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences and the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the Master track of Urbanism.	
Period of Education	Semester 1 and semester 2. The MSc3 Urbanism courses start both in September and February. The majority of students starts the MSc3 in September. In February just one studio will be on offer, the number of intensives will be reduced and the number of available mentors will be	

Concept Schedule	limited. The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 4 Urbanism

AR4U010	Graduation Lab Urbanism	30
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	B. Hausleitner	
Instructor	T. Kuzniecowa Bacchin	
Instructor	Dr. C. Furlan	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	1 hours per week starting from week 1 and ending in week 9.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	MSc1, 2 & 3 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Study Goals	At the end of this course, the student: - is able to describe and carry out research in the field of urbanism and process the research results as well as use drawings/maps/graphics as a means to research; - is able to describe the problem field of the selected topic and translate it into a field of graduation objectives (and design task) with associated research questions and research approach; - is able to describe a clear theoretical framework which is appropriate for the selected topic; - is able to carry out research by design in a methodological way focused on the research questions; - is able to process the research results in the final report adequately: i.e. formulated and/or imagined by means of analytical drawing(s); - is able to use drawings / maps / graphics as a means to research; - is able to define urban design methods, choices, aspects, effects and consequences by means of plan forms and design-instruments; - is able to define and visualised the own working method(s) and the (design) choices within the design process with sound arguments; - is able to define the spatial, functional, technical, and/or social aspects of the design adequately: clear, transparent and with a proper justification; - is able to use plan forms and design-instruments which suit his/her design task; - is able to describe, imagined and justified the effects and consequences of the design proposal(s) with respect to the aimed field; - is able to draw conclusions and define recommendations; - is able to evaluate the urban design and research aims in the conclusions; - is able to indicate clearly and logically which research questions are answered and how that has been processed in the urban design; - is able to define clear, concrete, specific recommendations based on the results of the urban research and/or design; - is able to specify for which questions is still additional (design) research necessary; - is able to show an analytical capacity to present a complex matter in a brief and concise way; - is able to describe the projects relevance, reflect on the products and present these; - is able to describe clearly the innovative (scientific and/or social) insights of the graduation project in text and images and, if necessary, concrete strategies and/or application possibilities for the field of the urbanism; - is able to position the own graduation project with respect to the field of the urbanism, as well as other adjacent scientific fields; - is able to discuss and present the thesis products (urban design & research) and the process (design & research) in the form of an epilogue or evaluation. In short, the student has to show that he/she is able to deliver a project of professional quality and of academic level in line with the end terms of the master track Urbanism.	
Education Method	In this semester students work individually, supervised by two mentors, within the frame of the studio and graduation lab of Urbanism. Meetings and lectures are less frequent than in the previous semester, the independency of the student is enlarged.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated three more times: at the third Evaluation: Compulsory Progress Review (P3) at the fourth Evaluation: Formal Assessment (P4) at the final Evaluation: Public Final Presentation (P5) A rubric (EMMA) will be used for grading.	

Special Information	The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities. On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.
Remarks	The maximum marking period is 15 working days. The final product of this course is an integral concise product that the student has to deliver at the end of Master 4, 1 week prior to the P5 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences, the 'Exit Qualifications' of the master Architecture, Urbanism and Building Sciences, the evaluation form 'Graduation Criteria Urbanism P3' and the assessment criteria of the 'Graduation Criteria Urbanism P3/P4/P5' of the Master track of Urbanism.
Period of Education	Semester 1 and semester 2. MSc3 starts in quarter 1 and 3. The AR4U010 course starts automatically after successfully finishing the MSc3 with a GO at the P2 presentation.
Concept Schedule	The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

Cross Domain City of the Future

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 3 Cross Domain City of the Future

AR3U105	Graduation Orientation	3
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	8 hours per week in week 1 and 2, and 4 hours per week starting in week 3 ending in week 9.	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Summary	Within this Graduation Orientation course, students will connect their graduation project research to the main research questions of the (cross-cutting) themes of the Urbanism research programme.	
Course Contents	Within the Graduation Orientation course, students are asked to develop a first, preliminary proposal for their graduation project and to propose their studio of preference and their 1st and 2nd mentor of preference. Students will be connected to the Urbanism graduation studios based on their individual proposal. Students will be asked to critically reflect on the variety of urbanism research themes and to position their intended graduation project within.	
Study Goals	At the end of this course, the student is able to: - develop a first proposal for his/her graduation project. - develop an overview of the different research themes in the department of Urbanism, and is able to state his/her own position within. - provide clear arguments of the connection between his/her own graduation topic, the Urbanism research program, and the preferred graduation studio(s). - communicate his/her first proposal for his/her graduation project in an academic way. - demonstrate skills of researching and academic writing.	
Education Method	Seminars and interactive peer-feedback (and expert-reflection) sessions. Combination of mainly individual but also partially group work.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment. A rubric will be used for grading. The rubric will be available in the year guide on the course specific Bright Space page. The assignment, developing a first proposal for his/her graduation project, will be assessed in two ways. The course coordinator team will assess the assignment according to the format (downloadable from Brightspace), based on the following criteria: - Critical reflection skills on the positioning within the Urbanism research portfolio - Academic writing and referencing - Scientific and societal relevance - Reflection on ethical issues The proposed studio coordinator or main mentor will assess the assignment based on the following criteria: - Clarity of motivation, problem (field) statement, aim of study, project approach - In depth analysis and underpinning of the project/thesis proposal - Appropriateness (of argumentation) for the proposed main mentor and studio As this assignment is a start to build towards P1/P2, the grading will be done by a Fail, Repair, or Pass. Students should achieve a minimum grade of Pass to pass this course. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before P2. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3 (intensive first two weeks).	
Concept Schedule	The sessions are scheduled on Mondays and Thursdays / Friday mornings or afternoons. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3U110	Graduation Exploration	12
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	12 hours per week in week 3 and 4 and week 6 and 7.	
Education Period	1 3	
Start Education	1 3	
Exam Period	1 3	
Course Language	English	
Course Contents	This course consists of 2 intensives/workshops. In the Graduation Orientation course students define their graduation topic, find a studio and 1st mentor. The studio choice defines which studio intensive the student will join. In this studio intensive the essential topics for the specific studio are addressed. Together with their 1st mentor and studio coordinator students select which other intensive is most fit for their graduation project. These intensives focus on a specific skill, technique, approach or method. Students used the gained knowledge of the intensives to sharpen their graduation project.	
Study Goals	At the end of this course, the student: - is able to analyse the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. - is able to evaluate the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. - is able to reflect on the essential topics addressed by the specific studio, the specific focus addressed in the selected focus intensive and the specific integrative approach addressed in the selected integrative intensive. - is able to use the content of the intensives to create and polish his/her graduation project. - shows social awareness and professional responsibility.	
Education Method	Intensives/workshops of 2 weeks each. Combination of individual and group work	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course.	
Assessment	Writing assignment, oral examination plus for some intensives/workshops design examination. A rubric will be used for grading. The rubric will be available in the year guide on the course specific Bright Space page. Each intensive results in an assignment. The combination of and reflection on these assignments results in a final assignment which supports the more precise definition of the graduation project. The assignments together result in 1 final grade. Students should achieve a minimum grade of 6,0 to pass this course. A Repair is provided when a specific part of the assignment is missing or insufficient. The student receives a well described repair assignment. The assignment must be handed in by the student before P2. The result of the repair assignment may only be Pass or Fail. In case of Pass a V is registered in Osiris. In case of a Fail the student needs to reenroll for the course.	
Remarks	The maximum marking period is 15 working days.	
Period of Education	Quarter 1 and quarter 3.	
Concept Schedule	The sessions might be scheduled on all day parts of the week. The actual schedule will be available via Bright Space > My Timetable.	
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.	
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.	
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).	

AR3U115	Graduation Lab Urbanism	15
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Hartevelde	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	B. Hausleitner	
Instructor	T. Kuzniecowa Bacchin	
Instructor	Dr. C. Furlan	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	8 hours per week starting from week 1 and ending in week 9.	
Education Period	2 4	
Start Education	2 4	
Exam Period	2 4	
Course Language	English	
Expected prior knowledge	MSc1 & 2 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the first part of the graduation laboratory; students (individually) develop their graduation project.	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the first part of the graduation laboratory; students (individually) develop their graduation project. After getting to know the main Urbanism research topics and research questions, the student works in content-driven studios with specialized researchers. The studios provide activities for the students, such as lectures, peer review sessions, master classes dedicated to improve students skills and lectures, symposiums and workshops organised by students themselves according to their own interests.	
Study Goals	At the end of this course, the student: - is able to describe and map the problem field of his graduation work on the basis of a motive, fascination, or question (Problem field). - is able to define a relevant field of graduation objectives, concerning research questions and design tasks (Field of graduation objectives). - is able to define an approach, with specific methods, techniques and design instruments for the graduation work (design and research), based on the results of the Master 3 AR3U013 course, which suits the objectives, the design task and the research questions. (Approach). - is able to present a consistent and adequately constructed theoretical framework for the graduation topic, based on the results of the Master 3 course AR3U023 (Theoretical framework). - is able to define and describe the project location and design task, together with an urban analysis, in line with the formulated problem field (Design and research location). - is able to define the in-between and end products appropriate for the aim of the graduation project (In-between and end products). - is able to put forward arguments on how the graduation work will provide a substantial contribution to society and science (Relevance). - is able to present a first concept or hypothesis, in which a first solution or direction for the design task or the main question is embedded (Concept);. - is able to provide the agreed time frame with the formulated in-between and end products (Planning). See the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the master track of Urbanism.	
Education Method	In this quarter, and according to their topic of interest, students will be assigned to work in studios where they will work closely with researchers specialized in their specific topic. Students will choose their first mentor from this studio, and a second mentor from a different section.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated two times: at the first Evaluation: Compulsory Progress Review (P1) at the second Evaluation: Formal Assessment (P2) A rubric (EMMA) will be used for grading. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities.	
Special Information	On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.	
Remarks	The maximum marking period is 15 working days. The final product of this course is the still growing thesis report, an integral product that the student has to deliver at the end of Master 3, 1 week prior to the P2 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences and the assessment criteria of the 'Graduation Criteria Urbanism P1/P2' of the Master track of Urbanism.	
Period of Education	Semester 1 and semester 2. The MSc3 Urbanism courses start both in September and February. The majority of students starts the MSc3 in September. In February just one studio will be on offer, the number of intensives will be reduced and the number of available mentors will be	

Concept Schedule	limited. The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

Year	2023/2024
Organization	Architecture and the Built Environment
Education	Master Architecture, Urbanism and Building Sciences

MSc 4 Cross Domain City of the Future

AR4U010	Graduation Lab Urbanism	30
Course Coordinator	Ir. K.P.M. Aalbers	
Course Coordinator	Dr. L.M. Calabrese	
Instructor	Dr.ir. M.G.A.D. Harteveld	
Instructor	Dr. D.A. Sepulveda Carmona	
Instructor	V.E. Balz	
Instructor	B. Hausleitner	
Instructor	T. Kuzniecowa Bacchin	
Instructor	Dr. C. Furlan	
Responsible for assignments	Dr. L.M. Calabrese	
Contact Hours / Week x/x/x/x	1 hours per week starting from week 1 and ending in week 9.	
Education Period	1 2 3 4	
Start Education	1 2 3 4	
Exam Period	1 2 3 4	
Course Language	English	
Expected prior knowledge	MSc1, 2 & 3 of the Master track Urbanism.	
Summary	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Course Contents	The second year of the Master track Urbanism program consists of two semesters, Master 3 (30 credits) and Master 4 (30 credits), both dedicated to the graduation project. These semesters give the student therefore a unique opportunity to do an in-depth research and design project in the field of urbanism. This course is the second part of the graduation laboratory; students continue (individually) developing their graduation project that was approved at the end of the previous semester (GO for P2).	
Study Goals	At the end of this course, the student: - is able to describe and carry out research in the field of urbanism and process the research results as well as use drawings/maps/graphics as a means to research; - is able to describe the problem field of the selected topic and translate it into a field of graduation objectives (and design task) with associated research questions and research approach; - is able to describe a clear theoretical framework which is appropriate for the selected topic; - is able to carry out research by design in a methodological way focused on the research questions; - is able to process the research results in the final report adequately: i.e. formulated and/or imagined by means of analytical drawing(s); - is able to use drawings / maps / graphics as a means to research; - is able to define urban design methods, choices, aspects, effects and consequences by means of plan forms and design-instruments; - is able to define and visualised the own working method(s) and the (design) choices within the design process with sound arguments; - is able to define the spatial, functional, technical, and/or social aspects of the design adequately: clear, transparent and with a proper justification; - is able to use plan forms and design-instruments which suit his/her design task; - is able to describe, imagined and justified the effects and consequences of the design proposal(s) with respect to the aimed field; - is able to draw conclusions and define recommendations; - is able to evaluate the urban design and research aims in the conclusions; - is able to indicate clearly and logically which research questions are answered and how that has been processed in the urban design; - is able to define clear, concrete, specific recommendations based on the results of the urban research and/or design; - is able to specify for which questions is still additional (design) research necessary; - is able to show an analytical capacity to present a complex matter in a brief and concise way; - is able to describe the projects relevance, reflect on the products and present these; - is able to describe clearly the innovative (scientific and/or social) insights of the graduation project in text and images and, if necessary, concrete strategies and/or application possibilities for the field of the urbanism; - is able to position the own graduation project with respect to the field of the urbanism, as well as other adjacent scientific fields; - is able to discuss and present the thesis products (urban design & research) and the process (design & research) in the form of an epilogue or evaluation. In short, the student has to show that he/she is able to deliver a project of professional quality and of academic level in line with the end terms of the master track Urbanism.	
Education Method	In this semester students work individually, supervised by two mentors, within the frame of the studio and graduation lab of Urbanism. Meetings and lectures are less frequent than in the previous semester, the independency of the student is enlarged.	
Literature and Study Materials	Mandatory and recommended literature will be mentioned on the specific Bright Space page at least one week prior to the start of the course and by the studio coordinator and mentors.	
Assessment	The assessment is imbedded in the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences. The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on the assessments of the graduation project. In this course the project is evaluated three more times: at the third Evaluation: Compulsory Progress Review (P3) at the fourth Evaluation: Formal Assessment (P4) at the final Evaluation: Public Final Presentation (P5) A rubric (EMMA) will be used for grading.	

Special Information	The 'Graduation manual Master of science Architecture, Urbanism & building Sciences' provides information on retake possibilities. On set conditions, Urbanism students have the possibility to carry out their graduation project at a company. Students who wish to do so are required to sign a standard internship agreement in advance, including a research proposal which has been approved by the main mentor. Additional conditions and requirements are stipulated in the internship agreement (master) which can be found in the year guide.
Remarks	The maximum marking period is 15 working days. The final product of this course is an integral concise product that the student has to deliver at the end of Master 4, 1 week prior to the P5 presentation. See the 'Graduation Regulations' of the Faculty of Architecture, Urbanism and Building Sciences, the 'Exit Qualifications' of the master Architecture, Urbanism and Building Sciences, the evaluation form 'Graduation Criteria Urbanism P3' and the assessment criteria of the 'Graduation Criteria Urbanism P3/P4/P5' of the Master track of Urbanism.
Period of Education	Semester 1 and semester 2. MSc3 starts in quarter 1 and 3. The AR4U010 course starts automatically after successfully finishing the MSc3 with a GO at the P2 presentation.
Concept Schedule	The sessions differ per week. The important dates and deadlines are represented in the academic graduation calendar and in the year guide available via Bright Space. The actual schedule will be available via Bright Space > My Timetable.
Minimum number of participants	For mandatory courses of the Master track of Urbanism the minimum number of participants is 1.
Maximum number of participants	For this course of the Master track of Urbanism the maximum number of participants is 100.
Course evaluation	For the course evaluations, see the Brightspace page 'Evaluations TUD' (requires enrolment).

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